WORLDWIDE INFLUENZA CENTRE

WHO CC for Reference & Research on Influenza

THE FRANCIS CRICK INSTITUTE

1 MIDLAND ROAD

LONDON NW1 1AT

Report prepared for the WHO annual consultation on the composition of influenza vaccine for the Northern Hemisphere 2022-2023

21st – 24th February 2022

Worldwide Influenza Centre (WIC)
WHO CC for Reference & Research on Influenza
The Francis Crick Institute
1Midland Road
London, NW1 1AT

Dr. John McCauley (Director)
Dr. Rodney Daniels (Deputy Director)
Dr. Ruth Harvey (Assistant Director)
Dr. Monica Galiano (Assistant Director)
Dr. Yi Pu Lin (Assistant Director)
Dr. Zheng Xiang
Ms. Lynne Whittaker
Ms. Burcu Ermetal
Dr. Karen Cross
Ms. Aine Rattigan
Ms. Alice Lilley
Mr. Michael Bennett
Ms. Chandrika Halai
Mr. Lorin Adams

Tel (1): 020 3796 1520 (J McC)
Tel (2): 020 3796 24444 (R D)
Tel (3): 020 3796 0563 (WIC)
Email: john.mccauley@crick.ac.uk; rod.daniels@crick.ac.uk; whocc@crick.ac.uk

Acknowledgements

Laboratories involved in Global Influenza Surveillance

We thank all who have contributed information, clinical specimens and viruses, and associated data, to the WHO Global Influenza Surveillance and Response System (GISRS), which provide the basis for our current understanding of recently circulating influenza viruses, and this summary.

Additional Francis Crick Institute Personnel

The work of the WIC is only made possible with the ongoing support and input from the Francis Crick Advanced Sequencing Facility (ASF), Biological Research Facility (BRF) and Bioinformatics Core-Science Technology Platform (BC-STP). We thank all members of staff in these platforms for their assistance.

Influenza Epidemiology in the WHO European Region

The epidemiological section of this report was prepared using data reported to The European Surveillance System (TESSy) and presented in Flu News Europe (http://www.flunewseurope.org) the joint ECDC-WHO Europe weekly influenza update, with the assistance of members of the Influenza and Other Respiratory Pathogens Group at the WHO European Office and the ECDC Influenza Surveillance Group:

Dr. Margaux M Mesle
Dr. Piers A N Mook
Dr. Karen Nahapetyan
Dr. Richard Pebody
Dr. Angeliki Melidou

CONTENTS

WHO Vaccine Recommendation Meeting for the 2022-2023 Northern Hemisphere influenza season

WHO CC, London – Final Report for the meeting: 20-24 February 2022

Epidemiology

Influenza activity in the WHO European Region, October 2021 – 10 February 2022

Note

The novel coronavirus disease 2019 (**COVID-19**) pandemic, caused by severe acute respiratory syndrome coronavirus 2 (**SARS-CoV-2**), continues to adversely affect healthcare presentations and testing capacities of some countries, territories and areas in the Region. Consequently, the analysis presented here, based on data reported to The European Surveillance System (TESSy), should be interpreted with care.

Summary

Table E1 shows a summary of influenza virus detections in the Region for the 2021-2022 season (weeks 40/2021-05/2022), compared to the 2020-2021 season. There was a 4.1-fold increase in the number of samples from patients fulfilling Influenza-Like Illness (ILI) and/or Acute Respiratory Infection (ARI) criteria being tested, even when compared with a more 'normal' season, 2019-2020 (3.3-fold), which led into the COVID-19 pandemic. With this increased testing, together with an increase in influenza activity compared to 2020-2021, there has been a 58.3-fold rise in the number of influenza-positive samples, though there was a reduction (56.7%) compared to the same period in 2019-2020. These data probably relate to at least two factors: (i) significant numbers of samples taken from patients fulfilling ILI and/or ARI criteria being infected with other agents, notably SARS-CoV-2; (ii) restrictions on travel and social/work-place gatherings, and public health measures (e.g. mask wearing and increased disinfection/hand-washing frequency) imposed to help curtail the spread of SARS-CoV-2. With these caveats, and mindful of the low number of detections during of the 2020-2021 season, for combined sentinel and non-sentinel data: the ratio of type A to type B detections has increased compared to the 2020-2021 season (1:1 to 24.5:1), with dominance of A(H3N2) over A(H1N1)pdm09 viruses, and the number of influenza B virus detections increased from 310 to 1,434 (4.6-fold) but only small numbers were ascribed to a lineage in both time periods (**Table E1**). Based on sequences available in GISAID, B/Yamagata-lineage viruses with collection dates after March 2020 have not been characterised genetically.

Table E1. Recent Influenza Seasons in the WHO European Region: summaries weeks 40-05

Virus type/subtype/lineage	Detections weeks 40/2021 - 05/2022			Totals*		Detectons weeks 40/2020 - 05/2022			Totals*		Detectons weeks 40/2019 - 05/2020			Totals*	
	Sentinel sources	Non-sentinel sources	Totals	%	Ratios	Sentinel sources	Non-sentinel sources	Totals	%	Ratios	Sentinel sources	Non-sentinel sources	Totals	%	Ratios
Influenza A	**2121**	**32975**	**35096**	**96.1**	**24.5:1**	**15**	**302**	**317**	**50.6**	**1:1**	**6645**	**60962**	**67607**	**80.0**	**4:1**
A(H1N1)pdm09	96	704	800	5.7		13	26	39	52.7		3731	8472	12203	48.6	
A(H3N2)	1399	11756	13155	94.3	16.4:1	2	33	35	47.3	0.9:1	2491	10406	12897	51.4	1.1:1
A not subtyped	626	20515	21141			0	243	243			423	42084	42507		
Influenza B	**29**	**1405**	**1434**	**3.9**		**7**	**303**	**310**	**49.4**		**3000**	**13848**	**16848**	**20.0**	
Victoria lineage	5	8	13	100.0		2	5	7	87.5	7:1	966	807	1773	96.6	28.6:1
Yamagata lineage	0	0	0			0	1	1	12.5		14	48	62	3.4	
Lineage not ascribed	24	1397	1421			5	297	302			2020	12993	15013		
Total detections	**2,150**	**34,380**	**36,530**			**22**	**605**	**627**			**9,645**	**74,810**	**84,455**		
Total tested	**30,627**	**>1,358,697**	**>1,389,324**			**21,173**	**>315,981**	**>337,154**			**29,308**	**>391,526**	**>420,834**		

Numbers taken from Flu News Europe for weeks 40-05 of the following year for the respective seasons

*** Percentages are shown for total detections (types A & B [in bold type], and for viruses ascribed to influenza A subtype and influenza B lineage). Ratios are given for type A:B [in bold type], A(H3N2):A(H1N1)pdm09 and Victoria:Yamagata lineages.**

Timing and spread

The 2021-2022 influenza epidemic in the WHO European Region started in week 49/2021, later than recent epidemics (weeks 45 to 48) (**Figures E1-A and B**). This was the first of three consecutive weeks when 10% of ILI and/or ARI specimens from sentinel primary care sites in the Region tested positive for influenza viruses. Between weeks 40/2021 and 05/2022 the proportion of specimens testing positive for influenza on a weekly basis reached a peak of 20% for the season in week 52/2021, which is lower than peaks in recent seasons (53-60%) but higher than in the 2020-2021 season when positivity did not exceed 2%. Activity in which ≥10% of sentinel specimens tested positive for influenza lasted for 5 weeks, compared to seasonal averages of ~10.5 weeks (range: 9-14 weeks) for recent epidemics (2015-2016 to 2018-2019 seasons), before falling below 10% in week 02/2022 (**Figure E1-A**).

Based on weekly reporting on the intensity of influenza activity, conducted by 49 countries and areas, there was more early activity in countries and areas in the East of the WHO European Region (**Figure E2**). In total, 29 of these reported activity above baseline, with 16 having reported medium activity or higher in some weeks (**Figure E2**). (Note: this indicator is typically based on ILI and/or ARI activity but at least one virus detection per week is required to meet the definition for an intensity level above baseline. Not all countries, territories or areas reporting intensity above baseline have reported detections.)

Distribution of viruses

In the Region as a whole and between weeks 40/2021 to 05/2022, a total of 36,530 influenza detections from sentinel and non-sentinel primary care sources were reported to The European Surveillance System (TESSy) (**Tables E1, E2 and E3-A**). This is compared to season totals of 947 and 165,194, respectively, for 2020-2021 and 2019-2020 (the last season where COVID-19 had limited impact on healthcare presentations and testing capacities in the Region) (**Table E2**). This increase is in the context of increased testing with 1,389,324 specimens tested compared, respectively, to 337,154 and 420,834 in the same periods of 2020-2021 and 2019-2020 (**Table E-1**). Combined sentinel and non-sentinel data for type A viruses in the Region showed that most viruses detected (96%) were type A and 4% were type B (a ratio of ~25:1) (**Tables E1 and E2**). This distribution was consistent across the Region with all subregions detecting at least 99% type A viruses, except for Northern Europe where there was a higher proportion of type B viruses (8%) (**Table E3-A**). Combined sentinel and non-sentinel data for type A viruses showed a majority (94%) to be A(H3N2) viruses, and 6% being A(H1N1)pdm09 viruses (a ratio of ~16:1) (**Tables E1 and E2**). This distribution was consistent across the Region except for Western Europe where A(H3N2) viruses accounted for 57% of detections and A(H1N1)pdm09 viruses accounted for 43% (**Table E3-A**). Only 13 of the type B viruses reported were ascribed to a lineage and all were B/Victoria. The subtype distribution proportions for influenza type A viruses detected, from non-sentinel sources, were relatively consistent for most weeks of the season to date (**Figure E1-C**).

When the analysis for weeks 40/2021 to 05/2022 was restricted to sentinel source data only, the regional distribution of influenza by type was similar: of the viruses detected (n=2,150), the majority were type A (99%). Of type A viruses, the majority were A(H3N2) subtype (94%) and of the type B viruses, all were B/Victoria-lineage (**Table E3-B**). These distributions were largely consistent across the subregions. Northern Europe was the only subregion reporting more than 10% of type B viruses. The distribution of type A viruses was consistent across the Region except for Western Europe where A(H3N2) viruses accounted for 66% of detections and A(H1N1)pdm09 viruses accounted for 34% of detections. Of the total detections, 31% were from West Asia and 29% were from South-West Europe (**Table E3-B**).

A dominant virus (sub)type (comprising at least 60% of influenza virus detections) could be assigned by 26 countries and areas within the WHO European Region between weeks 40/2021 and 05/2022 (25 had an insufficient number of detections (10 or fewer)). All countries and areas reported predominance of influenza type A viruses except for United Kingdom (Scotland) which reported co-dominance of type A and type B viruses (no lineage ascribed) (**Figure E3**). Twenty-one countries with at least 10 subtyped A viruses were able to ascribe dominance (i.e. the proportion of detections of an influenza subtype was 60% or greater) or co-dominance (between 41% and 59%) of influenza A virus subtypes for the season: 20 reported A(H3N2) dominance and France reported co-dominance of A(H1N1)pdm09 and A(H3N2) viruses (**Figure E3**).

Influenza virus characterisation

Up to week 05/2022, 816 A(H3N2) viruses had been characterized genetically, of which 722 were attributed to clade 3C.2a1b.2a.2 and one to clade 3C.2a1b.1a. Of 38 A(H1N1)pdm09 viruses that were characterized genetically, 25 were attributed to clade 6B.1A.5a.1 and nine to clade 6B.1A5a.2. Only eight B/Victoria-lineage viruses were characterized genetically, two belonging to clade V1A.3 and five to clade V1A.3a.2.

Neuraminidase inhibitor and baloxivir susceptibility

Neuraminidase inhibitor (NAI) susceptibility was assessed phenotypically for 229 influenza viruses (226 A(H3N2), two A(H1N1)pmd09 and three B/Victoria-lineage) with collection dates between week 40/2021 and 05/2022. All showed normal inhibition by the NAIs. Genotypically, 341 influenza viruses were assessed for NAI susceptibility (320 A(H3N2), 20 A(H1N1)pdm09 and one B/Victoria-lineage). None of the neuraminidase genes of these 570 viruses carried mutations that encode amino acid substitutions known to be associated with reduced susceptibility to NAIs.

Susceptibility to baloxavir marboxil was assessed genotypically for 341 viruses (320 A(H3N2), 20 A(H1N1)pdm09 and one B/Victoria-lineage). No virus carried mutations that encoded PA amino acid markers associated with reduced susceptibility.

Hospitalised cases

Only four countries reported case-based hospital data in 2021-2022. Between weeks 40/2021 and 05/2022, a total of 269 hospitalised laboratory-confirmed influenza cases were reported to TESSy. Of these, 236 cases were admitted to intensive care units (ICUs) in France (n=104, 44%), Sweden (n=76, 32%), United Kingdom (England) (n=55, 23%) and Czech Republic (n=1, <0.5%). The remaining cases were reported from other hospital wards in Ukraine (n=32, 97%) and Czech Republic (n=1, 3%). Among the 236 cases admitted to ICUs, 225 were infected with type A viruses (of 64 subtyped, 40 were A(H3N2) and 24 A(H1N1)pdm09) and 11 with type B virus (none was ascribed to a lineage). A total of 179 cases had a recorded age, falling within different age groups: 81 were 15-64 years (45%), 50 were 65 years or older (28%), 28 were under 5 years of age years (16%) and 20 were aged from 5-14 years (11%) (**Figure E4**).

A total of 19 countries and areas (Albania, Armenia, Belarus, Georgia, Germany, Kazakhstan, Kyrgyzstan, Lithuania, Malta, Montenegro, North Macedonia, Republic of Moldova, Russian Federation, Serbia, Spain, Turkey, Ukraine, Uzbekistan and Kosovo*) with surveillance systems for severe acute respiratory infection (SARI) reported a total of 84,122 cases between weeks 40/2021 and 05/2022 and, of these cases, 60,223 had known ages. In most countries and areas, majorities of cases were aged younger than 30 years; most SARI cases were among children aged less than 5 years in Spain, Uzbekistan, Kyrgyzstan, Kazakhstan, Turkey, Armenia (**Table E4**). Of 9,329 SARI cases tested for influenza virus over the period, 9% were positive (ranging from 30% in Armenia to <1% in Malta). The majority (n=858, 99%) were infected with type A viruses and only Russian Federation (n=3), Kyrgyzstan (n=1) and Turkey (n=1) reported type B viruses (none was ascribed to a lineage). Of 760 subtyped A viruses 758 were A(H3N2) and two were A(H1N1)pdm09).

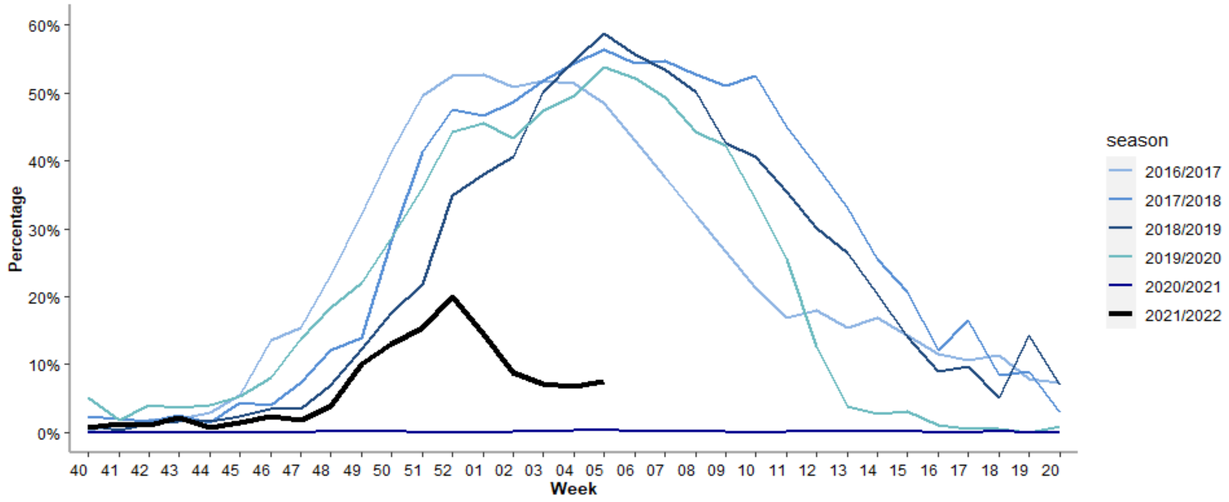
Mortality data

Pooled analyses of all-cause excess and influenza-attributable mortality from 29 European countries or sub-national regions participating in the EuroMOMO network (Austria, Belgium, Cyprus, Berlin (Germany), Denmark, England (UK), Estonia, Finland, France, Germany, Greece, Hesse (Germany), Hungary, Ireland, Israel, Italy, Luxemburg, Malta, the Netherlands, Northern Ireland (UK), Norway, Portugal, Scotland (UK), Slovenia, Spain, Sweden, Switzerland, Ukraine and Wales (UK)) showed overall a substantial increased excess all-cause mortality within the participating European countries probably related to the ongoing COVID-19 pandemic. The excess was observed mainly among older adults (65 years or older), but also among those aged 45 to 64 years of age.

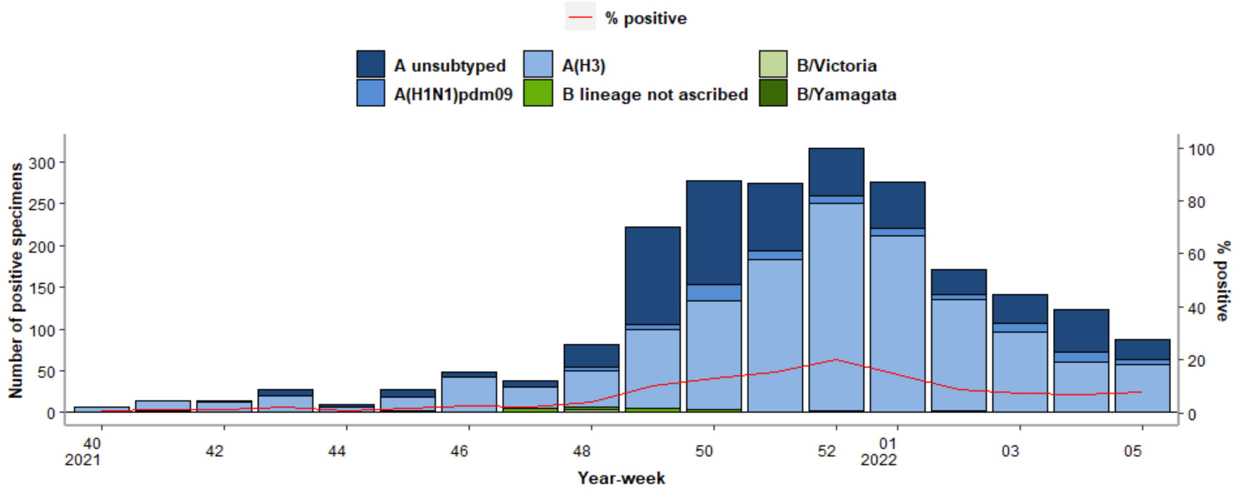
*In accordance with United Nations Security Council Resolution 1244 (1999).

Figure E1. Seasonal influenza activity in the WHO European Region: weeks 40/2021-05/2022

A. Percentage of sentinel ILI/ARI specimens testing positive for influenza by season



B. Sentinel specimens testing positive for influenza by week, (sub)type and percentage positive, 2021/22 season



C. Non-sentinel specimens testing positive for influenza by week, (sub)type and percentage positive, 2021/22 season

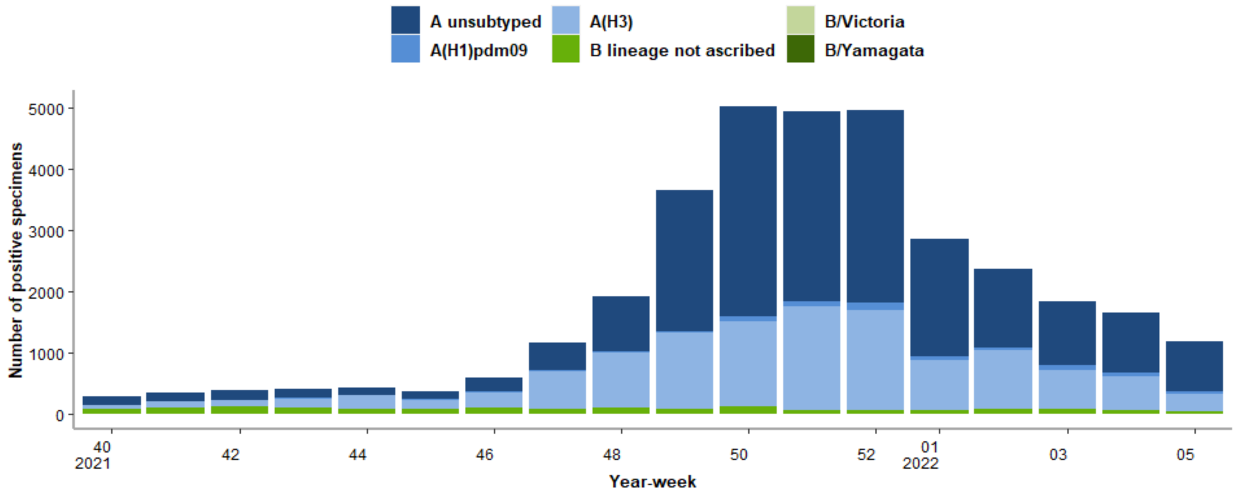
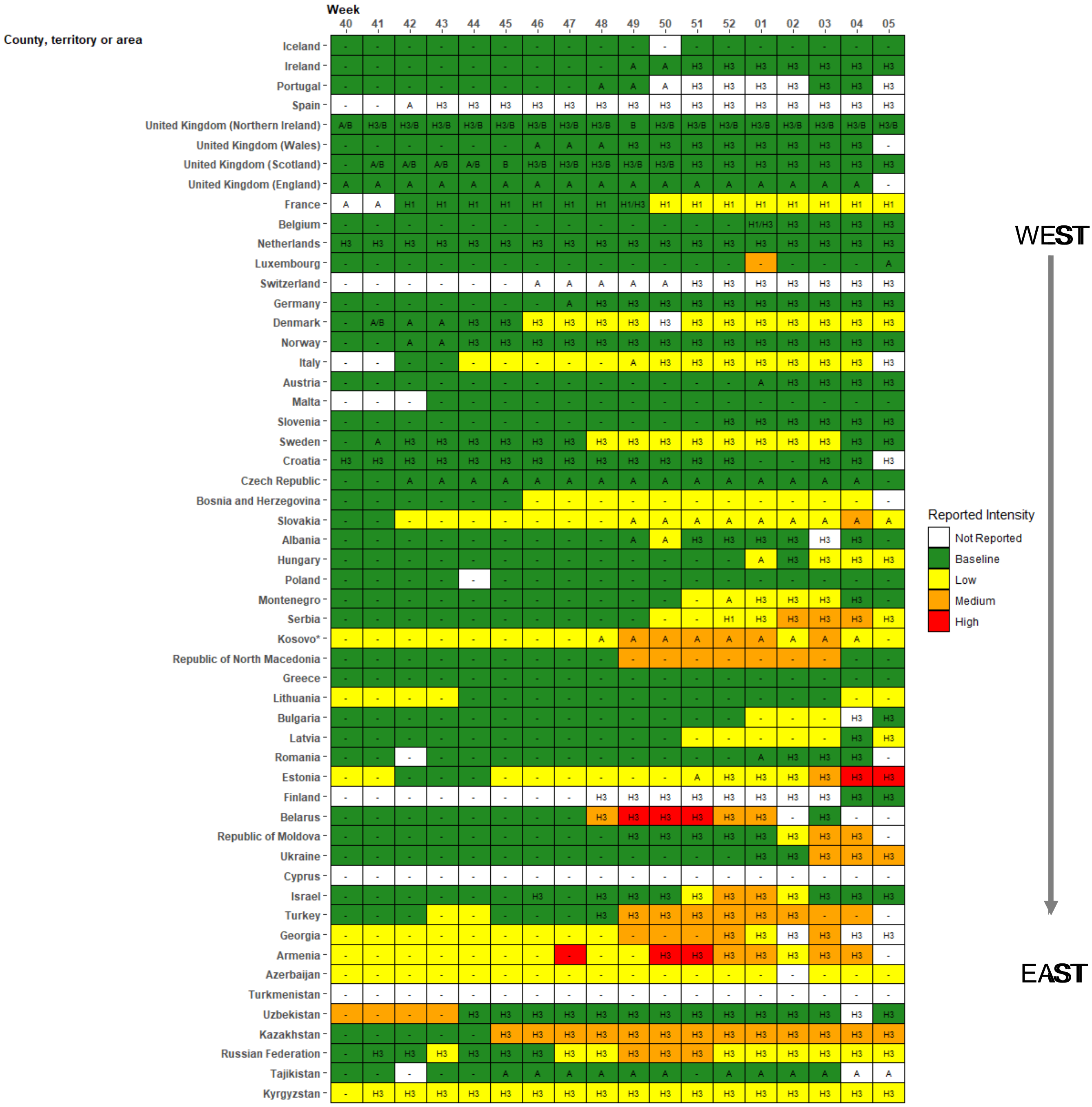


Figure E2. Intensity of influenza activity and dominant virus (sub)type in the WHO European Region by week and country, territory or area 2021-2022 season

Baseline intensity influenza activity is the level at which clinical influenza activity remains throughout the summer.



*In accordance with United Nations Security Council Resolution 1244 (1999).

Note

Luxembourg noted raised intensity for week 01/2022 but attributed this being largely due to SARS-CoV-2 circulation.

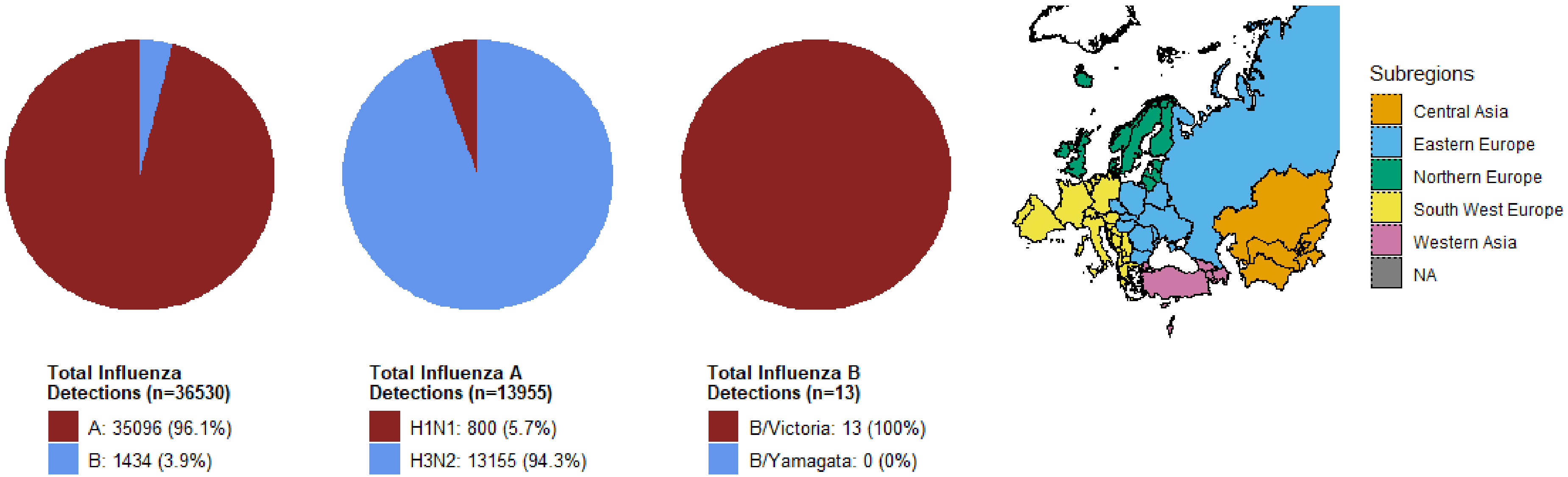
Table E2. 2021-2022 Influenza Season in the WHO European Region: weeks 40/2021-05/2022

	Sentinel sources			Non-sentinel sources			Totals		
Virus type/subtype/lineage	2021/22 season	2020/21 season	2019/20 season	2021/22 season	2020/21 season	2019/20 season	2021/22 season	2020/21 season	2019/20 season
Influenza A	**2,121 (99%)**	**42 (69%)**	**11,277 (64%)**	**32,975 (96%)**	**445 (50%)**	**108,445 (74%)**	**35,096 (96%, 24.5:1)**	**487 (51%,1.1:1)**	**119,722 (73%, 2.6:1)**
A(H1N1)pdm09	96 (6%)	13 (65%)	6,112 (59%)	704 (6%)	28 (35%)	20,180 (57%)	800 (6%)	41 (41%)	26,292 (56%)
A(H3N2)	1,399 (94%)	9 (35%)	4,171 (41%)	11,756 (94%)	52 (65%)	16,539 (43%)	13,155 (94%, 16.4:1)	61 (59%,1.4:1)	20,710 (44%, 0.8:1)
A not subtyped	626	20	994	20,515	365	71,726	21,141	385	72,720
Influenza B	**29 (1%)**	**18 (31%)**	**6,402 (36%)**	**1,405 (4%)**	**442 (50%)**	**39,070 (26%)**	**1,434 (4%)**	**460 (49%)**	**44,472 (27%)**
Victoria lineage	5	3	2,492	8	12	2,067	13 (100%, NA)	15 (94%,30.1:1)	4,559 (98%, 51.8:1)
Yamagata lineage	0	0	23	1	1	65	0 (0%)	1 (6%)	88 (2%)
Lineage not ascribed	24	15	3,887	1,397	429	36,938	1,421	444	40,825
Total detections	**2,150**	**60**	**17,679**	**34,380**	**887**	**147,515**	**36,530**	**947**	**165,194**
Total tested	**30,627**	**43,499**	**52,491**	**1,458,697**	**881,400**	**860,760**	**1,389,324**	**924,899**	**913,251**

Percentages are shown for total detections (types A & B [**in bold type**], and for viruses ascribed to influenza A subtype and influenza B lineage). Ratios are given for type A:B [**in bold type**], A(H3N2):A(H1N1)pdm09 and Victoria:Yamagata lineages.

Table E3. Distribution of Influenza Detections in the WHO European Region[a], 2021-2022 season

Laboratory confirmed influenza from sentinel and non-sentinel sources by type and subtype/lineage



[a] WHO European Region Member States grouped into sub-regions according to United Nations geographic composition. Available from: (http://unstats.un.org/UNSD/METHODS/M49/M49REGIN.HTM)

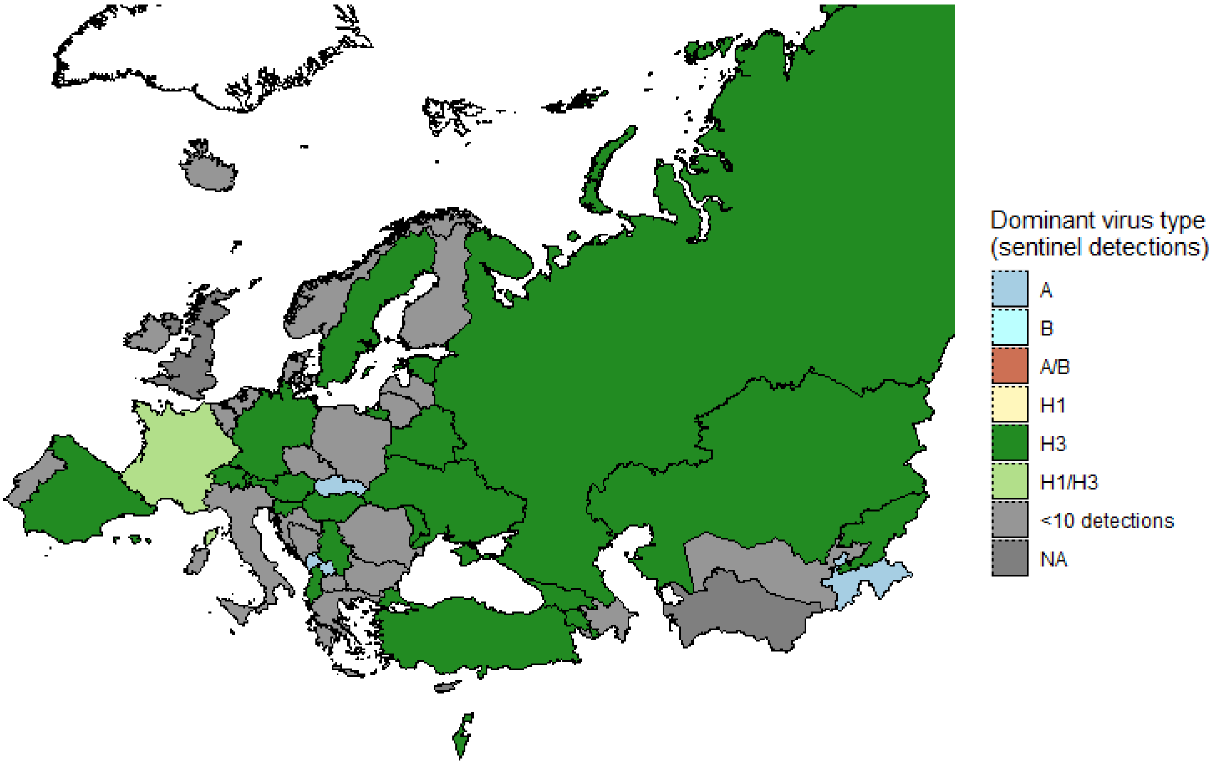
A. Laboratory confirmed influenza from sentinel and non-sentinel sources by geographic sub-region and influenza type and subtype/lineage

Influenza type	South West Europe	Western Europe	Eastern Europe	Northern Europe	West Asia	Central Asia	WHO European Region
Influenza A	**2,922 (97%)**	**7,912 (99%)**	**8,996 (99%)**	**13,862 (92%)**	**1,205 (100%)**	**199 (96%)**	**35,096 (96%)**
Influenza A subtyped	1,232	1646	8,094	1601	1,204	178	13,955
A(H1N1)pdm09	30 (2%)	701 (43%)	12 (0%)	57 (4%)	0 (0%)	0 (0%)	800 (6%)
A(H3N2)	1,202 (98%)	945 (57%)	8,082 (100%)	1,544 (96%)	1,204 (100%)	178 (100%)	13,155 (94%)
Influenza B	**76 (3%)**	**85 (1%)**	**119 (1%)**	**1,142 (8%)**	**3 (0%)**	**9 (4%)**	**1,434 (4%)**
Influenza B lineage determined	0	9	0	3	1	0	13
B/Yamagata-lineage	0	0	0	0	0	0	0 (0%)
B/Victoria-lineage	0	9	0	3	1	0	13 (100%)
Total	**2,998**	**7,997**	**9,115**	**15,004**	**1,208**	**208**	**36,530**

B. Laboratory confirmed influenza from sentinel sources by geographic sub-region and influenza type and subtype/lineage

Influenza type	South West Europe	Western Europe	Eastern Europe	Northern Europe	West Asia	Central Asia	WHO European Region
Influenza A	**633 (100%)**	**289 (99%)**	**306 (98%)**	**81 (87%)**	**661 (100%)**	**151 (99%)**	**2,121 (99%)**
Influenza A subtyped	129	256	249	70	661	130	2,150
A(H1N1)pdm09	2 (2%)	86 (34%)	2 (1%)	6 (9%)	0 (0%)	0 (0%)	96 (6%)
A(H3N2)	127 (98%)	170 (66%)	247 (99%)	64 (91%)	661 (100%)	130 (100%)	1,399 (94%)
Influenza B	**1 (0%)**	**4 (1%)**	**7 (2%)**	**12 (13%)**	**3 (0%)**	**2 (1%)**	**29 (1%)**
Influenza B lineage determined	0	4	0	0	1	0	5
B/Yamagata-lineage	0	0	0	0	0	0	0
B/Victoria-lineage	0	4	0	0	1	0	5
Total	**634**	**293**	**313**	**93**	**664**	**153**	**2,150**

Figure E3. Influenza circulation[a] in the WHO European Region by (sub)type and Country, territory or area, 2021/22 season



[a] Dominance of (sub)type is based on at least 60% of detections and co-dominance is based on between 40% and 60% of detections. Calculations are only considered valid if at least 10 viruses were detected.

Figure E4. Distribution of virus (sub)type in laboratory-confirmed hospitalised cases in intensive care units by age group in the WHO European Region, 2021/22 season

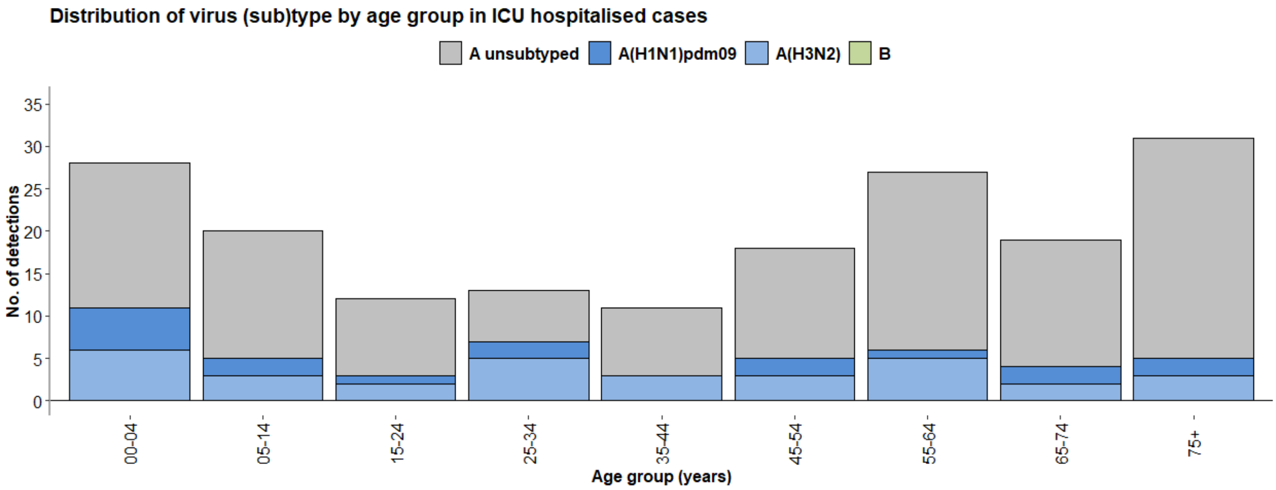


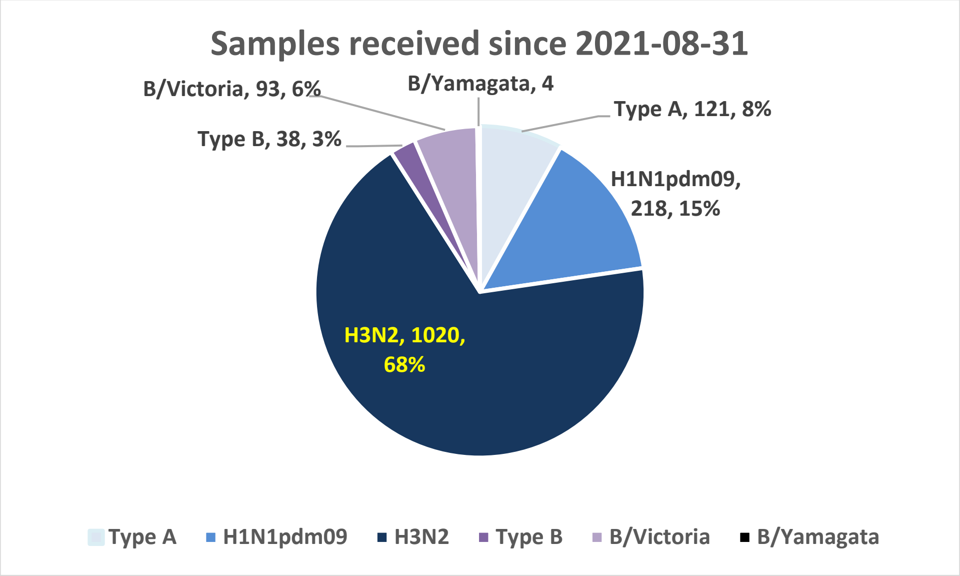
Table E4. Percentage distribution of severe acute respiratory infection (SARI) cases by age group and country, territory or area, and influenza positives, 2021/22 season

Country, territory or area	Total SARI cases	SARI cases with age data	Percentage (%) of total cases 0<5 years	5<15 years	15<30 years	30<65 years	>=65 years	SARI specimens	Positive detections (%)
Albania	4,947	2,279	16	5	7	72	0	1,919	76 (4%)
Armenia	2,925	2,925	55	14	6	16	9	473	140 (30%)
Belarus	7,240	7,115	2	1	39	29	29	228	16 (7%)
Georgia	2,647	2,542	32	10	14	28	15	1,123	36 (3%)
Germany	14,300	7,452	49	4	4	43	0	0	-
Kazakhstan	8,505	8,505	62	14	9	10	5	1,138	187 (16%)
Kyrgyzstan	3,871	3,871	77	7	4	8	4	645	105 (16%)
Lithuania	34	9	22	78	0	0	0	27	1 (4%)
Malta	466	163	0	0	11	89	0	201	0 (0%)
Montenegro	1,576	1,574	4	1	3	37	55	69	5 (7%)
North Macedonia	1,147	1,147	14	6	7	45	29	78	5 (6%)
Republic of Moldova	2,934	2,934	36	6	3	32	23	150	10 (7%)
Russian Federation	1,343	1,343	25	17	19	18	22	1,002	135 (13%)
Serbia	3,144	3,144	0	0	4	44	51	49	11 (22%)
Spain	15,018	2,422	87	13	0	0	0	0	-
Turkey	4,014	2,831	61	12	0	0	28	1,175	79 (7%)
Ukraine	9,714	9,714	12	6	7	38	37	760	23 (3%)
Uzbekistan	250	250	80	7	8	4	1	191	30 (16%)
Kosovo*	47	3	0	33	33	33	0	0	-
Total	**84,122**	**60,223**	**634 (36%)**	**234 (7%)**	**178 (10%)**	**546 (28%)**	**308 (19%)**	**9,329**	**863 (9%)**

*In accordance with United Nations Security Council Resolution 1244 (1999).

Influenza specimens received at the Francis Crick Institute.

Seasonal influenza virus samples with collection dates after 2021-08-31, totalling 1351, have been received from 40 countries in Europe, Africa and the Middle East (**Figure 1**). These samples comprised 111 clinical specimen/virus pairs, 881 clinical specimens, 321 viruses and 38 clinical specimens either in lysis mix (n = 19) or as RNA (n =19). An additional 143 samples, with collection dates prior to 2021-08-01, had also been shipped by some countries. Overall, Influenza A viruses represented 91% of the samples for viruses with collection dates pre-September and post-August, of which A(H3N2) viruses outnumbered A(H1N1)pdm09 viruses at a ratio of ~4.7:1 overall (~5.2:1 for viruses detected after August, while the ratio was ~2.1:1 for viruses collected before September). Influenza B viruses represented 9% of the samples received overall, 22% pre-September. All but four influenza B viruses ascribed to a lineage were B/Victoria (**Table 1**).



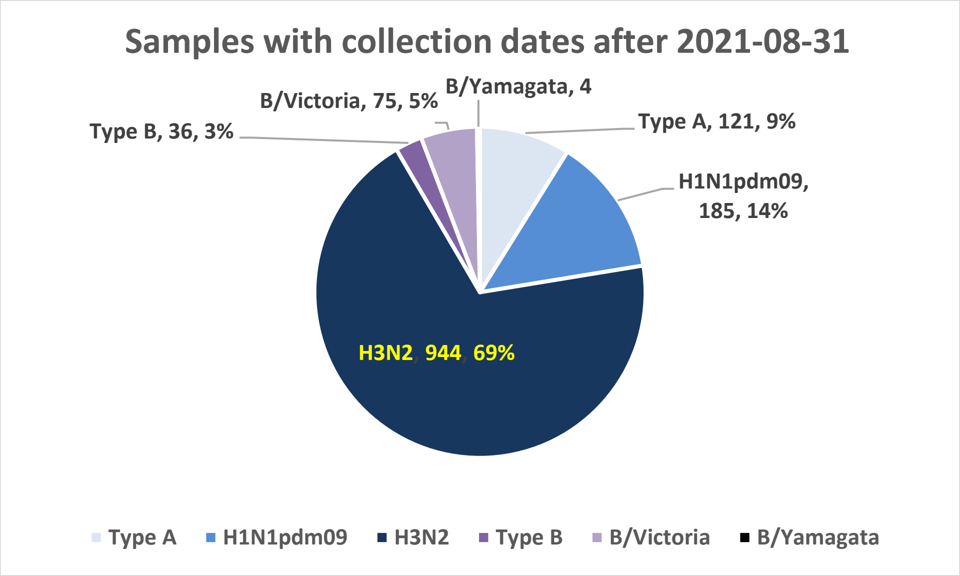


Table 2 shows the proportion of clinical samples that have been successfully characterised by next generation sequencing as of 2022-02-11. Approximately 33% of the clinical samples with collection dates after August 2021 yielded full length HA gene sequences.

Table 3 shows the recovery/isolation rates from the samples received after 2021-08-31. Recoveries for samples sent as clinical specimens that were taken for virus culture were 60% for H1N1pdm09, 85% for H3N2 and 65% for influenza B viruses, respectively. Recoveries for samples sent as virus isolates were 98% for H1N1pdm09, 96% for H3N2 and 97% for influenza B viruses, respectively. All A(H3N2) viruses showed sufficient agglutination of guinea pig red blood cells to allow assessment by HI assay in the presence of 20nM oseltamivir.

Table 4 summarises the reactivity of A(H1N1)pdm09, A(H3N2) and B/Victoria-lineage viruses collected after 2021-08-31 with post-infection ferret antisera raised against either vaccine viruses or reference viruses in HI assays and in plaque reduction neutralisation assays (PRNA). For HI assays numbers and percentages of viruses showing ≤2-, 4- and ≥8-fold reductions in titres relative to the titres of the antisera with their respective homologous reference/vaccine viruses are shown, for PRNA data is shown for ≤2-, >2≤4-, >4≤8- and >8-fold reductions.

For A(H1N1)pdm09 test viruses the HI and PRNA titres with antisera raised against the egg-propagated A/Guangdong-Maonan/SWL1536/2019 and A/Victoria/2570/2019 vaccine viruses, and against the cell culture-propagated cultivars of A/Guangdong-Maonan/SWL1536/2019 and A/Denmark/3280/2019 (A/Victoria/2570/2019-like) have been compared to the respective homologous titres.

For A(H3N2) test viruses in the HI assay, the results with antisera raised against egg-propagated A/Cambodia/e0826360/2020 and A/Darwin/9/2021 vaccine viruses and cell culture-propagated A/Cambodia/925256/2020 and A/Stockholm/5/2021 (A/Darwin/6/2021-like) are summarised.

For the A(H3N2) test viruses assessed by PRNA the 50% neutralising titres are summarised for antisera raised against cell culture-propagated A/Denmark/3264/2019, A/Bangladesh/4005/2020, A/Stockholm/5/2021 and A/Cambodia/925256/2020, and an antiserum raised against egg-propagated A/Darwin/9/2021 vaccine virus.

For Influenza B viruses of the B/Victoria-lineage the HI titres of test viruses are summarised for antisera raised against the egg-propagated cultivar of the vaccine virus B/Washington/02/2019 (clade V1A.3) and against an antiserum raised against the cell culture-propagated cultivar of B/Austria/1359417/2021 (subclade V1A.3a.2 with additional HA1 substitutions A127T, P144L and K203R). There are two distinct egg-propagated cultivars of B/Austria/1359417/2021 – one with an HA sequence identical to the cell culture-propagated cultivar and one with the HA1 substitution G141R. Antisera raised against both egg-propagated cultivars have been used in the HI assays and summarised in **Table 4**.

Figure 1. Countries Sharing Influenza-positive Samples Collected after 2021-08-31 with FCI-WIC.

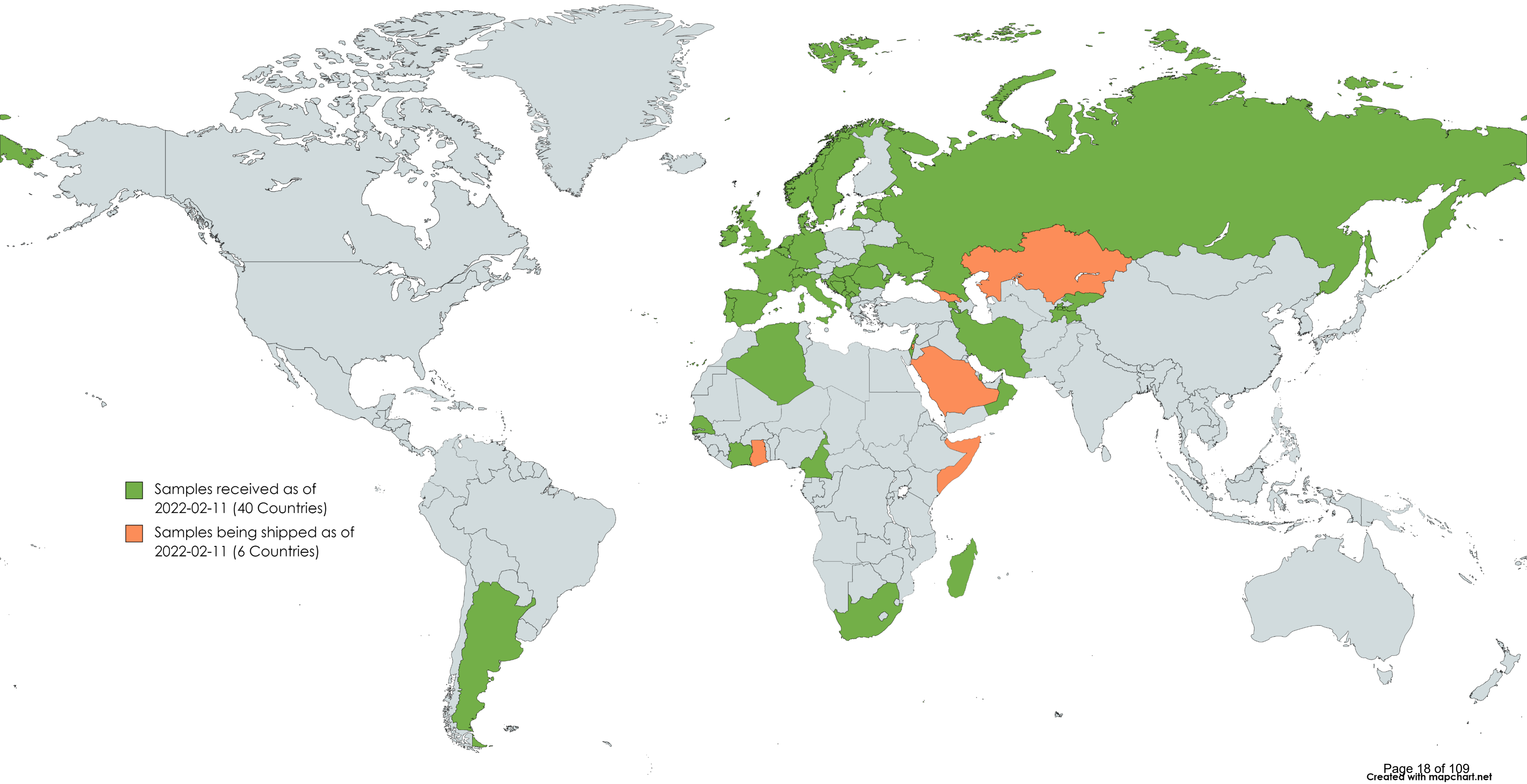


Table 1a. Summary of clinical samples and virus isolates received for virus characterisation, since the September 2021 VCM, by country*

MONTH	TOTAL RECEIVED Seasonal	A		H1N1pdm09		H3N2			B		B Victoria lineage		B Yamagata lineage	
Country		Number received	Number propagated[1]	Number received	Number propagated[1]	Number received	Number propagated[2]		Number received	Number propagated[1]	Number received	Number propagated[1]	Number received	Number propagated[1]
2021														
APRIL														
Ghana	3			3	2									
Qatar	3										3	3		
MAY														
Ghana	4			4	1									
Qatar	8								1	0	7	7		
JUNE														
Ghana	7			7	1									
Qatar	4								2	0	2	2		
JULY														
Cote d'Ivoire	3			3	in process									
Ghana	16			11	6	5	3	0						
Qatar	9					6	4	0			3	3		
South Africa	1										1	1		
AUGUST														
Belgium	1					1	1							
Cote d'Ivoire	3			3	in process									
Croatia	2					2	2							
Netherlands	1					1	1							
Norway	2					2	2							
Qatar	62			3	2	55	in process				4	in process		
Russia	1					1	0	0						
Senegal	2			2	0									
South Africa	9					1	1				8	8		
Sweden	2					2	2							
SEPTEMBER														
Belgium	1					1	1							
Cameroon	23			22	in process				1	0				
Croatia	3					3	2	0						
Denmark	5					5	5							
France	11			1	in process	10	9							
Israel	2					2	2							
Italy	1					1	1							
Lebanon	19					19	18	0						
Madagascar	8								7	in process	1	in process		
Netherlands	13					12	in process				1	1		
Oman	13			1	in process	12	in process							
Qatar	14			1	1	10	7				3	3		
Senegal	19			19	in process									
South Africa	27			6	6	5	5				16	16		
Spain	1					1	in process							
Sweden	2			1	1	1	1							
UK (England)	2					2	2							
OCTOBER														
Cameroon	4			3	1	1	in process							
Cote d'Ivoire	1			1	in process									
Denmark	2			1	1	1	1							
Estonia	1					1	in process							
France	12			9	in process	3	3							
Germany	2					2	in process							
Iran	3					3	3							
Ireland	1					1	in process							
Italy	5			3	3	2	2							
Kyrgyzstan	28					28	in process							
Lebanon	15					15	in process							
Madagascar	7								4	in process	3	in process		
Netherlands	37					37	in process							
Norway	7					7	in process							
Oman	33			10	in process	22	in process				1	1		
Portugal	3					2	0		1	0				
Russia	3					3	3							
Qatar	3			1	0						2	2		
Senegal	20			20	in process									
South Africa	24			13	13	7	6				4	4		
Spain	3					3	in process							
Sweden	2					2	2							
Tajikistan	7	6	in process			1	in process							
UK (England)	8					8	8							
UK (Scotland)	5										1	in process	4	in process

Table 1b. Summary of clinical samples and virus isolates received for virus characterisation, since the September 2021 VCM, by country*

MONTH	TOTAL RECEIVED Seasonal	A		H1N1pdm09		H3N2		B		B Victoria lineage		B Yamagata lineage	
Country		Number received	Number propagated[1]	Number received	Number propagated[1]	Number received	Number propagated[2]	Number received	Number propagated[1]	Number received	Number propagated[1]	Number received	Number propagated[1]
NOVEMBER													
Algeria	5					5	in process						
Armenia	2					2	in process						
Belgium	2					2	2						
Cameroon	3					1	in process			2	0		
Cote d'Ivoire	4			3	in process	1	in process						
Estonia	1					1	in process						
France	28			18	in process	10	in process						
Germany	5	2	in process			2	in process			1	in process		
Iran	10					10	10						
Ireland	4					3	in process			1	in process		
Israel	10					10	6						
Italy	5					5	5						
Kosova	2	2	in process										
Kyrgyzstan	22					22	in process						
Lebanon	58					58	in process						
Madagascar	32					11	in process	14	in process	7	in process		
Netherlands	23					23	in process						
Norway	8					8	in process						
Oman	39			15	in process	21	in process			3	1		
Qatar	3			1	1					2	1		
Romania	1									1	1		
Russia	36					36	35						
South Africa	13			6	6	1	0			6	6		
Spain	35			1	1	32	in process	1	in process	1	in process		
Sweden	5					5	5						
Switzerland	4					2	2			2	2		
Tajikistan	7	7	in process										
UK (N. Ireland)	3					2	in process	1	in process				
UK (Scotland)	2					2	in process						
DECEMBER													
Albania	39	39	in process										
Algeria	5					5	in process						
Argentina	9					9	in process						
Armenia	21					21	in process						
Belgium	15	1	in process	6	in process	8	in process						
Bosnia and Herzegovina	3					3	in process						
Cameroon	2							2	0				
Cote d'Ivoire	13					13	in process						
Estonia	7					7	in process						
France	1					1	1						
Germany	10					10	in process						
Hungary	2					2	2						
Iran	2					2	2						
Ireland	5			1	in process	4	in process						
Israel	30					30	in process						
Kosova	56	56	in process										
Latvia	5					5	in process						
Lebanon	76					76	in process						
Madagascar	18					11	8	3	0	4	3		
Montenegro	6					6	in process						
Netherlands	26			5	5	21	in process						
Norway	1					1	in process						
Portugal	17	1	in process			13	in process	1	in process	2	1		
Romania	5					5	5						
Russia	5					5	in process						
Serbia	7			5	in process	2	in process						
Spain	50					49	in process			1	in process		
Switzerland	10	1	in process			9	in process						
Ukraine	13					13	in process						
2022													
JANUARY													
Argentina	12					12	in process						
Armenia	2					2	in process						
Belgium	16			6	in process	10	in process						
Bosnia and Herzegovina	5	2	in process			3	in process						
Cote d'Ivoire	1					1	in process						
Estonia	4					4	4						
Germany	3					3	in process						
Hungary	2					2	2						
Ireland	4			1	in process	3	in process						
Israel	9					9	in process						
Kosova	2	2	in process										
Latvia	1					1	in process						
Montenegro	8	2	in process			6	in process						
Norway	9			1	in process	8	in process						
Romania	4			1	1	3	3						
Serbia	17					17	in process						
Switzerland	6					6	in process						
Ukraine	15					15	in process						
	1494	121	0	218	52	1020	189 0	38	0	93	66	4	0
40 Countries	1351	8.10%		14.59%		68.27%		2.54%		6.22%		0.27%	
				91.0%						9.0%			

1. Propagated to sufficient titre to perform HI assay (the totalled number does not include any from batches that are in process)
2. Propagated to sufficient titre to perform HI assay in the presence of 20nM oseltamivir (the totalled number does not include any from batches that are in process)
Numbers in red indicate viruses recovered but with insufficient HA titre to permit HI assay

Samples received were in lysis buffer - only genetic characterisation possible
Samples received (n = 19) were RNA - only genetic characterisation possible
* As of 2022-02-13

Table 2. Genetic characterisation[1] of samples supplied as clinical specimens only, since the September 2021 VCM*

MONTH	TOTAL RECEIVED Seasonal	SPECIMENS SUPPLIED IN VTM[2]			LM[3]			RNA[4]		
Country		Total	Good	Fail	Total	Good	Fail	Total	Good	Fail
APRIL - 2021										
Ghana	1	1	0	1						
Qatar	3	3	1	2						
MAY										
Ghana	4	4	0	4						
Qatar	8	8	5	3						
JUNE										
Ghana	7	7	3	4						
Qatar	4	4	1	3						
JULY										
Cote d'Ivoire	3	3	in process							
Ghana	16	16	6	10						
Qatar	9	9	7	2						
AUGUST										
Cote d'Ivoire	3	3	in process							
Norway	2	2	2	0						
Qatar	62	62	41	21						
Russia	1	1	1	0						
Senegal	2	2	1	1						
SEPTEMBER										
Cameroon	23	23	4	19						
France	1	1	0	1						
Lebanon	1	1	0	1						
Madagascar	8	8	1	7						
Oman	10	10	0	10						
Qatar	14	14	5	9						
Senegal	19	19	9	10						
Spain	1	1	1	0						
OCTOBER										
Cameroon	4	4	1	3						
Cote d'Ivoire	1	1	in process							
Estonia	1	1	0	1						
France	1	1	1	0						
Ireland	1	1	1	0						
Kyrgyzstan	28	28	6	22						
Lebanon	15	15	11	4						
Madagascar	7	7	3	4						
Norway	7	7	not done							
Oman	25	25	4	21						
Portugal	2	2	0	0						
Qatar	3	3	1	2						
Senegal	20	20	8	12						
Tajikistan	15	15	1	14						
UK (Scotland)	5				5	0	5			
NOVEMBER										
Algeria	5	5	2	3						
Armenia	2	2	0	2						
Cameroon	3	3	3	0						
Cote d'Ivoire	4	4	in process							
Estonia	1	1	0	1						
France	8	8	4	4						
Ireland	4	3	1	2	1	in process				
Israel	4							4	0	4
Kosova	2	2	in process							
Kyrgyzstan	22	22	3	19						
Lebanon	57	57	41	16						
Madagascar	32	32	9	23						
Norway	8	8	not done							
Oman	31	31	7	24						
Qatar	4	4	3	1						
Spain	24	24	6	18						
Tajikistan	11	11	1	10						
UK (N. Ireland)	3				3	0	3			
UK (Scotland)	2				2	0	2			
DECEMBER										
Albania	39	39	in process							
Algeria	5	5	2	3						
Argentina	7	7	in process							
Armenia	21	21	5	17						
Belgium	6	6	3	3						
Cameroon	2	2	0	2						
Cote d'Ivoire	13	13	in process							
Estonia	7	7	5	2						
Ireland	5	4	1	3	1	in process				
Israel	30	15	10	5				15	10	5
Kosova	56	56	in process							
Lebanon	76	76	15	61						
Madagascar	18	18	7	11						
Montenegro	6	6	1	5						
Norway	1	1	0	1						
Portugal	12	12	2	10						
Serbia	7	7	in process							
Spain	50	50	15	35						
Switzerland	5	5	2	3						
Ukraine	9	5	1	4	4	in process				
JANUARY - 2022										
Argentina	6	6	in process							
Armenia	2	2	2	0						
Belgium	10	10	4	6						
Cote d'Ivoire	1	1	in process							
Ireland	4	2	2	0	2	in process				
Israel	9	9	6	3						
Kosova	2	2	in process							
Montenegro	8	8	1	7						
Norway	9	9	3	6						
Serbia	17	17	in process							
Switzerland	5	5	4	1						
Ukraine	9	8	1	7	1	in process				
Pre-September total	125	125	68	51						
Post-August total	896	858	229	458	19	0	10	19	10	9
Total	1021	983	297	509	19	0	10	19	10	9

1. NGS yielded full-length HA gene sequence as a minimum
2. Influenza-positive specimens supplied in Virus Transport Medium
3. Influenza-positive specimens supplied in Lysis Mix
4. Influenza-positive specimens supplied as purified RNA
Not done: small volume so taken straight to virus isolation
* As of 2022-02-11

Table 3. Isolation rates for specimens collected after 2021-08-31

	Clinical specimen			Virus isolate		
	Number processed	Number propagated[1]	Percent recovered	Number processed	Number propagated[1]	Percent recovered
H1pdm09	25	15	60%	64	63	98%
H3	98	83	85%	221	213	96%
B/Victoria	20	13	65%	32	31	97%
B/Yamagata	0			0		

1. Propagated to sufficient HA titre to perform HI assay

As of 2022-02-13

Table 4. Summary of isolates with collection dates after 2021-08-31 showing reduction in reactivity with antisera raised against reference viruses in HI & plaque-reduction neutralisation assays (PRNA)

Virus type/subtype	Number tested	≤2-fold[a]	4-fold[a]	≥8-fold[a]	
H1[1a]	78	46 (59%)	8 (10%)	24 (31%)	
H1[1b]	78	46 (59%)	2 (3%)	30 (39%)	
H1[1c]	78	13 (17%)	4 (5%)	61 (78%)	
H1[1d]	78	14 (18%)	3 (4%)	61 (78%)	
H3[2a]	296	177 (60%)	81 (27%)	38 (13%)	
H3[2b]	296	61 (21%)	146 (49%)	89 (30%)	
H3[2c]	296	16 (5%)	31 (11%)	249 (84%)	
H3[2d]	296	1 (0.5%)	19 (6.5%)	276 (93%)	
B/Victoria[3a]	44	38 (86%)	1 (2%)	5 (11%)	
B/Victoria[3b]	44	39 (89%)	0	5 (11%)	
B/Victoria[3c]	44	0	0	44 (100%)	
B/Victoria[3d]	44	3 (7%)	2 (5%)	39 (89%)	
Virus type/subtype	**Number tested**	**≤2-fold[b]**	**>2≤4[b]**	**>4≤8[b]**	**>8-fold[b]**
H1[1a]	20	15 (75%)	2 (10%)		3 (15%)
H1[1b]	20	13 (65%)	4 (20%)		3 (15%)
H1[1c]	20	3 (15%)			17 (85%)
H1[1d]	20	3 (15%)			17 (85%)
H3[2f]	91	8 (9%)	3 (3%)	13 (14%)	67 (74%)
H3[2e]	91	13 (14%)	32 (35%)	36 (40%)	10 (11%)
H3[2a]	75	51 (68%)	12 (16%)	1 (1%)	11 (15%)
H3[2b]	75		3 (4%)	43 (57%)	29 (39%)
H3[2c]	91	8 (9%)	3 (3%)	5 (6%)	75 (82%)

a. Reduction in HI titre, compared to the homologous titre with post-infection ferret antisera indicated below:
b. Reduction in titre, compared to the homologous titre based on actual computer read values from a digitized image for PRNA:

1b. Compared to homologous titres for ferret antiserum raised against cell culture-propagated vaccine virus A/Guangdong-Maonan/SWL1536/2019 (genetic subgroup 6B.1A.5a.1)
1a. Compared to homologous titres for ferret antiserum raised against egg-propagated vaccine virus A/Guangdong-Maonan/SWL1536/2019 (genetic subgroup 6B.1A.5a.1)
1c. Compared to homologous titres for ferret antiserum raised against cell culture-propagated A/Denmark/3280/2019 (genetic subgroup 6B.1A.5a.2)
1d. Compared to homologous titres for ferret antiserum raised against egg-propagated vaccine virus IVR-215 A/Victoria//2570/2019 (genetic subgroup 6B.1A.5a.2)

2a. Compared to homologous titres for ferret antiserum raised against cell culture-propagated vaccine virus A/Stockholm/5/2021 (genetic subgroup 3C.2a1b.2a.2)
2b. Compared to homologous titres for ferret antiserum raised against egg-propagated A/Darwin/9/2021 (genetic subclade 3C.2a1b.2a.2)
2c. Compared to homologous titres for ferret antiserum raised against cell culture-propagated A/Cambodia/925256/2020 (genetic subgroup 3C.2a1b.2a.1)
2d. Compared to homologous titres for ferret antiserum raised against egg-propagated A/Cambodia/e0826360/2020 (genetic subgroup 3C.2a1b.2a.1)
2e. Compared to homologous titres for ferret antiserum raised against cell culture-propagated A/Bangladesh/4005/2020 (genetic subgroup 3C.2a1b.2a.2)
2f. Compared to homologous titres for ferret antiserum raised against cell culture-propagated A/Denmark/3264/2019 (genetic subgroup 3C.2a1b.1a)

3a. Compared to homologous titres for ferret antiserum raised against cell culture-propagated B/Austria/1359417/2021 - Clade V1A.3a.2
3b. Compared to homologous titres for ferret antiserum raised against egg-propagated B/Austria/1359417/2021 - Clade V1A.3a.2 (G141)
3c. Compared to homologous titres for ferret antiserum raised against egg-propagated B/Austria/1359417/2021 - Clade V1A3a.2 (G141R)
3d. Compared to homologous titres for ferret antiserum raised against egg-propagated vaccine virus B/Washingto/02/2019 - Clade V1A.3

As of 2022-02-14

H1N1pdm09

Influenza A(H1N1)pdm09 viruses with collection dates after 2021-08-31 were received from NICs in 17 countries in three WHO Regions: African (Cameroon, Cote d'Ivoire, Senegal and South Africa) Eastern Mediterranean (Oman and Qatar) and European (eleven countries) (**Table 1**). **Table 5** summarises antigenic (HI) results for the recognition of the viruses, propagated at the time of preparing this report, by antisera raised against the egg-propagated and cell culture-propagated cultivars of A/Guangdong-Maonan/SWL1536/2019 (the recommended vaccine virus for the Northern Hemisphere 2020-2021 influenza season), IVR-215 with its HA derived from egg-propagated A/Victoria/2570/2019 (the vaccine virus for the Southern Hemisphere 2021, the Northern Hemisphere 2021/2022 and the Southern Hemisphere 2022 seasons) and against A/Denmark/3280/2019, a cell culture-propagated virus similar to A/Victoria/2570/2019.

HI analysis

Tables 6-1 to 6-6 show the results of HI analysis of samples analysed after the September 2021 VCM. Viruses below the red line in **Tables 6-3**, **6-4, 6-5** and **6-6** had collection dates after 2021-08-31, and represented 95% of the total of 86 test viruses analysed. **Tables 6-2** and **6-5** shows HI results from some viruses shared by CDC, and **Tables 6-1** and **6-6** include two viruses shared by VIDRL. A summary of the HI results is shown in **Table 6-7**.

Approximately 20% of viruses had HA genes falling into subclade 6B.1A.5a.2, 80% in subclade 6B.1A.5a.1 and sequencing is pending on four of the test viruses.

Antisera raised against the vaccine virus IVR-215 (A/Victoria/2570/2019) or against cell culture-propagated A/Denmark/3280/2019, recognised all 17 of the identified 6B.1A.5a.2 viruses at titres within 4-fold of the homologous titres, and 82% and 76%, respectively, at titres within 2-fold of the homologous titres. An antiserum raised against egg-propagated A/Sydney/5/2021, also a subclade 6B.1A.5a.2 virus, was used in two tests (**Tables 6-5** and **6-6**) and recognised all six of the subclade 6B.1A.5a.2 test viruses at titres within 2-fold of the homologous titre. These three antisera recognised none of the test viruses in subclade 6B.1A.5a.1 at titres within 4-fold of the homologous titres of the antisera (**Table 6-7**). Antisera raised against reference viruses in subclades 6B.1A, 6B.1A.1, 6B.1A.5b and 6B.1A.5a.1 recognised none of the test viruses with HA genes in subclade 6B.1A.5a.2 at titres within 4-fold of the homologous titres.

The antisera raised against egg-propagated and cell culture-propagated cultivars of A/Guangdong-Maonan/SWL1536/2019 (subclade 6B.1A.5a.1) and other related reference viruses (A/Paris/1447/2017, subclade 6B.1A, A/Brisbane/02/2018, subclade 6B.1A.1, A/Ireland/87733/2019, subclade 6B.1A.5a.1, A/Switzerland/3330/2017 subclade 6B.1A.5b) recognised 68% to 92% of the test viruses that fell in subclade 6B.1A.5a.1 at titres within 4-fold of the homologous titres of the antisera, and 43% to 78% of viruses at titres within 2-fold of the homologous titres of the antisera. Antisera raised against A/Ghana/1894/2021 and A/North Carolina/01/2021, both subclade 6B.1A.5a.1, were used in **Tables 6-5** and **6-6**, and **Table 6-6** respectively. These showed similar patterns of recognition to the other antisera raised against viruses in subclade 6B.1A.5a.1.

In each case a sizable proportion of the test viruses in subclade 6B.1A.5a.1 were not recognised well by this set of antisera (between 7.7% and 32%). Notably, the antisera raised against A/Switzerland/3330/2017, A/Brisbane/2/2018, A/Ireland/87733/2019, A/Guangdong/Maonan/SWL1536/2019 and A/Ghana/1894/2021 egg-propagated viruses, and cell culture-propagated A/Paris/1447/2019 recognised 18% to 32% of these test viruses at titres over 4-fold reduced compared to the homologous titres. The majority of these test viruses showing low recognition, seen in **Tables 6-4** and **6-5,** had HA genes encoding **P137S** and **G155E** in **HA1**, or **G155E** in **HA1** with other amino acid substitutions (A/South Africa/R16455/2021, **Table 6-4**).

Of the fourteen viruses with HA genes in subclade 6B.1A.5a.1 encoding the substitutions **P137S** and **G155E** in **HA1**, ten showed polymorphism at additional residues in **HA1**.

Table 7 shows the results of a virus neutralisation assay on a set of H1N1pdm09 viruses. As in the HI assay, test viruses in subclade 6B.1A.5a.2 (n=3) were recognised well only by antisera raised against cell culture-propagated A/Denmark/3280/2019 and IVR-215 (A/Victoria/2570/2019). These two antisera recognised the test viruses in subclade 6B.1A.5a.1 poorly but these viruses were recognised better by antisera raised against reference viruses in subclades 6B.1A and 6B.1A.5a.1. Those viruses with substitutions **P137S** and **G155E** in **HA1** were recognised less well than the other test viruses lacking substitutions at **137** and **155** at relative absolute titres: they were recognised on average at titres approximately 4-fold lower than the absolute titres with test viruses lacking these substitutions by the antisera raised against A/Paris/1447/2017 and the egg-propagated cultivar of A/Guangdong-Manan/SWL1536/2019. In contrast, the antiserum raised against the cell culture-propagated cultivar of A/Guangdong-Maonan/SWL1536/2019 recognised test viruses with substitutions **P137S** and **G155E** in **HA1** at titres approximately 2-fold lower than titres with viruses lacking substitutions at **137** and **155**. Nevertheless, compared to the homologous titres of the antiserum raised against the cell culture-propagated or egg-propagated cultivars of A/Guangdong-Maonan/SWL1536/2019 this subset of 6B.1A.5a.1 test viruses were recognised on average at titres within 2-fold of the relative homologous titres.

Genetic analysis

All test viruses, for which HA gene sequences were known, fell into subclade 6B.1A which is defined by the amino acid substitutions **S74R**, **S164T** (altering a glycosylation motif introduced at residues **162-164** in clade 6B.1 viruses) and **I295V** in **HA1,** compared with the clade 6B.1 prototype virus, A/Michigan/45/2019. The vast majority of A(H1N1)pdm09-positive clinical specimens and viruses analysed had HA genes in group 6B.1A.5a in subclades 6B.1A.5a.1 or 6B.1A.5a.2.

Figure 2 shows a phylogenetic tree for HA genes of representative A(H1N1)pdm09 viruses most with collection dates from 2021-09-01, as deposited in GISAID at the time of preparing this report. **Figure 3** shows a tree for viruses which were sequenced at the Francis Crick Institute.

The HA gene of recent viruses fall within group 6B.1A.5a. All test viruses carried **N129D, T185I** and **N260D** amino acid substitutions in **HA1,** and the two major subclades in circulation recently are:

- 6B.1A.5a.1: the HA genes of viruses in this subgroup encode **D187A** and **Q189E** substitutions in **HA1**. Some viruses have the additional **HA1** substitutions **P137S**, **G155E**, of which some also have **S128T** in **HA1**; another group has the substitutions **G155E**, **A187S** and **R313K** in **HA1** and **D174N** in **HA2**, another has **R113K** in **HA1,** and another **I183T** in **HA2**.

- 6B.1A.5a.2: the HA genes of most viruses in this subgroup encode **HA**s with **K130N**, **N156K**, **L161I** and **V250A** substitutions in **HA1** and **E179D** substitution in **HA2**. The HA genes of a subgroup of viruses also encode **K54Q**, **A186T**, **Q189E**, **E224A**, **R259K** and **K308R** in **HA1**. Within this subclade some viruses have HA genes that encode **T216A** in **HA1** and some of these also encode **D94N** in **HA1**.

An amino acid sequence alignment of the HAs for representative A(H1N1)pdm09 viruses is shown in **Figure 4**; the vaccine viruses are shown in red, reference viruses are coloured blue, glycosylation motifs are highlighted, the genetic group/subgroup is shown in green and the HA1/HA2 proteolytic-processing site is highlighted.

Figure 5 shows H1pdm09-HA structures with the positions of amino acid substitutions in HA1 that define the two subclades, 6B.1A.5a.1 and 6B.1A.5a.2. Also marked are amino acid substitutions that are associated with emerging genetic subgroups within these two subclades.

Figure 6 shows a phylogenetic tree for the NA genes of the viruses shown in **Figure 2** and **Figure 7** a tree for viruses sequenced at the Francis Crick Institute (*cf* **Figure 3**). The NA trees and HA trees were largely congruent. Viruses with HA genes in both subclades 6B.1A.5a.1 and 6B.1A.5a.2 shared the NA substitutions **Q51K**, **F74S**, **I389K** and **T452I**. Many in the two subclades also shared **M19T**, **Y66F** and **N222K**. Viruses with HA genes in subclade 6B.1A.5a.1 had NA genes encoding **T16A**, some with the additional substitutions **S366N** or **K469N**. Recent viruses with HA genes in subclade 6B.1A.5a.2 had NA genes encoding the substitutions **V453M**, some also with **K469N** or **R2220K**.

Figure 8 shows a NA amino acid alignment for the same representative set of A(H1N1)pdm09 viruses as used in **Figure 4**.

Antiviral analysis

Susceptibility to the neuraminidase inhibitors (NAIs) oseltamivir and zanamivir was assessed phenotypically and by full NA gene sequencing, for 66 A(H1N1)pdm09 viruses detected and received within the course of the 2021-2022 influenza season (**Table 13**). All showed normal inhibition (NI: no reduced susceptibility) by both antivirals.

All of the 47 A(H1N1)pdm09 full-length PA gene sequences generated at the Francis Crick Institute, from individual viruses detected in the course of the 2021-2022 influenza season, encoded isoleucine at position 38 (I38) and showed no evidence of other substitutions associated with reduced susceptibility to baloxavir marboxil.

Conclusion

Viruses received at the Francis Crick Institute fell into two distinct antigenic and genetic groups. The majority fell into subclade 6B.1A.5a.1. These viruses were recognised generally well by antisera raised against viruses in the same subclade (e.g. A/Guangdong-Maonan/SWL1536/2019) but a sizable proportion of viruses in this subclade were recognised less well by these antisera. These viruses had HA genes encoding the substitutions **P137S** and **G155E** or **G155E** in **HA1**. Viruses in subclade 6B.1A.5a.2 were recognised well by antisera raised against reference viruses and vaccine viruses in the same subclade.

All viruses showed normal inhibition by oseltamivir and zanamivir and none showed amino acid substitutions indicative of reduced susceptibility to baloxavir marboxil.

Table 5. Summary of influenza A(H1N1)pdm09 viruses collected after 2021-08-31, analysed by HI

MONTH	H1N1pdm09													
			Post infection ferret antisera raised against											
			Clade 6B.1A.5a.1 virus			Clade 6B.1A.5a.1 virus			Clade 6B.1A.5a.2 virus			Clade 6B.1A.5a.2 virus		
Country	Number received	Number propagated[1,2]	≤2 fold[3]	4 fold[3]	≥8 fold[3]	≤2 fold[4]	4 fold[4]	≥8 fold[4]	≤2 fold[5]	4 fold[5]	≥8 fold[5]	≤2 fold[6]	4 fold[6]	≥8 fold[6]
2021														
SEPTEMBER														
Cameroon	22	3	2	1		2	1				3			3
France	1	0												
Oman	1	0												
Qatar	1	1			1			1		1		1		
Senegal	19	4	2	2		1		3			4			4
South Africa	6	6	6			6					6			6
Sweden	1	1			1			1	1			1		
OCTOBER														
Cameroon	3	1	1			1					1			1
Cote d'Ivoire	1	0												
Denmark	1	1			1			1		1			1	
France	9	8	4	2	2	4		4	1		7	1		7
Italy	3	3			3			3	2	1		3		
Oman	10	2			2			2	2			1	1	
Qatar	1	0												
Senegal	20	0												
South Africa	13	13	9	2	2	11		2	1		12	1		12
NOVEMBER														
Cote d'Ivoire	3	0												
France	18	13	9	1	3	9	1	3			13			13
Oman	15	3			3			3	3			3		
Qatar	1	1			1			1	1			1		
South Africa	6	6	4		2	4		2	1	1	4	1	1	4
Spain	1	1			1			1			1			1
DECEMBER														
Belgium	6	2	2			2					2			2
Ireland	1	0												
Netherlands	5	5	5			4		1			5			5
Serbia	5	0												
2022														
JANUARY														
Belgium	6	3	2		1	2		1			3			3
Norway	1	0												
Romania	1	1			1			1	1			1		
17 Countries	181	78	46	8	24	46	2	30	13	4	61	14	3	61
			59.0%	10.3%	30.8%	59.0%	2.6%	38.5%	16.7%	5.1%	78.2%	17.9%	3.8%	78.2%

1 Numbers shown in red indicate that not all of the specimens received had been propagated at the time this report was prepared
2 Propagated to sufficient titre in cell culture to perform HI assay
3 Reduction in HI titre with post-infection ferret antiserum raised against A/Guangdong-Maonan/SWL1536/2019 (cell-propagated vaccine virus)
4 Reduction in HI titre with post-infection ferret antiserum raised against A/Guangdong-Maonan/SWL1536/2019 (egg-propagated vaccine virus)
5 Reduction in HI titre with post-infection ferret antiserum raised against A/Denmark/3280/2019 (cell-propagated virus)
6 Reduction in HI titre with post-infection ferret antiserum raised against IVR-215 A/Victoria/2570/2019 (egg-propagated vaccine virus)

As of 2022-02-09

Figure 2. Phylogenetic comparison of A(H1N1)pdm09 HA genes (non-WIC sequences)

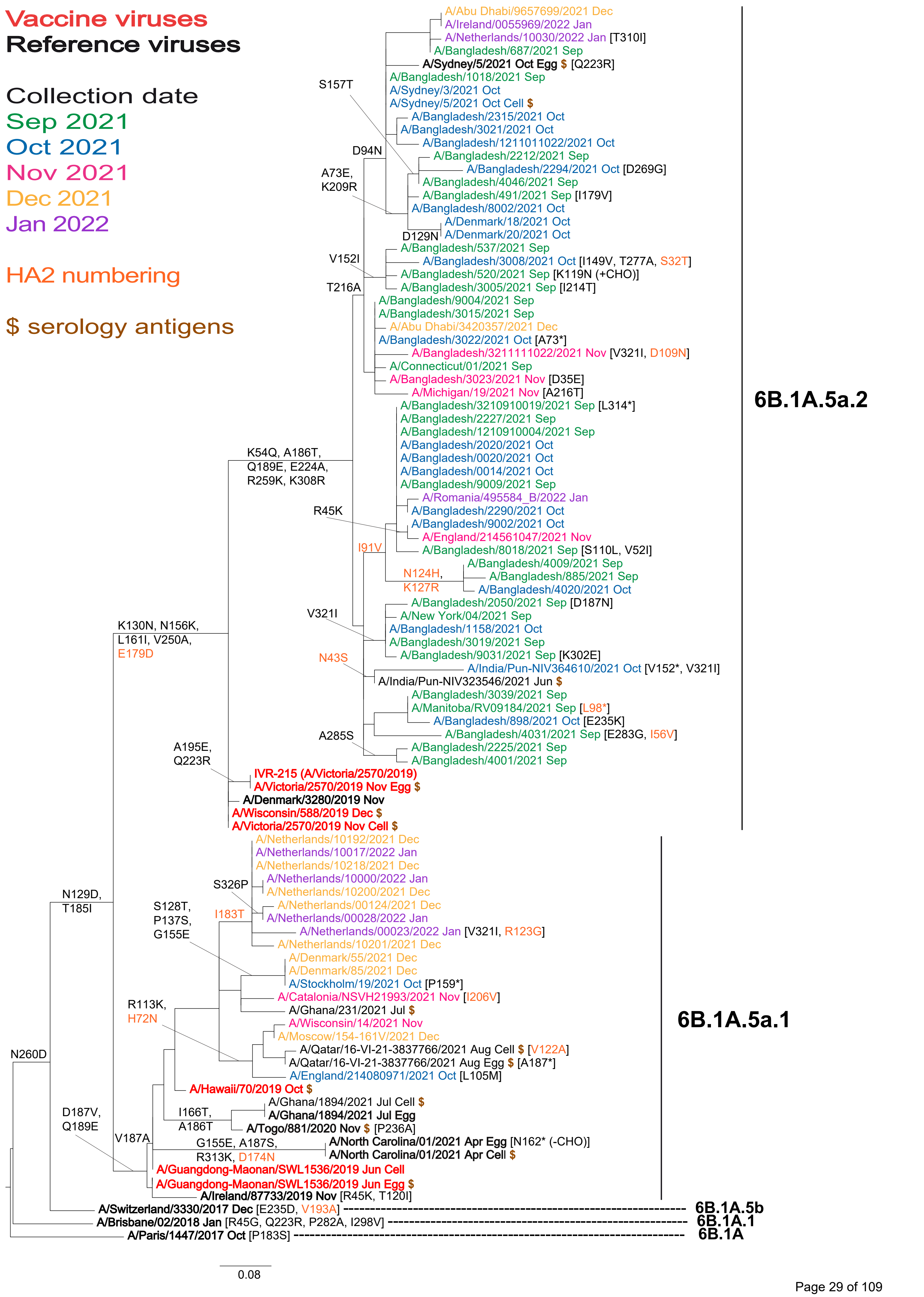


Figure 3. Phylogenetic comparison of A(H1N1)pdm09 HA genes (WIC sequences)

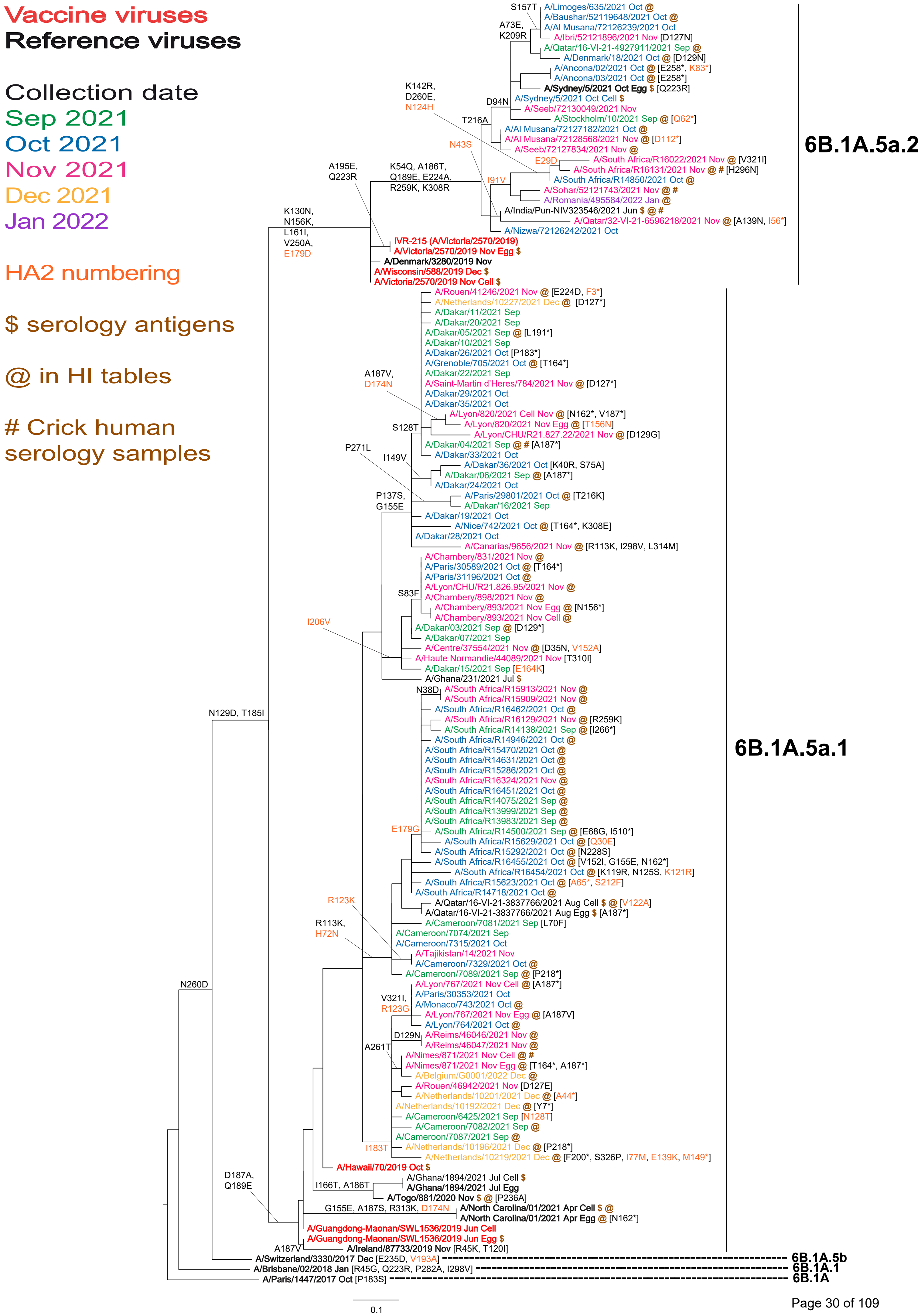


Figure 4. HA protein alignment for A(H1N1)pdm09 viruses

Vaccine virus: Reference virus: 6B Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

```
                                           10        20        30        40        50        60        70        80        90       100
                                  ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Paris/1447/2017                 DTLCIGYHANNSTDTVDTVLEKNVTVTHSVNLLEDKHNGKLCKLRGVAPLHLGKCNIAGWILGNPECESLSTARSWSYIVETSNSDNGTCYPGDFINYEE 6B.1A
A/Denmark/3280/2019               .................................................................................................... 6B.1A.5a.2
IVR-215 (A/Victoria/2570/2019)    .................................................................................................... 6B.1A.5a.2
A/Nizwa/72126242/2021             .....................................................Q.............................................. 6B.1A.5a.2
A/Qatar/32-VI-21-6596218/2021     .....................................................Q.............................................. 6B.1A.5a.2
A/South_Africa/R16131/2021        .....................................................Q.............................................. 6B.1A.5a.2
A/Romania/495584/2022             .....................................................Q.............................................. 6B.1A.5a.2
A/Al_Musana/72128568/2021         .....................................................Q.............................................. 6B.1A.5a.2
A/Seeb/72127834/2021              .....................................................Q.............................................. 6B.1A.5a.2
A/Stockholm/10/2021               .....................................................Q.......................................N...... 6B.1A.5a.2
A/Qatar/16-VI-21-4927911/2021     .....................................................Q..................E....................N...... 6B.1A.5a.2
A/Denmark/18/2021                 .....................................................Q..................E....................N...... 6B.1A.5a.2
A/Ibri/52121896/2021              .....................................................Q..................E....................N...... 6B.1A.5a.2
A/Limoges/635/2021                .....................................................Q..................E....................N...... 6B.1A.5a.2
A/Guangdong-Mao/SWL1536/2019E     .................................................................................................... 6B.1A.5a.1
A/Hawaii/70/2019                  .................................................................................................... 6B.1A.5a.1
A/Netherlands/10219/2021          .................................................................................................... 6B.1A.5a.1
A/Reims/46046/2021                .................................................................................................... 6B.1A.5a.1
A/Lyon/767/2021                   .................................................................................................... 6B.1A.5a.1
A/Cameroon/6425/2021              .................................................................................................... 6B.1A.5a.1
A/Cameroon/7081/2021              .....................................................................F.............................. 6B.1A.5a.1
A/Tajikistan/14/2021              .................................................................................................... 6B.1A.5a.1
A/South_Africa/R16455/2021        .................................................................................................... 6B.1A.5a.1
A/South_Africa/R16129/2021        .................................................................................................... 6B.1A.5a.1
A/South_Africa/R15913/2021        .....................................D.............................................................. 6B.1A.5a.1
A/Centre/37554/2021               ..................................N................................................................. 6B.1A.5a.1
A/Dakar/07/2021                   .................................................................................................... 6B.1A.5a.1
A/Chambery/893/2021               ..................................................................................F................. 6B.1A.5a.1
A/Paris/29801/2021                .................................................................................................... 6B.1A.5a.1
A/Canarias/9656/2021              .................................................................................................... 6B.1A.5a.1
A/Dakar/36/2021                   .......................................R..................................A......................... 6B.1A.5a.1
A/Netherlands/10227/2021          .................................................................................................... 6B.1A.5a.1

                                          110       120       130       140       150       160       170       180       190       200
                                  ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Paris/1447/2017                 LREQLSSVSSFERFEIFPKTSSWPNHDSNKGVTAACPHAGAKSFYKNLIWLVKKGNSYPKLNQTYINDKGKEVLVLWGIHHPSTTADQQSLYQNADAYVF
A/Denmark/3280/2019               ............................DN.........................K....I.....................P.I...............
IVR-215 (A/Victoria/2570/2019)    ............................DN.........................K....I.....................P.I.........E.....
A/Nizwa/72126242/2021             ............................DN.........................K....I.....................P.IT..E...........
A/Qatar/32-VI-21-6596218/2021     ............................DN........N................K....I.....................P.IT..E...........
A/South_Africa/R16131/2021        ............................DN...........R.............K....I.....................P.IT..E...........
A/Romania/495584/2022             ............................DN.........................K....I.....................P.IT..E...........
A/Al_Musana/72128568/2021         ............................DN.........................K....I.....................P.IT..E...........
A/Seeb/72127834/2021              ............................DN.........................K....I.....................P.IT..E...........
A/Stockholm/10/2021               ............................DN.........................K....I.....................P.IT..E...........
A/Qatar/16-VI-21-4927911/2021     ............................DN.........................K....I.....................P.IT..E...........
A/Denmark/18/2021                 .............................N.........................K....I.....................P.IT..E...........
A/Ibri/52121896/2021              ..........................N.DN.........................KT...I.....................P.IT..E...........
A/Limoges/635/2021                ............................DN.........................KT...I.....................P.IT..E...........
A/Guangdong-Mao/SWL1536/2019E     ............................D.....................................................P.I.V.E...........
A/Hawaii/70/2019                  ............................D.....................................................P.I.A.E...........
A/Netherlands/10219/2021          ............................D.....................................................P.I.A.E..........X
A/Reims/46046/2021                ..................................................................................P.I.A.E...........
A/Lyon/767/2021                   ............................D.....................................................P.I.X.E...........
A/Cameroon/6425/2021              ............................D.....................................................P.I.A.E...........
A/Cameroon/7081/2021              ............K...............D.....................................................P.I.A.E...........
A/Tajikistan/14/2021              ............K...............D.....................................................P.I.A.E...........
A/South_Africa/R16455/2021        ............K...............D......................I..E......X....................P.I.A.E...........
A/South_Africa/R16129/2021        ............K...............D.....................................................P.I.A.E...........
A/South_Africa/R15913/2021        ............K...............D.....................................................P.I.A.E...........
A/Centre/37554/2021               ............................D.....................................................P.I.A.E...........
A/Dakar/07/2021                   ............................D.....................................................P.I.A.E...........
A/Chambery/893/2021               ............................D.....................................................P.I.A.E...........
A/Paris/29801/2021                ............................D.......S.................E...........................P.I.A.E...........
A/Canarias/9656/2021              ............K...............D.......S.................E...........................P.I.A.E...........
A/Dakar/36/2021                   ............................D.......S...........V.....E...........................P.I.A.E...........
A/Netherlands/10227/2021          ..........................XTD.......S.................E...........................P.I.A.E...........

                                          210       220       230       240       250       260       270       280       290       300
                                  ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Paris/1447/2017                 VGTSRYSKKFKPEIATRPKVRDQEGRMNYYWTLVEPGDKITFEATGNLVVPRYAFTMERNAGSGIIISDTPVHDCNTTCQTPEGAINTSLPFQNVHPITI
A/Denmark/3280/2019               .................................................A.........D............X...........................
IVR-215 (A/Victoria/2570/2019)    ......................R..........................A.........D........................................
A/Nizwa/72126242/2021             .......................A.........................A........KD........................................
A/Qatar/32-VI-21-6596218/2021     .......................A.........................A........KD........................................
A/South_Africa/R16131/2021        .......................A.........................A........KE...................................N....
A/Romania/495584/2022             .......................A.........................A........KD........................................
A/Al_Musana/72128568/2021         ...............A.......A.........................A........KD........................................
A/Seeb/72127834/2021              ...............A.......A.........................A........KD........................................
A/Stockholm/10/2021               ...............A.......A.........................A........KD........................................
A/Qatar/16-VI-21-4927911/2021     ........R......A.......A.........................A........KD........................................
A/Denmark/18/2021                 ........R......A.......A.........................A........KD........................................
A/Ibri/52121896/2021              ........R......A.......A.........................A........KD........................................
A/Limoges/635/2021                ........R......A.......A.........................A........KD........................................
A/Guangdong-Mao/SWL1536/2019E     ...........................................................D........................................
A/Hawaii/70/2019                  ...........................................................D........................................
A/Netherlands/10219/2021          ...........................................................D........................................
A/Reims/46046/2021                ...........................................................D........................................
A/Lyon/767/2021                   ...........................................................D........................................
A/Cameroon/6425/2021              ...........................................................D........................................
A/Cameroon/7081/2021              ...........................................................D........................................
A/Tajikistan/14/2021              ...........................................................D........................................
A/South_Africa/R16455/2021        ...........................................................D........................................
A/South_Africa/R16129/2021        ..........................................................KD........................................
A/South_Africa/R15913/2021        ...........................................................D........................................
A/Centre/37554/2021               ...........................................................D........................................
A/Dakar/07/2021                   ...........................................................D........................................
A/Chambery/893/2021               ...........................................................D........................................
A/Paris/29801/2021                ...............K...........................................D..........L.............................
A/Canarias/9656/2021              ...........................................................D.....................................V..
A/Dakar/36/2021                   ...........................................................D........................................
A/Netherlands/10227/2021          ...........................................................D........................................
```

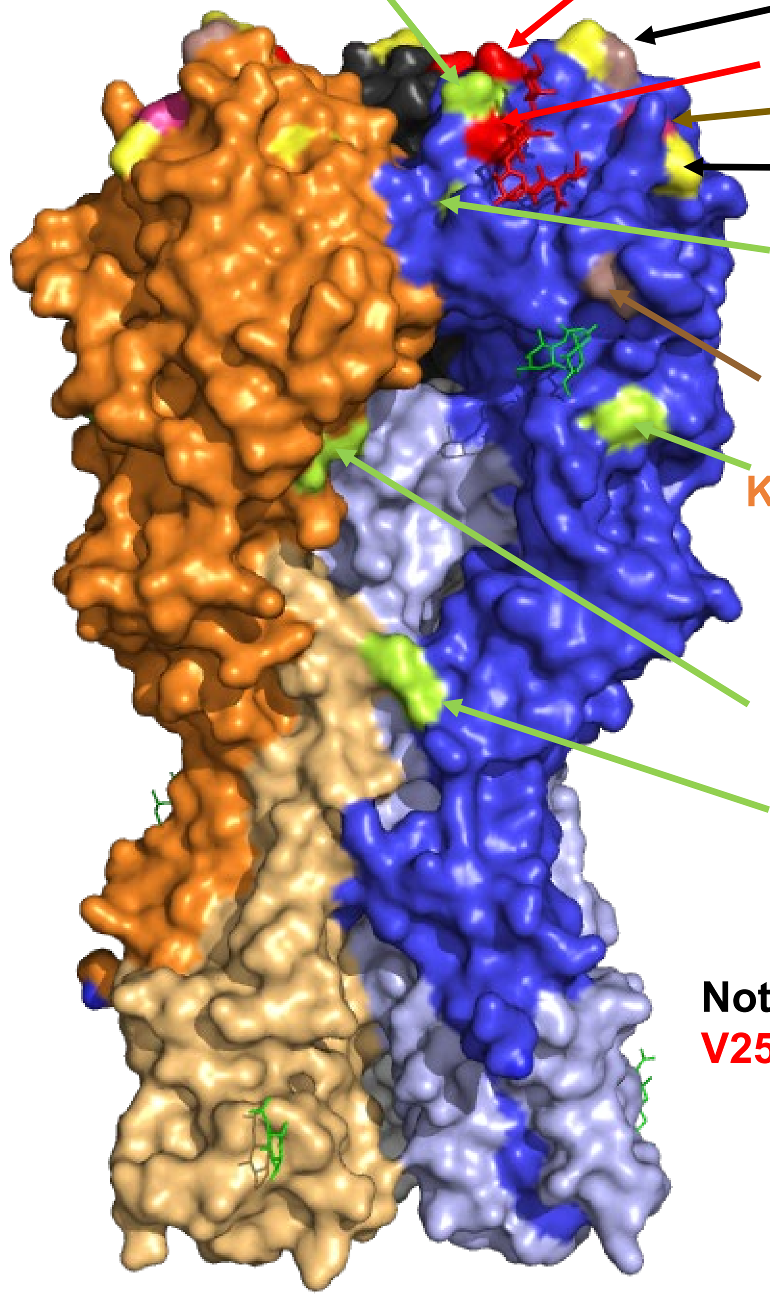
```
                                   310       320       330       340       350       360       370       380       390       400
                                   ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Paris/1447/2017                  GKCPKYVKSTKLRLATGLRNVPSIQSRGLFGAIAGFIEGGWTGMVDGWYGYHHQNEQGSGYAADLKSTQNAIDKITNKVNSVIEKMNTQFTAVGKEFNHL 6B.1A
A/Denmark/3280/2019                .................................................................................................... 6B.1A.5a.2
IVR-215 (A/Victoria/2570/2019)     .................................................................................................... 6B.1A.5a.2
A/Nizwa/72126242/2021              .......R............................................................................................ 6B.1A.5a.2
A/Qatar/32-VI-21-6596218/2021      .......R.............................................................S............X................. 6B.1A.5a.2
A/South_Africa/R16131/2021         .......R...............................................D............................................ 6B.1A.5a.2
A/Romania/495584/2022              .......R............................................................................................ 6B.1A.5a.2
A/Al_Musana/72128568/2021          .......R............................................................................................ 6B.1A.5a.2
A/Seeb/72127834/2021               .......R............................................................................................ 6B.1A.5a.2
A/Stockholm/10/2021                .......R................................................................................X........... 6B.1A.5a.2
A/Qatar/16-VI-21-4927911/2021      .......R............................................................................................ 6B.1A.5a.2
A/Denmark/18/2021                  .......R............................................................................................ 6B.1A.5a.2
A/Ibri/52121896/2021               .......R............................................................................................ 6B.1A.5a.2
A/Limoges/635/2021                 .......R............................................................................................ 6B.1A.5a.2
A/Guangdong-Mao/SWL1536/2019E      .................................................................................................... 6B.1A.5a.1
A/Hawaii/70/2019                   .................................................................................................... 6B.1A.5a.1
A/Netherlands/10219/2021           .........................P.......................................................................... 6B.1A.5a.1
A/Reims/46046/2021                 .................................................................................................... 6B.1A.5a.1
A/Lyon/767/2021                    ....................I............................................................................... 6B.1A.5a.1
A/Cameroon/6425/2021               .................................................................................................... 6B.1A.5a.1
A/Cameroon/7081/2021               ..................................................................................................N. 6B.1A.5a.1
A/Tajikistan/14/2021               ..................................................................................................N. 6B.1A.5a.1
A/South_Africa/R16455/2021         ..................................................................................................N. 6B.1A.5a.1
A/South_Africa/R16129/2021         ..................................................................................................N. 6B.1A.5a.1
A/South_Africa/R15913/2021         ..................................................................................................N. 6B.1A.5a.1
A/Centre/37554/2021                .................................................................................................... 6B.1A.5a.1
A/Dakar/07/2021                    .................................................................................................... 6B.1A.5a.1
A/Chambery/893/2021                .................................................................................................... 6B.1A.5a.1
A/Paris/29801/2021                 .................................................................................................... 6B.1A.5a.1
A/Canarias/9656/2021               .............M...................................................................................... 6B.1A.5a.1
A/Dakar/36/2021                    .................................................................................................... 6B.1A.5a.1
A/Netherlands/10227/2021           .................................................................................................... 6B.1A.5a.1

                                   410       420       430       440       450       460       470       480       490       500
                                   ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Paris/1447/2017                  EKRIENLNKKVDDGFLDIWTYNAELLVLLENERTLDYHDSNVKNLYEKVRNQLKNNAKEIGNGCFEFYHKCDNTCMESVKNGTYDYPKYSEEAKLNREKI
A/Denmark/3280/2019                ....................................................................................................
IVR-215 (A/Victoria/2570/2019)     ....................................................................................................
A/Nizwa/72126242/2021              ....................................................................................................
A/Qatar/32-VI-21-6596218/2021      ....................................................................................................
A/South_Africa/R16131/2021         .................V................................H.................................................
A/Romania/495584/2022              .................V..................................................................................
A/Al_Musana/72128568/2021          ......................................X.............................................................
A/Seeb/72127834/2021               ....................................................................................................
A/Stockholm/10/2021                ....................................................................................................
A/Qatar/16-VI-21-4927911/2021      ....................................................................................................
A/Denmark/18/2021                  ....................................................................................................
A/Ibri/52121896/2021               ....................................................................................................
A/Limoges/635/2021                 ....................................................................................................
A/Guangdong-Mao/SWL1536/2019E      ....................................................................................................
A/Hawaii/70/2019                   ....................................................................................................
A/Netherlands/10219/2021           ...M.............................................................K.........X........................
A/Reims/46046/2021                 ....................................................................................................
A/Lyon/767/2021                    .................................................G..................................................
A/Cameroon/6425/2021               ......................................................T.............................................
A/Cameroon/7081/2021               ....................................................................................................
A/Tajikistan/14/2021               .................................................K..................................................
A/South_Africa/R16455/2021         ....................................................................................................
A/South_Africa/R16129/2021         ....................................................................................................
A/South_Africa/R15913/2021         ....................................................................................................
A/Centre/37554/2021                ..............................................................................A.....................
A/Dakar/07/2021                    ....................................................................................................
A/Chambery/893/2021                ....................................................................................................
A/Paris/29801/2021                 ....................................................................................................
A/Canarias/9656/2021               ....................................................................................................
A/Dakar/36/2021                    ....................................................................................................
A/Netherlands/10227/2021           ....................................................................................................

                                   510       520       530       540
                                   ....|....|....|....|....|....|....|....|.........
A/Paris/1447/2017                  DGVKLESTRIYQILAIYSTVASSLVLVVSLGAISFWMCSNGSLQCRICI
A/Denmark/3280/2019                .....D...........................................
IVR-215 (A/Victoria/2570/2019)     .....D...........................................
A/Nizwa/72126242/2021              .....D...........................................
A/Qatar/32-VI-21-6596218/2021      .....D...........................................
A/South_Africa/R16131/2021         .....D...........................................
A/Romania/495584/2022              .....D...........................................
A/Al_Musana/72128568/2021          .....D...........................................
A/Seeb/72127834/2021               .....D...........................................
A/Stockholm/10/2021                .....D...........................................
A/Qatar/16-VI-21-4927911/2021      .....D...........................................
A/Denmark/18/2021                  .....D...........................................
A/Ibri/52121896/2021               .....D...........................................
A/Limoges/635/2021                 .....D...........................................
A/Guangdong-Mao/SWL1536/2019E      .................................................
A/Hawaii/70/2019                   .................................................
A/Netherlands/10219/2021           .........T.......................................
A/Reims/46046/2021                 .........T.......................................
A/Lyon/767/2021                    .........T.......................................
A/Cameroon/6425/2021               .........T.......................................
A/Cameroon/7081/2021               .................................................
A/Tajikistan/14/2021               .................................................
A/South_Africa/R16455/2021         .................................................
A/South_Africa/R16129/2021         .....G...........................................
A/South_Africa/R15913/2021         .....G...........................................
A/Centre/37554/2021                ................................V................
A/Dakar/07/2021                    ................................V................
A/Chambery/893/2021                ................................V................
A/Paris/29801/2021                 .................................................
A/Canarias/9656/2021               .................................................
A/Dakar/36/2021                    .................................................
A/Netherlands/10227/2021           .................................................
```

Vaccine virus: Reference virus: 6B Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

Figure 5. HA1 amino acid substitutions defining H1N1pdm09 6B.1A.5a viruses

6B.1A.5a.1 defining and major emergent subgroup

6B.1A.5a.2 defining and major emergent subgroup



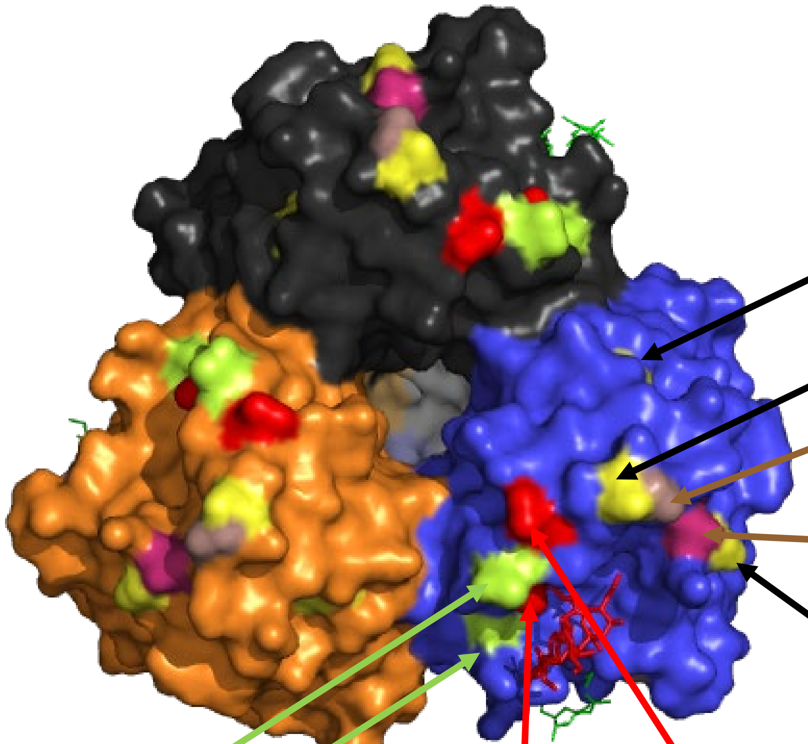


Figure 6. Phylogenetic comparison of A(H1N1)pdm09 NA genes (non-WIC sequences)

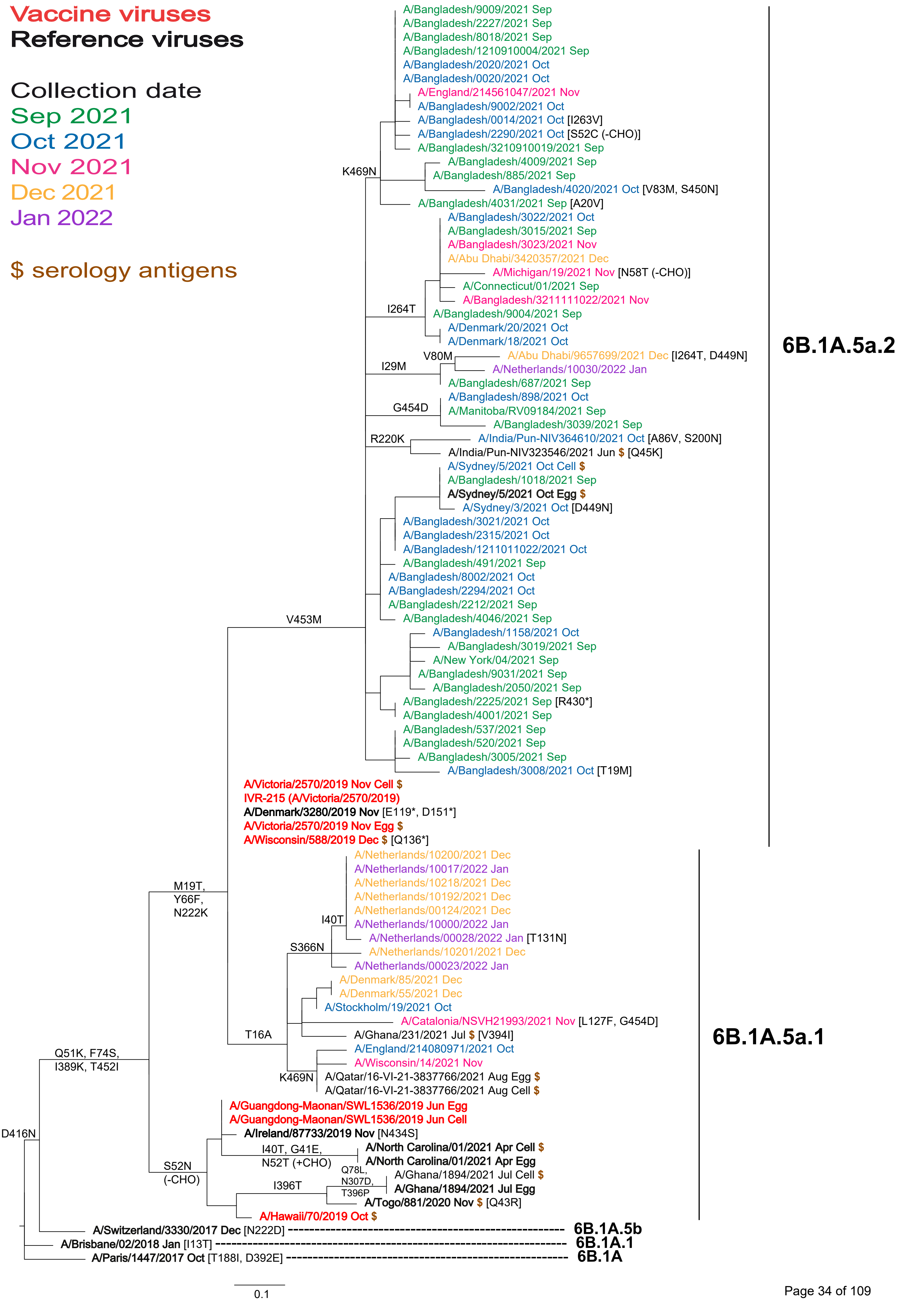


Figure 7. Phylogenetic comparison of A(H1N1)pdm09 NA genes (WIC sequences)

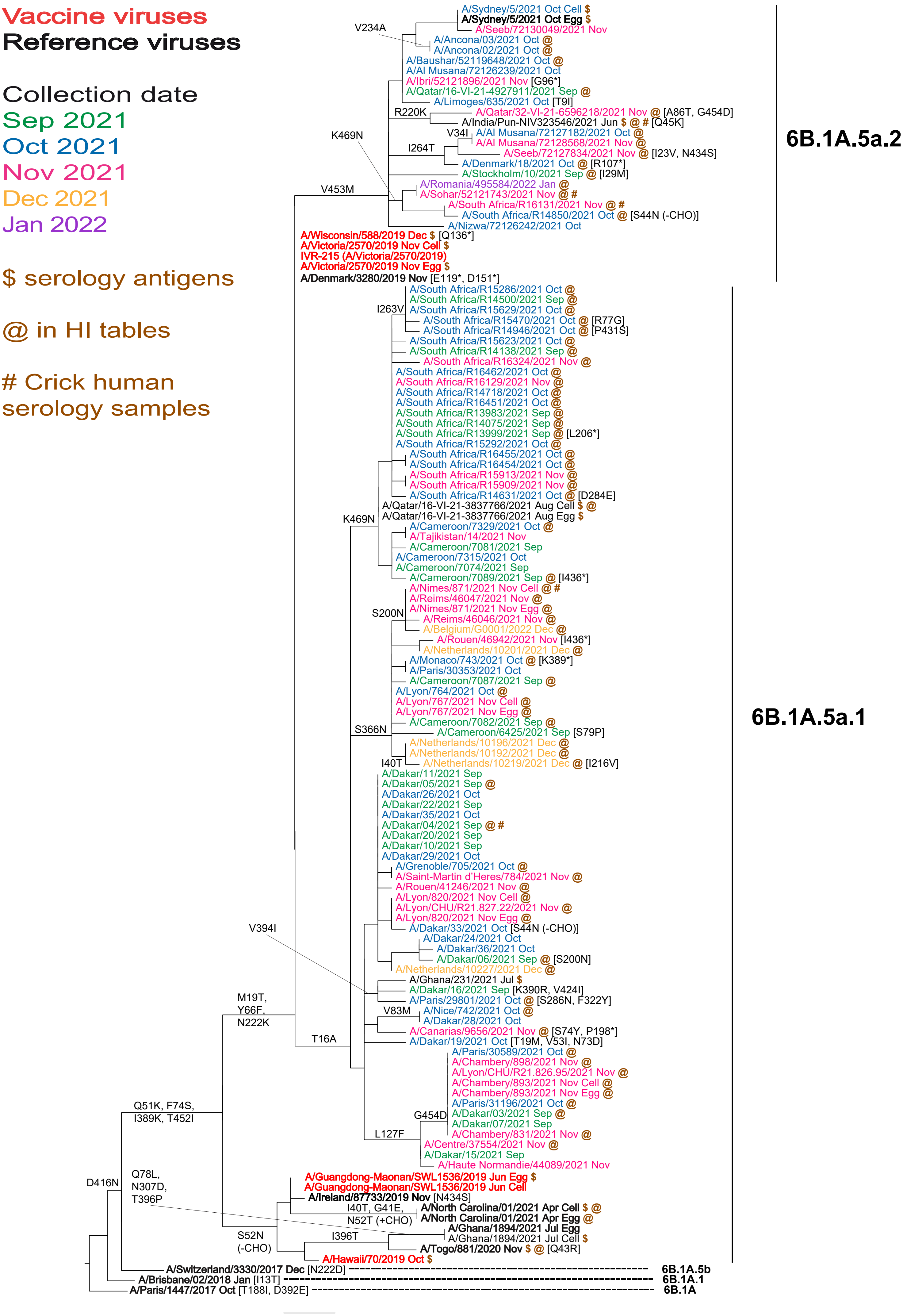


Figure 8. NA protein alignment for A(H1N1)pdm09 viruses

Vaccine virus: Reference virus: 6B Genetic group: Potential N-linked glycosylation site

```
                                             10        20        30        40        50        60        70        80        90       100
                                     ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Paris/1447/2017                    MNPNQKIITIGSICMTIGMANLILQIGNIISIWVSHSIQIGNQSQIETCNQSVITYENNTWVNQTYVNISNTNFAARQSVASVKLAGNSSLCPVSGWAIY 6B.1A
A/Denmark/3280/2019                  ..................T...............................K..............F.......S.......................... 6B.1A.5a.2
IVR-215 (A/Victoria/2570/2019)       ..................T...............................K..............F.......S.......................... 6B.1A.5a.2
A/Nizwa/72126242/2021                ..................T...............................K..............F.......S.......................... 6B.1A.5a.2
A/Qatar/32-VI-21-6596218/2021        ..................T...............................K..............F.......S...........T.............. 6B.1A.5a.2
A/South_Africa/R16131/2021           ..................T...............................K..............F.......S.......................... 6B.1A.5a.2
A/Romania/495584/2022                ..................T...............................K..............F.......S.......................... 6B.1A.5a.2
A/Al_Musana/72128568/2021            ..................T..............I................K..............F.......S.......................... 6B.1A.5a.2
A/Seeb/72127834/2021                 ..................T...V...........................K..............F.......S.......................... 6B.1A.5a.2
A/Stockholm/10/2021                  ..................T.........M.....................K..............F.......S.......................... 6B.1A.5a.2
A/Qatar/16-VI-21-4927911/2021        ..................T...............................K..............F.......S.......................... 6B.1A.5a.2
A/Denmark/18/2021                    ..................T...............................K..............F.......S.......................... 6B.1A.5a.2
A/Ibri/52121896/2021                 ..................T...............................K..............F.......S.....................X.... 6B.1A.5a.2
A/Limoges/635/2021                   ........I.........T...............................K..............F.......S.......................... 6B.1A.5a.2
A/Guangdong-Mao/SWL1536/2019E        ..................................................KN.....................S.......................... 6B.1A.5a.1
A/Hawaii/70/2019                     ..................................................KN.....................S.......................... 6B.1A.5a.1
A/Netherlands/10219/2021             ...............A..T....................T..........K..............F.......S.......................... 6B.1A.5a.1
A/Reims/46046/2021                   ...............A..T...............................K..............F.......S.......................... 6B.1A.5a.1
A/Lyon/767/2021                      ...............A..T...............................K..............F.......S.......................... 6B.1A.5a.1
A/Cameroon/6425/2021                 ...............A..T...............................K..............F.......S....P..................... 6B.1A.5a.1
A/Cameroon/7081/2021                 ...............A..T...............................K..............F.......S.......................... 6B.1A.5a.1
A/Tajikistan/14/2021                 ...............A..T...............................K..............F.......S.......................... 6B.1A.5a.1
A/South_Africa/R16455/2021           ...............A..T...............................K..............F.......S.......................... 6B.1A.5a.1
A/South_Africa/R16129/2021           ...............A..T...............................K..............F.......S.......................... 6B.1A.5a.1
A/South_Africa/R15913/2021           ...............A..T...............................K..............F.......S.......................... 6B.1A.5a.1
A/Centre/37554/2021                  ...............A..T...............................K..............F.......S.......................... 6B.1A.5a.1
A/Dakar/07/2021                      ...............A..T...............................K..............F.......S.......................... 6B.1A.5a.1
A/Chambery/893/2021                  ...............A..T...............................K..............F.......S.......................... 6B.1A.5a.1
A/Paris/29801/2021                   ...............A..T...............................K..............F.......S.......................... 6B.1A.5a.1
A/Canarias/9656/2021                 ...............A..T...............................K..............F.......Y.......................... 6B.1A.5a.1
A/Dakar/36/2021                      ...............A..T...............................K..............F.......S.......................... 6B.1A.5a.1
A/Netherlands/10227/2021             ...............A..T...............................K..............F.......S.......................... 6B.1A.5a.1

                                            110       120       130       140       150       160       170       180       190       200
                                     ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Paris/1447/2017                    SKDNSVRIGSKGDVFVIREPFISCSPLECRTFFLTQGALLNDKHSNGTIKDRSPYRTLMSCPIGEVPSPYNSRFESVAWSASACHDGINWLTIGISGPDS
A/Denmark/3280/2019                  ..................X...............................X....................................T............
IVR-215 (A/Victoria/2570/2019)       .......................................................................................T............
A/Nizwa/72126242/2021                .......................................................................................T............
A/Qatar/32-VI-21-6596218/2021        .......................................................................................T............
A/South_Africa/R16131/2021           .......................................................................................T............
A/Romania/495584/2022                .......................................................................................T............
A/Al_Musana/72128568/2021            .......................................................................................T............
A/Seeb/72127834/2021                 .......................................................................................T............
A/Stockholm/10/2021                  .......................................................................................T............
A/Qatar/16-VI-21-4927911/2021        .......................................................................................T............
A/Denmark/18/2021                    ......X................................................................................T............
A/Ibri/52121896/2021                 .......................................................................................T............
A/Limoges/635/2021                   .......................................................................................T............
A/Guangdong-Mao/SWL1536/2019E        .......................................................................................T............
A/Hawaii/70/2019                     .......................................................................................T............
A/Netherlands/10219/2021             .......................................................................................T............
A/Reims/46046/2021                   .......................................................................................T...........N
A/Lyon/767/2021                      .......................................................................................T............
A/Cameroon/6425/2021                 .......................................................................................T............
A/Cameroon/7081/2021                 .......................................................................................T............
A/Tajikistan/14/2021                 .......................................................................................T............
A/South_Africa/R16455/2021           .......................................................................................T............
A/South_Africa/R16129/2021           .......................................................................................T............
A/South_Africa/R15913/2021           .......................................................................................T............
A/Centre/37554/2021                  ..........................F............................................................T............
A/Dakar/07/2021                      ..........................F............................................................T............
A/Chambery/893/2021                  ..........................F............................................................T............
A/Paris/29801/2021                   .......................................................................................T............
A/Canarias/9656/2021                 .......................................................................................T.........X..
A/Dakar/36/2021                      .......................................................................................T............
A/Netherlands/10227/2021             .......................................................................................T............

                                            210       220       230       240       250       260       270       280       290       300
                                     ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Paris/1447/2017                    GAVAVLKYNGIITDTIKSWRNNILRTQESECACVNGSCFTIMTDGPSDGQASYKIFRIEKGKIIKSVEMKAPNYHYEECSCYPDSSEITCVCRDNWHGSN
A/Denmark/3280/2019                  .....................K..............................................................................
IVR-215 (A/Victoria/2570/2019)       .....................K..............................................................................
A/Nizwa/72126242/2021                .....................K..............................................................................
A/Qatar/32-VI-21-6596218/2021        ...................K.K..............................................................................
A/South_Africa/R16131/2021           .....................K..............................................................................
A/Romania/495584/2022                .....................K..............................................................................
A/Al_Musana/72128568/2021            .....................K.........................................T....................................
A/Seeb/72127834/2021                 .....................K.........................................T....................................
A/Stockholm/10/2021                  .....................K..............................................................................
A/Qatar/16-VI-21-4927911/2021        .....................K..............................................................................
A/Denmark/18/2021                    .....................K.........................................T....................................
A/Ibri/52121896/2021                 .....................K..............................................................................
A/Limoges/635/2021                   .....................K..............................................................................
A/Guangdong-Mao/SWL1536/2019E        ....................................................................................................
A/Hawaii/70/2019                     ....................................................................................................
A/Netherlands/10219/2021             ...............V.....K..............................................................................
A/Reims/46046/2021                   .....................K..............................................................................
A/Lyon/767/2021                      .....................K..............................................................................
A/Cameroon/6425/2021                 .....................K.................................................................X............
A/Cameroon/7081/2021                 .....................K..............................................................................
A/Tajikistan/14/2021                 .....................K..............................................................................
A/South_Africa/R16455/2021           .....................K..............................................................................
A/South_Africa/R16129/2021           .....................K..............................................................................
A/South_Africa/R15913/2021           .....................K..............................................................................
A/Centre/37554/2021                  .....................K..............................................................................
A/Dakar/07/2021                      .....................K..............................................................................
A/Chambery/893/2021                  .....................K..............................................................................
A/Paris/29801/2021                   .....................K...............................................................N..............
A/Canarias/9656/2021                 .....................K..............................................................................
A/Dakar/36/2021                      .....................K..............................................................................
A/Netherlands/10227/2021             .....................K..............................................................................
```

```
                                       310       320       330       340       350       360       370       380       390       400
                                 ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Paris/1447/2017                RPWVSFNQNLEYQMGYICSGVFGDNPRPNDKTGSCGPVSSNGANGVKGFSFKYGNGVWIGRTKSISSRKGFEMIWDPNGWTGTDNKFSIKQEIVGINEWS 6B.1A
A/Denmark/3280/2019              ........................................................................................K..D........ 6B.1A.5a.2
IVR-215 (A/Victoria/2570/2019)   ........................................................................................K..D........ 6B.1A.5a.2
A/Nizwa/72126242/2021            ........................................................................................K..D........ 6B.1A.5a.2
A/Qatar/32-VI-21-6596218/2021    ........................................................................................K..D........ 6B.1A.5a.2
A/South_Africa/R16131/2021       ........................................................................................K..D........ 6B.1A.5a.2
A/Romania/495584/2022            ........................................................................................K..D........ 6B.1A.5a.2
A/Al_Musana/72128568/2021        ........................................................................................K..D........ 6B.1A.5a.2
A/Seeb/72127834/2021             ........................................................................................K..D........ 6B.1A.5a.2
A/Stockholm/10/2021              ........................................................................................K..D........ 6B.1A.5a.2
A/Qatar/16-VI-21-4927911/2021    ........................................................................................K..D........ 6B.1A.5a.2
A/Denmark/18/2021                ........................................................................................K..D........ 6B.1A.5a.2
A/Ibri/52121896/2021             ........................................................................................K..D........ 6B.1A.5a.2
A/Limoges/635/2021               ........................................................................................K..D........ 6B.1A.5a.2
A/Guangdong-Mao/SWL1536/2019E    ........................................................................................K..D........ 6B.1A.5a.1
A/Hawaii/70/2019                 ........................................................................................K..D........ 6B.1A.5a.1
A/Netherlands/10219/2021         .................................................................N......................K..D........ 6B.1A.5a.1
A/Reims/46046/2021               .................................................................N......................K..D........ 6B.1A.5a.1
A/Lyon/767/2021                  .................................................................N......................K..D........ 6B.1A.5a.1
A/Cameroon/6425/2021             .................................................................N......................K..D........ 6B.1A.5a.1
A/Cameroon/7081/2021             ........................................................................................K..D........ 6B.1A.5a.1
A/Tajikistan/14/2021             ........................................................................................K..D........ 6B.1A.5a.1
A/South_Africa/R16455/2021       ........................................................................................K..D........ 6B.1A.5a.1
A/South_Africa/R16129/2021       ........................................................................................K..D........ 6B.1A.5a.1
A/South_Africa/R15913/2021       ........................................................................................K..D........ 6B.1A.5a.1
A/Centre/37554/2021              ........................................................................................K..D........ 6B.1A.5a.1
A/Dakar/07/2021                  ........................................................................................K..D........ 6B.1A.5a.1
A/Chambery/893/2021              ........................................................................................K..D........ 6B.1A.5a.1
A/Paris/29801/2021               .....................Y..................................................................K..D.I...... 6B.1A.5a.1
A/Canarias/9656/2021             ........................................................................................K..D........ 6B.1A.5a.1
A/Dakar/36/2021                  ........................................................................................K..D........ 6B.1A.5a.1
A/Netherlands/10227/2021         ........................................................................................K..D........ 6B.1A.5a.1

                                       410       420       430       440       450       460
                                 ....|....|....|....|....|....|....|....|....|....|....|....|....|....
A/Paris/1447/2017                GYSGSFVQHPELTGLDCIRPCFWVELIRGRPEENTIWTSGSSISFCGVDSDTVGWSWPDGAELPFTIDK
A/Denmark/3280/2019              ...............N...................................I.................
IVR-215 (A/Victoria/2570/2019)   ...............N...................................I.................
A/Nizwa/72126242/2021            ...............N...................................IM................
A/Qatar/32-VI-21-6596218/2021    ...............N...................................IMD...............
A/South_Africa/R16131/2021       ...............N...................................IM...............N
A/Romania/495584/2022            ...............N...................................IM...............N
A/Al_Musana/72128568/2021        ...............N...................................IM................
A/Seeb/72127834/2021             ...............N.................S.................IM................
A/Stockholm/10/2021              ...............N...................................IM................
A/Qatar/16-VI-21-4927911/2021    ...............N...................................IM................
A/Denmark/18/2021                ...............N...................................IM................
A/Ibri/52121896/2021             ...............N...................................IM................
A/Limoges/635/2021               ...............N...................................IM................
A/Guangdong-Mao/SWL1536/2019E    ...............N...................................I.................
A/Hawaii/70/2019                 ...............N...................................I.................
A/Netherlands/10219/2021         ...............N...................................I.................
A/Reims/46046/2021               ...............N...................................I.................
A/Lyon/767/2021                  ...............N...................................I.................
A/Cameroon/6425/2021             ...............N...................................I.................
A/Cameroon/7081/2021             ...............N...................................I................N
A/Tajikistan/14/2021             ...............N...................................I................N
A/South_Africa/R16455/2021       ...............N...................................I................N
A/South_Africa/R16129/2021       ...............N...................................I................N
A/South_Africa/R15913/2021       ...............N...................................I................N
A/Centre/37554/2021              ...............N...................................I.................
A/Dakar/07/2021                  ...............N...................................I.D...............
A/Chambery/893/2021              ...............N...................................I.D...............
A/Paris/29801/2021               ...............N...................................I.................
A/Canarias/9656/2021             ...............N...................................I.................
A/Dakar/36/2021                  ...............N...................................I.................
A/Netherlands/10227/2021         ...............N...................................I.................
```

Vaccine virus: Reference virus: 6B Genetic group: Potential N-linked glycosylation site

Table 6-1. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2021-09-14)

					Haemagglutination inhibition titre							
					Post-infection ferret antisera							
Viruses	Other information		Collection date	Passage history	A/Paris 1447/17	A/Bris 02/18	A/Swit 3330/17	A/Ire 87733/19	A/G-M SWL1536/19	A/G-M SWL1536/19	A/Denmark 3280/19	A/Vict 2570/19
		Passage history			MDCK	Egg	Egg	Egg	Egg	MDCK	MDCK	Egg
		Ferret number			F03/18[*2]	F09/19[*1]	F23/18[*1]	St Jude's F18/20[*1]	F12/20[*1]	F09/20[*1]	F08/20[*1]	F26/20[*1]
		Genetic group			6B.1A	6B.1A.1	6B.1A.5b	6B.1A.5a.1	6B.1A.5a.1	6B.1A.5a.1	6B.1A.5a.2	6B.1A.5a.2
REFERENCE VIRUSES												
A/Paris/1447/2017		6B.1A	2017-10-20	MDCK1/MDCK3	2560	320	320	320	1280	640	<	<
A/Brisbane/02/2018		6B.1A.1	2018-01-04	E3/E2	2560	640	1280	1280	2560	2560	40	160
A/Switzerland/3330/2017	clone 35	6B.1A.5b	2017-12-20	E6/E2	1280	320	1280	320	1280	320	<	40
A/Ireland/87733/2019		6B.1A.5a.1	2019-11-03	E4	2560	640	1280	640	2560	2560	40	160
A/Guangdong-Maonan/SWL1536/2019		6B.1A.5a.1	2019-06-17	E3/E2	1280	320	320	320	1280	640	<	80
A/Guangdong-Maonan/SWL1536/2019		6B.1A.5a.1	2019-06-17	C2/MDCK1	1280	640	640	640	1280	1280	<	80
A/Denmark/3280/2019		6B.1A.5a.2	2019-11-10	MDCK4/MDCK5	80	<	80	40	40	160	1280	1280
A/Victoria/2570/2019		6B.1A.5a.2	2018-11-22	E4/E2	80	40	320	80	80	160	1280	1280
IVR-215 (A/Victoria/2570/2019)		6B.1A.5a.2	2018-11-22	E4/D7/E1	<	40	160	40	ND	160	1280	2560
TEST VIRUSES												
A/Victoria/2454/2019		6B.1A.5a.1	2019-08-28	E4/E1	1280	160	160	320	640	640	<	40
A/Ghana/97/2021		6B.1A.5a.1	2021-05-27	MDCK1	1280	320	640	640	1280	1280	40	80
A/Ghana/1243/2021		6B.1A.5a.1	2021-06-29	MDCK2	2560	640	640	1280	2560	2560	40	160
A/Qatar/16-VI-21-3837766/2021		6B.1A.5a.1	2021-08-04	MDCK2	1280	640	640	640	2560	2560	40	80
A/Qatar/16-VI-21-3837777/2021		6B.1A.5a.1	2021-08-04	MDCK1	1280	640	640	640	2560	2560	40	160
						Vaccine NH 2019-20 SH 2020			Vaccine NH 2020-21			Vaccine SH 2021 NH 2021-22 SH 2022

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 < = <40; 2 < = <80; ND = Not Done

Sequences in phylogenetic trees

< 4-fold | 4-fold | 8-fold | > 8-fold | < not recognised by the antiserum | ≥ 160 (when homologous titre ≥ 2560) | ≥ 160 (no homologous titre)

Table 6-2. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2021-09-29)

Haemagglutination inhibition titre — Post-infection ferret antisera

Viruses	Other information	Passage history / Ferret number / Genetic group	Collection date	Passage history	A/Paris 1447/17 MDCK F03/18[*2] 6B.1A	A/Bris 02/18 Egg F09/19[*1] 6B.1A.1	A/Swit 3330/17 Egg F23/18*[1] 6B.1A.5b	A/Ire 87733/19 Egg St Jude's F18/20[*1] 6B.1A.5a.1	A/G-M SWL1536/19 Egg F12/20[*1] 6B.1A.5a.1	A/G-M SWL1536/19 MDCK F09/20[*1] 6B.1A.5a.1	A/Denmark 3280/19 MDCK F08/20[*1] 6B.1A.5a.2	A/Vict 2570/19 Egg F26/20[*1] 6B.1A.5a.2	A/Togo 881/20 MDCK CDC F2021-057[*1] 6B.1A.5a.1	A/Nor Carol 01/21 Egg CDC F2021-103[*1] 6B.1A.5a.1	A/Nor Carol 01/21 MDCK CDC F2021-105[*1] 6B.1A.5a.1
REFERENCE VIRUSES															
A/Paris/1447/2017		6B.1A	2017-10-20	MDCK1/MDCK3	2560	320	320	320	1280	640	<	<	320	80	80
A/Brisbane/02/2018		6B.1A.1	2018-01-04	E3/E2	2560	1280	1280	1280	2560	2560	40	160	640	80	80
A/Switzerland/3330/2017	clone 35	6B.1A.5b	2017-12-20	E6/E2	1280	320	1280	320	1280	320	<	40	320	80	80
A/Ireland/87733/2019		6B.1A.5a.1	2019-11-03	E4	2560	640	1280	640	2560	2560	40	160	640	320	160
A/Guangdong-Maonan/SWL1536/2019		6B.1A.5a.1	2019-06-17	E3/E2	1280	320	320	320	1280	640	<	80	320	160	80
A/Guangdong-Maonan/SWL1536/2019		6B.1A.5a.1	2019-06-17	C2/MDCK1	1280	640	640	640	1280	1280	<	80	640	320	320
A/Denmark/3280/2019		6B.1A.5a.2	2019-11-10	MDCK4/MDCK5	80	<	80	40	40	160	1280	1280	<	40	40
IVR-215 (A/Victoria/2570/2019)		6B.1A.5a.2	2018-11-22	E4/D7/E2	<	40	160	40	ND	160	1280	2560	80	160	160
TEST VIRUSES															
A/Togo/837/2020	I166T, A186T	6B.1A.5a.1	2020-11-04	E2/E1	1280	320	320	320	1280	1280	<	40	320	160	160
A/Togo/881/2020	I166T, A186T	6B.1A.5a.1	2020-11-14	SIAT3/MDCK1	640	320	320	160	640	640	<	40	320	320	160
A/North Carolina/01/2021	G155E, A187S	6B.1A.5a.1	2021-04-04	E2/E1	80	40	40	40	80	320	<	40	80	640	320
A/North Carolina/01/2021	G155E, A187S	6B.1A.5a.1	2021-04-04	SIAT3/MDCK1	80	80	40	40	80	320	<	40	160	640	320

[1] Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)
1 < = <40; 2 < = <80; ND = Not Done

Vaccine NH 2019-20 SH 2020 (A/Bris 02/18)

Vaccine NH 2020-21 (A/G-M SWL1536/19 Egg)

Vaccine SH 2021 NH 2021-22 SH 2022 (A/Vict 2570/19)

Sequences in phylogenetic trees

< 4-fold; 4-fold; 8-fold; > 8-fold; < not recognised by the antiserum; ≥ 160 (when homologous titre ≥ 2560); ≥ 160 (no homologous titre)

Table 6-3. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2021-12-17)

					Haemagglutination inhibition titre							
					Post-infection ferret antisera							
Viruses	Other information		Collection date	Passage history	A/Paris 1447/17	A/Bris 02/18	A/Swit 3330/17	A/Ire 87733/19	A/G-M SWL1536/19	A/G-M SWL1536/19	A/Denmark 3280/19	IVR-215 A/Vic/2570/19
		Passage history			MDCK	Egg	Egg	Egg	Egg	MDCK	MDCK	Egg
		Ferret number			F03/18[*2]	F09/19[*1]	F23/18[*1]	St Jude's F18/20[*1]	F12/20[*1]	F09/20[*1]	F08/20[*1]	F37/21[*1]
		Genetic group			6B.1A	6B.1A.1	6B.1A.5b	6B.1A.5a.1	6B.1A.5a.1	6B.1A.5a.1	6B.1A.5a.2	6B.1A.5a.2
REFERENCE VIRUSES												
A/Paris/1447/2017		6B.1A	2017-10-20	MDCK1/MDCK3	2560	1280	1280	1280	2560	2560	40	80
A/Brisbane/02/2018		6B.1A.1	2018-01-04	E3/E2	2560	1280	640	640	2560	1280	40	160
A/Switzerland/3330/2017	clone 35	6B.1A.5b	2017-12-20	E6/E2	1280	320	640	320	1280	1280	<	80
A/Ireland/87733/2019		6B.1A.5a.1	2019-11-03	E4	1280	640	640	640	1280	1280	<	80
A/Guangdong-Maonan/SWL1536/2019		6B.1A.5a.1	2019-06-17	E3/E2	640	320	320	640	1280	1280	40	80
A/Guangdong-Maonan/SWL1536/2019		6B.1A.5a.1	2019-06-17	C2/MDCK1	1280	640	640	1280	2560	2560	40	80
A/Denmark/3280/2019		6B.1A.5a.2	2019-11-10	MDCK4/MDCK5	160	40	80	80	80	160	1280	2560
IVR-215 (A/Victoria/2570/2019)		6B.1A.5a.2	2018-11-22	E4/D7/E1	<	80	40	80	160	160	1280	2560
TEST VIRUSES												
A/Baushar/52119648/2021	A73E, D94N, S157T, K209R, T216A	6B.1A.5a.2	2021-10-10	SIAT1/MDCK1	<	<	<	<	<	<	1280	640
A/Seeb/72127834/2021	T216A	6B.1A.5a.2	2021-11-01	SIAT1/MDCK1	<	<	<	<	40	40	640	1280
A/Al Musana/72128568/2021	T216A	6B.1A.5a.2	2021-11-14	SIAT1/MDCK1	<	<	<	40	40	40	1280	1280
A/Denmark/18/2021	A73E, D94N, D129N, K209R, T216A	6B.1A.5a.2	2021-10-21	MDCK2/MDCK1	<	<	40	40	40	40	320	640
A/Al Musana/72127182/2021	T216A	6B.1A.5a.2	2021-10-24	SIAT1/MDCK1	<	<	40	<	40	40	640	1280
A/Sohar/52121743/2021		6B.1A.5a.2	2021-11-09	SIAT1/MDCK1	<	<	40	80	80	80	2560	2560
						Vaccine NH 2019-20 SH 2020			Vaccine NH 2020-21			Vaccine SH 2021 NH 2021-22 SH 2022

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)
1 < = <40; 2 < = <80; ND = Not Done

Sequences in phylogenetic trees

< 4-fold | 4-fold | 8-fold | > 8-fold | < not recognised by the antiserum | ≥ 160 (when homologous titre ≥ 2560) | ≥ 160 (no homologous titre)

Table 6-4. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2022-01-18)

Haemagglutination inhibition titre — Post-infection ferret antisera

Viruses	Other information	Genetic group	Collection date	Passage history	A/Paris 1447/17 MDCK F03/18[*2] 6B.1A	A/Bris 02/18 Egg F09/19[*1] 6B.1A.1	A/Swit 3330/17 Egg F23/18[*1] 6B.1A.5b	A/Ire 87733/19 Egg St Jude's F18/20[*1] 6B.1A.5a.1	A/G-M SWL1536/19 Egg F12/20[*1] 6B.1A.5a.1	A/G-M SWL1536/19 MDCK F09/20[*1] 6B.1A.5a.1	A/Denmark 3280/19 MDCK F08/20[*1] 6B.1A.5a.2	IVR-215 A/Vic/2570/19 Egg F37/21[*1] 6B.1A.5a.2
REFERENCE VIRUSES												
A/Paris/1447/2017		6B.1A	2017-10-20	MDCK1/MDCK3	1280	640	640	640	1280	1280	<	40
A/Brisbane/02/2018		6B.1A.1	2018-01-04	E3/E2	2560	1280	1280	1280	2560	2560	<	160
A/Switzerland/3330/2017	clone 35	6B.1A.5b	2017-12-20	E6/E2	640	640	1280	640	1280	1280	<	80
A/Ireland/87733/2019		6B.1A.5a.1	2019-11-03	E4	1280	640	1280	1280	2560	2560	<	160
A/Guangdong-Maonan/SWL1536/2019		6B.1A.5a.1	2019-06-17	E3/E2	1280	640	640	640	2560	2560	<	160
A/Guangdong-Maonan/SWL1536/2019		6B.1A.5a.1	2019-06-17	C2/MDCK1	1280	640	640	1280	2560	2560	<	160
A/Denmark/3280/2019		6B.1A.5a.2	2019-11-10	MDCK4/MDCK5	80	40	80	80	160	160	1280	2560
IVR-215 (A/Victoria/2570/2019)		6B.1A.5a.2	2018-11-22	E4/D7/E2	<	40	40	80	80	160	1280	2560
TEST VIRUSES												
A/Limoges/635/2021	A73E, D94N, S157T, K209R, T216A	6B.1A.5a.2	2021-10-06	MDCK2/MDCK1	<	<	<	<	<	40	640	1280
A/South Africa/R14850/2021	K142R, D260E	6B.1A.5a.2	2021-10-11	MDCK2/MDCK1	<	<	<	<	<	<	640	1280
A/South Africa/R16131/2021	K142R, D260E, H296N	6B.1A.5a.2	2021-11-08	MDCK1/MDCK1	<	<	<	<	40	40	1280	1280
A/Stockholm/10/2021	D94N, T216A	6B.1A.5a.2	2021-09-13	SIAT1/MDCK1	<	<	40	40	40	40	1280	2560
A/South Africa/R16022/2021	K142R, D260E, V321I	6B.1A.5a.2	2021-11-04	MDCK1/MDCK1	80	80	80	80	160	80	320	640
A/South Africa/R13474/2021		no sequence	2021-09-08	MDCK1/MDCK1	1280	640	640	640	2560	2560	<	80
A/South Africa/R13999/2021	R113K	6B.1A.5a.1	2021-09-17	MDCK1/MDCK1	2560	640	640	1280	2560	2560	<	80
A/South Africa/R13983/2021	R113K	6B.1A.5a.1	2021-09-17	MDCK1/MDCK1	2560	640	1280	1280	2560	2560	<	80
A/South Africa/R14075/2021	R113K	6B.1A.5a.1	2021-09-20	MDCK1/MDCK1	1280	320	640	640	1280	1280	40	80
A/South Africa/R14138/2021	R113K, I266X	6B.1A.5a.1	2021-09-21	MDCK1/MDCK1	1280	320	640	640	1280	2560	<	80
A/South Africa/R14500/2021	E68G, R113K	6B.1A.5a.1	2021-09-30	MDCK1/MDCK1	1280	640	640	1280	2560	2560	<	80
A/South Africa/R14718/2021	R113K	6B.1A.5a.1	2021-10-07	MDCK1/MDCK1	1280	640	640	1280	2560	1280	<	80
A/South Africa/R14946/2021	R113K	6B.1A.5a.1	2021-10-13	MDCK1/MDCK1	2560	640	640	1280	2560	2560	<	80
A/Paris/30589/2021	S83F, T164X(-CHO)	6B.1A.5a.1	2021-10-18	MDCK2/MDCK1	1280	640	640	1280	2560	2560	<	160
A/South Africa/R16462/2021	R113K	6B.1A.5a.1	2021-10-18	MDCK2/MDCK1	1280	640	640	640	2560	1280	<	80
A/South Africa/R16451/2021	R113K	6B.1A.5a.1	2021-10-18	MDCK2/MDCK1	1280	320	320	320	1280	640	<	80
A/Paris/31196/2021	S83F	6B.1A.5a.1	2021-10-19	MDCK2/MDCK1	2560	640	640	640	2560	2560	<	80
A/South Africa/R15292/2021	R113K, N228S	6B.1A.5a.1	2021-10-19	MDCK1/MDCK1	2560	640	1280	1280	2560	1280	<	160
A/South Africa/R15286/2021	R113K	6B.1A.5a.1	2021-10-19	MDCK1/MDCK1	1280	640	640	640	1280	1280	<	80
A/South Africa/R15470/2021	R113K	6B.1A.5a.1	2021-10-21	MDCK1/MDCK1	2560	640	640	640	2560	1280	<	80
A/South Africa/R15623/2021	R113K	6B.1A.5a.1	2021-10-25	MDCK1/MDCK1	1280	320	640	640	1280	1280	<	80
A/South Africa/R16454/2021	R113K, K119R, N125S	6B.1A.5a.1	2021-10-25	MDCK1/MDCK1	1280	320	640	640	1280	1280	<	80
A/Monaco/743/2021	V321I	6B.1A.5a.1	2021-10-26	MDCK2/MDCK1	1280	640	640	640	2560	1280	40	80
A/Lyon/764/2021	V321I	6B.1A.5a.1	2021-10-27	MDCK2/MDCK1	1280	320	320	640	1280	1280	<	80
A/South Africa/R15629/2021	R113K	6B.1A.5a.1	2021-10-27	MDCK1/MDCK1	1280	320	640	640	2560	640	<	80
A/Lyon/767/2021	A187X, V321I	6B.1A.5a.1	2021-11-01	MDCK2/MDCK1	1280	320	640	640	1280	1280	<	80
A/South Africa/R15913/2021	N38D, R113K	6B.1A.5a.1	2021-11-03	MDCK1/MDCK1	1280	640	640	640	2560	2560	<	80
A/South Africa/R15909/2021	N38D, R113K	6B.1A.5a.1	2021-11-03	MDCK1/MDCK1	1280	320	640	640	2560	1280	<	80
A/South Africa/R16324/2021	N38D, R113K	6B.1A.5a.1	2021-11-10	MDCK1/MDCK1	1280	640	640	640	2560	1280	<	80
A/Chambery/831/2021	S83F	6B.1A.5a.1	2021-11-16	MDCK2/MDCK1	1280	320	640	640	1280	1280	<	80
A/Chambery/898/2021	S83F	6B.1A.5a.1	2021-11-19	MDCK2/MDCK1	1280	640	640	1280	2560	1280	<	80
A/Nimes/871/2021	A261T	6B.1A.5a.1	2021-11-19	MDCK3/MDCK1	1280	320	320	640	1280	1280	<	80
A/Reims/46047/2021	D129N	6B.1A.5a.1	2021-11-28	MDCK1/MDCK1	1280	640	640	640	2560	1280	<	80
A/South Africa/R14631/2021	R113K	6B.1A.5a.1	2021-10-05	MDCK1/MDCK1	640	160	320	320	1280	1280	<	40
A/Centre/37554/2021	D35N	6B.1A.5a.1	2021-11-02	MDCK2/MDCK1	640	640	640	640	1280	1280	<	160
A/South Africa/R16129/2021	R113K, R259K	6B.1A.5a.1	2021-11-03	MDCK1/MDCK1	640	320	640	640	1280	1280	<	80
A/Lyon/CHU/R21.826.95/2021	S83F	6B.1A.5a.1	2021-11-19	MDCK2/MDCK1	640	320	320	640	1280	1280	<	80
A/Reims/46046/2021	D129N	6B.1A.5a.1	2021-11-21	MDCK2/MDCK1	640	320	640	320	1280	1280	<	40
A/Chambery/893/2021	S83F	6B.1A.5a.1	2021-11-23	MDCK2/MDCK1	640	320	320	320	1280	1280	<	80
A/Grenoble/705/2021	D127X, S128T, P137S, G155E	6B.1A.5a.1	2021-10-23	MDCK2/MDCK1	320	160	160	320	320	640	<	80
A/Nice/742/2021	P137S, G155E, T164X(-CHO), K308E	6B.1A.5a.1	2021-10-24	MDCK2/MDCK1	160	80	80	160	320	640	<	80
A/Saint-Martin d'Heres/784/2021	D127X, S128T, P137S, G155E	6B.1A.5a.1	2021-11-02	MDCK2/MDCK1	160	160	160	160	160	640	<	80
A/Canarias/9656/2021	R113K, P137S, G155E, I298V, L314M	6B.1A.5a.1	2021-11-08	SIAT1/MDCK1	160	80	80	160	160	320	<	80
A/Rouen/41246/2021	S128T, P137S, G155E, E224D	6B.1A.5a.1	2021-11-15	MDCK2/MDCK1	160	80	160	160	160	320	<	80
A/Lyon/820/2021	S128T, P137S, G155E, N162X(-CHO), A187X	6B.1A.5a.1	2021-11-16	MDCK2/MDCK1	160	80	80	160	160	640	<	80
A/Lyon/CHU/R21.827.22/2021	S128T, D129G, P137S, G155E	6B.1A.5a.1	2021-11-20	MDCK2/MDCK1	160	80	80	160	160	320	<	40
A/Paris/29801/2021	P137S, G155E, T216K, P271L	6B.1A.5a.1	2021-10-10	MDCK1/MDCK1	80	80	80	80	160	320	<	40
A/South Africa/R16455/2021	R113K, V152I, G155E, N162X(-CHO)	6B.1A.5a.1	2021-10-22	MDCK1/MDCK1	<	40	40	80	160	320	<	80
						Vaccine NH 2019-20 SH 2020			Vaccine NH 2020-21			Vaccine SH 2021 NH 2021-22 SH 2022

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)
1 < = <40; 2 < = <80; ND = Not Done

Sequences in phylogenetic trees

< 4-fold | 4-fold | 8-fold | > 8-fold | < not recognised by the antiserum | ≥ 160 (when homologous titre ≥ 2560) | ≥ 160 (no homologous titre)

Table 6-5. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2022-02-01)

Viruses	Other information		Collection date	Passage history	A/Paris 1447/17	A/Bris 02/18	A/Swit 3330/17	A/Ire 87733/19	A/G-M SWL1536/19	A/G-M SWL1536/19	NEW A/Ghana 1894/21	A/Denmark 3280/19	IVR-215 A/Vic/2570/19	NEW A/Sydney 5/21
		Haemagglutination inhibition titre — Post-infection ferret antisera												
		Passage history			MDCK	Egg	Egg	Egg	Egg	MDCK	Egg	MDCK	Egg	Egg
		Ferret number			F03/18*2	F09/19*1	F23/18*1	St Jude's F18/20*1	F12/20*1	F09/20*1	F02/22*1	F08/20*1	F37/21*1	F04/22*1
		Genetic group			6B.1A	6B.1A.1	6B.1A.5b	6B.1A.5a.1	6B.1A.5a.1	6B.1A.5a.1	6B.1A.5a.1	6B.1A.5a.2	6B.1A.5a.2	6B.1A.5a.2
REFERENCE VIRUSES														
A/Paris/1447/2017		6B.1A	2017-10-20	MDCK1/MDCK3	2560	1280	640	640	2560	1280	640	<	80	40
A/Brisbane/02/2018		6B.1A.1	2018-01-04	E3/E2	2560	1280	640	1280	2560	2560	1280	40	160	80
A/Switzerland/3330/2017	clone 35	6B.1A.5b	2017-12-20	E6/E2	1280	640	640	640	1280	1280	640	<	80	80
A/Ireland/87733/2019		6B.1A.5a.1	2019-11-03	E4	2560	1280	640	1280	2560	2560	1280	<	160	80
A/Guangdong-Maonan/SWL1536/2019		6B.1A.5a.1	2019-06-17	E3/E2	2560	640	640	1280	2560	2560	1280	<	160	80
A/Guangdong-Maonan/SWL1536/2019		6B.1A.5a.1	2019-06-17	C2/MDCK1	1280	640	320	640	2560	1280	1280	<	80	80
A/Ghana/1894/2021		6B.1A.5a.1	2021-07-21	E2	2560	1280	640	1280	>5120	>5120	5120	40	320	160
A/Denmark/3280/2019		6B.1A.5a.2	2019-11-10	MDCK4/MDCK5	160	80	40	80	80	160	40	2560	2560	2560
IVR-215 (A/Victoria/2570/2019)		6B.1A.5a.2	2018-11-22	E4/D7/E2	80	80	40	80	160	160	40	1280	2560	2560
A/Sydney/5/2021		6B.1A.5a.2	2021-10-16	E3/E1	80	40	40	40	80	80	80	640	1280	1280
TEST VIRUSES														
A/Lousiana/01/2020		6B.1A.7	2020-01-04	SIAT2/MDCK1	>5120	2560	>5120	2560	>5120	>5120	>5120	80	40	80
A/Maine/01/2021		6B.1A.5a1	2021-07-09	SIAT2/MDCK1	1280	640	320	640	1280	1280	1280	<	40	80
A/Qatar/16-VI-21-4927911/2021	A73E, D94N, K209R, T216A	6B.1A.5a.2	2021-09-14	MDCK1	<	<	<	<	40	40	<	640	1280	1280
A/Ancona/01/2021		no sequence	2021-10-19	MDCK3/MDCK1	<	<	<	<	<	<	<	640	1280	640
A/Ancona/02/2021	D94N, T216A, E258X	6B.1A.5a.2	2021-10-24	MDCK3/MDCK1	40	40	40	<	80	80	80	1280	2560	2560
A/Ancona/03/2021	D94N, T216A, E258X	6B.1A.5a.2	2021-10-26	MDCK3/MDCK1	40	40	40	40	40	40	<	1280	2560	2560
A/Qatar/32-VI-21-6596218/2021	A139N	6B.1A.5a.2	2021-11-24	MDCK1	<	<	40	40	40	40	40	1280	2560	2560
A/Romania/495584/2022		6B.1A.5a.2	2022-01-04	SIAT1/MDCK1	40	40	40	80	80	80	40	2560	>5120	>5120
A/Dakar/03/2021	D129X	6B.1A.5a.1	2021-09-21	MDCK1	2560	640	640	1280	2560	2560	1280	<	80	80
A/Cameroon/7087/2021		6B.1A.5a.1	2021-09-27	MDCK2	1280	640	640	640	1280	2560	1280	<	80	80
A/Cameroon/7089/2021		6B.1A.5a.1	2021-09-29	MDCK3	2560	1280	1280	1280	2560	2560	2560	<	80	80
A/Cameroon/7329/2021	R113K	6B.1A.5a.1	2021-10-15	MDCK2	2560	1280	1280	1280	2560	2560	2560	<	80	80
A/Nimes/871/2021	A261T	6B.1A.5a.1	2021-11-19	E1/E1	1280	640	640	1280	2560	1280	1280	<	160	80
A/Chambery/893/2021	S83F	6B.1A.5a.1	2021-11-23	E1/E1	1280	640	640	1280	2560	2560	2560	80	320	160
A/Netherlands/10192/2021	Y7X	6B.1A.5a.1	2021-12-19	MDCK-MIX2/MDCK1	2560	640	640	1280	2560	2560	2560	<	80	160
A/Netherlands/10196/2021	P218X	6B.1A.5a.1	2021-12-20	MDCK-MIX2/MDCK1	2560	1280	1280	1280	2560	2560	2560	<	160	80
A/Netherlands/10201/2021		6B.1A.5a.1	2021-12-26	MDCK-MIX2/MDCK1	2560	1280	1280	1280	2560	2560	2560	80	160	80
A/Netherlands/10219/2021	F200X, S326P	6B.1A.5a.1	2021-12-29	MDCK-MIX2/MDCK1	2560	1280	1280	2560	>5120	>5120	2560	40	160	160
A/Lyon/767/2021	A187X, V321I	6B.1A.5a.1	2021-11-01	E2/E1	640	320	320	640	1280	1280	640	<	80	80
A/Cameroon/7082/2021		6B.1A.5a.1	2021-09-27	MDCK2	320	160	160	160	640	320	640	<	<	40
A/Dakar/04/2021	S128T, P137S, G155E, A187X	6B.1A.5a.1	2021-09-20	MDCK1	320	160	160	160	320	640	640	<	80	80
A/Netherlands/10227/2021	D127X, S128T, P137S, G155E	6B.1A.5a.1	2021-12-30	MDCK-MIX2/MDCK1	320	160	160	160	320	640	640	<	80	80
A/Dakar/06/2021	P137S, I149V, G155E, A187X	6B.1A.5a.1	2021-09-21	MDCK1	160	80	80	80	160	320	160	<	<	40
A/Dakar/05/2021	S128T, P137S, G155E, L191X	6B.1A.5a.1	2021-09-24	MDCK1	160	80	80	80	160	320	320	<	<	40
A/Lyon/820/2021	S128T, P137S, G155E, N162X(-CHO), A187X	6B.1A.5a.1	2021-11-16	E1/E1	160	160	160	160	320	640	320	<	80	80

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)
1 < = <40; 2 < = <80; ND = Not Done
Titre >5120 taken as 5120 for heat-mapping

Vaccine NH 2019-20 SH 2020 (A/Bris 02/18)

Vaccine NH 2020-21 (A/G-M SWL1536/19 Egg)

Vaccine SH 2021 NH 2021-22 SH 2022 (IVR-215 A/Vic/2570/19)

Sequences in phylogenetic trees

< 4-fold | 4-fold | 8-fold | > 8-fold | < not recognised by the antiserum | ≥ 160 (when homologous titre ≥ 2560) | ≥ 160 (no homologous titre)

Table 6-6. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2022-02-08)

Viruses	Other information	Passage history / Ferret number / Genetic group	Collection date	Passage history	Haemagglutination inhibition titre — Post-infection ferret antisera: A/Paris 1447/17 MDCK F03/18[*2] 6B.1A	A/Bris 02/18 Egg F09/19[*1] 6B.1A.1	A/Swit 3330/17 Egg F23/18[*1] 6B.1A.5b	A/Ire 87733/19 Egg St Jude's F18/20[*1] 6B.1A.5a.1	A/G-M SWL1536/19 Egg F12/20[*1] 6B.1A.5a.1	A/G-M SWL1536/19 MDCK F09/20[*1] 6B.1A.5a.1	A/Ghana 1894/21 Egg F02/22[*1] 6B.1A.5a.1	A/N Carolina 01/21 Cell CDC F2021-105 6B.1A.5a.1	A/Denmark 3280/19 MDCK F08/20[*1] 6B.1A.5a.2	IVR-215 A/Vic/2570/19 Egg F37/21[*1] 6B.1A.5a.2	A/Sydney 5/21 Egg F04/22[*1] 6B.1A.5a.2
REFERENCE VIRUSES															
A/Paris/1447/2017		6B.1A	2017-10-20	MDCK1/MDCK3	2560	1280	640	640	2560	1280	640	80	<	80	40
A/Brisbane/02/2018		6B.1A.1	2018-01-04	E3/E2	2560	1280	640	1280	2560	2560	1280	80	40	160	80
A/Switzerland/3330/2017	clone 35	6B.1A.5b	2017-12-20	E6/E2	1280	640	1280	640	1280	1280	640	80	<	80	40
A/Ireland/87733/2019		6B.1A.5a.1	2019-11-03	E4	2560	1280	640	640	2560	2560	640	160	<	160	40
A/Guangdong-Maonan/SWL1536/2019		6B.1A.5a.1	2019-06-17	E3/E2	2560	640	640	1280	1280	2560	640	160	<	160	40
A/Guangdong-Maonan/SWL1536/2019		6B.1A.5a.1	2019-06-17	C2/MDCK1	1280	640	320	640	2560	1280	1280	160	<	80	80
A/Ghana/1894/2021		6B.1A.5a.1	2021-07-21	E2	2560	1280	1280	1280	>5120	>5120	2560	1280	ND	ND	160
A/North Carolina/01/2021		6B.1A.5a.1		SIAT3/MDCK1	160	80	80	80	160	640	640	640	<	<	40
A/Denmark/3280/2019		6B.1A.5a.2	2019-11-10	MDCK4/MDCK6	160	80	40	80	80	160	40	40	2560	2560	2560
IVR-215 (A/Victoria/2570/2019)		6B.1A.5a.2	2018-11-22	E4/D7/E2	80	80	40	80	160	160	80	160	1280	2560	1280
A/Sydney/5/2021		6B.1A.5a.2	2021-10-16	E3/E1	80	40	40	40	80	80	40	80	1280	2560	2560
TEST VIRUSES															
A/India/Pun-NIV323546/2021		6B.1A.5a.2	2021-06-22	X3/SIAT2/MDCK1	<	<	<	<	<	40	<	40	640	640	1280
A/Belgium/S0063/2022		seq pending	2021-12-07	MDCK1	1280	320	640	640	1280	1280	1280	320	<	80	80
A/Belgium/G0001/2022	A261T	6B.1A.5a.1	2021-12-17	MDCK1	1280	320	320	640	1280	1280	1280	640	<	40	80
A/Belgium/H0007/2022		seq pending	2022-01-11	MDCK1	160	80	80	160	160	640	640	640	<	40	40
A/Belgium/H0038/2022		seq pending	2022-01-17	MDCK1	640	320	320	640	1280	1280	640	320	<	40	40
A/Belgium/H0042/2022		seq pending	2022-01-20	MDCK1	1280	320	640	640	1280	1280	1280	640	<	80	80
						Vaccine NH 2019-20 SH 2020			Vaccine NH 2020-21					Vaccine SH 2021 NH 2021-22 SH 2022	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)
1 < = <40; 2 < = <80; ND = Not Done

Sequences in phylogenetic trees

< 4-fold | 4-fold | 8-fold | > 8-fold | < not recognised by the antiserum | ≥ 160 (when homologous titre ≥ 2560) | ≥ 160 (no homologous titre)

Table 6-7. Antigenic analyses of influenza A(H1N1)pdm09 viruses with collection dates after 2021-08-31 (Summary)

		Haemagglutination inhibition titre										
		Post-infection ferret antisera										
Viruses		A/Paris 1447/17	A/Bris 02/18	A/Swit 3330/17	A/Ire 87733/19	A/G-M SWL1536/19	A/G-M SWL1536/19	A/Ghana 1894/21	A/N Carolina 01/21	A/Denmark 3280/19	IVR-215 A/Vic/2570/19	A/Sydney 5/21
	Passage history	MDCK	Egg	Egg	Egg	Egg	MDCK	Egg	Cell	MDCK	Egg	Egg
	Ferret number	F03/18*2	F09/19*1	F23/18*1	St Jude's F18/20*1	F12/20*1	F09/20*1	F02/22*1	CDC F2021-105	F08/20*1	F37/21*1	F04/22*1
	Genetic group	6B.1A	6B.1A.1	6B.1A.5b	6B.1A.5a.1	6B.1A.5a.1	6B.1A.5a.1	6B.1A.5a.1	6B.1A.5a.1	6B.1A.5a.2	6B.1A.5a.2	6B.1A.5a.2
REFERENCE VIRUSES												
A/Paris/1447/2017	6B.1A	2560	1280	640	640	2560	1280	640	80	<	80	40
A/Brisbane/02/2018	6B.1A.1	2560	1280	640	1280	2560	2560	1280	80	40	160	80
A/Switzerland/3330/2017	6B.1A.5b	1280	640	1280	640	1280	1280	640	80	<	80	40
A/Ireland/87733/2019	6B.1A.5a.1	2560	1280	640	640	2560	2560	640	160	<	160	40
A/Guangdong-Maonan/SWL1536/2019	6B.1A.5a.1	2560	640	640	1280	1280	2560	640	160	<	160	40
A/Guangdong-Maonan/SWL1536/2019	6B.1A.5a.1	1280	640	320	640	2560	1280	1280	160	<	80	80
A/Ghana/1894/2021	6B.1A.5a.1	2560	1280	1280	1280	>5120	>5120	2560	1280	ND	ND	160
A/North Carolina/01/2021	6B.1A.5a.1	160	80	80	80	160	640	640	640	<	<	40
A/Denmark/3280/2019	6B.1A.5a.2	160	80	40	80	80	160	40	40	2560	2560	2560
IVR-215 (A/Victoria/2570/2019)	6B.1A.5a.2	80	80	40	80	160	160	80	160	1280	2560	1280
A/Sydney/5/2021	6B.1A.5a.2	80	40	40	40	80	80	40	80	1280	2560	2560
TEST VIRUSES												
Total Number tested		82	82	82	82	82	82	28	5	82	82	28
No with titre reduction ≤2-fold		47	28	41	45	49	51	10	5	13	14	6
%		*57.3*	*34.1*	*50.0*	*54.9*	*59.8*	*62.2*	*35.7*	*100*	*15.9*	*17.1*	*21.4*
No with titre reduction =4-fold		4	20	12	6	1	9	5		4	3	
%		*4.9*	*24.4*	*14.6*	*7.3*	*1.2*	*11.0*	*17.9*		*4.8*	*3.6*	
No with titre reduction ≥8-fold		31	34	29	31	32	22	13		65	65	22
%		*37.8*	*41.5*	*35.4*	*37.8*	*39.0*	*26.8*	*46.4*		*79.3*	*79.3*	*78.6*
Number tested	6B.1A.5a.1	65	65	65	65	65	65	22	5	65	65	22
No with titre reduction ≤2-fold		47	28	41	45	49	51	10	5			
%		*72.3*	*43.1*	*63*	*69.2*	*75.4*	*78.5*	*45.5*	*100*			
No with titre reduction =4-fold		4	20	12	6	1	9	5				
%		*6.2*	*30.8*	*18.5*	*9.3*	*1.5*	*13.8*	*22.7*				
No with titre reduction ≥8-fold		14	17	12	14	15	5	7		65	65	22
%		*21.5*	*26.1*	*18.5*	*21.5*	*23.1*	*7.7*	*31.8*		*100*	*100*	*100*
Number tested	6B.1A.5a.2	17	17	17	17	17	17	6	0	17	17	6
No with titre reduction ≤2-fold										13	14	6
%										*76.5*	*82.4*	*100*
No with titre reduction =4-fold										4	3	
%										*23.5*	*17.6*	
No with titre reduction ≥8-fold		17	17	17	17	17	17	6				
%		*100*	*100*	*100*	*100*	*100*	*100*	*100*				
			Vaccine NH 2019-20 SH 2020			Vaccine NH 2020-21					Vaccine SH 2021 NH 2021-22 SH 2022	

Reference virus results are taken from an individual table as an example. Summaries for each antiserum are based on fold-reductions observed on the days that HI assays were performed.

Table 7. Antigenic analyses of influenza A(H1N1)pdm09 viruses - Plaque Reduction Neutralisation (MDCK)

Viruses	Passage history / Ferret number / Genetic group	Collection date	Passage history	Haemagglutination inhibition titre / Post-infection fereret antisera: A/Paris 1447/17 MDCK F03/18[*2] 6B.1A		A/Guangdong-Mao SWL1536/19 MDCK F09/20[*1] 6B.1A.5a.1		A/Guangdong-Mao SWL1536/19 Egg F12/20[*1] 6B.1A.5a.1		A/Denmark 3280/19 MDCK F08/20[*1] 6B.1A.5a.2		IVR-215 A/Victoria/2570/19 Egg F37/21[*1] 6B.1A.5a.2		HA1 substitutions compared to A/Paris/1447/2017: Egg adaptation HA substitutions compared to the corresponding cell culture propagated virus
				2-fold	Read*	2-fold	Read*	2-fold	Read*	2-fold	Read*	2-fold	Read*	
REFERENCE VIRUSES														
A/Paris/1447/2017	6B.1A	2017-10-20	MDCK1/MDCK3	5120	6772	2560	3093	2560	3796	<	<	160	139	
A/Guangdong-Maonan/SWL1536/2019	6B.1A.5a.1	2019-06-17	C2/MDCK1	5120	4034	5120	5013	5120	5025	<	<	320	384	N129D, S183P, T185I, D187A. Q189E, N260D
A/Guangdong-Maonan/SWL1536/2019	6B.1A.5a.1	2019-06-17	E3/E2	2560	3182	2560	2347	2560	2880	<	<	80	106	N129D, S183P, T185I, A187V. Q189E, N260D
A/Denmark/3280/2019	6B.1A.5a.2	2019-11-10	MDCK4/MDCK5	320	264	320	387	320	287	2560	3200	5120	4070	N129D, K130N, N156K, L161I, S183P, T185I, V250A, N260D, H273X
IVR-215 (A/Victoria/2570/2019)	6B.1A.5a.2	2018-11-22	E4/D7/E2	<	<	640	617	160	176	5120	4346	>5120	>5120	N129D, K130N, N156K, L161I, S183P, T185I, A195E, Q223R, V250A, N260D
TEST VIRUSES														
A/Limoges/635/2021	6B.1A.5a.2	2021-10-06	MDCK2/MDCK1	<	<	40	52	80	63	>5120	>5120	>5120	>5120	K54Q, A73E, D94N, N129D, K130N, N156K, S157T, L161I, S183P, T185I, A186T, Q189E, K209R, T216A, E224A, V250A, R259K, N260D, K308R
A/South Africa/R14850/2021	6B.1A.5a.2	2021-10-11	MDCK2/MDCK1	<	<	<	<	40	52	2560	2147	5120	4693	K54Q, N129D, K130N, K142R, N156K, L161I, S183P, T185I, A186T, Q189E, E224A, V250A, R259K, N260E, K308R
A/South Africa/R16131/2021	6B.1A.5a.2	2021-11-08	MDCK1/MDCK1	<	<	<	<	<	<	2560	3638	5120	6737	K54Q, N129D, K130N, K142R, N156K, L161I, S183P, T185I, A186T, Q189E, E224A, V250A, R259K, D260E, H296N, K308R
A/Reims/46047/2021	6B.1A.5a.1	2021-11-28	MDCK1/MDCK1	>5120	>5120	5120	7111	5120	7529	40	55	160	146	D129N, S183P, T185I, D187A, Q189E, N260D
A/South Africa/R15623/2021	6B.1A.5a.1	2021-10-25	MDCK1/MDCK1	5120	5044	5120	6400	5120	7314	<	<	160	160	R113K, N129D, S183P, T185I, D187A, Q189E, N260D
A/South Africa/R16324/2021	6B.1A.5a.1	2021-11-10	MDCK1/MDCK1	5120	4442	5120	5225	5120	4897	<	<	160	140	R113K, N129D, S183P, T185I, D187A, Q189E, N260D
A/South Africa/R15292/2021	6B.1A.5a.1	2021-10-19	MDCK1/MDCK1	5120	4594	5120	5225	5120	5004	<	<	160	140	R113K, N129D, S183P, T185I, D187A, Q189E, N228S, N260D
A/South Africa/R16454/2021	6B.1A.5a.1	2021-10-25	MDCK1/MDCK1	5120	5120	5120	6244	5120	6095	<	<	160	160	R113K, K119R, N125S, N129D, S183P, T185I, D187A, Q189E, N260D
A/South Africa/R16455/2021	6B.1A.5a.1	2021-10-22	MDCK1/MDCK1	<	<	1280	1862	640	901	<	<	160	215	R113K, N129D, V152I, G155E, N162X(-cho), S183P, T185I, D187A, Q189E, N260D
A/Monaco/743/2021	6B.1A.5a.1	2021-10-26	MDCK2/MDCK1	>5120	>5120	>5120	>5120	>5120	8533	40	56	320	314	N129D, S183P, T185I, D187A, Q189E, N260D, V321I
A/Nimes/871/2021	6B.1A.5a.1	2021-11-19	MDCK3/MDCK1	5120	5734	5120	7111	5120	6737	<	<	160	158	N129D, S183P, T185I, D187A, Q189E, N260D, A261T
A/Lyon/CHU/R21.826.95/2021	6B.1A.5a.1	2021-11-19	MDCK2/MDCK1	5120	4338	5120	6564	5120	6564	<	<	320	301	S83F, N129D, S183P, T185I, D187A, Q189E, N260D
A/Paris/30589/2021	6B.1A.5a.1	2021-10-18	MDCK2/MDCK1	5120	4172	>5120	7757	5120	4978	<	<	320	262	S83F, N129D, T164X(-cho), S183P, T185I, D187A, Q189E, N260D
A/Paris/29801/2021	6B.1A.5a.1	2021-10-10	MDCK1/MDCK1	640	528	2560	2964	1280	1177	80	75	320	347	N129D, P137S, G155E, S183P, T185I, D187A, Q189E, T216K, N260D, P271L
A/Nice/742/2021	6B.1A.5a.1	2021-10-24	MDCK2/MDCK1	1280	1174	2560	3218	1280	1097	<	<	160	205	N129D, P137S, G155E, T164X(-cho), S183P, T185I, D187A, Q189E, N260D, K308E
A/Lyon/CHU/R21.827.22/2021	6B.1A.5a.1	2021-11-20	MDCK2/MDCK1	2560	2216	5120	5333	1280	1807	80	65	320	421	S128T, N129G, P137S, G155E, S183P, T185I, D187A, Q189E, N260D
A/Rouen/41246/2021	6B.1A.5a.1	2021-11-15	MDCK2/MDCK1	640	668	2560	1920	640	764	40	51	160	204	S128T, N129D, P137S, G155E, S183P, T185I, D187A, Q189E, E224D, N260D
A/Grenoble/705/2021	6B.1A.5a.1	2021-10-23	MDCK2/MDCK1	2560	2048	5120	7111	2560	2271	80	96	640	560	S128T, N129D, P137S, G155E, T164X(-cho), S183P, T185I, D187A, Q189E, N260D
A/Lyon/820/2021	6B.1A.5a.1	2021-11-16	MDCK2/MDCK1	1280	1100	2560	3413	1280	1422	<	<	160	227	S128T, N129D, P137S, G155E, N162X(-cho), S183P, T185I, D187X, Q189E, N260D
A/Togo/881/2020	6B.1A.5a.1	2020-11-14	SIAT3/MDCK1	5120	4480	5120	4577	5120	4442	<	<	160	132	N129D, I166T, S183P, T185I, A186T, D187A, Q189E, P236A, N260D
								Vaccine NH 2020-21				Vaccine SH 2021 NH 2021-22 SH 2022		

[1] Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

* The software cutoff for reliable estimation of a titre is approximately 7680, readings marked as >5120 fall beyond this range and indicate too few/no plaques pesent to give a reliable count

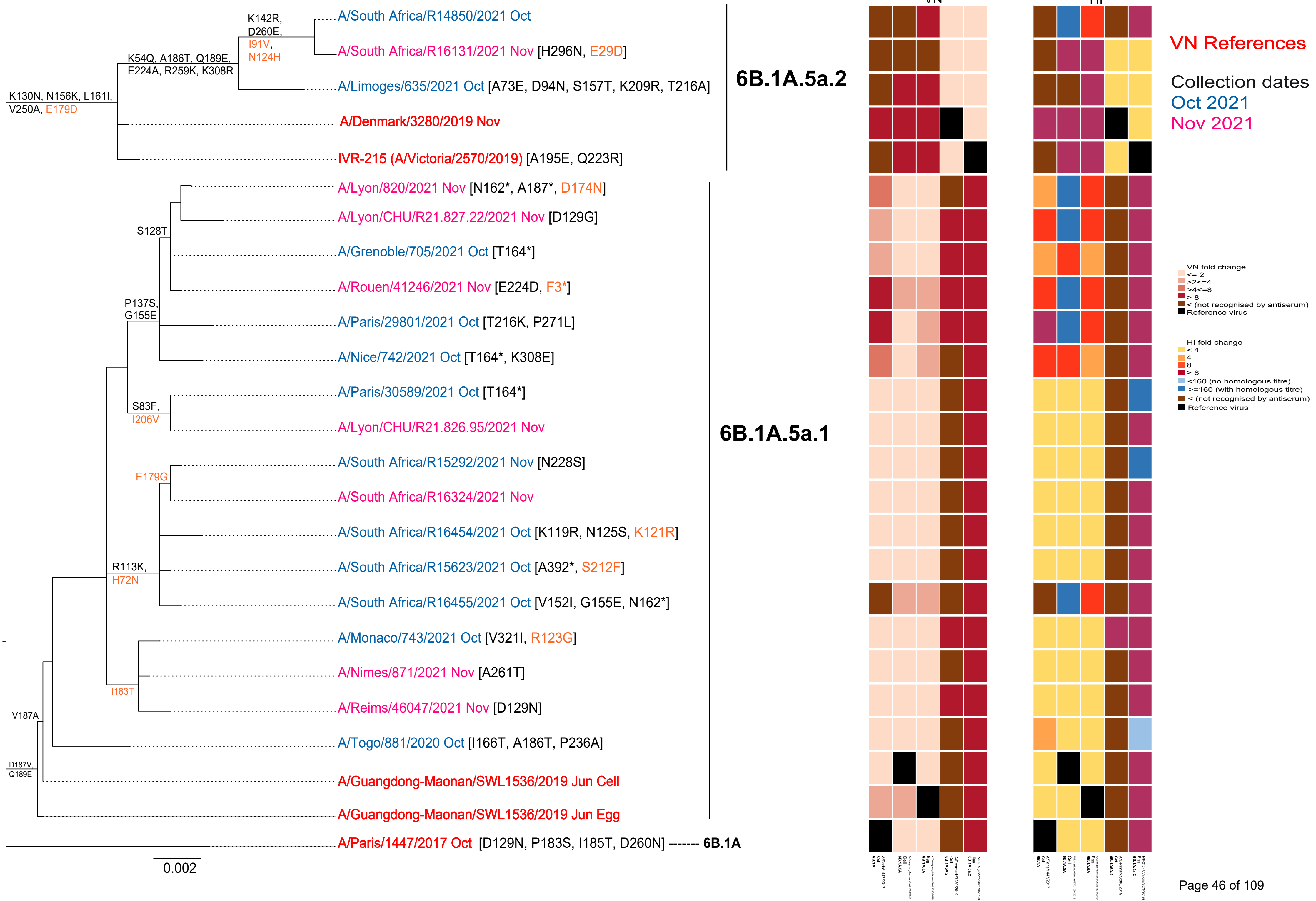
Substitutions compared to cell culture-propagated A/Victoria/2570/2019 for 6B.1A.5a.2 viruses

Substitutions compared to cell culture-propagated A/Guangdong-Maonan/SWL1536/2019 for 6B.1A.5a.1 viruses

Polymorphisms resulting in potential loss of the 162-164 glycosylation site

Modification of subclade-defining residues

Figure 9. HA phylogeny of A(H1N1)pdm09 viruses used in VN assay



Influenza A(H3N2) virus analyses.

Influenza A(H3N2) viruses collected since 2021-08-31 were received from NICs in 37 countries or administrative regions in four WHO Regions: African (Algeria, Cameroon, Cote d'Ivoire, Madagascar and South Africa), American (Argentina), Eastern Mediterranean (Iran, Lebanon, Oman and Qatar), and European (27 countries) (**Table 1**).

Influenza A(H3N2) viruses had been difficult to characterise antigenically by HI assay due to either the variable agglutination of red blood cells (RBCs) from guinea pigs, turkeys and humans or the loss of the ability of viruses to agglutinate any of these RBCs. However, since the September 2021 VCM HI analysis was able to be done on 296 cell culture-propagated viruses that agglutinated guinea pig RBCs (**Tables 9-1 to 9-11**). **Tables 9-1, 9-7** and **9-10** include four viruses from CDC and **Table 9-10** also includes two viruses from NIID, Tokyo. Virus neutralisation by plaque reduction neutralisation assay (PRNA) was done on 91 test viruses, including one from CDC (**Tables 10-1 to 10-7**).

HI analyses.

A total of 296 viruses were analysed by HI assay and summaries of the HI results are shown in **Tables 4** and **8**. Individual HI results are shown in **Tables 9-1** to **9-11**. Test viruses in each table are sorted by date of collection and genetic clade/subclade/group/subgroup for those viruses that were sequenced, with those below the red horizontal lines in the tables being viruses with collection dates after 2021-08-31 (**Tables 9-2 to 9-11**). The summary shown in **Table 4** is of the HI results for viruses collected since 2021-08-31 with antisera raised against egg-propagated vaccine viruses A/Cambodia/e0826360/2020 (subclade 3C.2a1b.2a.1) and A/Darwin/9/2021 (subclade 3C.2a1b.2a.2), and cell culture-propagated A/Cambodia/925256/2020 (subclade 3C.2a1b.2a.1) and A/Stockholm/5/2021 (A/Darwin/6/2021-like, subclade 3C.2a1b.2a.2).

The viruses sequenced and analysed by HI fell into clade 3C.2a subclades 3C.2a1b.2a.2 (89% of the sequenced viruses), 3C.2a1b.2a1 (only two viruses, just under 1% of the total number of sequenced viruses), 3C.2a1b.1a (six viruses, 2.5% of the total sequenced) and 3C.2a1b.1b (16 viruses, 7% of the sequenced viruses).

Of the viruses in subclade 3C.2a1b.2a.2 (n = 202), characterised by **HA1** amino acid substitutions **Y159N**, **T160I** (resulting in loss of a glycosylation site), **L164Q**, **G186D** and **D190N**, most had an HA with the additional amino acid substitution **H156S** in **HA1**, as seen in A/Darwin/9/2021 and A/Stockholm/5/2021:

- an antiserum raised against egg-propagated A/Darwin/9/2021 recognised 71% of these 3C.2a1b.2a.2 viruses at titres within 4-fold of the homologous titre, but only 15% at titres within 2-fold;
- an antiserum raised against cell culture-propagated A/Stockholm/5/2021 recognised 95% of these 3C.2a1b.2a.2 viruses at titres within 4-fold of the homologous titre, and 65% at titres within 2-fold;

- an antiserum raised against cell culture-propagated A/Bangladesh/4005/2020 (also subclade 3C.2a1b.2a.2) recognised 94% of the 3C.2a1b.2a.2 viruses at titres within 4-fold of the homologous titre, and 44% at titres within 2-fold;
- an antiserum raised against cell culture-propagated A/Cambodia/925256/2020 (subclade 3C.2a1b.2a.1) recognised only 25 viruses in subclade 3C.2a1b.2a.2 (25%) at titres within 4-fold of the homologous titre, one within 2-fold;
- an antiserum raised against egg-propagated A/Cambodia/e0826360/2020 (subclade 3C.2a1b.2a.1) recognised only 13 viruses in subclade 3C.2a1b.2a.2 (6.5%) at titres within 4-fold of the homologous titre, none within 2-fold;
- an antiserum raised against cell culture-propagated A/Hong Kong/2671/2019 (subgroup 3C.2a1b.1b) recognised only 4 (2%) of the subgroup 3C.2a1b.2a.2 test viruses at titres within 4-fold of the homologous titre, one within 2-fold;
- an antiserum raised against A/Denmark/3264/2019 (subgroup 3C.2a1b.1a) recognised 48% of the subgroup 3C.2a1b.2a.2 test viruses at titres within 4-fold of the homologous virus, 28% within 2-fold;
- an antiserum raised against the cell culture-propagated clade 3C.3a virus A/Kansas/14/2017 recognised only 14% viruses in subclade 3C.2a1b.2a.2 at titres within 4-fold of the homologous titres, and only two at titres within 2-fold.

The test viruses in subclade 3C.2a1b.1b (n = 16) have **HA1** amino acid substitutions of **E50K**, **T135K (-CHO)**, **S137F** and **K310R**, and:

- were recognised poorly by antisera raised against three 3C.2a1b.2a.2 reference/vaccine viruses: cell culture-propagated A/Bangladesh/4005/2020, A/Stockholm/5/2021, and egg-propagated A/Darwin/9/2021 at titres greater than 4-fold lower than the titres with the homologous viruses;
- the antiserum raised against cell-propagated A/Cambodia/925256/2020 recognised 88% of the test viruses in subclade 3C.2a1b.1b at titres within 4-fold of the homologous titre and eight of the 16 at titres within 2-fold, but the antiserum raised against egg-propagated A/Cambodia/e0826360/2020 recognised all these test viruses poorly;
- antisera raised against both cell culture-propagated A/Hong Kong/2671/2017 (subclade 3C.2a1b.1b) and cell culture-propagated A/Denmark/3264/2019 (subclade 3C.2a1b.1a) recognised the majority of subclade 3C.2a1b.1b viruses at titres within 4-fold of their homologous titres (88% and 81% respectively) and recognised 56% and 44%, respectively, of these viruses at titres within 2-fold of the homologous titres;
- the antiserum raised against cell culture-propagated A/Kansas/14/2017 recognised none of the viruses in subclade 3C.2a1b.1b at titres within 4-fold of the homologous titre.

A small number of viruses (n=6) with HA genes in subclade 3C.2a1b.1a were analysed:

- they were recognised best by the antiserum raised against A/Denmark/3264/2019 with all six viruses being recognised at titres within 4-fold of the homologous titre, three within 2-fold;

- were also recognised well by the antiserum raised against cell culture-propagated A/Cambodia/925256/2020, five of the six at titres within 4-fold of the homologous titre, but the antiserum raised against egg-propagated A/Cambodia/e0826360/2020 recognised five of these six test viruses poorly;
- the antiserum raised against cell culture-propagated A/Bangladesh/4005/2020 recognised five of the six test viruses at titres within 4-fold of the homologous titres, but none within 2-fold. In contrast, the antiserum raised against cell culture-propagated A/Stockholm/5/2021 (with the **HA1** substitution **H156S**) recognised only one of these six test viruses at a titre within 4-fold of the homologous titre. The antiserum raised against egg-propagated A/Darwin/9/2021 recognised none of the test viruses at titres within 4-fold of the homologous titres;
- the antiserum raised against cell culture-propagated A/Kansas/14/2017 recognised none of the viruses in subclade 3C.2a1b.1a at titres within 4-fold of the homologous titres.

Two viruses were received that had HA genes in subclade 3C.2a1b.2a.1: they were not recognised well by antisera raised against egg-propagated A/Cambodia/e0826360/2020 and egg-propagated A/Darwin/9/2021, but they were recognised well by antisera raised against cell culture-propagated A/Denmark/3264/2019, A/Bangladesh/4005/2020, and somewhat less well by the other antisera.

Virus Neutralisation results

Results of virus neutralisation, assessed by PRNA, are shown in **Tables 10-1 to 10-7**. A total of 90 test viruses from NICs were analysed. The majority of viruses analysed (88%) were in subgroup 3C.2a1b.2a.2, nine (10%) were in subgroup 3C.2a1b.1b and two viruses were in subgroup 3C.2a1b.1a. A summary of the results is shown in **Table 4** and a phylogenetic tree of the HA gene of the viruses analysed by PRNA is shown in **Figure 17** with panels summarising the reductions in antiserum titres compared to homologous titres in PRNA and HI assay for the same panel of test viruses.

For viruses in subclade 3C.2a1b.2a.2 (n = 79):

- an antiserum raised against cell culture-propagated A/Stockholm/5/2021 recognised all of the 3C.2a1b.2a.2 viruses at titres within 4-fold of the homologous titre, and 81% at titres within 2-fold;
- an antiserum raised against cell culture-propagated A/Bangladesh/4005/2020 recognised 57% of the 3C.2a1b.2a.2 viruses at titres within 4-fold of the homologous titre, and 16% at titres within 2-fold;
- an antiserum raised against egg-propagated A/Darwin/9/2021 (an antiserum raised against NIB-126A was used in **Table 10-1**) recognised only 5% of the 3C.2a1b.2a.2 viruses at titres within 4-fold of the homologous titre, none within 2-fold;
- antisera raised against cell culture-propagated A/Norway/3275/2018 (subclade 3C.2a1b.2), A/Cambodia/925256/2020 (3C.2a1b.2a.1) and A/Hong Kong/2671/2019 (3C.2a1b.1b), A/Denmark/3264/2019 (3C.2a1b.1a) and the 3C.3a.1 virus A/Kansas/14/2017 recognised very few 3C.2a1b.2a.2 viruses at titres within 4-fold of the respective homologous titres, ranging from none being recognised at titres within

4-fold of the homologous titres, with most viruses not being recognised at all by the antisera, to 7.5% by the antiserum raised against A/Kansas/14/2017.

Viruses in subclade 3C.2a1b.1b (n = 9):

- were not recognised by antisera raised against 3C.2a1b.2a.2 viruses (A/Bangladesh/4005/2020, A/Stockholm/5/2021 and A/Darwin/9/2021);
- were not recognised by antisera raised against the 3C.3a.1 virus A/Kansas/14/2017;
- were recognised poorly by the antiserum raised against A/Norway/3275/2018;
- were recognised well by an antiserum raised against cell culture-propagated A/Cambodia/925256/2020 with eight of the nine test viruses being recognised at titres within 4-fold of the homologous titre, seven within 2-fold;
- were recognised well by an antiserum raised against cell culture-propagated A/Denmark/3264/2019 with seven of the nine test viruses being recognised at titres within 4-fold of the homologous titre (all within 2-fold).

A/South Africa/R15681/2021 in this subclade was not recognised by any of the antisera.

Viruses in subclade 3C.2a1b.1a (n = 2). These were A/South Africa/R16482/2021 (**Table 10-6**) and A/Switzerland/51230/2021 (**Table 10-7**).

A/South Africa/R16482/2021 was recognised well by antisera raised against A/Cambodia/925256/2020 and A/Denmark/3264/2019 at titres similar to the homologous titres. A/Switzerland/51230/2021 was recognised less well with the antiserum raised against A/Cambodia/925256/2020 yielding at a titre within 4-fold of the homologous titre while the antiserum raised against A/Denmark/3264/2019 gave a titre somewhat lower than 4-fold reduced over the homologous titre. The remaining antisera in the panel recognised A/South Africa/R16482/2021 and A/Switzerland/51230/2021 less well.

There are two sets of heat maps shown adjacent to the phylogenetic tree in **Figure 17.** These summarise the VN results and the HI results for the same viruses, where available, and a selection from the panel of reference antisera used in the HI assay. The heat map for the VN results illustrates the better recognition of test viruses by antisera raised against viruses within the same subclade.

Genetic analyses.

Phylogenetic analysis of the HA genes of representative recently circulating A(H3N2) viruses (with collection dates after 2021-08-31 as available in GISAID) is shown in **Figure 10** and a phylogenetic tree of those analysed at the Francis Crick Institute is shown in **Figure 11**. The HA genes of the vast majority fell within clade 3C.2a, with viruses in subclade 3C.2a1b.2a.2 predominating.

Major HA clades/subclades

- The HA genes of recent Clade **3C.2a** viruses share a series of amino acid substitutions **E62G**, **N91S**, **N121K**, **K144S**, **S159Y**, **K160T** (+CHO), **Q311H** and **R326K** in **HA1** and **I77V, M149I** and **G155E** in **HA2** and define subclade **3C.2a1b**.

- Subclade **3C.2a1b.1** is defined by viruses having an additional **T135K** in **HA1** substitution**,** resulting in the loss of a glycosylation site, and splits into two further subclades**:**
    - Subclade **3C.2a1b.1a** is defined by the additional substitutions **G186D**, **D190N** and **S198P** in **HA1** (e.g. A/Denmark/3264/2019), with some viruses also having **S205F** or **I192F** substitutions in **HA1**.
    - Subclade **3C.2a1b.1b** is defined by the substitution **S137F** in **HA1** (e.g. A/Hong Kong/6271/2019), with newer viruses also having substitutions **E50K**, **R261Q** and **K310R** in **HA1** and **A201V** in **HA2**.
- Subclade **3C.2a1b.2** is defined by substitutions **A128T** (+CHO), **T131K** and **S138A** in **HA1**, and **V200I** in **HA2** (e.g. A/Norway/3275/2019). Most recent viruses also carry **Y83E**, **Y94N**, **Y195F** in **HA1** **and I193M** in **HA2**.
    - Subclade **3C.2a1b.2a.1** viruses (e.g. A/Cambodia/e0826360/2020) have HA glycoproteins with the additional substitutions **K171N**, **G186S** and **S198P** in **HA1**.
    - Subclade **3C.2a1b.2a.2** viruses have HA glycoproteins with the substitutions **Y159N**, **T160I** (resulting in the loss of a glycosylation site), **L164Q**, **G186D** and **D190N** in **HA1**. The majority of viruses within subclade **3C.2a1b.2a.2** now also have **HA1 H156S** substitution (e.g. A/Stockholm/5/2021 and A/Darwin/9/2021), many with **D53N** or **D53G** in **HA1.** Some of the subgroup with **D53G** (e.g. A/Darwin/6/2021) also have HA glycoproteins with **D104G** and **K276R** substitutions in **HA1**.

Figure 12 shows an HA amino acid sequence alignment for a representative selection of the viruses used to generate the phylogenies. The vaccine viruses A/Hong Kong/45/2019 (subclade **3C.2a1b.1b**), A/Cambodia/e0826360/2020 (subclade **3C.2a1b.2a.1**) and A/Darwin/9/20121 (subclade **3C.2a1b.2a.2**) are shown in red, glycosylation motifs are highlighted, reference viruses are coloured blue and the HA1/HA2 proteolytic cleavage site is highlighted. The clade/subclade/group/subgroup to which each of the HA sequences belongs is also shown.

Figure 13 show the locations on a 2005 H3 HA structure of amino acid substitutions that define subclade 3C.2a1b viruses, and its subclades (3C.2a1b.1, 1a, .1b, 2a.1 and 2a.2).

Figure 14 shows a NA phylogenetic tree for representative viruses with their HA genetic groups indicated, as in **Figure 10**, and **Figure 15** shows a phylogenetic tree of viruses with gene sequences determined at the Francis Crick Institute (*cf*. **Figure 11**). **Figure 16** shows a NA amino acid sequence alignment for a representative selection of these viruses. As above, the vaccine viruses are shown in red, glycosylation motifs are highlighted, the reference viruses are coloured blue and the clade/subclade/group/subgroup to which the corresponding HA gene sequences belong is shown.

The NA gene sequences of viruses with HA genes within subclade **3C.2a1b** show generally similar subclade/group clustering. Viruses with HA genes in subclade **3C.2a1b.2** have the substitutions **S315R** and **G346D** in **NA**. Viruses with HA genes in subclade **3C.2a1b.2a.1** and subclade **3C.2a1b.2a.2** share the substitutions **D463N** and **N465S** in **NA**, creating a potential glycosylation site.

Antiviral resistance

Susceptibility to the neuraminidase inhibitors (NAIs) oseltamivir and zanamivir was assessed phenotypically and by full NA gene sequencing, for 272 A(H3N2) viruses detected and received within the course of the 2021-2022 influenza season (**Table 13**). All but one virus showed normal inhibition (NI: no reduced susceptibility) by both antivirals. A/Tripoli/5646/2021 showed reduced inhibition by zanamivir, associated with **NA A246V** substitution, within a glycosylation site (NAT → NVT).

All 246 A(H3N2) full-length PA gene sequences generated at the Crick, from individual viruses detected during the 2021-2022 influenza season, encoded isoleucine at position 38 (I38) and showed no evidence of other substitutions associated with reduced susceptibility to baloxavir marboxil.

Conclusion.

Received A(H3N2) samples with collection dates after 2021-08-31 have HAs in subclade **3C.2a1b** with the large majority in subclade **3C.2a1b.2a.2**. The viruses falling into this subclade were recognised well in HI and PRNA by antisera raised against cell culture-propagated A/Stockholm/5/2021 (subclade **3C.2a1b.2a.2** with the additional substitution **H156S** in **HA1**)

All A(H3N2) viruses characterised genetically showed no markers of reduced susceptibility to baloxavir marboxil and all but one of those analysed phenotypically showed normal inhibition by both oseltamivir and zanamivir.

Table 8. Summary of influenza A(H3N2) viruses collected after 2021-08-31, analysed by HI

MONTH	H3N2													
			3C.2a1b.2a.2: Egg-propagated A/Darwin/9/2021			3C.2a1b.2a.2: Cell-propagated A/Stockholm/5/2021			3C.2a1b.2a.1: Egg-propagated A/Cambodia/e0826360/2020			3C.3a1b.2a.1: Cell-propagated A/Cambodia/925256/2020		
Country	Number received	Number propagated[1,2]	≤2 fold[3]	4 fold[3]	≥8 fold[3]	≤2 fold[4]	4 fold[4]	≥8 fold[4]	≤2 fold[5]	4 fold[5]	≥8 fold[5]	≤2 fold[6]	4 fold[6]	≥8 fold6[6]
2021 SEPTEMBER														
Belgium	1	1		1				1			1			1
Croatia	3	2		1	1	2					2			2
Denmark	5	5		1	4	3	2				5			5
France	10	9	1	3	5	4	1	4		1	8	3		6
Israel	2	2		2		2				2				2
Italy	1	1		1		1					1		1	
Lebanon	19	18	1	17		18					18			18
Netherlands	12	10	8	2		9	1				10		1	9
Oman	12	3		2	1	2		1			3	1		2
Qatar	10	7			7	6	1				7			7
South Africa	5	5	1		4	2	1	2		2	3	2	1	2
Spain	1	0												
Sweden	1	1			1			1			1	1		
UK (England)	2	2		2		1	1				2			2
OCTOBER														
Cameroon	1	0												
Denmark	1	1		1		1				1			1	
Estonia	1	0												
France	3	3	1	1	1	2		1		2	1		2	1
Germany	2	0												
Iran	3	3		2	1	2	1				3			3
Ireland	1	1		1		1				1				1
Italy	2	2			2			2			2			2
Kyrgyzstan	28	0												
Lebanon	15	0												
Netherlands	37	8	2	4	2	3	5				8			8
Norway	7	3		3		3					3			3
Oman	22	6	1	3	2	5	1				6			6
Portugal	2	0												
Russia	3	3	3			3					3		1	2
South Africa	7	6		2	4	3	1	2		3	3	1	3	2
Spain	3	1		1		1					1			1
Tajikistan	1	0												
Sweden	2	2		2		2					2			2
UK (England)	8	8		5	3	5	3				8			8
NOVEMBER														
Algeria	5	0												
Armenia	2	0												
Belgium	2	2	1	1		1		1			2			2
Cameroon	1	0												
Cote d'Ivoire	1	0												
Estonia	1	0												
France	10	7	1	1	5	2	1	4			7	2	2	3
Germany	2	0												
Iran	10	10	2	8		7	3				10		2	8
Ireland	2	0												
Israel	6	6		6		3	3				6		1	5
Italy	5	5		1	4	1	3	1		2	3	2	2	1
Kyrgyzstan	22	0												
Lebanon	58	11	3	8		11					11			11
Madagascar	11	0												
Netherlands	23	3			3		3				3			3
Norway	8	4	2	2		4					4		4	
Oman	21	2		2		2					2			2
Russia	36	35	12	13	10	17	14	4		1	34	1	4	30
South Africa	1	0												
Spain	32	5		4	1	2	3				5			5
Sweden	5	5	3	2		5					5			5
Switzerland	2	2		1	1		1	1			2			2
DECEMBER														
Algeria	5	0												
Argentina	9	0												
Armenia	21	9	3	4	2	5	4			1	8		1	8
Belgium	8	0												
Bosnia and Herzegovina	3	0												
Cote d'Ivoire	13	0												
Estonia	7	0												
France	1	1		1		1					1			1
Germany	10	9	1	7	1	7	1	1	1		8		1	8
Hungary	2	2			2			2			2			2
Iran	2	2		1	1	2					2		2	
Ireland	3	2		1	1	1	1				2			2
Israel	15	0												
Latvia	5	0												
Lebanon	76	5	5			3	2				5			5
Madagascar	11	8	1		7	1	4	3		3	5	1	2	5
Montenegro	6	0												
Netherlands	21	3			3	2	1				3			3
Norway	1	0												
Portugal	13	1			1			1			1			1
Romania	5	5		3	2		5				5			5
Russia	5	4	3	1		4					4			4
Serbia	2	0												
Spain	49	6	2	4			5	1			6			6
Switzerland	9	2			2		1	1			2	1		1
Ukraine	9	0												
2022 JANUARY														
Argentina	12	0												
Armenia	2	1			1			1			1	1		
Belgium	10	0												
Bosnia and Herzegovina	3	0												
Cote d'Ivoire	1	0												
Estonia	4	4	2	2		1	3				4			4
Germany	3	3		3		3					3			3
Hungary	2	2		2			2				2			2
Ireland	2	0												
Israel	9	0												
Latvia	1	0												
Montenegro	6	0												
Norway	8	3		2	1		3				3			3
Romania	3	3			3			3			3			3
Serbia	17	5		5		5					5			5
Switzerland	6	0												
Ukraine	14	6	2	4		6					6			6
	913	296	61	146	89	177	81	38	1	19	276	16	31	249
37 Countries			20.6%	49.3%	30.1%	59.8%	27.4%	12.8%	0.3%	6.4%	93.2%	5.4%	10.5%	84.1%

1. Numbers shown in red indicate that not all of the specimens received had been propagated at the time this report was prepared
2. Propagated to sufficient titre to perform HI assay in presence of 20nM oseltamivir
3. Compared to homologous titres for ferret antisera raised egg-propagated A/Darwin/9/2021 (3C.2a1b.2a.2)
4. Compared to homologous titres for ferret antisera raised against tissue culture-propagated A/Stockholm/5/2021 (3C.2a1b.2a.2)
5. Compared to homologous titres for ferret antisera raised against egg-propagated A/Cambodia/e0826360/2020 (3C.2a1b.2a.1)
6. Compared to homologous titres for ferret antisera raised against tissue culture-propagated A/Cambodia/925256/2020 (3C.2a1b.2a.1)

As of 2022-02-14

Figure 10. Phylogenetic comparison of A(H3N2) HA genes (non-WIC sequences)

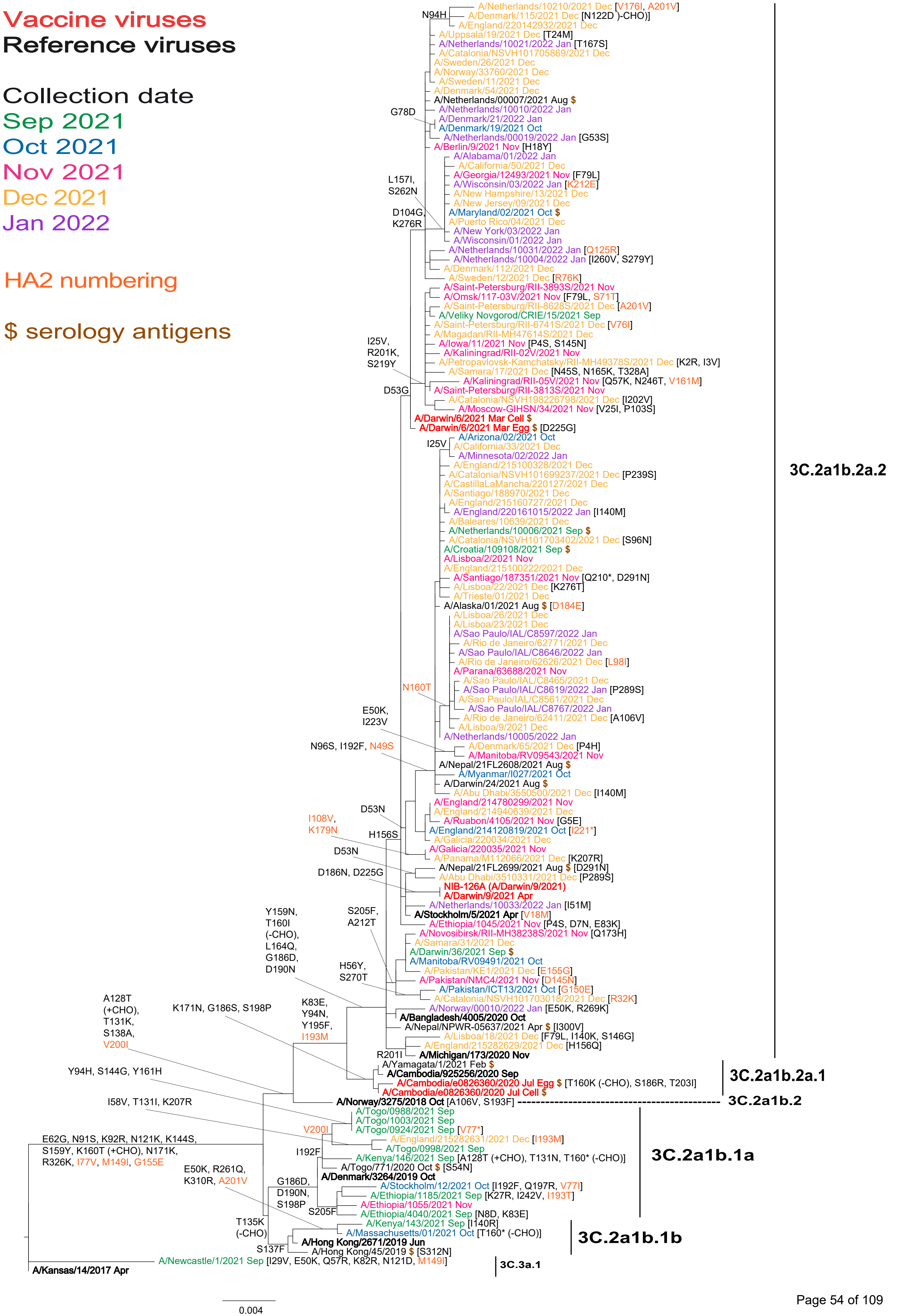


Figure 11. Phylogenetic comparison of A(H3N2) HA genes (WIC sequences)

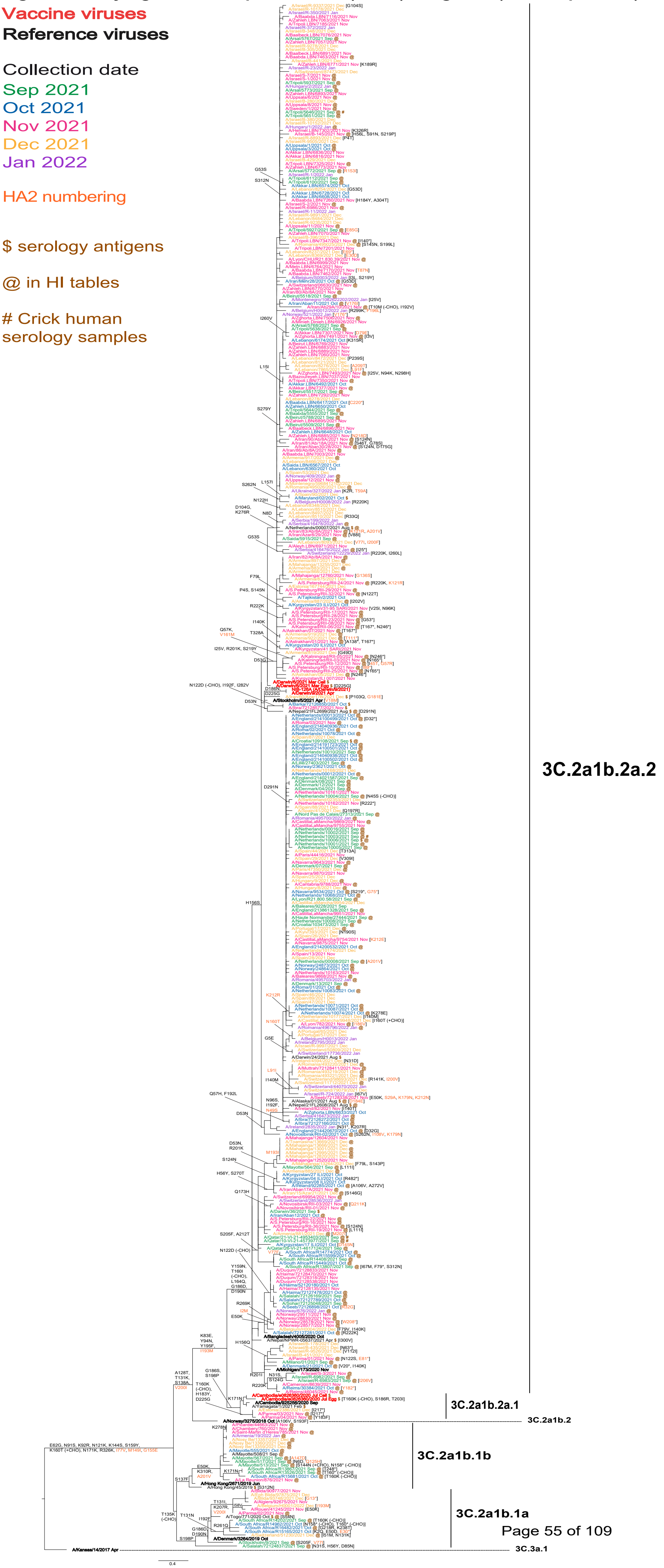


Figure 12. HA protein alignment for A(H3N2) viruses

Vaccine virus: Reference virus: 6B Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

```
                                   10        20        30        40        50        60        70        80        90       100
                          ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Norway/3275/2018        QKIPGNDNSTATLCLGHHAVPNGTIVKTITNDRIEVTNATELVQNSSIGEICDSPHQILDGGNCTLIDALLGDPQCDGFQNKKWDLFVERSRAYSNCYPY 3C.2a1b.2
A/Salalah/72124837/2021   ..............................S.........................Y............................N............... 1a
A/Switzerland/51230/2021  ..................................................M................................................. 1a
A/Rouen/41245/2021        .................................................K.......V.......................................... 1a
A/Hong Kong/45/2019       .................................................................................................... 1b
A/South_Africa/R15681/2021 .................................................K.................................................. 1b
A/Chambery/760/2021       .................................................K.................................................. 1b
A/Parma/04/2021           ..................................................................................E..........N...... 2a.1
A/Cambodia/e0826360/2020  ..................................................................................E..........N...... 2a.1
A/Bangladesh/4005/2020    ..................................................................................E..........N...... 2a.2
A/Sohar/72125048/2021     .................................................K................................E..........N...... 2a.2
A/Israel/R-6983/2021      ..............................S...................................................E..........N...... 2a.2
A/Armenia/881/2021        ..................................................................................E..........N...... 2a.2
A/Mahajanga/12820/2021    ....................................................N.............................E..........N...... 2a.2
A/Ireland/92285/2021      .......................................................Y..........................E..........N...... 2a.2
A/England/214420670/2021  ...............................G....................N.............................E..........N...... 2a.2
A/Romania/493220/2021     ....................................................N.............................E..........N.S.... 2a.2
A/Portugal/65/2021        ....................................................N.............................E..........N.S.... 2a.2
A/Switzerland/02353/2021  ....................................................N.............................E..........N.S.... 2a.2
A/Paris/44416/2021        ....................................................N.............................E..........N.S.... 2a.2
A/Barka/72126850/2021     ....................................................N.............................E..........N...... 2a.2
A/Darwin/9/2021E          ..................................................................................E..........N...... 2a.2
A/Armenia/919/2021        ........................V...........................G.............................E..........N...... 2a.2
A/Kyrgyzstan/23_ILI/2021  ...S....................V...........................G.............................E..........N...... 2a.2
A/S.Petersburg/RII-28/2021 ........................V...........................G.............................E..........N...... 2a.2
A/Iran/83/Ab/8A/2021      ....................................................G.............................E..........N...... 2a.2
A/Lebanon/8348/2021       ....................................................G.............................E..........N...... 2a.2
A/Akkar.LBN/6492/2021     ..............I.....................................G.............................E..........N...... 2a.2
A/Zghorta.LBN/7491/2021   ..V.................................................G.............................E..........N...... 2a.2
A/Spain/90/2021           ....................................................G.............................E..........N...... 2a.2
A/Akkar.LBN/6574/2021     ....................................................S.............................E..........N...... 2a.2
A/Hungary/1/2022 Jan      ....................................................G.............................E..........N...... 2a.2
```

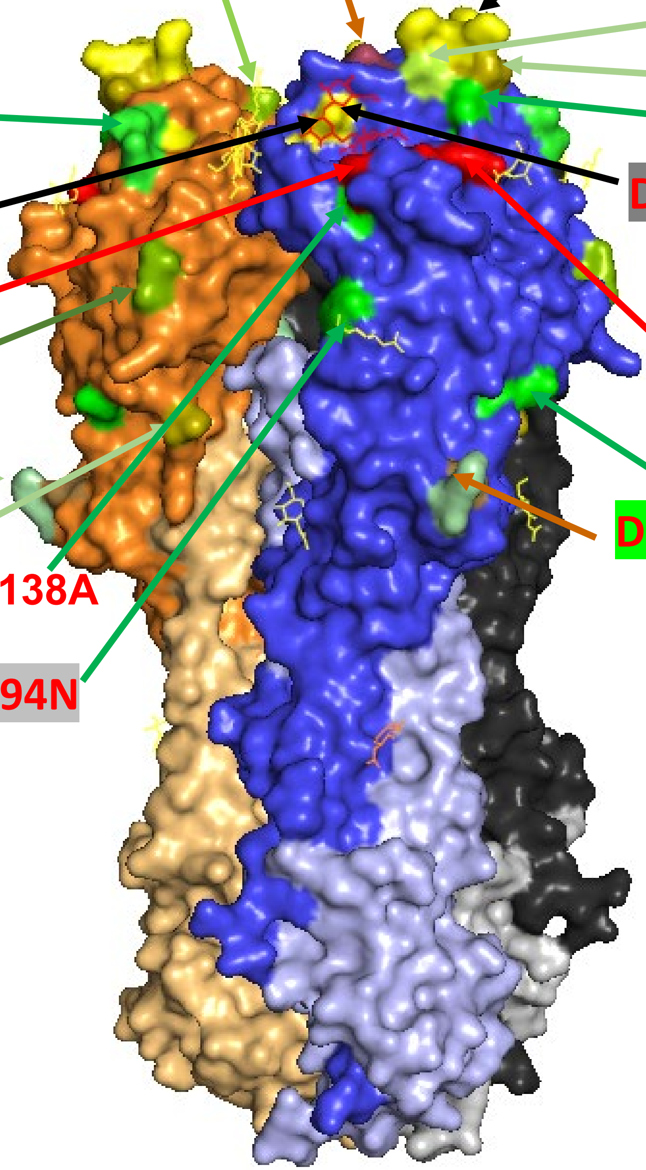
```
                                  110       120       130       140       150       160       170       180       190       200
                          ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Norway/3275/2018        DVPDYVSLRSLVASSGTLEFKNESFNWTGVKQNGTSSACIRGSSSSFFSRLNWLTHLNYTYPALNVTMPNKEQFDKLYIWGVHHPGTDKDQIFLYAQSSG
A/Salalah/72124837/2021   .....A.....................A..T...K..S......................................................S....P..
A/Switzerland/51230/2021  .....A.....................A......K..S...............................................D...N.FS....P..
A/Rouen/41245/2021        .....A.....................A..I...K..S...............................................D...N.FS....P..
A/Hong Kong/45/2019       .....A.....................A..T...K.FS......................................................S.......
A/South_Africa/R15681/2021 .....A.....................A..T...K.FS.....................K..........N.....................S.......
A/Chambery/760/2021       .....A.....................A..T...K.FS......................................................S.......
A/Parma/04/2021           .....A.....................................................K......................F..S......S.F..P..
A/Cambodia/e0826360/2020  .....A..............................................................N..............S......S.F..P..
A/Bangladesh/4005/2020    .....A....................................................NI...Q.....................D...N..S.F.....
A/Sohar/72125048/2021     .....A...............D....................................NI...Q.....................D...N..S.F.....
A/Israel/R-6983/2021      .....A.................G..................................NI...Q.....................D...N..S.F.....
A/Armenia/881/2021        .....A....................................................NI...Q.....................D...N..S.F.....
A/Mahajanga/12820/2021    .....A....................................................NI...Q.....................D...N..S.F.....
A/Ireland/92285/2021      ..........................................................NI...Q.....................D...N..S.F.....
A/England/214420670/2021  .....A.................................................S..NI...Q.....................D...N..S.F.....
A/Romania/493220/2021     .....A.................................M...............S..NI...Q.....................D...N.FS.F.....
A/Portugal/65/2021        .....A.................................................S..NI...Q.....................D...N.FS.F.....
A/Switzerland/02353/2021  .....A.................................................S..NI...Q.....................D...N.FS.F.....
A/Paris/44416/2021        .....A.................................................S..NI...Q.....................D...N.FS.F.....
A/Barka/72126850/2021     .....A...............D.................................S..NI...Q.....................D...N.FS.F.....
A/Darwin/9/2021E          .....A.................................................S..NI...Q.....................N...N..S.F.....
A/Armenia/919/2021        .....A.................................K...............S..NI...Q.....................D...N..S.F.....
A/Kyrgyzstan/23_ILI/2021  .....A......................................N..........S..NI...Q.....................D...N..S.F.....
A/S.Petersburg/RII-28/2021 .....A.................................................S..NI...Q.....................D...N..S.F.....
A/Iran/83/Ab/8A/2021      ...G.A.................................................S..NI...Q.....................D...N..S.F.....
A/Lebanon/8348/2021       ...G.A...............H.................................S..NI...Q.....................D...N..S.F.....
A/Akkar.LBN/6492/2021     ...G.A.................................................S..NI...Q.....................D...N..S.F.....
A/Zghorta.LBN/7491/2021   ...G.A.................................................S..NI...Q.....................D...N..S.F.....
A/Spain/90/2021           ...G.A.................................................SI.NI...Q.....................D...N..S.F.....
A/Akkar.LBN/6574/2021     ...G.A.................................................S..NI...Q.....................D...N..S.F.....
A/Hungary/1/2022 Jan      ...G.A.................................................S..NI...Q.....................D...N..S.F.....
```

```
                                  210       220       230       240       250       260       270       280       290       300
                          ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Norway/3275/2018        RITVSTKRSQQAVIPNIGSRPRIRDIPSRISIYWTIVKPGDILLINSTGNLIAPRGYFKIRSGKSSIMRSDAPIGKCKSECITPNGSIPNDKPFQNVNRI
A/Salalah/72124837/2021   ....................................................................................................
A/Switzerland/51230/2021  ....................................................................................................
A/Rouen/41245/2021        ......R.............................................................................................
A/Hong Kong/45/2019       ....................................................................................................
A/South_Africa/R15681/2021 ....................................................................................................
A/Chambery/760/2021       .............................................................................N......................
A/Parma/04/2021           ........................G...........................................................................
A/Cambodia/e0826360/2020  ....................................................................................................
A/Bangladesh/4005/2020    ....................................................................................................
A/Sohar/72125048/2021     ....................................................................................................
A/Israel/R-6983/2021      I..................K................................................................................
A/Armenia/881/2021        ....F......T........................................................................................
A/Mahajanga/12820/2021    K...F......T........................................................................................
A/Ireland/92285/2021      ....F......T.........................................................T.V............................
A/England/214420670/2021  ....................................................................................................
A/Romania/493220/2021     ....................................................................................................
A/Portugal/65/2021        ....................................................................................................
A/Switzerland/02353/2021  ..........................................................................................N.........
A/Paris/44416/2021        ....................................................................................................
A/Barka/72126850/2021     .................................................................................V..................
A/Darwin/9/2021E          ........................G...........................................................................
A/Armenia/919/2021        K.................Y.................................................................................
A/Kyrgyzstan/23_ILI/2021  K.................Y.................................................................................
A/S.Petersburg/RII-28/2021 K.................Y..K..............................................................................
A/Iran/83/Ab/8A/2021      ...........................................................................R........................
A/Lebanon/8348/2021       ...........................................................................R........................
A/Akkar.LBN/6492/2021     ...........................................................................R..Y.....................
A/Zghorta.LBN/7491/2021   ...........................................................V...............R..Y.....................
A/Spain/90/2021           .............................................................N.............R........................
A/Akkar.LBN/6574/2021     ...........................................................................R........................
A/Hungary/1/2022 Jan      ...........................................................................R........................
```

```
                                    310       320       330       340       350       360       370       380       390       400
                            ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Norway/3275/2018          TYGACPRYVKQSTLKLATGMRNVPEKQTRGIFGAIAGFIENGWEGMVDGWYGFRHQNSEGRGQAADLKSTQAAIDQINGKLNRLIGKTNEKFHQIEKEFS 3C.2a1b.2
A/Salalah/72124837/2021     .................................................................................................... 1a
A/Switzerland/51230/2021    .................................................................................................... 1a
A/Rouen/41245/2021          .................................................................................................... 1a
A/Hong Kong/45/2019         ...........N........................................................................................ 1b
A/South_Africa/R15681/2021  .........R.......................................................................................... 1b
A/Chambery/760/2021         .........R.......................................................................................... 1b
A/Parma/04/2021             .................................................................................................... 2a.1
A/Cambodia/e0826360/2020    .................................................................................................... 2a.1
A/Bangladesh/4005/2020      .................................................................................................... 2a.2
A/Sohar/72125048/2021       .................................................................................................... 2a.2
A/Israel/R-6983/2021        .................................................................................................... 2a.2
A/Armenia/881/2021          .................................................................................................... 2a.2
A/Mahajanga/12820/2021      .................................................................................................... 2a.2
A/Ireland/92285/2021        .................................................................................................... 2a.2
A/England/214420670/2021    .................................................................................................... 2a.2
A/Romania/493220/2021       .............................................................................S...................... 2a.2
A/Portugal/65/2021          .............................................................................S...................... 2a.2
A/Switzerland/02353/2021    .............................................................................S...................... 2a.2
A/Paris/44416/2021          .............................................................................S...................... 2a.2
A/Barka/72126850/2021       .................................................................................................... 2a.2
A/Darwin/9/2021E            .................................................................................................... 2a.2
A/Armenia/919/2021          .................................................................................................... 2a.2
A/Kyrgyzstan/23_ILI/2021    .................................................................................................... 2a.2
A/S.Petersburg/RII-28/2021  .................................................................................................... 2a.2
A/Iran/83/Ab/8A/2021        .................................................................................................... 2a.2
A/Lebanon/8348/2021         .................................................................................................... 2a.2
A/Akkar.LBN/6492/2021       .................................................................................................... 2a.2
A/Zghorta.LBN/7491/2021     .................................................................................................... 2a.2
A/Spain/90/2021             .................................................................................................... 2a.2
A/Akkar.LBN/6574/2021       .................................................................................................... 2a.2
A/Hungary/1/2022 Jan        .................................................................................................... 2a.2

                                    410       420       430       440       450       460       470       480       490       500
                            ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Norway/3275/2018          EVEGRVQDLEKYVEDTKIDLWSYNAELLVALENQHTIDLTDSEMNKLFEKTKKQLRENAEDMGNGCFKIYHKCDNACIGSIRNETYDHNVYRDEALNNRF
A/Salalah/72124837/2021     ....................................................................................................
A/Switzerland/51230/2021    ....................................................................................................
A/Rouen/41245/2021          ....................................................................................................
A/Hong Kong/45/2019         ....................................................................................................
A/South_Africa/R15681/2021  ....................................................................................................
A/Chambery/760/2021         ....................................................................................................
A/Parma/04/2021             ....................................................................................................
A/Cambodia/e0826360/2020    ....................................................................................................
A/Bangladesh/4005/2020      ....................................................................................................
A/Sohar/72125048/2021       ....................................................................................................
A/Israel/R-6983/2021        ....................................................................................................
A/Armenia/881/2021          ....................................................................................................
A/Mahajanga/12820/2021      ....................................................................................................
A/Ireland/92285/2021        ....................................................................................................
A/England/214420670/2021    ....................................................................................................
A/Romania/493220/2021       ....................................................................................................
A/Portugal/65/2021          ........................................................................................T...........
A/Switzerland/02353/2021    ....................................................................................................
A/Paris/44416/2021          ....................................................................................................
A/Barka/72126850/2021       ....................................................................................................
A/Darwin/9/2021E            ....................................................................................................
A/Armenia/919/2021          ....................................................................................................
A/Kyrgyzstan/23_ILI/2021    ....................................................................................................
A/S.Petersburg/RII-28/2021  ....................................................................................................
A/Iran/83/Ab/8A/2021        .................................................R..................................................
A/Lebanon/8348/2021         ....................................................................................................
A/Akkar.LBN/6492/2021       ....................................................................................................
A/Zghorta.LBN/7491/2021     ....................................................................................................
A/Spain/90/2021             ....................................................................................................
A/Akkar.LBN/6574/2021       ....................................................................................................
A/Hungary/1/2022 Jan        ....................................................................................................

                                    510       520       530       540       550
                            ....|....|....|....|....|....|....|....|....|....|
A/Norway/3275/2018          QIKGVELKSGYKDWILWISFAISCFLLCIALLGFIMWACQKGNIRCNICI
A/Salalah/72124837/2021     ............................V.....................
A/Switzerland/51230/2021    ............................V.....................
A/Rouen/41245/2021          ..................................................
A/Hong Kong/45/2019         ............................V.....................
A/South_Africa/R15681/2021  ............................VV....................
A/Chambery/760/2021         ............................VV....................
A/Parma/04/2021             .....................M............................
A/Cambodia/e0826360/2020    .....................M............................
A/Bangladesh/4005/2020      .....................M............................
A/Sohar/72125048/2021       .....................M............................
A/Israel/R-6983/2021        .....................M............V...............
A/Armenia/881/2021          .....................M.............I..............
A/Mahajanga/12820/2021      ..................................................
A/Ireland/92285/2021        .....................M............................
A/England/214420670/2021    .....................M............................
A/Romania/493220/2021       .....................M............................
A/Portugal/65/2021          .....................M............................
A/Switzerland/02353/2021    .....................M............................
A/Paris/44416/2021          .....................M............................
A/Barka/72126850/2021       .....................M............................
A/Darwin/9/2021E            .....................M............................
A/Armenia/919/2021          .....................M............................
A/Kyrgyzstan/23_ILI/2021    .....................M............................
A/S.Petersburg/RII-28/2021  .....................M............................
A/Iran/83/Ab/8A/2021        .....................M.......V....................
A/Lebanon/8348/2021         .....................M............................
A/Akkar.LBN/6492/2021       .....................M............................
A/Zghorta.LBN/7491/2021     .....................M............................
A/Spain/90/2021             .....................M............................
A/Akkar.LBN/6574/2021       .....................M............................
A/Hungary/1/2022 Jan        .....................M............................
```

Vaccine virus: Reference virus: 6B Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

Figure 13. HA1 amino acid substitutions defining H3N2 3C.2a1b viruses



Not visible: D104G, Y195F

Substitutions defining subgroups .1, .1a and .1b are indicated in black, those defining 2a.1 in blue, and those defining 2a.2 in red. The 2a.2 substitutions are coded to match the order in which they appear in the H3N2 HA phylogenetic trees.

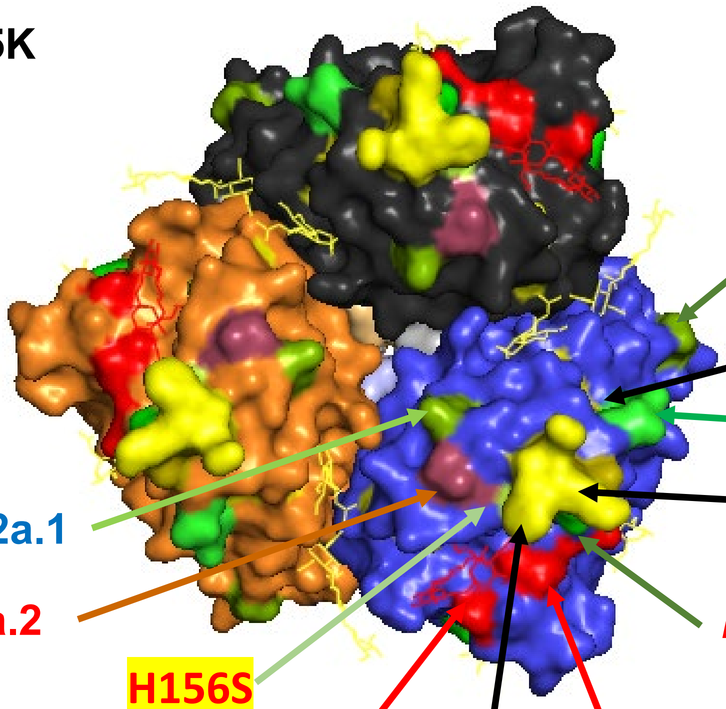


Figure 14. Phylogenetic comparison of A(H3N2) NA genes (non-WIC sequences)

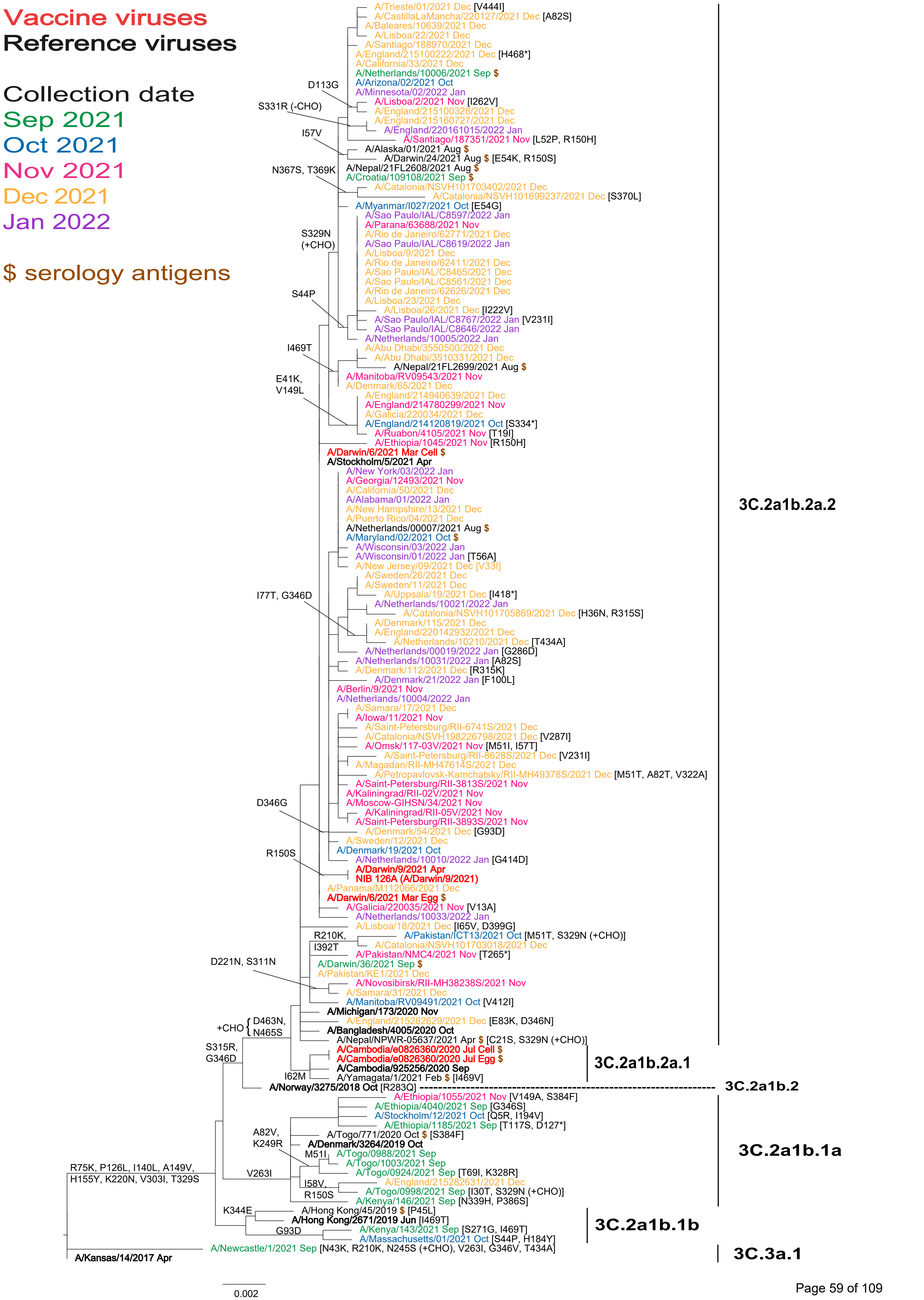


Figure 15. Phylogenetic comparison of A(H3N2) NA genes (WIC sequences)

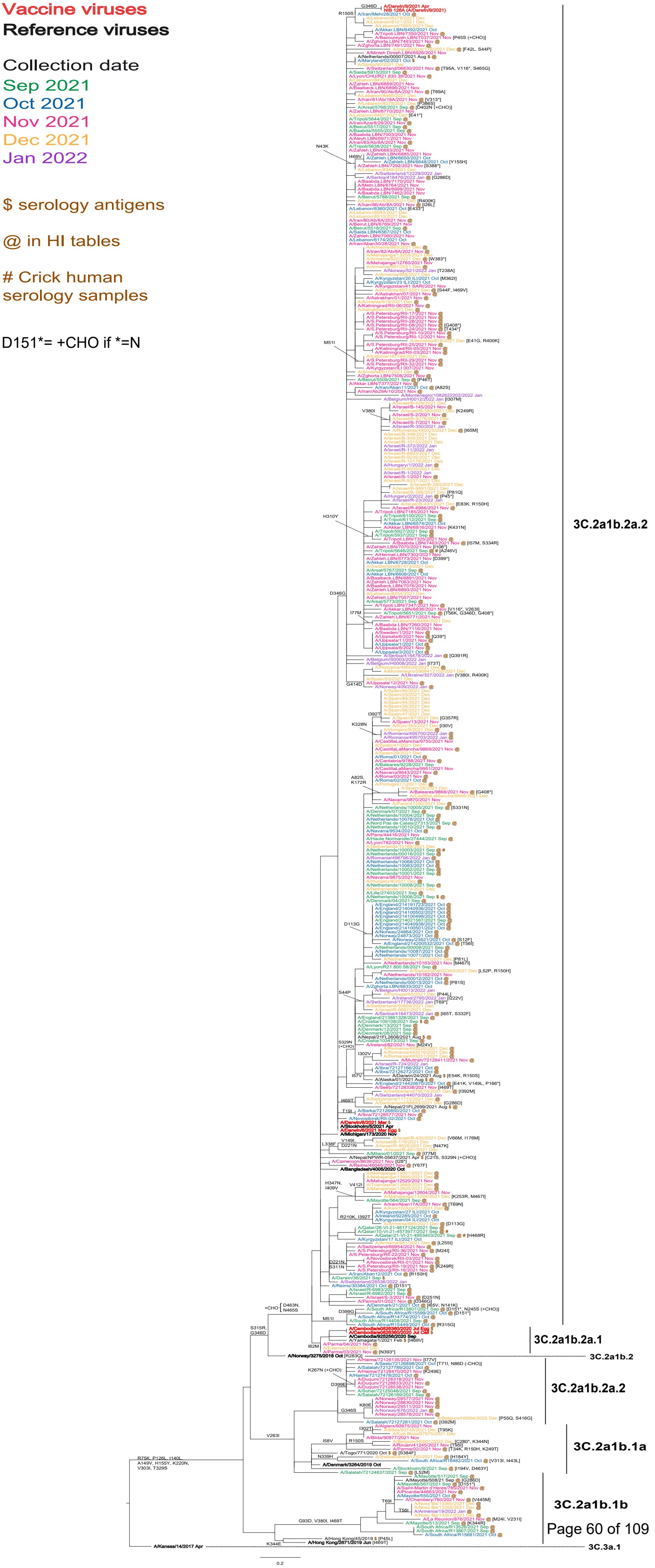


Figure 16. NA protein alignment for A(H3N2) viruses

Vaccine virus: Reference virus: 6B Genetic group: Potential N-linked glycosylation site

```
                                    10        20        30        40        50        60        70        80        90       100
                           ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Norway/3275/2018         MNPNQKIITIGSVSLTISTICFFMQIAILITTVTLHFKQYEFNSPPNNQVMLCEPTIIERNITEIVYLTNTTIEKEICPKPAEYRNWSKPQCGITGFAPF 3C.2a1b.2
A/Salalah/72124837/2021    ...................................................M................................................ 1a
A/Switzerland/51230/2021   .................................................................................................... 1a
A/Rouen/41245/2021         .........................................................V....................................I..... 1a
A/Hong Kong/45/2019        ............................................L....................................................... 1b
A/South Africa/R15681/2021 ............................................................................................D....... 1b
A/Chambery/760/2021        ....................................................................I.......................D....... 1b
A/Parma/04/2021            .............................................................M...................................... 2a.1
A/Cambodia/e0826360/2020   .............................................................M...................................... 2a.1
A/Bangladesh/4005/2020     .................................................................................................... 2a.2
A/Sohar/72125048/2021      .................................................................................................... 2a.2
A/Israel/R-6983/2021       .................................................................................................... 2a.2
A/Armenia/881/2021         .................................................................................................... 2a.2
A/Mahajanga/12820/2021     .................................................................................................... 2a.2
A/Ireland/92285/2021       .................................................................................................... 2a.2
A/England/214420670/2021   ........................................K........................................................... 2a.2
A/Romania/493220/2021      ........................................................V........................................... 2a.2
A/Portugal/65/2021         ...........................................L........................................................ 2a.2
A/Switzerland/02353/2021   ...................................................P................................................ 2a.2
A/Paris/44416/2021         .................................................................................................... 2a.2
A/Barka/72126850/2021      ..................I................................................................................. 2a.2
A/Darwin/9/2021            .................................................................................................... 2a.2
A/Armenia/919/2021         .................................................................................................... 2a.2
A/Kyrgyzstan/23 ILI/2021   .................................................................................................... 2a.2
A/S.Petersburg/RII-28/2021 .................................................................................................... 2a.2
A/Iran/83/Ab/8A/2021       .................................................................................................... 2a.2
A/Lebanon/8348/2021        ..........................................K......................................................... 2a.2
A/Akkar.LBN/6492/2021      .................................................................................................... 2a.2
A/Zghorta.LBN/7491/2021    .................................................................................................... 2a.2
A/Spain/90/2021            .................................................................................................... 2a.2
A/Akkar.LBN/6574/2021      .................................................................................................... 2a.2
A/Hungary/1/2022           .................................................................................................... 2a.2

                                   110       120       130       140       150       160       170       180       190       200
                           ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Norway/3275/2018         SKDNSIRLSAGGDIWVTREPYVSCDLDKCYQFALGQGTTLNNVHSNNTVRDRTPYRTLLMNELGVPFHLGTKQVCIAWSSSSCHDGKAWLHVCITGDDKN
A/Salalah/72124837/2021    ....................................................................................................
A/Switzerland/51230/2021   ...................................................................................Y................
A/Rouen/41245/2021         .................................................S..................................................
A/Hong Kong/45/2019        ....................................................................................................
A/South Africa/R15681/2021 ....................................................................................................
A/Chambery/760/2021        ....................................................................................................
A/Parma/04/2021            ....................................................................................................
A/Cambodia/e0826360/2020   ....................................................................................................
A/Bangladesh/4005/2020     ....................................................................................................
A/Sohar/72125048/2021      ....................................................................................................
A/Israel/R-6983/2021       ....................................................................................................
A/Armenia/881/2021         ....................................................................................................
A/Mahajanga/12820/2021     ....................................................................................................
A/Ireland/92285/2021       ....................................................................................................
A/England/214420670/2021   ................................................L................X..................................
A/Romania/493220/2021      ....................................................................................................
A/Portugal/65/2021         ....................................................................................................
A/Switzerland/02353/2021   .................................................H..................................................
A/Paris/44416/2021         ....................................................................................................
A/Barka/72126850/2021      ....................................................................................................
A/Darwin/9/2021            .................................................S..................................................
A/Armenia/919/2021         ....................................................................................................
A/Kyrgyzstan/23 ILI/2021   ....................................................................................................
A/S.Petersburg/RII-28/2021 ....................................................................................................
A/Iran/83/Ab/8A/2021       ....................................................................................................
A/Lebanon/8348/2021        ....................................................................................................
A/Akkar.LBN/6492/2021      ....................................................................................................
A/Zghorta.LBN/7491/2021    ....................................................................................................
A/Spain/90/2021            ....................................................................................................
A/Akkar.LBN/6574/2021      ....................................................................................................
A/Hungary/1/2022           ....................................................................................................

                                   210       220       230       240       250       260       270       280       290       300
                           ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Norway/3275/2018         ATASFIYNGRLVDSVVSWSNDILRTQESECVCINGTCTVVMTDGNATGKADTKILFIEEGKIVHTSKLSGSAQHVEECSCYPQYPGVRCVCRDNWKGSNR
A/Salalah/72124837/2021    ..............................................................I...................R.................
A/Switzerland/51230/2021   ..............................................................I...................R.................
A/Rouen/41245/2021         ..............................................................I...................R.................
A/Hong Kong/45/2019        ..................................................................................R.................
A/South Africa/R15681/2021 ..................................................................................R.................
A/Chambery/760/2021        ..................................................................................R.................
A/Parma/04/2021            ..................................................................................R.................
A/Cambodia/e0826360/2020   ..................................................................................R.................
A/Bangladesh/4005/2020     ..................................................................................R.................
A/Sohar/72125048/2021      ..............................................................I...................R.................
A/Israel/R-6983/2021       ..................................................................................R.................
A/Armenia/881/2021         ....................N.................................I...........................R.................
A/Mahajanga/12820/2021     ..................................................................................R.................
A/Ireland/92285/2021       .........K........................................................................R.................
A/England/214420670/2021   ..................................................................................R.................
A/Romania/493220/2021      ..................................................................................R.................
A/Portugal/65/2021         ..................................................................................R.................
A/Switzerland/02353/2021   ..................................................................................R.................
A/Paris/44416/2021         ..................................................................................R.................
A/Barka/72126850/2021      ..................................................................................R.................
A/Darwin/9/2021            ..................................................................................R.................
A/Armenia/919/2021         ..................................................................................R.................
A/Kyrgyzstan/23 ILI/2021   ..................................................................................R.................
A/S.Petersburg/RII-28/2021 ..................................................................................R.................
A/Iran/83/Ab/8A/2021       ..................................................................................R.................
A/Lebanon/8348/2021        ..................................................................................R.................
A/Akkar.LBN/6492/2021      ..................................................................................R.................
A/Zghorta.LBN/7491/2021    ..................................................................................R.................
A/Spain/90/2021            ..................................................................................R.................
A/Akkar.LBN/6574/2021      ..................................................................................R.................
A/Hungary/1/2022           ..................................................................................R.................
```

```
                                    310       320       330       340       350       360       370       380       390       400
                            ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Norway/3275/2018          PIIDINIKDHSIVSRYVCSGLVGDTPRKSDSSSSSHCLNPNNEKGDHGVKGWAFDDGNDVWMGRTINETSRLGYETFKVVEGWSNPKSKLQINRQVIVDR 3C.2a1b.2
A/Salalah/72124837/2021     ..............S..............................G...................................................... 1a
A/Switzerland/51230/2021    ..............S.......................H......G...................................................... 1a
A/Rouen/41245/2021          ..............S..............................G...................................................... 1a
A/Hong Kong/45/2019         ..............S............................E.G...................................................... 1b
A/South Africa/R15681/2021  ..............S..............................G.................................I.................... 1b
A/Chambery/760/2021         ..............S..............................G.................................I.................... 1b
A/Parma/04/2021             .................................................................................................... 2a.1
A/Cambodia/e0826360/2020    .................................................................................................... 2a.1
A/Bangladesh/4005/2020      .................................................................................................... 2a.2
A/Sohar/72125048/2021       ..............S..............................G....................................................E. 2a.2
A/Israel/R-6983/2021        .................................................................................................... 2a.2
A/Armenia/881/2021          ..........N......................................................................................... 2a.2
A/Mahajanga/12820/2021      ..............................................N..................................................... 2a.2
A/Ireland/92285/2021        ...........................................................................................T........ 2a.2
A/England/214420670/2021    .................................................................................................... 2a.2
A/Romania/493220/2021       .V..........................N....................................................................... 2a.2
A/Portugal/65/2021          ............................N....................................................................... 2a.2
A/Switzerland/02353/2021    ............................N....................................................................... 2a.2
A/Paris/44416/2021          ............................N....................................................................... 2a.2
A/Barka/72126850/2021       .................................................................................................... 2a.2
A/Darwin/9/2021             .................................................................................................... 2a.2
A/Armenia/919/2021          .............................................G...................................................... 2a.2
A/Kyrgyzstan/23 ILI/2021    .............................................G...................................................... 2a.2
A/S.Petersburg/RII-28/2021  .............................................G...................................................... 2a.2
A/Iran/83/Ab/8A/2021        .............................................G...................................................... 2a.2
A/Lebanon/8348/2021         .............................................G...................................................... 2a.2
A/Akkar.LBN/6492/2021       .............................................G...................................................... 2a.2
A/Zghorta.LBN/7491/2021     .............................................G...................................................... 2a.2
A/Spain/90/2021             .............................................G...................................................... 2a.2
A/Akkar.LBN/6574/2021       .............................................G...................................................... 2a.2
A/Hungary/1/2022            .............................................G...................................................... 2a.2
```

```
                                    410       420       430       440       450       460
                            ....|....|....|....|....|....|....|....|....|....|....|....|....|....
A/Norway/3275/2018          GDRSGYSGIFSVEGKSCINRCFYVELIRGRKEETEVLWTSNSIVVFCGTSGTYGTGSWPDGADLNLMHI
A/Salalah/72124837/2021     .....................................................................
A/Switzerland/51230/2021    .....................................................................
A/Rouen/41245/2021          .....................................................................
A/Hong Kong/45/2019         .....................................................................
A/South Africa/R15681/2021  ....................................................................T
A/Chambery/760/2021         ............................................M.......................T
A/Parma/04/2021             ..............................................................N.S....
A/Cambodia/e0826360/2020    ..............................................................N.S....
A/Bangladesh/4005/2020      ..............................................................N.S....
A/Sohar/72125048/2021       .....................................................................
A/Israel/R-6983/2021        ..............................................................N.S....
A/Armenia/881/2021          ..............................................................N.S....
A/Mahajanga/12820/2021      ........V..I..................................................N.S....
A/Ireland/92285/2021        ..............................................................N.S....
A/England/214420670/2021    ..............................................................N.S....
A/Romania/493220/2021       ..............................................................N.S....
A/Portugal/65/2021          ..............................................................N.S....
A/Switzerland/02353/2021    ..............................................................N.S....
A/Paris/44416/2021          ..............................................................N.S....
A/Barka/72126850/2021       ..............................................................N.S....
A/Darwin/9/2021             ..............................................................N.S....
A/Armenia/919/2021          ..............................................................N.S....
A/Kyrgyzstan/23 ILI/2021    ..............................................................N.S....
A/S.Petersburg/RII-28/2021  ..............................................................N.S....
A/Iran/83/Ab/8A/2021        ..............................................................N.S....
A/Lebanon/8348/2021         ..............................................................N.S....
A/Akkar.LBN/6492/2021       ..............................................................N.S....
A/Zghorta.LBN/7491/2021     ..............................................................N.S....
A/Spain/90/2021             ..............................................................N.S....
A/Akkar.LBN/6574/2021       ..............................................................N.S....
A/Hungary/1/2022            ..............................................................N.S....
```

Vaccine virus: Reference virus: 6B Genetic group: Potential N-linked glycosylation site

Table 9-1. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2021-10-08

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre / Post-infection ferret antisera								
	Passage history			A/Denmark 3264/19 SIAT	A/HK 2671/19 Cell St Judes	A/Camb e0826360/20 Egg	A/Camb 925256/20 SIAT	A/Bang 4005/20 SIAT	A/Darwin 9/21 Egg CDC	A/Stock 5/21 SIAT	A/Mich 173/20 SIAT CDC	A/Kansas 14/17 SIAT
	Ferret number			F19/20[*1]	F21/20[*1]	F10/21[*1]	F03/21[*1]	F07/21[*1]	10043883[*1]	F35/21[*1]	F2021-108[*1]	F17/19[*1]
	Genetic group			3C.2a1b.1a	3C.2a1b.1b	3C.2a1b.2a.1	3C.2a1b.2a.1	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.3a.1
REFERENCE VIRUSES												
A/Denmark/3264/2019	3C.2a1b.1a	2019-10-25	SIAT5	320	160	160	640	320	160	320	160	160
A/Hong Kong/2671/2019	3C.2a1b.1b	2019-06-17	MDCK1/SIAT4	160	320	80	320	160	40	160	160	160
A/Cambodia/e0826360/2020	3C.2a1b.2a.1	2020-07-16	E5/E2	320	<	2560	320	320	320	320	1280	160
A/Cambodia/925256/2020	3C.2a1b.2a.1	2020-09-25	SIAT4	160	80	80	1280	160	ND	160	ND	80
A/Bangladesh/4005/2020	3C.2a1b.2a.2	2020-10-04	SIAT2	640	160	320	320	640	80	1280	1280	320
A/Darwin/9/2021	3C.2a1b.2a.2	2021-04-17	E3/E2	40	40	40	40	40	320	80	80	40
A/Stockholm/5/2021	3C.2a1b.2a.2	2021-04-16	S0/S3	160	40	80	160	320	40	1280	320	80
A/Michigan/173/2020	3C.2a1b.2a.2	2020-11-19	SIAT2/SIAT1	320	160	320	320	640	320	640	640	320
A/Kansas/14/2017	3C.3a.1	2017-12-14	SIAT3/SIAT2	80	80	80	160	80	160	160	320	640
TEST VIRUSES												
A/Togo/771/2020	3C.2a1b.1a	2020-10-16	SIAT4/SIAT1	160	80	40	160	40	<	40	80	<
A/Qatar/16-VI-21-3809350/2021	no sequence	2021-08-03	SIAT2	160	40	320	160	320	160	160	ND	160
A/Qatar/16-VI-21-3801619/2021	no sequence	2021-08-03	SIAT1	160	40	80	80	320	160	160	ND	80
A/Delaware/01/2021	3C.2a1b.2a.2	2021-04-16	SIAT2/SIAT1	80	40	80	80	160	160	640	160	40
A/Qatar/34-VI-21-3664968/2021	3C.2a1b.2a.2	2021-07-27	SIAT0	80	40	80	80	160	160	640	160	80
A/Qatar/16-VI-21-3673175/2021	3C.2a1b.2a.2	2021-07-27	SIAT0	160	40	320	160	320	160	640	640	160
A/Qatar/16-VI-21-3738537/2021	3C.2a1b.2a.2	2021-07-30	SIAT2	40	40	80	40	80	40	80	80	40
A/Qatar/16-VI-21-3744895/2021	3C.2a1b.2a.2	2021-07-31	SIAT1	320	40	320	160	320	80	320	640	160
A/Qatar/24-VI-21-3752470/2021	3C.2a1b.2a.2	2021-08-01	SIAT1	320	160	320	160	640	320	640	640	160
A/Qatar/16-VI-21-3820344/2021	3C.2a1b.2a.2	2021-08-02	SIAT1	160	40	320	160	320	80	320	ND	160
A/Qatar/16-VI-21-3782139/2021	3C.2a1b.2a.2	2021-08-02	SIAT2	160	40	320	160	320	160	320	640	160
A/Qatar/16-VI-21-3778452/2021	3C.2a1b.2a.2	2021-08-02	SIAT2	160	40	320	160	320	80	320	640	160
A/Qatar/16-VI-21-3773931/2021	3C.2a1b.2a.2	2021-08-02	SIAT2	320	40	320	160	320	160	320	640	160
A/Qatar/16-VI-21-3798042/2021	3C.2a1b.2a.2	2021-08-03	SIAT1	160	40	320	160	320	80	320	ND	160
A/Qatar/10-VI-21-3813895/2021	3C.2a1b.2a.2	2021-08-04	SIAT1	160	40	160	80	320	80	40	ND	160
A/Qatar/10-VI-21-3843467/2021	3C.2a1b.2a.2	2021-08-05	SIAT1	160	40	320	160	320	160	40	ND	160
A/Qatar/24-VI-21-3868954/2021	3C.2a1b.2a.2	2021-08-06	SIAT1	<	80	40	40	80	40	40	ND	40
A/Qatar/10-VI-21-3886441/2021	3C.2a1b.2a.2	2021-08-07	SIAT1	160	40	320	160	320	160	80	ND	160
A/Qatar/24-VI-21-3912649/2021	3C.2a1b.2a.2	2021-08-08	SIAT1	160	40	320	80	320	80	80	ND	160
A/Qatar/10-VI-21-3904627/2021	3C.2a1b.2a.2	2021-08-08	SIAT2	160	40	320	160	320	80	80	ND	160
A/Qatar/24-VI-21-3942034/2021	3C.2a1b.2a.2	2021-08-09	SIAT1	160	40	320	160	320	80	160	ND	160
A/Qatar/24-VI-21-3929575/2021	3C.2a1b.2a.2	2021-08-09	SIAT2	160	40	320	80	320	80	160	ND	160
A/Qatar/47-VI-21-3971441/2021	3C.2a1b.2a.2	2021-08-10	SIAT1	160	40	160	160	320	80	320	640	160
A/Qatar/47-VI-21-3968065/2021	3C.2a1b.2a.2	2021-08-10	SIAT1	160	40	160	160	320	80	320	640	160
A/Qatar/24-VI-21-3962572/2021	3C.2a1b.2a.2	2021-08-10	SIAT1	160	80	320	160	320	80	320	640	160
A/Qatar/16-VI-21-3950368/2021	3C.2a1b.2a.2	2021-08-10	SIAT1	160	80	320	160	320	160	640	640	160
A/Qatar/47-VI-21-3995023/2021	3C.2a1b.2a.2	2021-08-11	SIAT1	320	40	320	160	320	160	640	640	160
A/Qatar/24-VI-21-3985321/2021	3C.2a1b.2a.2	2021-08-11	SIAT2	160	40	160	80	320	80	320	640	80
A/Qatar/24-VI-21-3980766/2021	3C.2a1b.2a.2	2021-08-11	SIAT2	40	40	40	80	320	80	80	80	40
A/Qatar/16-VI-21-3994833/2021	3C.2a1b.2a.2	2021-08-11	SIAT1	320	40	320	160	320	160	640	640	160
A/Qatar/16-VI-21-3994707/2021	3C.2a1b.2a.2	2021-08-11	SIAT1	160	40	160	80	320	80	320	320	160
A/Qatar/16-VI-21-3999123/2021	3C.2a1b.2a.2	2021-08-12	SIAT1	320	40	320	160	320	80	640	640	160
A/Qatar/24-VI-21-4041206/2021	3C.2a1b.2a.2	2021-08-13	SIAT1	160	40	320	160	320	160	320	640	80
A/Qatar/24-VI-21-4031255/2021	3C.2a1b.2a.2	2021-08-13	SIAT2	320	40	320	160	320	160	640	640	160
A/Qatar/16-VI-21-4035811/2021	3C.2a1b.2a.2	2021-08-13	SIAT1	160	40	40	80	160	160	640	160	40
A/Qatar/16-VI-21-4035218/2021	3C.2a1b.2a.2	2021-08-13	SIAT1	160	40	80	160	320	160	640	320	80
A/Qatar/39-VI-21-4147440/2021	3C.2a1b.2a.2	2021-08-17	SIAT1	160	40	320	160	320	160	640	640	80
A/Qatar/39-VI-21-4154225/2021	3C.2a1b.2a.2	2021-08-17	SIAT1	80	<	80	80	320	160	640	320	40
						Vaccine NH 2021-22			Vaccine SH 2022			

*Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)
1 < = <40, ND = Not Done

Sequences in phylogenetic trees

< 4-fold | 4-fold | 8-fold | > 8-fold | < not recognised by the antiserum | ≥ 160 (when homologous titre ≥ 2560) | ≥ 160 (no homologous titre)

Table 9-2. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2021-10-22

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre Post-infection ferret antisera							
				A/Denmark	A/HK	A/Camb	A/Camb	A/Bang	A/Darwin	A/Stock	A/Kansas
				3264/19	2671/19	e0826360/20	925256/20	4005/20	9/21	5/21	14/17
	Passage history			SIAT	Cell	Egg	SIAT	SIAT	Egg	SIAT	SIAT
	Ferret number			F19/20[*1]	St Judes F21/20[*1]	F10/21[*1]	F03/21[*1]	F07/21[*1]	F38/21[*1]	F35/21[*1]	F17/19[*1]
	Genetic group			3C.2a1b.1a	3C.2a1b.1b	3C.2a1b.2a.1	3C.2a1b.2a.1	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.3a.1
REFERENCE VIRUSES											
A/Denmark/3264/2019	3C.2a1b.1a	2019-10-25	SIAT5	320	160	80	640	160	320	160	80
A/Hong Kong/2671/2019	3C.2a1b.1b	2019-06-17	MDCK1/SIAT4	320	320	80	320	160	320	160	80
A/Cambodia/e0826360/2020	3C.2a1b.2a.1	2020-07-16	E5/E2	160	<	1280	160	160	640	320	80
A/Cambodia/925256/2020	3C.2a1b.2a.1	2020-09-25	SIAT4	160	80	80	640	160	320	160	80
A/Bangladesh/4005/2020	3C.2a1b.2a.2	2020-10-04	SIAT2	160	40	160	160	320	640	640	160
A/Darwin/9/2021	3C.2a1b.2a.2	2021-04-17	E3/E2	80	<	160	160	160	2560	640	40
A/Stockholm/5/2021	3C.2a1b.2a.2	2021-04-16	S0/S3	80	<	40	80	160	1280	640	40
A/Kansas/14/2017	3C.3a.1	2017-12-14	SIAT3/SIAT2	40	<	40	40	40	80	80	640
TEST VIRUSES											
A/Baabda/5555/2021	3C.2a1b.2a.2	2021-09-01	SIAT1	<	<	<	40	80	640	320	<
A/Beirut/5518/2021	3C.2a1b.2a.2	2021-09-01	SIAT1	<	<	<	<	80	640	320	<
A/Beirut/5517/2021	3C.2a1b.2a.2	2021-09-01	SIAT1	40	<	<	<	80	640	320	<
A/Beirut/5509/2021	3C.2a1b.2a.2	2021-09-01	SIAT0	40	<	<	<	80	640	320	<
A/Tripoli/5651/2021	3C.2a1b.2a.2	2021-09-07	SIAT1	80	<	<	40	160	640	640	<
A/Tripoli/5646/2021	3C.2a1b.2a.2	2021-09-07	SIAT1	<	<	<	40	80	640	320	<
A/Tripoli/5644/2021	3C.2a1b.2a.2	2021-09-07	SIAT1	<	<	<	40	80	640	320	<
A/Tripoli/5638/2021	3C.2a1b.2a.2	2021-09-07	SIAT1	<	<	<	40	80	640	320	<
A/Arsal/5773/2021	3C.2a1b.2a.2	2021-09-13	SIAT1	40	<	<	<	80	640	320	<
A/Arsal/5772/2021	3C.2a1b.2a.2	2021-09-13	SIAT1	40	<	<	<	80	640	320	<
A/Arsal/5768/2021	3C.2a1b.2a.2	2021-09-13	SIAT1	40	<	<	40	80	640	640	<
A/Arsal/5767/2021	3C.2a1b.2a.2	2021-09-13	SIAT1	80	<	<	40	160	640	320	<
A/Beirut/5788/2021	3C.2a1b.2a.2	2021-09-14	SIAT1	40	<	<	<	80	640	320	<
A/Tripoli/5937/2021	3C.2a1b.2a.2	2021-09-21	SIAT1	40	<	<	<	80	640	320	<
A/Tripoli/5927/2021	3C.2a1b.2a.2	2021-09-21	SIAT1	80	<	160	40	160	1280	640	40
A/Saida/5915/2021	3C.2a1b.2a.2	2021-09-21	SIAT1	40	<	<	40	80	640	320	<
A/Tripoli/6100/2021	3C.2a1b.2a.2	2021-09-30	SIAT1	40	<	<	40	80	640	320	<
A/Tripoli/6112/2021	3C.2a1b.2a.2	2021-09-30	SIAT1	40	<	<	40	80	640	320	<
						Vaccine NH 2021-22			Vaccine SH 2022		

[*] Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)
1 < = <40, ND = Not Done

Sequences in phylogenetic trees

< 4-fold | 4-fold | 8-fold | > 8-fold | < not recognised by the antiserum | ≥ 160 (when homologous titre ≥ 2560) | ≥ 160 (no homologous titre)

Table 9-3. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2021-11-05

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre							
				Post-infection ferret antisera							
				A/Denmark 3264/19	A/HK 2671/19	A/Camb e0826360/20	A/Camb 925256/20	A/Bang 4005/20	A/Darwin 9/21	A/Stock 5/21	A/Kansas 14/17
	Passage history			SIAT	Cell	Egg	SIAT	SIAT	Egg	SIAT	SIAT
	Ferret number			F19/20[*1]	St Judes F21/20[*1]	F10/21[*1]	F03/21[*1]	F07/21[*1]	F38/21[*1]	F35/21[*1]	F17/19[*1]
	Genetic group			3C.2a1b.1a	3C.2a1b.1b	3C.2a1b.2a.1	3C.2a1b.2a.1	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.3a.1
REFERENCE VIRUSES											
A/Denmark/3264/2019	3C.2a1b.1a	2019-10-25	SIAT5	320	320	80	640	160	320	160	160
A/Hong Kong/2671/2019	3C.2a1b.1b	2019-06-17	MDCK1/SIAT4	320	320	160	640	160	320	320	160
A/Cambodia/e0826360/2020	3C.2a1b.2a.1	2020-07-16	E5/E2	160	<	2560	160	320	640	320	80
A/Cambodia/925256/2020	3C.2a1b.2a.1	2020-09-25	SIAT4	160	80	80	640	160	320	160	80
A/Bangladesh/4005/2020	3C.2a1b.2a.2	2020-10-04	SIAT2	640	160	160	640	1280	2560	1280	320
A/Darwin/9/2021	3C.2a1b.2a.2	2021-04-17	E3/E2	160	<	320	80	640	2560	1280	80
A/Stockholm/5/2021	3C.2a1b.2a.2	2021-04-16	S0/S3	160	40	40	160	640	2560	1280	80
A/Kansas/14/2017	3C.3a.1	2017-12-14	SIAT3/SIAT2	80	40	40	80	160	160	160	640
TEST VIRUSES											
A/Netherlands/00007/2021	3C.2a1b.2a.2	2021-08-02	hCK/SIAT1	80	<	160	40	320	1280	640	<
A/Netherlands/10003/2021	3C.2a1b.2a.2	2021-09-02	MDCK-MIX2/SIAT1	80	<	160	80	320	1280	1280	40
A/Netherlands/10001/2021	3C.2a1b.2a.2	2021-09-06	MDCK-MIX2/SIAT1	40	<	160	80	320	640	640	40
A/Netherlands/10002/2021	3C.2a1b.2a.2	2021-09-06	MDCK-MIX2/SIAT1	40	<	160	80	320	1280	640	40
A/Netherlands/00016/2021	3C.2a1b.2a.2	2021-09-06	hCK/SIAT1	160	<	160	80	320	2560	1280	40
A/Netherlands/10004/2021	3C.2a1b.2a.2	2021-09-07	MDCK-MIX2/SIAT1	80	<	160	80	320	1280	640	40
A/Netherlands/10005/2021	3C.2a1b.2a.2	2021-09-07	MDCK-MIX2/SIAT2	80	<	160	80	320	1280	640	40
A/Netherlands/10006/2021	3C.2a1b.2a.2	2021-09-07	MDCK-MIX2/SIAT2	80	<	160	160	320	1280	1280	40
A/Netherlands/00008/2021	3C.2a1b.2a.2	2021-09-08	hCK/SIAT1	80	<	160	80	320	1280	640	40
A/Netherlands/10008/2021	3C.2a1b.2a.2	2021-09-09	MDCK-MIX2/SIAT1	40	<	160	40	160	640	320	<
A/Netherlands/10010/2021	3C.2a1b.2a.2	2021-09-11	MDCK-MIX2/SIAT1	40	<	160	80	160	1280	640	<
A/Netherlands/00012/2021	3C.2a1b.2a.2	2021-10-04	hCK/SIAT1	80	<	160	80	320	1280	640	40
A/Netherlands/00013/2021	3C.2a1b.2a.2	2021-10-08	hCK/SIAT1	160	<	160	160	320	2560	1280	40
						Vaccine NH 2021-22			Vaccine SH 2022		

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)
1 < = <40, ND = Not Done

Sequences in phylogenetic trees

< 4-fold | 4-fold | 8-fold | > 8-fold | < not recognised by the antiserum | ≥ 160 (when homologous titre ≥ 2560) | ≥ 160 (no homologous titre)

Table 9-4. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2021-11-12

				Haemagglutination inhibition titre									
				Post-infection ferret antisera									
								NEW				NEW	
Viruses	Other information	Collection date	Passage history	A/Denmark 3264/19	A/HK 2671/19	A/Camb e0826360/20	A/Camb 925256/20	A/Camb 925256/20	A/Bang 4005/20	A/Darwin 9/21	A/Stock 5/21	A/Stock 5/21	A/Kansas 14/17
	Passage history			SIAT	Cell	Egg	SIAT	SIAT	SIAT	Egg	SIAT	SIAT	SIAT
	Ferret number			F19/20[*1]	St Judes F21/20[*1]	F10/21[*1]	F03/21[*1]	F43/21[*1]	F07/21[*1]	F38/21[*1]	F35/21[*1]	F42/21[*1]	F17/19[*1]
	Genetic group			3C.2a1b.1a	3C.2a1b.1b	3C.2a1b.2a.1	3C.2a1b.2a.1	3C.2a1b.2a.1	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.3a.1
REFERENCE VIRUSES													
A/Denmark/3264/2019	3C.2a1b.1a	2019-10-25	SIAT5	320	320	160	640	320	320	320	160	160	80
A/Hong Kong/2671/2019	3C.2a1b.1b	2019-06-17	MDCK1/SIAT4	320	320	160	640	160	320	320	160	80	80
A/Cambodia/e0826360/2020	3C.2a1b.2a.1	2020-07-16	E5/E2	160	<	1280	160	40	320	320	160	40	80
A/Cambodia/925256/2020	3C.2a1b.2a.1	2020-09-25	SIAT4	160	160	160	640	320	320	320	160	160	80
A/Bangladesh/4005/2020	3C.2a1b.2a.2	2020-10-04	SIAT3	160	80	320	160	80	640	640	640	160	160
A/Darwin/9/2021	3C.2a1b.2a.2	2021-04-17	E3/E2	160	<	640	160	80	640	2560	640	320	40
A/Stockholm/5/2021	3C.2a1b.2a.2	2021-04-16	SIAT0/SIAT3	80	<	80	80	<	320	1280	640	320	40
A/Kansas/14/2017	3C.3a.1	2017-12-14	SIAT3/SIAT2	40	<	80	40	<	40	80	80	<	320
TEST VIRUSES													
A/Croatia/97333/2021	3C.2a1b.2a.2	2021-08-17	MDCKx/SIAT1	<	<	80	80	<	160	640	320	160	<
A/Croatia/99546/2021	3C.2a1b.2a.2	2021-08-25	MDCKx/SIAT1	<	<	80	40	<	160	640	320	160	<
A/Croatia/103473/2021	3C.2a1b.2a.2	2021-09-07	MDCKx/SIAT1	<	<	80	40	<	160	640	320	160	<
A/Croatia/109108/2021	3C.2a1b.2a.2	2021-09-20	MDCKx/SIAT2	<	<	80	40	<	160	320	320	160	<
						Vaccine NH 2021-22				Vaccine SH 2022			

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 < = <40, ND = Not Done

Sequences in phylogenetic trees

< 4-fold | 4-fold | 8-fold | > 8-fold | < not recognised by the antiserum | ≥ 160 (when homologous titre ≥ 2560) | ≥ 160 (no homologous titre)

Table 9-5. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2021-12-15

Viruses	Other information		Collection date	Passage history	Haemagglutination inhibition titre							
					Post-infection ferret antisera							
					A/Denmark	A/HK	A/Camb	A/Camb	A/Bang	A/Darwin	A/Stock	A/Kansas
					3264/19	2671/19	e0826360/20	925256/20	4005/20	9/21	5/21	14/17
		Passage history			SIAT	Cell	Egg	SIAT	SIAT	Egg	SIAT	SIAT
		Ferret number			F19/20*1	St Judes F21/20*1	F10/21*1	F03/21*1	F07/21*1	F38/21*1	F35/21*1	F17/19*1
		Genetic group			3C.2a1b.1a	3C.2a1b.1b	3C.2a1b.2a.1	3C.2a1b.2a.1	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.3a.1
REFERENCE VIRUSES												
A/Denmark/3264/2019		3C.2a1b.1a	2019-10-25	SIAT5	320	320	160	640	160	160	80	80
A/Hong Kong/2671/2019		3C.2a1b.1b	2019-06-17	MDCK1/SIAT4	320	320	160	640	160	160	80	80
A/Cambodia/e0826360/2020		3C.2a1b.2a.1	2020-07-16	E5/E2	160	<	1280	160	160	320	160	80
A/Cambodia/925256/2020		3C.2a1b.2a.1	2020-09-25	SIAT4	160	160	160	640	320	320	160	80
A/Bangladesh/4005/2020		3C.2a1b.2a.2	2020-10-04	SIAT3	160	<	160	160	640	640	640	160
A/Darwin/9/2021		3C.2a1b.2a.2	2021-04-17	E3/E4	160	<	320	80	320	2560	640	40
A/Stockholm/5/2021		3C.2a1b.2a.2	2021-04-16	S0/S3	80	<	80	80	160	1280	640	40
A/Kansas/14/2017		3C.3a.1	2017-12-14	SIAT3/SIAT2	40	<	40	80	80	80	80	640
TEST VIRUSES												
A/Denmark/04/2021		3C.2a1b.2a.2	2021-09-15	SIAT3/SIAT1	40	<	80	40	160	640	640	40
A/Denmark/08/2021		3C.2a1b.2a.2	2021-09-17	SIAT3/SIAT1	40	<	40	40	160	320	160	<
A/Denmark/07/2021		3C.2a1b.2a.2	2021-09-17	SIAT2/SIAT1	40	<	40	40	160	320	160	<
A/Denmark/13/2021		3C.2a1b.2a.2	2021-09-21	SIAT3/SIAT1	40	<	40	40	160	320	320	<
A/Denmark/12/2021		3C.2a1b.2a.2	2021-09-25	SIAT3/SIAT1	40	<	40	40	160	320	320	<
A/Denmark/21/2021	I140K, R201I	3C.2a1b.2a.2	2021-10-29	SIAT3/SIAT1	160	<	320	160	640	640	640	160

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)
1 < = <40, ND = Not Done

Vaccine NH 2021-22 (A/Camb e0826360/20)

Vaccine SH 2022 (A/Darwin 9/21)

Sequences in phylogenetic trees

< 4-fold | 4-fold | 8-fold | > 8-fold | < not recognised by the antiserum | ≥ 160 (when homologous titre ≥ 2560) | ≥ 160 (no homologous titre)

Table 9-6. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2022-01-21

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre – Post-infection ferret antisera							
	Passage history			A/Denmark 3264/19 SIAT	A/HK 2671/19 Cell	A/Camb e0826360/20 Egg	A/Camb 925256/20 SIAT	A/Bang 4005/20 SIAT	A/Darwin 9/21 Egg	A/Stock 5/21 SIAT	A/Kansas 14/17 SIAT
	Ferret number			F19/20*1	St Judes F21/20*1	F10/21*1	F03/21*1	F07/21*1	F38/21*1	F35/21*1	F17/19*1
	Genetic group			3C.2a1b.1a	3C.2a1b.1b	3C.2a1b.2a.1	3C.2a1b.2a.1	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.3a.1
REFERENCE VIRUSES											
A/Denmark/3264/2019	3C.2a1b.1a	2019-10-25	SIAT5	640	640	160	640	320	320	160	160
A/Hong Kong/2671/2019	3C.2a1b.1b	2019-06-17	MDCK1/SIAT4	640	640	160	640	320	320	160	160
A/Cambodia/e0826360/2020	3C.2a1b.2a.1	2020-07-16	E5/E2	320	<	2560	160	320	320	320	80
A/Cambodia/925256/2020	3C.2a1b.2a.1	2020-09-25	SIAT4	160	160	160	640	320	320	160	80
A/Bangladesh/4005/2020	3C.2a1b.2a.2	2020-10-04	SIAT3	320	80	320	160	640	640	640	320
A/Darwin/9/2021	3C.2a1b.2a.2	2021-04-17	E3/E2	320	40	320	80	640	2560	640	80
A/Stockholm/5/2021	3C.2a1b.2a.2	2021-04-16	S0/S3	160	40	80	80	320	1280	640	40
A/Kansas/14/2017	3C.3a.1	2017-12-14	SIAT3/SIAT2	80	40	40	80	80	160	160	640
TEST VIRUSES											
A/Salalah/72124837/2021	3C.2a1b.1a	2021-09-05	SIAT1/SIAT1	160	160	80	320	80	80	40	40
A/Mayotte/517/2021	3C.2a1b.1b	2021-09-11	MDCK2/SIAT1	160	320	<	320	80	<	40	<
A/Mayotte/513/2021	3C.2a1b.1b	2021-09-15	MDCK2/SIAT1	80	40	<	40	<	40	40	<
A/Mayotte/508/21	3C.2a1b.1b	2021-09-18	MDCK2/SIAT1	320	320	<	320	80	40	40	<
A/Mayotte/567/2021	3C.2a1b.1b	2021-09-26	MDCK3/SIAT1	320	320	<	320	80	40	40	<
A/Mayotte/555/2021	3C.2a1b.1b	2021-10-01	MDCK2/SIAT1	160	160	<	160	80	<	40	<
A/Chambery/760/2021	3C.2a1b.1b	2021-11-01	MDCK2/SIAT1	640	320	<	320	40	80	<	<
A/Saint-Martin d'Heres/785/2021	3C.2a1b.1b	2021-11-03	MDCK2/SIAT1	160	320	<	160	40	<	<	<
A/La Reunion/876/2021	3C.2a1b.1b	2021-11-14	MDCK2/SIAT1	160	320	<	160	40	<	40	<
A/Picardie/44663/2021	3C.2a1b.1b	2021-11-24	MDCK1/SIAT1	160	160	<	320	40	40	<	<
A/Mayotte/616/2021	no sequence	2021-10-09	MDCK2/SIAT1	320	<	640	80	640	640	320	160
A/Sohar/72125048/2021	3C.2a1b.2a.2	2021-09-08	SIAT1/SIAT1	320	40	80	80	640	640	320	80
A/Nord Pas de Calais/27313/2021	3C.2a1b.2a.2	2021-09-09	MDCK1/SIAT1	40	<	40	40	160	640	320	<
A/Haute Normandie/27444/2021	3C.2a1b.2a.2	2021-09-11	MDCK1/SIAT1	80	<	80	40	160	640	320	<
A/Lille/27403/2021	3C.2a1b.2a.2	2021-09-13	MDCK1/SIAT1	80	<	80	80	160	1280	320	<
A/Lyon/R21.800.58/2021	3C.2a1b.2a.2	2021-09-16	MDCK2/SIAT1	40	<	40	40	160	320	160	<
A/Mayotte/564/2021	3C.2a1b.2a.2	2021-09-28	MDCK2/SIAT1	320	<	640	80	640	640	320	160
A/Salalah/72126169/2021	3C.2a1b.2a.2	2021-09-29	SIAT1/SIAT1	160	80	160	80	640	640	640	160
A/Ibra/72126272/2021	3C.2a1b.2a.2	2021-10-03	SIAT1/SIAT1	160	40	80	80	320	1280	640	<
A/Reims/30384/2021	3C.2a1b.2a.2	2021-10-08	MDCK2/SIAT2	640	40	640	160	1280	1280	1280	320
A/Norway/23621/2021	3C.2a1b.2a.2	2021-10-13	SIAT1	40	<	40	40	160	640	320	<
A/Navarra/9534/2021	3C.2a1b.2a.2	2021-10-16	SIAT1/SIAT1	80	<	80	40	160	640	320	<
A/Barka/72126850/2021	3C.2a1b.2a.2	2021-10-17	SIAT1/SIAT1	40	<	40	40	160	320	160	<
A/Norway/24873/2021	3C.2a1b.2a.2	2021-10-24	SIAT1	40	<	40	80	160	640	320	<
A/Salalah/72127281/2021	3C.2a1b.2a.2	2021-10-24	SIAT1/SIAT1	160	<	80	40	320	320	320	80
A/Ibra/72127166/2021	3C.2a1b.2a.2	2021-10-24	SIAT1/SIAT1	40	<	40	40	160	640	320	<
A/Norway/24864/2021	3C.2a1b.2a.2	2021-10-26	SIAT1	40	<	40	40	160	640	320	<
A/Netherlands/10071/2021	3C.2a1b.2a.2	2021-10-26	MDCK-MIX2/SIAT1	<	<	40	40	160	640	160	<
A/Haima/72127478/2021	3C.2a1b.2a.2	2021-10-26	SIAT1/SIAT1	320	40	80	80	640	640	320	160
A/Netherlands/10074/2021	3C.2a1b.2a.2	2021-10-28	MDCK-MIX2/SIAT1	<	<	40	40	80	640	160	<
A/Novosibirsk/RII-02/2021	3C.2a1b.2a.2	2021-10-30	SIAT2/SIAT1	80	<	80	40	320	1280	640	<
A/Netherlands/10087/2021	3C.2a1b.2a.2	2021-10-31	MDCK-MIX2/SIAT1	40	<	40	40	160	640	160	<
A/Salalah/72127789/2021	3C.2a1b.2a.2	2021-10-31	SIAT1/SIAT1	320	40	80	80	640	640	320	160
A/Novosibirsk/RII-03/2021	3C.2a1b.2a.2	2021-11-02	SIAT2/SIAT1	160	40	320	160	640	640	640	160
A/Novosibirsk/RII-01/2021	3C.2a1b.2a.2	2021-11-03	SIAT2/SIAT1	160	<	320	80	320	320	320	80
A/S.Petersburg/RII-10/2021	3C.2a1b.2a.2	2021-11-04	MDCK3/SIAT1	320	80	320	160	640	2560	1280	160
A/S.Petersburg/RII-12/2021	3C.2a1b.2a.2	2021-11-04	SIAT2/SIAT1	320	80	320	160	640	1280	640	160
A/Lyon/782/2021	3C.2a1b.2a.2	2021-11-06	MDCK2/SIAT1	40	<	40	40	160	320	160	<
A/Navarra/9643/2021	3C.2a1b.2a.2	2021-11-06	SIAT1/SIAT1	40	<	40	40	160	640	160	<
A/Muttrah/72128411/2021	3C.2a1b.2a.2	2021-11-09	SIAT1/SIAT2	40	<	40	40	160	640	320	<
A/S.Petersburg/RII-08/2021	3C.2a1b.2a.2	2021-11-13	SIAT2/SIAT1	160	40	160	40	320	640	320	80
A/Ibra/72128577/2021	3C.2a1b.2a.2	2021-11-14	SIAT1/SIAT1	40	<	40	40	160	640	320	<
A/Cantabria/9788/2021	3C.2a1b.2a.2	2021-11-15	SIAT1/SIAT1	40	<	40	40	160	640	320	<
A/Castilla La Mancha/9869/2021	3C.2a1b.2a.2	2021-11-16	SIAT1/SIAT1	40	<	40	40	80	640	160	<
A/Norway/28578/2021	3C.2a1b.2a.2	2021-11-19	SIAT1	320	40	160	160	640	640	640	160
A/Norway/28830/2021	3C.2a1b.2a.2	2021-11-21	SIAT1	320	40	160	160	640	640	1280	160
A/Norway/28577/2021	3C.2a1b.2a.2	2021-11-21	SIAT1	320	80	160	160	1280	1280	1280	160
A/Lyon/CHU- R21.830.39/2021	3C.2a1b.2a.2	2021-11-22	MDCK2/SIAT1	80	<	40	40	160	1280	640	<
A/Norway/29511/2021	3C.2a1b.2a.2	2021-11-25	SIAT1	320	80	160	160	640	1280	640	160
A/Baleares/9868/2021	3C.2a1b.2a.2	2021-11-26	SIAT1/SIAT1	40	<	40	40	160	640	320	<
A/Reims/46045/2021	3C.2a1b.2a.2	2021-11-30	MDCK1/SIAT1	160	40	320	80	640	640	320	160
A/Paris/47350/2021	3C.2a1b.2a.2	2021-12-02	MDCK1/SIAT1	40	<	40	80	160	640	320	<
						Vaccine NH 2021-22			Vaccine SH 2022		

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)
1 < = <40, ND = Not Done

Sequences in phylogenetic trees

< 4-fold | 4-fold | 8-fold | > 8-fold | < not recognised by the antiserum | ≥ 160 (when homologous titre ≥ 2560) | ≥ 160 (no homologous titre)

Table 9-7. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2022-01-26

Viruses	Other information	Genetic group	Collection date	Passage history	A/Denmark 3264/19 SIAT F19/20[1] 3C.2a1b.1a	A/HK 2671/19 Cell St Judes F21/20[1] 3C.2a1b.1b	A/Camb e0826360/20 Egg F10/21[1] 3C.2a1b.2a.1	A/Camb 925256/20 SIAT F03/21[1] 3C.2a1b.2a.1	A/Bang 4005/20 SIAT F07/21[1] 3C.2a1b.2a.2	A/Darwin 9/21 Egg F38/21[1] 3C.2a1b.2a.2	A/Stock 5/21 SIAT F35/21[1] 3C.2a1b.2a.2	A/Kansas 14/17 SIAT F17/19[1] 3C.3a.1
					Haemagglutination inhibition titre – Post-infection ferret antisera							
REFERENCE VIRUSES												
A/Denmark/3264/2019		3C.2a1b.1a	2019-10-25	SIAT5	640	640	320	640	320	320	320	160
A/Hong Kong/2671/2019		3C.2a1b.1b	2019-06-17	MDCK1/SIAT4	320	640	160	640	160	320	160	160
A/Cambodia/e0826360/2020		3C.2a1b.2a.1	2020-07-16	E5/E2	320	<	2560	160	320	320	320	80
A/Cambodia/925256/2020		3C.2a1b.2a.1	2020-09-25	SIAT4	160	160	160	640	320	320	160	160
A/Bangladesh/4005/2020		3C.2a1b.2a.2	2020-10-04	SIAT3	320	80	320	320	640	640	640	320
A/Darwin/9/2021		3C.2a1b.2a.2	2021-04-17	E3/E2	320	40	320	80	640	2560	640	80
A/Stockholm/5/2021		3C.2a1b.2a.2	2021-04-16	S0/S3	160	40	80	80	320	1280	640	40
A/Kansas/14/2017		3C.3a.1	2017-12-14	SIAT3/SIAT2	80	40	40	80	80	160	160	640
TEST VIRUSES												
A/South Africa/R12902/2021		3C.2a1b.1a	2021-08-27	MDCK3/SIAT1	160	160	160	320	160	160	80	80
A/Qatar/24-VI-21-4532193/2021		no sequence	2021-08-30	SIAT1	320	<	640	160	640	640	320	160
A/Alaska/01/2021		3C.2a1b.2a.2	2021-08-20	SIAT1	40	<	80	80	160	640	160	40
A/Stockholm/8/2021		3C.2a1b.2a.2	2021-08-10	SIAT1/SIAT1	80	<	80	80	160	1280	320	40
A/Orebro/1/2021		3C.2a1b.2a.2	2021-08-21	SIAT1/SIAT1	80	<	80	40	160	640	320	40
A/Qatar/47-VI-21-4548906/2021		3C.2a1b.2a.2	2021-08-31	SIAT1	160	<	320	80	320	320	160	80
A/Stockholm/9/2021		3C.2a1b.1a	2021-09-06	SIAT1/SIAT1	320	160	160	640	160	160	80	80
A/South Africa/R14202/2021		3C.2a1b.1a	2021-09-23	MDCK1/SIAT1	160	<	640	80	160	160	160	80
A/South Africa/R16482/2021		3C.2a1b.1a	2021-10-18	MDCK2/SIAT1	320	80	320	320	160	160	80	80
A/South Africa/R13526/2021		3C.2a1b.1b	2021-09-07	MDCK1/SIAT1	80	160	40	160	<	<	40	40
A/South Africa/R13867/2021		3C.2a1b.1b	2021-09-16	MDCK1/SIAT1	160	160	40	640	40	<	<	40
A/South Africa/R15681/2021	T160K(-cho)	3C.2a1b.1b	2021-10-26	MDCK1/SIAT1	40	80	80	80	40	<	40	80
A/Qatar/47-VI-21-4610963/2021		no sequence	2021-09-02	SIAT1	40	<	80	40	160	320	160	40
A/Qatar/16-VI-21-4634794/2021		no sequence	2021-09-03	SIAT1	160	<	160	80	320	320	320	80
A/Qatar/15-VI-21-4746170/2021		no sequence	2021-09-07	SIAT1	160	<	160	80	320	320	320	80
A/Qatar/24-VI-21-4774653/2021		no sequence	2021-09-08	SIAT1	160	<	160	80	320	320	320	80
A/Minieh Dinieh.LBN/7496/2021		no sequence	2021-11-23	SIAT1	160	<	80	80	320	1280	640	40
A/Saida.LBN/7412/2021		no sequence	2021-11-23	SIAT1	80	<	40	80	160	640	320	40
A/Qatar/10-VI-21-4573977/2021		3C.2a1b.2a.2	2021-09-01	SIAT1	160	<	320	80	320	320	320	80
A/Qatar/26-VI-21-4617124/2021		3C.2a1b.2a.2	2021-09-02	SIAT1	160	<	160	80	320	320	320	80
A/South Africa/R13807/2021		3C.2a1b.2a.2	2021-09-13	MDCK1/SIAT1	320	160	640	320	640	1280	640	320
A/Qatar/21-VI-21-4953403/2021		3C.2a1b.2a.2	2021-09-15	SIAT1	160	<	160	80	320	320	320	80
A/England/213861328/2021		3C.2a1b.2a.2	2021-09-19	MDCK1/SIAT1	40	<	80	80	160	640	160	40
A/South Africa/R14408/2021		3C.2a1b.2a.2	2021-09-20	MDCK1/SIAT1	160	<	320	80	320	320	320	160
A/England/214021587/2021		3C.2a1b.2a.2	2021-09-30	SIAT1/SIAT1	40	<	80	80	160	640	320	40
A/England/214040938/2021		3C.2a1b.2a.2	2021-10-03	SIAT1/SIAT1	40	<	80	80	160	640	320	40
A/England/214040936/2021		3C.2a1b.2a.2	2021-10-04	SIAT1/SIAT1	40	<	80	80	160	640	320	40
A/England/214100502/2021		3C.2a1b.2a.2	2021-10-06	SIAT1/SIAT1	40	<	80	80	160	320	160	40
A/England/214100501/2021		3C.2a1b.2a.2	2021-10-06	SIAT1/SIAT1	40	<	80	80	160	320	160	40
A/South Africa/R14774/2021		3C.2a1b.2a.2	2021-10-07	MDCK2/SIAT1	320	<	640	160	640	640	320	160
A/England/214100499/2021		3C.2a1b.2a.2	2021-10-07	MDCK1/SIAT1	40	<	80	40	80	320	160	40
A/England/214191723/2021		3C.2a1b.2a.2	2021-10-12	MDCK1/SIAT1	40	<	80	80	160	640	320	40
A/England/214200532/2021		3C.2a1b.2a.2	2021-10-13	SIAT1/SIAT1	40	<	80	80	160	640	320	40
A/Uppsala/1/2021		3C.2a1b.2a.2	2021-10-21	SIAT1/SIAT1	80	<	80	80	160	640	320	40
A/South Africa/R15449/2021		3C.2a1b.2a.2	2021-10-21	MDCK1/SIAT1	320	<	640	160	320	640	640	160
A/Netherlands/10068/2021		3C.2a1b.2a.2	2021-10-24	MDCK-MIX2/SIAT2	40	<	40	40	40	320	160	40
A/Uppsala/3/2021		3C.2a1b.2a.2	2021-10-26	SIAT1/SIAT1	80	<	80	80	160	640	320	40
A/South Africa/R15599/2021		3C.2a1b.2a.2	2021-10-26	MDCK1/MDCK1	320	<	320	160	320	320	320	160
A/Netherlands/10083/2021		3C.2a1b.2a.2	2021-10-28	MDCK-MIX2/SIAT2	40	<	80	40	160	320	160	40
A/England/214420670/2021		3C.2a1b.2a.2	2021-10-28	MDCK1/SIAT1	80	<	80	80	160	640	320	40
A/Netherlands/10078/2021		3C.2a1b.2a.2	2021-10-29	MDCK-MIX2/SIAT2	40	<	80	80	160	640	320	40
A/Uppsala/6/2021		3C.2a1b.2a.2	2021-11-01	SIAT1/SIAT1	80	<	80	80	160	1280	320	40
A/Sweden/1/2021		3C.2a1b.2a.2	2021-11-03	SIAT1/SIAT1	80	<	80	80	320	1280	640	40
A/Uppsala/8/2021		3C.2a1b.2a.2	2021-11-03	SIAT1/SIAT1	80	<	80	80	320	2560	640	40
A/Uppsala/11/2021		3C.2a1b.2a.2	2021-11-09	SIAT1/SIAT1	80	<	80	80	160	640	320	80
A/Uppsala/12/2021		3C.2a1b.2a.2	2021-11-10	SIAT1/SIAT1	80	<	80	40	160	640	320	40
A/Zghorta.LBN/7506/2021		3C.2a1b.2a.2	2021-11-23	SIAT1	40	<	40	40	160	640	640	40
A/Zghorta.LBN/7491/2021		3C.2a1b.2a.2	2021-11-23	SIAT1	80	<	40	80	160	640	320	40
A/Akkar.LBN/7377/2021		3C.2a1b.2a.2	2021-11-23	SIAT1	80	<	40	40	160	640	320	40
A/Tripoli.LBN/7350/2021		3C.2a1b.2a.2	2021-11-23	SIAT1	80	<	40	80	320	1280	640	40
A/Tripoli.LBN/7347/2021		3C.2a1b.2a.2	2021-11-23	SIAT1	80	<	40	80	320	1280	640	40
A/Tripoli.LBN/7325/2021		3C.2a1b.2a.2	2021-11-23	SIAT1	80	<	40	40	160	640	320	40
A/Zghorta.LBN/7493/2021		3C.2a1b.2a.2	2021-11-24	SIAT1	40	<	40	40	160	640	640	40
A/Baabda.LBN/7463/2021		3C.2a1b.2a.2	2021-11-24	SIAT1	40	<	40	40	320	640	640	40
A/Baabda.LBN/7462/2021		3C.2a1b.2a.2	2021-11-24	SIAT1	40	<	40	40	160	640	320	40
A/Navarra/9870/2021		3C.2a1b.2a.2	2021-11-24	SIAT1/SIAT2	40	<	80	40	160	320	160	40
A/Netherlands/10162/2021		3C.2a1b.2a.2	2021-11-26	MDCK-MIX2/SIAT1	40	<	40	80	80	320	160	40
A/Netherlands/10161/2021		3C.2a1b.2a.2	2021-11-28	MDCK-MIX2/SIAT1	40	<	80	80	80	320	160	40
A/Netherlands/10163/2021		3C.2a1b.2a.2	2021-11-30	MDCK-MIX2/SIAT1	40	<	80	40	80	320	160	40
A/Netherlands/10174/2021		3C.2a1b.2a.2	2021-12-01	MDCK-MIX2/SIAT1	40	<	80	80	80	320	320	40
A/Netherlands/10168/2021		3C.2a1b.2a.2	2021-12-01	MDCK-MIX2/SIAT1	40	<	80	40	80	320	160	40
A/Netherlands/10177/2021		3C.2a1b.2a.2	2021-12-02	MDCK-MIX2/SIAT1	40	<	80	80	160	320	320	40
							Vaccine NH 2021-22			Vaccine SH 2022		

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)
1 < = <40, ND = Not Done

Sequences in phylogenetic trees

< 4-fold | 4-fold | 8-fold | > 8-fold | < not recognised by the antiserum | ≥ 160 (when homologous titre ≥ 2560) | ≥ 160 (no homologous titre)

Table 9-8. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2022-01-28

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre Post-infection ferret antisera A/Denmark 3264/19	A/HK 2671/19	A/Camb e0826360/20	A/Camb 925256/20	A/Bang 4005/20	A/Darwin 9/21	A/Stock 5/21	A/Kansas 14/17
	Passage history			SIAT	Cell	Egg	SIAT	SIAT	Egg	SIAT	SIAT
	Ferret number			F19/20*1	St Judes F21/20*1	F10/21*1	F03/21*1	F07/21*1	F38/21*1	F35/21*1	F17/19*1
	Genetic group			3C.2a1b.1a	3C.2a1b.1b	3C.2a1b.2a.1	3C.2a1b.2a.1	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.3a.1
REFERENCE VIRUSES											
A/Denmark/3264/2019	3C.2a1b.1a	2019-10-25	SIAT5	320	320	160	1280	320	320	320	160
A/Hong Kong/2671/2019	3C.2a1b.1b	2019-06-17	MDCK1/SIAT4	160	320	160	640	160	160	160	160
A/Cambodia/e0826360/2020	3C.2a1b.2a.1	2020-07-16	E5/E2	160	<	2560	160	320	320	320	160
A/Cambodia/925256/2020	3C.2a1b.2a.1	2020-09-25	SIAT5	160	80	320	640	320	320	320	160
A/Bangladesh/4005/2020	3C.2a1b.2a.2	2020-10-04	SIAT3	160	40	160	320	640	640	640	160
A/Darwin/9/2021	3C.2a1b.2a.2	2021-04-17	E3/E2	160	<	640	160	640	2560	1280	80
A/Stockholm/5/2021	3C.2a1b.2a.2	2021-04-16	S0/S3	80	<	80	80	320	640	1280	40
A/Kansas/14/2017	3C.3a.1	2017-12-14	SIAT3/SIAT2	40	<	40	80	80	80	160	640
TEST VIRUSES											
A/Switzerland/51230/2021	3C.2a1b.1a	2021-12-02	MDCK1/SIAT1	80	80	160	320	160	160	160	40
A/Nosy Be/13359/2021	3C.2a1b.1b	2021-12-06	SIAT1	160	80	<	160	40	<	40	40
A/Nosy Be/13357/2021	3C.2a1b.1b	2021-12-06	SIAT1	160	160	<	320	80	40	80	40
A/Nosy Be/13360/2021	3C.2a1b.1b	2021-12-07	SIAT1	160	160	<	160	80	40	40	40
A/Armenia/921/2021	no sequence	2021-12-28	SIAT1	160	<	320	80	320	1280	640	80
A/Switzerland/06630/2021	3C.2a1b.2a.2	2021-11-02	MDCK1/SIAT1	40	<	40	40	160	640	320	<
A/Astrakhan/01/2021	3C.2a1b.2a.2	2021-11-03	MDCK4/SIAT1	40	<	80	<	80	160	160	40
A/Switzerland/69954/2021	3C.2a1b.2a.2	2021-11-20	MDCK1/SIAT1	160	<	320	80	640	320	320	80
A/S.Petersburg/RII-25/2021	3C.2a1b.2a.2	2021-11-22	MDCK3/SIAT1	160	<	320	80	320	640	320	80
A/S.Petersburg/RII-16/2021	3C.2a1b.2a.2	2021-11-22	SIAT2/SIAT1	160	<	320	80	320	320	320	80
A/S.Petersburg/RII-19/2021	3C.2a1b.2a.2	2021-11-23	SIAT1/SIAT1	160	<	320	40	320	320	320	80
A/S.Petersburg/RII-17/2021	3C.2a1b.2a.2	2021-11-23	SIAT2/SIAT1	160	<	320	40	160	640	320	40
A/Kaliningrad/RII-03/2021	3C.2a1b.2a.2	2021-11-24	MDCK3/SIAT1	80	<	160	40	320	320	320	40
A/S.Petersburg/RII-22/2021	3C.2a1b.2a.2	2021-11-25	SIAT2/SIAT1	160	<	320	80	320	320	320	80
A/Kaliningrad/RII-06/2021	3C.2a1b.2a.2	2021-11-25	MDCK3/SIAT1	160	<	160	40	160	320	160	80
A/Kaliningrad/RII-05/2021	3C.2a1b.2a.2	2021-11-25	MDCK3/SIAT1	80	<	160	40	160	320	160	40
A/Astrakhan/07/2021	3C.2a1b.2a.2	2021-11-25	MDCK3/SIAT1	320	40	640	80	640	640	640	160
A/S.Petersburg/RII-24/2021	3C.2a1b.2a.2	2021-11-27	SIAT2/SIAT1	80	<	320	40	320	640	320	40
A/S.Petersburg/RII-23/2021	3C.2a1b.2a.2	2021-11-27	SIAT2/SIAT1	80	<	320	40	160	320	320	40
A/S.Petersburg/RII-28/2021	3C.2a1b.2a.2	2021-11-29	SIAT2/SIAT1	160	<	320	80	320	640	320	80
A/S.Petersburg/RII-36/2021	3C.2a1b.2a.2	2021-11-30	MDCK2/SIAT1	160	<	320	80	320	320	320	80
A/S.Petersburg/RII-32/2021	3C.2a1b.2a.2	2021-11-30	MDCK2/SIAT1	160	<	160	80	320	1280	640	80
A/S.Petersburg/RII-29/2021	3C.2a1b.2a.2	2021-11-30	MDCK2/SIAT1	80	<	320	80	320	640	640	40
A/Mahajanga/12820/2021	3C.2a1b.2a.2	2021-12-01	SIAT1	160	<	640	80	320	320	320	80
A/Astrakhan/05/2021	3C.2a1b.2a.2	2021-12-02	MDCK3/SIAT1	320	<	320	80	320	640	640	80
A/Mahajanga/13001/2021	3C.2a1b.2a.2	2021-12-03	SIAT1	160	<	320	80	320	320	320	80
A/Mahajanga/12995/2021	3C.2a1b.2a.2	2021-12-06	SIAT1	160	<	640	80	320	320	320	80
A/Toamasina/13669/2021	3C.2a1b.2a.2	2021-12-15	SIAT1	160	<	320	80	320	320	320	80
A/Portugal/17/2021	3C.2a1b.2a.2	2021-12-15	SIAT1/SIAT1	<	<	40	40	160	320	160	<
A/Switzerland/11712/2021	3C.2a1b.2a.2	2021-12-22	MDCK1/SIAT1	40	<	80	80	160	320	320	<
A/Armenia/887/2021	3C.2a1b.2a.2	2021-12-24	SIAT1	320	<	640	160	320	1280	640	80
A/Armenia/885/2021	3C.2a1b.2a.2	2021-12-24	SIAT1	160	<	320	80	320	320	320	80
A/Armenia/883/2021	3C.2a1b.2a.2	2021-12-24	SIAT1	160	<	320	80	320	1280	640	80
A/Armenia/881/2021	3C.2a1b.2a.2	2021-12-24	SIAT1	160	<	160	40	320	320	320	80
A/Armenia/922/2021	3C.2a1b.2a.2	2021-12-28	SIAT1	160	<	320	40	320	640	320	80
A/Armenia/919/2021	3C.2a1b.2a.2	2021-12-28	SIAT1	160	<	320	80	320	640	320	40
A/Armenia/917/2021	3C.2a1b.2a.2	2021-12-28	SIAT1	80	<	80	40	160	640	640	<
A/Armenia/897/2021	3C.2a1b.2a.2	2021-12-28	SIAT1	160	<	320	80	320	640	640	80
						Vaccine NH 2021-22			Vaccine SH 2022		

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)
1 < = <40, ND = Not Done

Sequences in phylogenetic trees

< 4-fold | 4-fold | 8-fold | > 8-fold | < not recognised by the antiserum | ≥ 160 (when homologous titre ≥ 2560) | ≥ 160 (no homologous titre)

Table 9-9. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2022-01-28

Viruses	Other information		Collection date	Passage history	Haemagglutination inhibition titre / Post-infection ferret antisera: A/Denmark 3264/19	A/HK 2671/19	A/Camb e0826360/20	A/Camb 925256/20	A/Bang 4005/20	A/Darwin 9/21	A/Stock 5/21	A/Kansas 14/17
		Passage history			SIAT	Cell St Judes	Egg	SIAT	SIAT	Egg	SIAT	SIAT
		Ferret number			F19/20*1	F21/20*1	F10/21*1	F03/21*1	F07/21*1	F38/21*1	F35/21*1	F17/19*1
		Genetic group			3C.2a1b.1a	3C.2a1b.1b	3C.2a1b.2a.1	3C.2a1b.2a.1	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.3a.1
REFERENCE VIRUSES												
A/Denmark/3264/2019		3C.2a1b.1a	2019-10-25	SIAT5	320	320	160	1280	320	320	320	160
A/Hong Kong/2671/2019		3C.2a1b.1b	2019-06-17	MDCK1/SIAT4	160	320	160	640	160	160	160	160
A/Cambodia/e0826360/2020		3C.2a1b.2a.1	2020-07-16	E5/E2	160	<	2560	160	320	320	320	160
A/Cambodia/925256/2020		3C.2a1b.2a.1	2020-09-25	SIAT5	160	80	320	640	320	320	320	160
A/Bangladesh/4005/2020		3C.2a1b.2a.2	2020-10-04	SIAT3	160	40	160	320	640	640	640	160
A/Darwin/9/2021		3C.2a1b.2a.2	2021-04-17	E3/E2	160	<	640	160	640	2560	1280	80
A/Stockholm/5/2021		3C.2a1b.2a.2	2021-04-16	S0/S3	80	<	80	80	320	640	1280	40
A/Kansas/14/2017		3C.3a.1	2017-12-14	SIAT3/SIAT2	40	<	40	80	80	80	160	640
TEST VIRUSES												
A/Parma/02/2021		3C.2a1b.1a	2021-11-03	SIAT2/SIAT1	160	160	160	320	160	320	160	80
A/Parma/03/2021		3C.2a1b.2a.1	2021-11-14	SIAT2/SIAT1	320	80	640	320	320	320	320	160
A/Parma/04/2021		3C.2a1b.2a.1	2021-11-22	SIAT2/SIAT1	160	40	640	160	320	320	320	80
A/Israel/R-6982/2021		3C.2a1b.2a.2	2021-09-12	Px/SIAT1	320	<	640	160	640	640	640	160
A/Israel/R-6983/2021		3C.2a1b.2a.2	2021-09-19	Px/SIAT1	320	40	640	160	640	640	640	160
A/Milano/01/2021		3C.2a1b.2a.2	2021-09-26	SIAT3/SIAT1	160	<	320	160	320	640	640	160
A/Iran/Mehr28/2021		3C.2a1b.2a.2	2021-10-14	SIAT3/SIAT1	80	<	80	80	320	640	640	40
A/Roma/01/2021		3C.2a1b.2a.2	2021-10-20	SIAT3/SIAT1	40	<	80	80	160	320	160	40
A/Roma/02/2021		3C.2a1b.2a.2	2021-10-27	SIAT3/SIAT1	40	<	80	80	160	320	160	40
A/Iran/Aban12/2021		3C.2a1b.2a.2	2021-10-28	SIAT3/SIAT1	160	<	320	80	320	320	320	80
A/Iran/Aban11/2021		3C.2a1b.2a.2	2021-10-30	SIAT3/SIAT1	80	80	80	80	320	640	640	40
A/Iran/83/Ab/8A/2021		3C.2a1b.2a.2	2021-11-03	SIAT4/SIAT1	80	<	80	40	160	640	320	<
A/Iran/82/Ab/8A/2021		3C.2a1b.2a.2	2021-11-03	SIAT4/SIAT2	320	40	320	160	640	1280	1280	80
A/Iran/81/Ab/18A/2021		3C.2a1b.2a.2	2021-11-03	SIAT4/SIAT1	80	<	80	80	320	640	640	40
A/Iran/80/Ab/8A/2021		3C.2a1b.2a.2	2021-11-03	SIAT4/SIAT1	80	<	80	80	320	1280	640	40
A/Roma/03/2021		3C.2a1b.2a.2	2021-11-03	SIAT2/SIAT1	40	<	80	80	160	320	320	<
A/Iran/90/Ab/8A/2021		3C.2a1b.2a.2	2021-11-04	SIAT5/SIAT1	80	<	80	80	320	640	640	40
A/Iran/Aban17A/2021		3C.2a1b.2a.2	2021-11-06	SIAT5/SIAT1	160	<	320	160	640	640	640	160
A/Iran/86/Ab/8A/2021		3C.2a1b.2a.2	2021-11-09	SIAT4/SIAT1	80	<	80	80	320	640	640	40
A/Israel/R-6986/2021		3C.2a1b.2a.2	2021-11-09	Px/SIAT1	80	<	80	40	320	640	320	40
A/Israel/S-1/2021		3C.2a1b.2a.2	2021-11-09	Px/SIAT1	80	<	80	40	320	640	320	40
A/Israel/S-2/2021		3C.2a1b.2a.2	2021-11-09	Px/SIAT1	80	<	80	40	320	640	640	40
A/Israel/S-3/2021		3C.2a1b.2a.2	2021-11-09	Px/SIAT1	160	<	320	160	640	640	640	160
A/Israel/S-7/2021		3C.2a1b.2a.2	2021-11-09	Px/SIAT1	80	<	80	40	320	640	320	40
A/Iran/Aban30/28/2021		3C.2a1b.2a.2	2021-11-17	SIAT5/SIAT1	80	<	80	80	320	640	320	40
A/Parma/01/2021		3C.2a1b.2a.2	2021-11-17	SIAT4/SIAT1	320	80	320	160	640	640	640	160
A/Iran/Ab29A/10/2021		3C.2a1b.2a.2	2021-11-18	SIAT5/SIAT1	40	<	40	40	160	640	320	<
A/Israel/B-145/2021		3C.2a1b.2a.2	2021-11-22	Px/SIAT1	160	<	320	80	320	640	640	80
A/Iran/Azar8/29/2021		3C.2a1b.2a.2	2021-11-27	SIAT5/SIAT1	80	<	80	80	320	640	640	40
A/Iran/60/Azar17B/2021		3C.2a1b.2a.2	2021-12-06	SIAT5/SIAT1	160	<	160	160	320	640	640	80
A/Iran/15/Azar21/2021		3C.2a1b.2a.2	2021-12-06	SIAT5/SIAT1	160	40	320	160	320	320	640	160
A/Hungary/9/2021		3C.2a1b.2a.2	2021-12-07	MDCK1/SIAT1	40	<	40	40	160	320	160	40
A/Hungary/8/2021		3C.2a1b.2a.2	2021-12-07	MDCK1/SIAT1	40	<	40	40	160	320	160	40
A/Romania/493221/2021		3C.2a1b.2a.2	2021-12-15	SIAT2/SIAT1	40	<	80	40	160	320	320	40
A/Romania/493220/2021		3C.2a1b.2a.2	2021-12-15	SIAT2/SIAT1	40	<	80	40	160	320	320	40
A/Romania/493219/2021		3C.2a1b.2a.2	2021-12-16	SIAT2/SIAT1	40	<	80	80	160	640	320	40
A/Romania/495026/2021		3C.2a1b.2a.2	2021-12-29	SIAT2/SIAT1	80	<	80	40	160	640	320	<
A/Romania/495025/2021		3C.2a1b.2a.2	2021-12-29	SIAT2/SIAT1	40	<	<	<	160	640	320	<
A/Hungary/1/2022		3C.2a1b.2a.2	2022-01-03	MDCK1/SIAT1	40	<	40	40	160	640	320	<
A/Romania/495703/2022		3C.2a1b.2a.2	2022-01-07	SIAT1/SIAT1	<	<	40	40	160	320	160	<
A/Romania/495700/2022		3C.2a1b.2a.2	2022-01-07	SIAT1/SIAT1	40	<	80	40	160	320	160	<
A/Hungary/2/2022		3C.2a1b.2a.2	2022-01-11	MDCK1/SIAT1	80	<	80	40	160	640	320	40
A/Romania/496796/2022		3C.2a1b.2a.2	2022-01-13	SIAT1/SIAT1	40	<	40	40	160	320	160	40
							Vaccine NH 2021-22			Vaccine SH 2022		

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)
1 < = <40, ND = Not Done

Sequences in phylogenetic trees

< 4-fold | 4-fold | 8-fold | > 8-fold | < not recognised by the antiserum | ≥ 160 (when homologous titre ≥ 2560) | ≥ 160 (no homologous titre)

Table 9-10. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2022-02-04

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre							
				Post-infection ferret antisera							
				A/Denmark 3264/19	A/HK 2671/19	A/Camb e0826360/20	A/Camb 925256/20	A/Bang 4005/20	A/Darwin 9/21	A/Stock 5/21	A/Kansas 14/17
	Passage history			SIAT	Cell St Judes	Egg	SIAT	SIAT	Egg	SIAT	SIAT
	Ferret number			F19/20[*1]	F21/20[*1]	F10/21[*1]	F03/21[*1]	F07/21[*1]	F38/21[*1]	F35/21[*1]	F17/19[*1]
	Genetic group			3C.2a1b.1a	3C.2a1b.1b	3C.2a1b.2a.1	3C.2a1b.2a.1	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.3a.1
REFERENCE VIRUSES											
A/Denmark/3264/2019	3C.2a1b.1a	2019-10-25	SIAT5	320	640	320	1280	320	640	320	160
A/Hong Kong/2671/2019	3C.2a1b.1b	2019-06-17	MDCK1/SIAT4	320	640	320	1280	160	640	160	160
A/Cambodia/e0826360/2020	3C.2a1b.2a.1	2020-07-16	E5/E2	80	<	2560	160	160	320	160	80
A/Cambodia/925256/2020	3C.2a1b.2a.1	2020-09-25	SIAT5	160	160	160	1280	160	320	160	160
A/Bangladesh/4005/2020	3C.2a1b.2a.2	2020-10-04	SIAT3	160	80	320	320	640	1280	640	320
A/Darwin/9/2021	3C.2a1b.2a.2	2021-04-17	E3/E2	160	40	640	160	320	2560	640	80
A/Stockholm/5/2021	3C.2a1b.2a.2	2021-04-16	S0/S3	80	40	160	160	320	2560	1280	80
A/Kansas/14/2017	3C.3a.1	2017-12-14	SIAT3/SIAT2	80	40	80	80	40	160	80	640
TEST VIRUSES											
A/Wisconsin/02/2021	3C.2a1b.2a.2	2021-01-23	SIAT3/SIAT1	160	80	320	160	640	2560	1280	160
A/Delaware/01/2021	3C.2a1b.2a.2	2021-04-16	SIAT2/SIAT1	80	40	160	80	320	1280	640	80
A/Nepal/21FL2608/2021	3C.2a1b.2a.2	2021-08-10	Hck3/SIAT1	40	<	80	40	160	640	320	40
A/Nepal/21FL2699/2021	3C.2a1b.2a.2	2021-08-11	Hck3/SIAT1	80	<	80	80	160	1280	320	40
A/Alaska/01/2021	3C.2a1b.2a.2	2021-08-20	SIAT1/SIAT1	40	40	80	40	160	640	160	40
A/Belgium/H0007/2021	pending	2021-08-06	P1/SIAT1	80	40	80	40	160	1280	640	40
A/Belgium/H0006/2021	pending	2021-09-04	P1/SIAT1	40	40	40	40	80	640	160	40
A/Yaroslavl/6/2021	pending	2021-10-25	SIAT2/SIAT1	320	40	320	160	320	2560	640	80
A/Vladivostok/1/2021	pending	2021-10-31	MDCK1/SIAT1/SIAT1	160	80	320	320	320	1280	640	160
A/Vladimir/9/2021	pending	2021-11-01	SIAT3/SIAT1	160	40	320	160	320	1280	640	160
A/Vladivostok/2/2021	pending	2021-11-01	MDCK1/SIAT1/SIAT1	320	40	320	320	320	2560	640	160
A/Yaroslavl/24/2021	pending	2021-11-08	SIAT3/SIAT1	80	<	160	40	160	640	160	80
A/Yaroslavl/19/2021	pending	2021-11-08	SIAT2/SIAT1	160	40	320	160	320	1280	640	160
A/Uglich/22/2021	pending	2021-11-12	SIAT2/SIAT1	160	80	320	160	320	2560	640	160
A/Belgium/H0010/2021	pending	2021-11-15	P1/SIAT1	80	40	160	80	160	1280	640	80
A/Vladivostok/11/2021	pending	2021-11-16	MDCK1/SIAT1/SIAT1	160	40	320	80	320	640	320	160
A/Yaroslavl/32/2021	pending	2021-11-18	SIAT2/SIAT1	160	80	320	640	320	1280	640	320
A/Belgium/G0127/2021	pending	2021-11-22	P1/SIAT1	40	<	80	80	160	640	160	40
A/Vladivostok/12/2021	pending	2021-11-23	MDCK1/SIAT1/SIAT1	160	40	320	80	320	640	320	80
A/Velikiy Novgorod/46/2021	pending	2021-11-24	SIAT2/SIAT1	160	80	320	160	320	1280	640	80
A/Orenburg/44/2021	pending	2021-11-24	SIAT3/SIAT1	160	80	320	160	320	2560	640	160
A/Vladivostok/14/2021	pending	2021-11-26	MDCK1/SIAT1/SIAT1	160	40	320	80	320	640	320	80
A/Vladimir/40/2021	pending	2021-11-29	SIAT2/SIAT1	160	80	320	160	320	1280	640	160
A/Orenburg/41/2021	pending	2021-11-30	SIAT2/SIAT1	160	40	320	80	320	640	320	160
A/Moscow/38/2021	pending	2021-11-30	SIAT2/SIAT1	160	80	320	160	320	1280	640	160
A/Orenburg/45/2021	pending	2021-12-01	SIAT3/SIAT1	160	80	320	160	320	1280	640	160
A/Moscow/43/2021	pending	2021-12-06	SIAT2/SIAT1	160	80	320	160	320	1280	640	160
A/Velikiy Novgorod/49/2021	pending	2021-12-08	SIAT2/SIAT1	160	80	320	160	320	1280	640	80
A/Mahajanga/13255/2021	pending	2021-12-09	SIAT2	320	80	640	160	320	2560	1280	160
A/Spain/89/2021	pending	2021-12-22	SIAT1	80	40	80	80	160	1280	320	40
A/Spain/87/2021	pending	2021-12-23	SIAT1	40	40	80	80	160	640	320	40
A/Spain/84/2021	pending	2021-12-25	SIAT1	40	<	80	80	160	640	320	40
A/Spain/83/2021	pending	2021-12-26	SIAT1	40	<	80	40	160	640	320	40
A/Spain/82/2021	pending	2021-12-27	SIAT1	<	<	80	80	160	1280	320	40
A/Spain/81/2021	pending	2021-12-28	SIAT1	<	40	80	80	80	640	160	40
A/Lebanon/8515/2021	pending	2021-12-29	SIAT1	80	40	80	40	160	1280	320	40
A/Lebanon/8510/2021	pending	2021-12-29	SIAT1	80	40	80	40	160	1280	640	40
A/Lebanon/8505/2021	pending	2021-12-29	SIAT1	80	40	80	40	160	1280	640	40
A/Lebanon/8500/2021	pending	2021-12-29	SIAT1	80	40	80	80	160	1280	640	80
A/Lebanon/8497/2021	pending	2021-12-29	SIAT1	80	40	80	40	160	1280	320	40
A/Estonia/168139/2022	pending	2022-01-03	SIAT1/SIAT1	160	40	160	80	320	1280	640	80
A/Armenia/19/2022	3C.2a1b.1b	2022-01-05	SIAT2	320	640	80	640	80	160	80	160
A/Estonia/168142/2022	pending	2022-01-05	SIAT1/SIAT1	80	40	80	40	160	1280	320	40
A/Estonia/168141/2022	pending	2022-01-05	SIAT1/SIAT1	80	40	80	40	160	640	320	40
A/Estonia/168131/2022	pending	2022-01-06	MDCKx/SIAT1	80	40	160	80	160	640	320	80
						Vaccine NH 2021-22			Vaccine SH 2022		

Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)
1 < = <40, ND = Not Done

Sequences in phylogenetic trees

< 4-fold | 4-fold | 8-fold | > 8-fold | < not recognised by the antiserum | ≥ 160 (when homologous titre ≥ 2560) | ≥ 160 (no homologous titre)

Table 9-11. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2022-02-11

				Haemagglutination inhibition titre							
				Post-infection ferret antisera							
Viruses	Other information	Collection date	Passage history	A/Denmark 3264/19	A/HK 2671/19	A/Camb e0826360/20	A/Camb 925256/20	A/Bang 4005/20	A/Darwin 9/21	A/Stock 5/21	A/Kansas 14/17
	Passage history			SIAT	Cell St Judes	Egg	SIAT	SIAT	Egg	SIAT	SIAT
	Ferret number			F19/20*1	F21/20*1	F10/21*1	F03/21*1	F07/21*1	F38/21*1	F35/21*1	F17/19*1
	Genetic group			3C.2a1b.1a	3C.2a1b.1b	3C.2a1b.2a.1	3C.2a1b.2a.1	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.3a.1
REFERENCE VIRUSES											
A/Denmark/3264/2019	3C.2a1b.1a	2019-10-25	SIAT3/SIAT4	320	640	320	1280	320	640	320	160
A/Hong Kong/2671/2019	3C.2a1b.1b	2019-06-17	MDCK1/SIAT4	320	320	160	640	320	320	160	160
A/Cambodia/e0826360/2020	3C.2a1b.2a.1	2020-07-16	E5/E2	160	<	2560	160	320	640	320	160
A/Cambodia/925256/2020	3C.2a1b.2a.1	2020-09-25	SIAT5	160	160	160	1280	160	320	160	160
A/Bangladesh/4005/2020	3C.2a1b.2a.2	2020-10-04	SIAT3	160	80	320	320	640	1280	640	320
A/Darwin/9/2021	3C.2a1b.2a.2	2021-04-17	E3/E2	160	40	320	160	640	5120	1280	160
A/Stockholm/5/2021	3C.2a1b.2a.2	2021-04-16	S0/S3	80	40	160	160	320	2560	1280	80
A/Kansas/14/2017	3C.3a.1	2017-12-14	SIAT3/SIAT2	80	40	80	160	80	160	160	640
TEST VIRUSES											
A/South Africa/R14962/2021	pending	2021-10-12	MDCK3/MDCK2/SIAT1	320	160	640	160	320	640	320	160
A/Ireland/92285/2021	3C.2a1b.2a.2	2021-10-21	SIAT1	160	40	640	160	640	1280	640	320
A/Ireland/2707/2021	pending	2021-12-05	SIAT1	40	<	80	80	160	640	320	40
A/Berlin/4/2021	pending	2021-12-16	P2/SIAT1	160	<	80	80	320	1280	640	40
A/Ireland/6979/2021	pending	2021-12-20	SIAT1	80	<	80	80	160	1280	640	40
A/Nordrhein-Westfalen/1/2021	pending	2021-12-20	P1/SIAT1	1280	320	1280	320	640	2560	2560	320
A/Rheinland-Pfalz/2/2021	pending	2021-12-22	P1/SIAT1	160	<	80	80	320	1280	640	40
A/Nordrhein-Westfalen/2/2021	pending	2021-12-22	P1/SIAT1	80	<	80	40	160	640	160	40
A/Berlin/5/2021	pending	2021-12-23	P1/SIAT1	80	<	80	40	160	1280	320	40
A/Baden-Wurttemberg/2/2021	pending	2021-12-27	P1/SIAT1	160	80	160	160	320	1280	640	160
A/Nordrhein-Westfalen/3/2021	pending	2021-12-28	P1/SIAT1	80	<	80	80	160	1280	640	40
A/Berlin/6/2021	pending	2021-12-29	P1/SIAT1	80	<	80	80	160	1280	640	40
A/Berlin/7/2021	pending	2021-12-31	P1/SIAT1	160	40	80	80	320	1280	640	40
A/Saarland/1/2022	pending	2022-01-03	P1/SIAT1	80	<	80	80	160	1280	640	40
A/Thuringen/1/2022	pending	2022-01-03	P1/SIAT1	160	40	80	80	320	1280	640	40
A/Kyiv/41/2022	pending	2022-01-04	SIATx/SIAT1	160	80	320	160	640	2560	640	160
A/Norway/409/2022	3C.2a1b.2a.2	2022-01-04	SIAT1	40	<	40	40	80	640	320	<
A/Hessen/1/2022	pending	2022-01-04	P1/SIAT1	80	<	80	80	160	1280	640	40
A/Kyiv/7/2022	pending	2022-01-05	SIATx/SIAT1	160	40	160	80	320	2560	640	40
A/Kyiv/5/2022	pending	2022-01-05	SIATx/SIAT1	160	40	160	80	320	1280	640	40
A/Kyiv/4/2022	pending	2022-01-05	SIATx/SIAT1	80	<	80	80	320	1280	640	40
A/Kyiv/3/2022	pending	2022-01-05	SIATx/SIAT1	160	<	160	80	320	1280	640	40
A/Kyiv/1/2022	pending	2022-01-05	SIATx/SIAT1	160	<	160	80	320	1280	640	40
A/Norway/530/2022	pending	2022-01-06	SIAT1	80	<	80	40	160	1280	320	<
A/Norway/676/2022	3C.2a1b.2a.2	2022-01-11	SIAT1	320	160	320	160	640	1280	320	320
A/Serbia/206/2022	pending	2022-01-13	SIAT1	160	40	160	80	320	1280	640	40
A/Serbia/416478/2022	3C.2a1b.2a.2	2022-01-17	SIAT1	160	40	160	80	320	1280	640	40
A/Serbia/416476/2022	3C.2a1b.2a.2	2022-01-17	SIAT1	160	40	160	80	320	1280	640	40
A/Serbia/416473/2022	3C.2a1b.2a.2	2022-01-17	SIAT1	80	<	160	80	320	1280	640	40
A/Serbia/336/2022	pending	2022-01-17	SIAT1	160	40	80	80	320	1280	640	40
						Vaccine NH 2021-22			Vaccine SH 2022		

Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 < = <40, ND = Not Done

Sequences in phylogenetic trees

< 4-fold | 4-fold | 8-fold | > 8-fold | < not recognised by the antiserum | ≥ 160 (when homologous titre ≥ 2560) | ≥ 160 (no homologous titr

Table 9-12. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) Summary

Viruses		Haemagglutination inhibition titre							
		Post-infection ferret antisera							
		A/Denmark	A/HK	A/Camb	A/Camb	A/Bang	A/Darwin	A/Stock	A/Kansas
		3264/19	2671/19	e0826360/20	925256/20	4005/20	9/21	5/21	14/17
	Passage history	SIAT	Cell	Egg	SIAT	SIAT	Egg	SIAT	SIAT
	Ferret number	F19/20[*1]	St Judes F21/20[*1]	F10/21[*1]	F03/21[*1]	F07/21[*1]	F38/21[*1]	F35/21[*1]	F17/19[*1]
	Genetic group	3C.2a1b.1a	3C.2a1b.1b	3C.2a1b.2a.1	3C.2a1b.2a.1	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.3a.1
REFERENCE VIRUSES									
A/Denmark/3264/2019	3C.2a1b.1a	320	640	320	1280	320	640	320	160
A/Hong Kong/2671/2019	3C.2a1b.1b	320	320	160	640	320	320	160	160
A/Cambodia/e0826360/2020	3C.2a1b.2a.1	160	<	2560	160	320	640	320	160
A/Cambodia/925256/2020	3C.2a1b.2a.1	160	160	160	1280	160	320	160	160
A/Bangladesh/4005/2020	3C.2a1b.2a.2	160	80	320	320	640	1280	640	320
A/Darwin/9/2021	3C.2a1b.2a.2	160	40	320	160	640	>5120	1280	160
A/Stockholm/5/2021	3C.2a1b.2a.2	80	40	160	160	320	2560	1280	80
A/Kansas/14/2017	3C.3a.1	80	40	80	160	80	160	160	640
TEST VIRUSES									
Total Number tested		296	296	296	296	296	296	296	296
No with titre reduction ≤2-fold		104	13	1	16	131	61	177	6
%		*35.1*	*4.4*	*0.4*	*5.4*	*44.3*	*20.6*	*59.8*	*2.0*
No with titre reduction =4-fold		75	14	19	34	133	146	82	44
%		*25.4*	*4.7*	*6.4*	*11.5*	*44.9*	*49.3*	*27.7*	*14.9*
No with titre reduction ≥8-fold		117	269	276	246	32	89	37	246
%		*39.5*	*90.9*	*93.2*	*83.1*	*10.8*	*30.1*	*12.5*	*83.1*
Number tested	3C.2a1b.2a.2	202	202	202	202	202	202	202	202
No with titre reduction ≤2-fold		56	1		1	88	31	131	4
%		*27.7*	*0.5*		*0.5*	*43.6*	*15.4*	*64.9*	*2.0*
No with titre reduction =4-fold		42	3	13	24	101	113	60	25
%		*20.8*	*1.5*	*6.4*	*11.9*	*50.0*	*55.9*	*29.7*	*12.4*
No with titre reduction ≥8-fold		104	198	189	177	13	58	11	173
%		*51.5*	*98.0*	*93.6*	*87.6*	*6.4*	*28.7*	*5.4*	*85.6*
Number tested	3C.2a1b.1b	16	16	16	16	16	16	16	16
No with titre reduction ≤2-fold		7	9		8				
%		*43.8*	*56.3*		*50.0*				
No with titre reduction =4-fold		6	5		6				
%		*37.5*	*31.3*		*37.6*				
No with titre reduction ≥8-fold		3	2	16	2	16	16	16	16
%		*18.7*	*12.4*	*100*	*12.4*	*100*	*100*	*100*	*100*
Number tested	3C.2a1b.1a	6	6	6	6	6	6	6	6
No with titre reduction ≤2-fold		3	1		5				
No with titre reduction =4-fold		3	3	1		5		1	
No with titre reduction ≥8-fold			2	5	1	1	6	5	6
Number tested	3C.2a1b.2a.1	2	2	2	2	2	2	2	2
No with titre reduction ≤2-fold		2			1	2			
No with titre reduction =4-fold			1		1			2	1
No with titre reduction ≥8-fold			1	2			2		1
				Vaccine NH 2021-22			Vaccine SH 2022		

Reference virus results are taken from an individual table as an example. Summaries for each antiserum are based on fold-reductions observed on the days that HI assays were performed.

Table 10-1. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2021-11-04

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation Titre[1] Post-infection ferret antisera A/Norway 3275/2018 SIAT F03/19 3C.2a1b.2 2-fold	Read	A/Cambodia 925256/2020 SIAT F03/21 3C.2a1b.2a.1 2-fold	Read	A/Bangladesh 4005/2020 SIAT F07/21 3C.2a1b.2a.2 2-fold	Read	NIB-126A A/Dar 09/2021 Egg F33/21 3C.2a1b.2a.2 2-fold	Read	A/Denmark 3264/19 SIAT F19/20 3C.2a1b.1a 2-fold	Read	A/Hong Kong 2671/19 SIAT F33/19 3C.2a1b.1b 2-fold	Read	A/Kansas 14/2017 SIAT F17/19 3C.3a.1 2-fold	Read	HA1 substitutions for 3C.2a/3a viruses compared to A/Norway/3275/2018: Egg adaptation HA substitutions compared to the corresponding cell culture propagated virus
REFERENCE VIRUSES																		
A/Norway/3275/2018	3C.2a1b.2	2018-10-04	SIAT2 10^{-3}	320	371	80	70	<	<	<	<	80	90	40	45	<	<	
A/Cambodia/925256/2020	3C.2a1b.2a.1	2020-09-25	SIAT3 10^{-3}	80	70	320	243	<	<	<	<	80	85	80	89	<	<	K83E, Y94N, V106A, K171N, G186S, F193S, Y195F, S198P
A/Bangladesh/4005/2020	3C.2a1b.2a.2	2020-10-04	SIAT2 10^{-3}	<	<	40	41	320	350	640	512	<	<	<	<	40	54	K83E, Y94N, V106A, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F
NIB 126A (A/Darwin/9/2021)	3C.2a1b.2a.2	2021-04-17	E3/E1/E5	<	<	160	173	640	567	2560	1994	160	124	<	<	40	49	K83E, Y94N, V106A, H156S, Y159N, T160I(-cho), L164Q, G186N, D190N, F193S, Y195F, D225G
A/Denmark/3264/2019	3C.2a1b.1a	2019-10-25	SIAT3/SIAT2 10^{-3}	40	52	320	254	<	<	40	40	320	273	160	163	<	<	V106A, T128A(-cho), K131T, T135K(-cho), A138S, G186D, D190N, F193S, S198P
A/Hong Kong/2671/2019	3C.2a1b.1b	2019-06-17	MDCK1/SIAT4 10^{-3}	40	50	320	345	<	<	<	<	320	310	320	290	<	<	V106A, T128A(-cho), K131T, T135K(-cho), S137F, A138S, F193S
A/Kansas/14/2017	3C.3a.1	2017-12-14	SIAT3/SIAT2 10^{-3}	<	<	<	<	<	<	<	<	<	<	<	<	160	132	G62E, S91N, R92K, V106A, K121N, T128A(-cho), K131T, A138S, S144K, Y159S, T160K(-cho), K171N, F193S, K326R
TEST VIRUSES																		
A/Beirut/5518/2021	3C.2a1b.2a.2	2021-09-01	SIAT1	<	<	<	<	80	65	320	258	<	<	<	<	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), Q164L, G186D, D190N, F193S, Y195F, K276R
A/Tripoli/5651/2021	3C.2a1b.2a.2	2021-09-07	SIAT1	<	<	<	<	40	58	320	258	<	<	<	<	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), Q164L, G186D, D190N, F193S, Y195F, K276R
A/Arsal/5773/2021	3C.2a1b.2a.2	2021-09-13	SIAT1	<	<	<	<	80	91	320	430	<	<	<	<	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), Q164L, G186D, D190N, F193S, Y195F, K276R
A/Arsal/5772/2021	3C.2a1b.2a.2	2021-09-13	SIAT1	<	<	<	<	80	76	320	320	<	<	<	<	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), Q164L, G186D, D190N, F193S, Y195F, K276R
A/Arsal/5767/2021	3C.2a1b.2a.2	2021-09-13	SIAT1	<	<	<	<	80	71	320	410	<	<	<	<	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), Q164L, G186D, D190N, F193S, Y195F, K276R
A/Tripoli/5937/2021	3C.2a1b.2a.2	2021-09-21	SIAT1	<	<	<	<	80	78	320	457	<	<	<	<	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), Q164L, G186D, D190N, F193S, Y195F, K276R
A/Saida/5915/2021	3C.2a1b.2a.2	2021-09-21	SIAT1	<	<	<	<	80	90	320	286	<	<	<	<	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), Q164L, G186D, D190N, F193S, Y195F, K276R
A/Tripoli/6112/2021	3C.2a1b.2a.2	2021-09-30	SIAT1	<	<	<	<	80	71	320	341	<	<	<	<	<	<	D53S, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), Q164L, G186D, D190N, F193S, Y195F, K276R
A/Tripoli/6100/2021	3C.2a1b.2a.2	2021-09-30	SIAT1	<	<	<	<	80	74	320	320	<	<	<	<	<	<	D53S, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), Q164L, G186D, D190N, F193S, Y195F, K276R
A/Tripoli/5646/2021	3C.2a1b.2a.2	2021-09-07	SIAT1	<	<	<	<	80	83	320	311	<	<	<	<	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Tripoli/5927/2021	3C.2a1b.2a.2	2021-09-21	SIAT1	<	<	<	<	80	96	320	285	40	52	<	<	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Baabda/5555/2021	3C.2a1b.2a.2	2021-09-01	SIAT1	<	<	<	<	80	72	320	255	<	<	<	<	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), Q164L, G186D, D190N, F193S, Y195F, K276R, S279Y
A/Tripoli/5644/2021	3C.2a1b.2a.2	2021-09-07	SIAT1	<	<	<	<	80	104	640	506	<	<	<	<	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), Q164L, G186D, D190N, F193S, Y195F, K276R, S279Y
A/Beirut/5788/2021	3C.2a1b.2a.2	2021-09-14	SIAT1	<	<	<	<	80	77	320	316	<	<	<	<	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), Q164L, G186D, D190N, F193S, Y195F, K276R, S279Y
A/Tripoli/5638/2021	3C.2a1b.2a.2	2021-09-07	SIAT1	<	<	<	<	80	88	320	466	<	<	<	<	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), Q164L, G186D, D190N, F193S, Y195F, I260V, K276R, S279Y
A/Arsal/5768/2021	3C.2a1b.2a.2	2021-09-13	SIAT1	<	<	<	<	80	84	640	512	<	<	<	<	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), Q164L, G186D, D190N, F193S, Y195F, I260V, K276R, S279Y

[1] Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Table 10-2. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2021-11-25

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation Titre[1] Post-infection ferret antisera A/Norway 3275/2018 SIAT F03/19 3C.2a1b.2 2-fold	Read	A/Cambodia 925256/2020 SIAT F03/21 3C.2a1b.2a.1 2-fold	Read	A/Bangladesh 4005/2020 SIAT F07/21 3C.2a1b.2a.2 2-fold	Read	A/Stockholm 5/2021 SIAT F35/21 3C.2a1b.2a.2 2-fold	Read	A/Darwin 09/2021 Egg F38/21 3C.2a1b.2a.2 2-fold	Read	A/Denmark 3264/19 SIAT F19/20 3C.2a1b.1a 2-fold	Read	A/Kansas 14/2017 SIAT F17/19 3C.3a.1 2-fold	Read	HA1 substitutions for 3C.2a/3a viruses compared to A/Norway/3275/2018: Egg adaptation HA substitutions compared to the corresponding cell culture propagated virus
REFERENCE VIRUSES																		
A/Norway/3275/2018	3C.2a1b.2	2018-10-04	SIAT2 10^{-3}	320	371	80	70	<	<	<	<	80	72	80	90	<	<	
A/Cambodia/925256/2020	3C.2a1b.2a.1	2020-09-25	SIAT3 10^{-3}	80	70	320	243	<	<	<	<	80	99	80	85	<	<	K83E, Y94N, V106A, K171N, G186S, F193S, Y195F, S198P
A/Bangladesh/4005/2020	3C.2a1b.2a.2	2020-10-04	SIAT2 10^{-3}	<	<	40	41	320	350	640	727	1280	1030	<	<	40	54	K83E, Y94N, V106A, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F
A/Stockholm/5/2021	3C.2a1b.2a.2	2021-04-16	SIAT0/SIAT3 10^{-3}	<	<	<	<	80	68	320	465	1280	1044	<	<	<	<	K83E, Y94N, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F
A/Darwin/9/2021	3C.2a1b.2a.2	2021-04-17	E3/E2 10^{-3}	<	<	80	84	320	320	1280	1609	5120	4994	160	138	<	<	K83E, Y94N, V106A, H156S, Y159N, T160I(-cho), L164Q, G186N, D190N, F193S, Y195F, D225G
A/Denmark/3264/2019	3C.2a1b.1a	2019-10-25	SIAT3/SIAT2 10^{-3}	40	52	320	254	<	<	<	<	<	<	320	273	<	<	V106A, T128A(-cho), K131T, T135K(-cho), A138S, G186D, D190N, F193S, S198P
A/Kansas/14/2017	3C.3a.1	2017-12-14	SIAT3/SIAT2 10^{-3}	<	<	<	<	<	<	<	<	<	<	<	<	160	132	G62E, S91N, R92K, V106A, K121N, T128A(-cho), K131T, A138S, S144K, Y159S, T160K(-cho), K171N, F193S, K326R
TEST VIRUSES																		
A/Beirut/5509/2021	3C.2a1b.2a.2	2021-09-01	SIAT1	<	<	<	<	80	60	320	313	640	728	<	<	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R, S279Y
A/Beirut/5517/2021	3C.2a1b.2a.2	2021-09-01	SIAT1	<	<	<	<	80	78	320	356	1280	1034	<	<	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R, S279Y
A/Netherlands/10003/2021	3C.2a1b.2a.2	2021-09-02	MM2/SIAT1	<	<	<	<	80	84	320	345	640	931	<	<	<	<	D53N, K83E, Y94N, N96S(+cho), V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, I192F, F193S, Y195F
A/Netherlands/10001/2021	3C.2a1b.2a.2	2021-09-06	MM2/SIAT1	<	<	<	<	80	85	320	467	1280	1280	<	<	<	<	D53N, K83E, Y94N, N96S(+cho), V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, I192F, F193S, Y195F
A/Netherlands/10002/2021	3C.2a1b.2a.2	2021-09-06	MM2/SIAT1	<	<	<	<	80	75	320	450	1280	1195	<	<	<	<	D53N, K83E, Y94N, N96S(+cho), V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, I192F, F193S, Y195F
A/Netherlands/00016/2021	3C.2a1b.2a.2	2021-09-06	hCKx/SIAT1	<	<	<	<	80	75	320	296	640	837	<	<	<	<	D53N, K83E, Y94N, N96S(+cho), V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, I192F, F193S, Y195F
A/Netherlands/10005/2021	3C.2a1b.2a.2	2021-09-07	MM2/SIAT1	<	<	<	<	40	59	160	223	640	571	<	<	<	<	D53N, K83E, Y94N, N96S(+cho), V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, I192F, F193S, Y195F
A/Netherlands/10006/2021	3C.2a1b.2a.2	2021-09-07	MM2/SIAT1	<	<	<	<	80	73	320	320	1280	1024	<	<	<	<	D53N, K83E, Y94N, N96S(+cho), V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, I192F, F193S, Y195F
A/Netherlands/00008/2021	3C.2a1b.2a.2	2021-09-08	hCKx/SIAT1	<	<	<	<	80	108	320	291	640	632	<	<	<	<	D53N, K83E, Y94N, N96S(+cho), V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, I192F, F193S, Y195F
A/Netherlands/00012/2021	3C.2a1b.2a.2	2021-10-04	hCKx/SIAT1	<	<	<	<	80	73	320	306	640	667	<	<	<	<	D53N, K83E, Y94N, N96S(+cho), V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, I192F, F193S, Y195F
A/Netherlands/00013/2021	3C.2a1b.2a.2	2021-10-08	hCKx/SIAT1	<	<	<	<	80	72	320	387	640	785	<	<	<	<	D53N, K83E, Y94N, N96S(+cho), V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, I192F, F193S, Y195F
A/Netherlands/10004/2021	3C.2a1b.2a.2	2021-09-07	MM2/SIAT1	<	<	<	<	80	113	320	281	640	832	<	<	<	<	N45S(-cho), D53N, K83E, Y94N, N96S(+cho), V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, I192F, F193S, Y195F

[1] Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Table 10-3. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2021-12-09

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation Titre[1] Post-infection ferret antisera A/Norway 3275/2018 SIAT F03/19 3C.2a1b.2		A/Cambodia 925256/2020 SIAT F03/21 3C.2a1b.2a.1		A/Bangladesh 4005/2020 SIAT F07/21 3C.2a1b.2a.2		A/Stockholm 5/2021 SIAT F35/21 3C.2a1b.2a.2		A/Darwin 09/2021 Egg F38/21 3C.2a1b.2a.2		A/Denmark 3264/19 SIAT F19/20 3C.2a1b.1a		A/Kansas 14/2017 SIAT F17/19 3C.3a.1		HA1 substitutions for 3C.2a/3a viruses compared to A/Norway/3275/2018: Egg adaptation HA substitutions compared to the corresponding cell culture propagated virus
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	
REFERENCE VIRUSES																		
A/Norway/3275/2018	3C.2a1b.2	2018-10-04	SIAT2 10^{-3}	320	371	80	70	<	<	<	<	80	72	80	90	<	<	
A/Cambodia/925256/2020	3C.2a1b.2a.1	2020-09-25	SIAT3 10^{-3}	80	70	320	243	<	<	<	<	80	99	80	85	<	<	K83E, Y94N, V106A, K171N, G186S, F193S, Y195F, S198P
A/Bangladesh/4005/2020	3C.2a1b.2a.2	2020-10-04	SIAT2 10^{-3}	<	<	40	41	320	350	640	727	1280	1030	<	<	40	54	K83E, Y94N, V106A, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F
A/Stockholm/5/2021	3C.2a1b.2a.2	2021-04-16	SIAT0/SIAT3 10^{-3}	<	<	<	<	80	68	320	465	1280	1044	<	<	<	<	K83E, Y94N, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F
A/Darwin/9/2021	3C.2a1b.2a.2	2021-04-17	E3/E2 10^{-3}	<	<	80	84	320	320	1280	1609	5120	4994	160	138	<	<	K83E, Y94N, V106A, H156S, Y159N, T160I(-cho), L164Q, G186N, D190N, F193S, Y195F, D225G
A/Denmark/3264/2019	3C.2a1b.1a	2019-10-25	SIAT3/SIAT2 10^{-3}	40	52	320	254	<	<	<	<	<	<	320	273	<	<	V106A, T128A(-cho), K131T, T135K(-cho), A138S, G186D, D190N, F193S, S198P
A/Kansas/14/2017	3C.3a.1	2017-12-14	SIAT3/SIAT2 10^{-3}	<	<	<	<	<	<	<	<	<	<	<	<	160	132	G62E, S91N, R92K, V106A, K121N, T128A(-cho), K131T, A138S, S144K, Y159S, T160K(-cho), K171N, F193S, K326R
TEST VIRUSES																		
A/Croatia/103473/2021	3C.2a1b.2a.2	2021-09-07	MDCKx/SIAT1	<	<	<	<	40	43	160	207	640	539	<	<	<	<	D53N, K83E, Y94N, N96S(+cho), V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, I192F, F193S, Y195F
A/Croatia/109108/2021	3C.2a1b.2a.2	2021--09-20	MDCKx/SIAT1	<	<	<	<	40	58	320	254	640	620	<	<	<	<	D53N, K83E, Y94N, N96S(+cho), V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, I192F, F193S, Y195F

[1] Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Table 10-4. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2022-01-20

Viruses		Collection Date	Passage History	Neutralisation Titre[1] Post-infection ferret antisera														HA1 substitutions for 3C.2a/3a viruses compared to A/Norway/3275/2018: Egg adaptation HA substitutions compared to the corresponding cell culture propagated virus
				A/Norway 3275/2018		A/Cambodia 925256/2020		A/Bangladesh 4005/2020		A/Stockholm 5/2021		A/Darwin 09/2021		A/Denmark 3264/19		A/Kansas 14/2017		
	Passage history			SIAT		SIAT		SIAT		SIAT		Egg		SIAT		SIAT		
	Ferret number			F03/19		F03/21		F07/21		F35/21		F38/21		F19/20		F17/19		
	Genetic group			3C.2a1b.2		3C.2a1b.2a.1		3C.2a1b.2a.2		3C.2a1b.2a.2		3C.2a1b.2a.2		3C.2a1b.1a		3C.3a.1		
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	
REFERENCE VIRUSES																		
A/Norway/3275/2018	3C.2a1b.2	2018-10-04	SIAT2 10^{-3}	320	371	80	70	<	<	<	<	80	72	80	90	<	<	
A/Cambodia/925256/2020	3C.2a1b.2a.1	2020-09-25	SIAT3 10^{-3}	80	70	320	243	<	<	<	<	80	99	80	85	<	<	K83E, Y94N, V106A, K171N, G186S, F193S, Y195F, S198P
A/Bangladesh/4005/2020	3C.2a1b.2a.2	2020-10-04	SIAT2 10^{-3}	<	<	40	41	320	350	640	727	1280	1030	<	<	40	54	K83E, Y94N, V106A, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F
A/Stockholm/5/2021	3C.2a1b.2a.2	2021-04-16	SIAT0/SIAT3 10^{-3}	<	<	<	<	80	68	320	465	1280	1044	<	<	<	<	K83E, Y94N, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F
A/Darwin/9/2021	3C.2a1b.2a.2	2021-04-17	E3/E2 10^{-3}	<	<	80	84	320	320	1280	1609	5120	4994	160	138	<	<	K83E, Y94N, V106A, H156S, Y159N, T160I(-cho), L164Q, G186N, D190N, F193S, Y195F, D225G
A/Denmark/3264/2019	3C.2a1b.1a	2019-10-25	SIAT3/SIAT2 10^{-3}	40	52	320	254	<	<	<	<	<	<	320	273	<	<	V106A, T128A(-cho), K131T, T135K(-cho), A138S, G186D, D190N, F193S, S198P
A/Kansas/14/2017	3C.3a.1	2017-12-14	SIAT3/SIAT2 10^{-3}	<	<	<	<	<	<	<	<	<	<	<	<	160	132	G62E, S91N, R92K, V106A, K121N, T128A(-cho), K131T, A138S, S144K, Y159S, T160K(-cho), K171N, F193S, K326R
TEST VIRUSES																		
A/Denmark/04/2021	3C.2a1b.2a.2	2021-09-15	SIAT3/SIAT1	<	<	<	<	80	70	320	351	1280	983	<	<	<	<	D53N, K83E, Y94N, N96S(+cho), V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, I192F, F193S, Y195F
A/Denmark/07/2021	3C.2a1b.2a.2	2021-09-17	SIAT2/SIAT1	<	<	<	<	80	72	320	302	640	877	<	<	<	<	D53N, K83E, Y94N, N96S(+cho), V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, I192F, F193S, Y195F
A/Denmark/08/2021	3C.2a1b.2a.2	2021-09-17	SIAT3/SIAT1	<	<	<	<	80	109	320	474	1280	1100	<	<	<	<	D53N, K83E, Y94N, N96S(+cho), V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, I192F, F193S, Y195F
A/Denmark/13/2021	3C.2a1b.2a.2	2021-09-21	SIAT3/SIAT1	<	<	<	<	80	89	640	480	1280	1189	<	<	<	<	D53N, K83E, Y94N, N96S(+cho), V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, I192F, F193S, Y195F
A/Denmark/12/2021	3C.2a1b.2a.2	2021-09-25	SIAT3/SIAT1	<	<	<	<	80	92	320	423	1280	1109	<	<	<	<	D53N, K83E, Y94N, N96S(+cho), V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, I192F, F193S, Y195F
A/Ibra/72126272/2021	3C.2a1b.2a.2	2021-10-03	SIAT1/SIAT1	<	<	<	<	80	103	640	508	1280	1200	<	<	<	<	D53N, Q57H, K83E, Y94N, N96S(+cho), V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, I192L, F193S, Y195F
A/Ibra/72127166/2021	3C.2a1b.2a.2	2021-10-24	SIAT1/SIAT1	<	<	<	<	80	66	320	250	640	544	<	<	<	<	D53N, Q57H, K83E, Y94N, N96S(+cho), V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, I192L, F193S, Y195F
A/S.Petersburg/RII-10/2021	3C.2a1b.2a.2	-	MDCK3/SIAT1	<	<	<	<	160	164	640	526	1280	1280	80	60	<	<	I25V, D53G, K83E, Y94N, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, R201K, S219Y
A/Salalah/72127281/2021	3C.2a1b.2a.2	2021-10-24	SIAT1/SIAT1	<	<	<	<	80	91	160	135	160	215	<	<	<	<	E50K, K83E, Y94N, V106A, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F
A/Sohar/72125048/2021	3C.2a1b.2a.2	2021-09-08	SIAT1/SIAT1	<	<	40	43	320	289	320	261	320	356	40	45	<	<	E50K, K83E, Y94N, V106A, N122D(-cho), Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F
A/Salalah/72126169/2021	3C.2a1b.2a.2	2021-09-29	SIAT1/SIAT1	<	<	<	<	320	280	320	350	640	702	<	<	<	<	E50K, K83E, Y94N, V106A, N122D(-cho), Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F
A/Haima/72127478/2021	3C.2a1b.2a.2	2021-10-26	SIAT1/SIAT1	<	<	<	<	320	341	320	469	640	715	40	59	40	56	E50K, K83E, Y94N, V106A, N122D(-cho), Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F
A/Salalah/72127789/2021	3C.2a1b.2a.2	2021-10-31	SIAT1/SIAT1	<	<	<	<	320	421	640	538	1280	1013	40	56	40	54	E50K, K83E, Y94N, V106A, N122D(-cho), Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F
A/Novosibirsk/RII-01/2021	3C.2a1b.2a.2	-	SIAT2/SIAT1	<	<	<	<	80	118	160	121	160	202	40	53	<	<	K83E, Y94N, V106A, Y159N, T160I(-cho), L164Q, Q173H, G186D, D190N, F193S, Y195F, S205F, A212T
A/Denmark/21/2021	3C.2a1b.2a.2	2021-10-29	SIAT3/SIAT1	<	<	40	44	240	263	320	351	640	565	80	104	40	58	V20X, K83E, Y94N, V106A, I140K, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, R201I
A/Salalah/72124837/2021	3C.2a1b.1b	2021-09-05	SIAT1/SIAT1	40	43	320	354	<	<	<	<	<	<	160	208	<	<	N31S, H56Y, D85N, V106A, T128A(-cho), K131T, T135K(-cho), A138S, F193S, S198P

[1] Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Table 10-5. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2022-01-24

Viruses	Passage history / Ferret number / Genetic group	Collection Date	Passage History	Neutralisation Titre[1] Post-infection ferret antisera: A/Norway 3275/2018 SIAT F03/19 3C.2a1b.2		A/Cambodia 925256/2020 SIAT F03/21 3C.2a1b.2a.1		A/Bangladesh 4005/2020 SIAT F07/21 3C.2a1b.2a.2		A/Stockholm 5/2021 SIAT F35/21 3C.2a1b.2a.2		A/Darwin 09/2021 Egg F38/21 3C.2a1b.2a.2		A/Denmark 3264/19 SIAT F19/20 3C.2a1b.1a		A/Kansas 14/2017 SIAT F17/19 3C.3a.1		HA1 substitutions for 3C.2a/3a viruses compared to A/Norway/3275/2018: Egg adaptation HA substitutions compared to the corresponding cell culture propagated virus
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	
REFERENCE VIRUSES																		
A/Norway/3275/2018	3C.2a1b.2	2018-10-04	SIAT2 10^{-3}	320	371	80	70	<	<	<	<	80	72	80	90	<	<	
A/Cambodia/925256/2020	3C.2a1b.2a.1	2020-09-25	SIAT3 10^{-3}	80	70	320	243	<	<	<	<	80	99	80	85	<	<	K83E, Y94N, V106A, K171N, G186S, F193S, Y195F, S198P
A/Bangladesh/4005/2020	3C.2a1b.2a.2	2020-10-04	SIAT2 10^{-3}	<	<	40	41	320	350	640	727	1280	1030	<	<	40	54	K83E, Y94N, V106A, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F
A/Stockholm/5/2021	3C.2a1b.2a.2	2021-04-16	SIAT0/SIAT3 10^{-3}	<	<	<	<	80	68	320	465	1280	1044	<	<	<	<	K83E, Y94N, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F
A/Darwin/9/2021	3C.2a1b.2a.2	2021-04-17	E3/E2 10^{-3}	<	<	80	84	320	320	1280	1609	5120	4994	160	138	<	<	K83E, Y94N, V106A, H156S, Y159N, T160I(-cho), L164Q, G186N, D190N, F193S, Y195F, D225G
A/Denmark/3264/2019	3C.2a1b.1a	2019-10-25	SIAT3/SIAT2 10^{-3}	40	52	320	254	<	<	<	<	<	<	320	273	<	<	V106A, T128A(-cho), K131T, T135K(-cho), A138S, G186D, D190N, F193S, S198P
A/Kansas/14/2017	3C.3a.1	2017-12-14	SIAT3/SIAT2 10^{-3}	<	<	<	<	<	<	<	<	<	<	<	<	160	132	G62E, S91N, R92K, V106A, K121N, T128A(-cho), K131T, A138S, S144K, Y159S, T160K(-cho), K171N, F193S, K326R
TEST VIRUSES																		
A/Novosibirsk/RII-03/2021	3C.2a1b.2a.2	-	SIAT2/SIAT1	<	<	<	<	160	133	160	136	160	218	40	44	<	<	K83E, Y94N, V106A, Y159N, T160I(-cho), L164Q, Q173H, G186D, D190N, F193S, Y195F, S205F, A212T
A/Reims/30384/2021	3C.2a1b.2a.2	2021-10-08	MDCK2/SIAT1	<	<	80	61	320	397	320	438	640	581	80	97	40	40	K83E, Y94N, V106A, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, R201I, R220K
A/Novosibirsk/RII-02/2021	3C.2a1b.2a.2	-	SIAT2/SIAT1	<	<	<	<	80	80	320	432	1280	1054	<	<	<	<	D53N, K83E, Y94N, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, S262N
A/Alaska/01/2021	3C.2a1b.2a.2	2021-08-20	SIAT1/SIAT1	<	<	<	<	80	74	320	378	1280	985	<	<	<	<	D53N, K83E, Y94N, N96S(+cho), V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, I192F, F193S, Y195F
A/Nord Pas de Calais/27313/2021	3C.2a1b.2a.2	2021-09-09	MDCK1/SIAT1	<	<	<	<	80	91	320	344	1280	1024	<	<	<	<	D53N, K83E, Y94N, N96S(+cho), V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, I192F, F193S, Y195F
A/Lille/27403/2021	3C.2a1b.2a.2	2021-09-13	MDCK1/SIAT1	<	<	<	<	80	88	320	367	640	921	<	<	<	<	D53N, K83E, Y94N, N96S(+cho), V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, I192F, F193S, Y195F
A/Norway/23621/2021	3C.2a1b.2a.2	2021-10-13	SIAT1	<	<	<	<	80	69	320	280	640	736	<	<	<	<	D53N, K83E, Y94N, N96S(+cho), V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, I192F, F193S, Y195F
A/Norway/24864/2021	3C.2a1b.2a.2	2021-10-26	SIAT1	<	<	40	50	320	130	640	501	1280	1259	<	<	<	<	D53N, K83E, Y94N, N96S(+cho), V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, I192F, F193S, Y195F
A/Haute Normandie/27444/2021	3C.2a1b.2a.2	2021-09-11	MDCK1/SIAT1	<	<	<	<	80	99	320	392	640	935	<	<	<	<	D53N, L59X, K83E, Y94N, N96S(+cho), V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, I192F, F193S, Y195F
A/Lyon/CHU/ R21.830.39/2021	3C.2a1b.2a.2	2021-11-22	MDCK2/SIAT1	<	<	<	<	80	106	320	468	1280	1184	<	<	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Norway/28578/2021	3C.2a1b.2a.2	2021-11-19	SIAT1	<	<	40	52	640	485	640	690	1280	1001	80	66	80	64	E50K, K83E, Y94N, V106A, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, R269K
A/Norway/28577/2021	3C.2a1b.2a.2	2021-11-21	SIAT1	<	<	40	51	640	480	640	560	640	841	80	67	80	62	E50K, K83E, Y94N, V106A, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, R269K
A/Norway/29511/2021	3C.2a1b.2a.2	2021-11-25	SIAT1	<	<	<	<	640	532	640	524	640	800	80	61	40	59	E50K, K83E, Y94N, V106A, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, R269K
A/Picardie/44663/2021	3C.2a1b.1b	2021-11-24	MDCK1/SIAT1	<	<	320	410	<	<	<	<	<	<	160	187	<	<	N31S, H56Y, D85N, V106A, T128A(-cho), K131T, T135K(-cho), A138S, F193S, S198P

[1] Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Table 10-6. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2022-01-31

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation Titre[1] Post-infection ferret antisera A/Norway 3275/2018 SIAT F03/19 3C.2a1b.2		A/Cambodia 925256/2020 SIAT F03/21 3C.2a1b.2a.1		A/Bangladesh 4005/2020 SIAT F07/21 3C.2a1b.2a.2		A/Stockholm 5/2021 SIAT F35/21 3C.2a1b.2a.2		A/Darwin 09/2021 Egg F38/21 3C.2a1b.2a.2		A/Denmark 3264/19 SIAT F19/20 3C.2a1b.1a		A/Kansas 14/2017 SIAT F17/19 3C.3a.1		HA1 substitutions for 3C.2a/3a viruses compared to A/Norway/3275/2018: Egg adaptation HA substitutions compared to the corresponding cell culture propagated virus
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	
REFERENCE VIRUSES																		
A/Norway/3275/2018	3C.2a1b.2	2018-10-04	SIAT2 10^{-3}	320	371	80	70	<	<	<	<	80	72	80	90	<	<	
A/Cambodia/925256/2020	3C.2a1b.2a.1	2020-09-25	SIAT3 10^{-3}	80	70	320	243	<	<	<	<	80	99	80	85	<	<	K83E, Y94N, V106A, K171N, G186S, F193S, Y195F, S198P
A/Bangladesh/4005/2020	3C.2a1b.2a.2	2020-10-04	SIAT2 10^{-3}	<	<	40	41	320	350	640	727	1280	1030	<	<	40	54	K83E, Y94N, V106A, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F
A/Stockholm/5/2021	3C.2a1b.2a.2	2021-04-16	SIAT0/SIAT3 10^{-3}	<	<	<	<	80	68	320	465	1280	1044	<	<	<	<	K83E, Y94N, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F
A/Darwin/9/2021	3C.2a1b.2a.2	2021-04-17	E3/E2 10^{-3}	<	<	80	84	320	320	1280	1609	5120	4994	160	138	<	<	K83E, Y94N, V106A, H156S, Y159N, T160I(-cho), L164Q, G186N, D190N, F193S, Y195F, D225G
A/Denmark/3264/2019	3C.2a1b.1a	2019-10-25	SIAT3/SIAT2 10^{-3}	40	52	320	254	<	<	<	<	<	<	320	273	<	<	V106A, T128A(-cho), K131T, T135K(-cho), A138S, G186D, D190N, F193S, S198P
A/Kansas/14/2017	3C.3a.1	2017-12-14	SIAT3/SIAT2 10^{-3}	<	<	<	<	<	<	<	<	<	<	<	<	160	132	G62E, S91N, R92K, V106A, K121N, T128A(-cho), K131T, A138S, S144K, Y159S, T160K(-cho), K171N, F193S, K326R
TEST VIRUSES																		
A/South Africa/R16482/2021	3C.2a1b.1a	2021-10-18	MDCK2/SIAT1	<	<	320	246	40	53	<	<	80	73	160	157	<	<	V106A, T128A(-cho), K131N, T135K(-cho), A138S, G186D, D190N, I192F, F193S, S198P, G218R, K238T, R261Q
A/La Reunion/876/2021	3C.2a1b.1b	2021-11-14	MDCK2/SIAT1	<	<	320	341	<	<	<	<	<	<	320	320	<	<	E50K, V106A, T128A(-cho), K131T, T135K(-cho), S137F, A138S, F193S, K310R
A/Saint-Martin d'Heres/785/2021	3C.2a1b.1b	2021-11-03	MDCK2/SIAT1	<	<	320	448	<	<	<	<	<	<	320	263	<	<	E50K, V106A, T128A(-cho), K131T, T135K(-cho), S137F, A138S, F193S, K278N, K310R
A/Chambery/ 760/2021	3C.2a1b.1b	2021-11-01	MDCK2/SIAT1	<	<	320	400	<	<	<	<	<	<	320	301	<	<	E50K, V106A, T128A(-cho), K131T, T135K(-cho), S137F, A138S, F193S, K278N, K310R
A/South Africa/R15681/2021	3C.2a1b.1b	2021-10-26	MDCK1/SIAT1	<	<	<	<	<	<	<	<	<	<	<	<	<	<	E50K, V106A, T128A(-cho), K131T, T135K(-cho), S137F, A138S, T160K(-cho), K171N, F193S, K310R
A/Baleares/9868/2021	3C.2a1b.2a.2	2021-11-26	SIAT1/SIAT1	<	<	<	<	80	82	320	302	640	803	<	<	<	<	D53N, K83E, Y94N, N96S(+cho), V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, I192F, F193S, Y195F
A/Navarra/9870/2021	3C.2a1b.2a.2	2021-11-24	SIAT1/SIAT1	<	<	<	<	80	102	640	512	1280	968	<	<	<	<	D53N, K83E, Y94N, N96S(+cho), V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, I192F, F193S, Y195F
A/Castilla La Mancha/9869/2021	3C.2a1b.2a.2	2021-11-16	SIAT1/SIAT1	<	<	<	<	80	91	640	480	640	931	<	<	<	<	D53N, K83E, Y94N, N96S(+cho), V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, I192F, F193S, Y195F
A/Navarra/9643/2021	3C.2a1b.2a.2	2021-11-06	SIAT1/SIAT1	<	<	<	<	80	118	640	588	1280	1096	<	<	<	<	D53N, K83E, Y94N, N96S(+cho), V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, I192F, F193S, Y195F
A/Lyon/782/2021	3C.2a1b.2a.2	2021-11-06	MDCK2/SIAT1	<	<	<	<	80	87	320	297	640	813	<	<	<	<	D53N, K83E, Y94N, N96S(+cho), V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, I192F, F193S, Y195F
A/Uppsala/12/2021	3C.2a1b.2a.2	2021-11-10	SIAT1/SIAT1	<	<	<	<	80	83	640	602	1280	1166	<	<	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Uppsala/11/2021	3C.2a1b.2a.2	2021-11-09	SIAT1/SIAT1	<	<	<	<	80	115	640	664	1280	1095	<	<	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Sweden/1/2021	3C.2a1b.2a.2	2021-11-03	SIAT1/SIAT1	<	<	<	<	80	94	320	473	1280	1205	<	<	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Uppsala/8/2021	3C.2a1b.2a.2	2021-11-03	SIAT1/SIAT1	<	<	<	<	80	97	640	552	1280	1061	<	<	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/South Africa/R15449/2021	3C.2a1b.2a.2	2021-10-21	MDCK1/SIAT1	<	<	<	<	160	228	160	215	320	292	80	73	<	<	K83E, Y94N, V106A, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, S205F, A212T

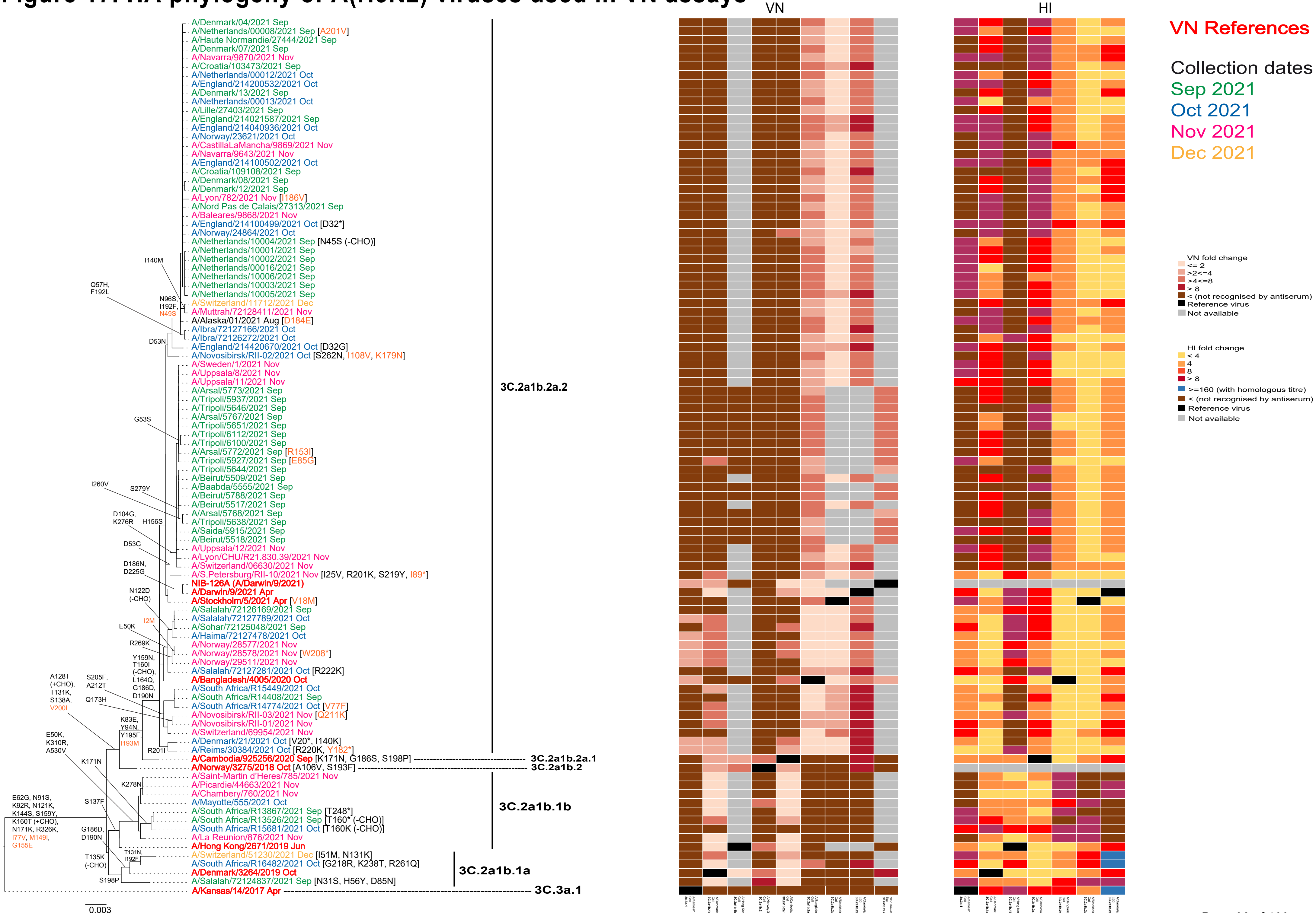
[1] Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Table 10-7. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2022-02-07

Viruses	Passage history / Ferret number / Genetic group	Collection Date	Passage History	Neutralisation Titre[1] Post-infection ferret antisera: A/Norway 3275/2018 SIAT F03/19 3C.2a1b.2		A/Cambodia 925256/2020 SIAT F03/21 3C.2a1b.2a.1		A/Bangladesh 4005/2020 SIAT F07/21 3C.2a1b.2a.2		A/Stockholm 5/2021 SIAT F35/21 3C.2a1b.2a.2		A/Darwin 09/2021 Egg F38/21 3C.2a1b.2a.2		A/Denmark 3264/19 SIAT F19/20 3C.2a1b.1a		A/Kansas 14/2017 SIAT F17/19 3C.3a.1		HA1 substitutions for 3C.2a/3a viruses compared to A/Norway/3275/2018: Egg adaptation HA substitutions compared to the corresponding cell culture propagated virus
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	
REFERENCE VIRUSES																		
A/Norway/3275/2018	3C.2a1b.2	2018-10-04	SIAT2 10^{-3}	320	371	80	70	<	<	<	<	80	72	80	90	<	<	
A/Cambodia/925256/2020	3C.2a1b.2a.1	2020-09-25	SIAT3 10^{-3}	80	70	320	243	<	<	<	<	80	99	80	85	<	<	K83E, Y94N, V106A, K171N, G186S, F193S, Y195F, S198P
A/Bangladesh/4005/2020	3C.2a1b.2a.2	2020-10-04	SIAT2 10^{-3}	<	<	40	41	320	350	640	727	1280	1030	<	<	40	54	K83E, Y94N, V106A, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F
A/Stockholm/5/2021	3C.2a1b.2a.2	2021-04-16	SIAT0/SIAT3 10^{-3}	<	<	<	<	80	68	320	465	1280	1044	<	<	<	<	K83E, Y94N, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F
A/Darwin/9/2021	3C.2a1b.2a.2	2021-04-17	E3/E2 10^{-3}	<	<	80	84	320	320	1280	1609	5120	4994	160	138	<	<	K83E, Y94N, V106A, H156S, Y159N, T160I(-cho), L164Q, G186N, D190N, F193S, Y195F, D225G
A/Denmark/3264/2019	3C.2a1b.1a	2019-10-25	SIAT3/SIAT2 10^{-3}	40	52	320	254	<	<	<	<	<	<	320	273	<	<	V106A, T128A(-cho), K131T, T135K(-cho), A138S, G186D, D190N, F193S, S198P
A/Kansas/14/2017	3C.3a.1	2017-12-14	SIAT3/SIAT2 10^{-3}	<	<	<	<	<	<	<	<	<	<	<	<	160	132	G62E, S91N, R92K, V106A, K121N, T128A(-cho), K131T, A138S, S144K, Y159S, T160K(-cho), K171N, F193S, K326R
TEST VIRUSES																		
A/South Africa/R14408/2021	3C.2a1b.2a.2	2021-09-20	MDCK1/SIAT1	<	<	<	<	320	291	160	156	160	229	<	<	<	<	K83E, Y94N, V106A, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, S205F, A212T
A/South Africa/R14774/2021	3C.2a1b.2a.2	2021-10-07	MDCK2/SIAT1	<	<	<	<	320	440	160	202	320	315	40	57	<	<	K83E, Y94N, V106A, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, S205F, A212T
A/Switzerland/69954/2021	3C.2a1b.2a.2	2021-11-20	MDCK1/SIAT1	<	<	<	<	160	199	80	84	160	194	<	<	<	<	K83E, Y94N, V106A, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, S205F, A212T
A/England/214420670/2021	3C.2a1b.2a.2	2021-10-28	MDCK1/SIAT1	<	<	<	<	80	119	160	196	640	533	<	<	<	<	D32G, D53N, K83E, Y94N, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F
A/England/214021587/2021	3C.2a1b.2a.2	2021-09-30	SIAT1/SIAT1	<	<	<	<	80	73	160	177	640	499	<	<	<	<	D53N, K83E, Y94N, N96S(+cho), V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, I192F, F193S, Y195F
A/England/214040936/2021	3C.2a1b.2a.2	2021-10-04	SIAT1/SIAT1	<	<	<	<	80	70	160	170	640	480	<	<	<	<	D53N, K83E, Y94N, N96S(+cho), V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, I192F, F193S, Y195F
A/England/214100502/2021	3C.2a1b.2a.2	2021-10-06	SIAT1/SIAT1	<	<	<	<	80	111	320	269	640	665	<	<	<	<	D53N, K83E, Y94N, N96S(+cho), V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, I192F, F193S, Y195F
A/England/214200532/2021	3C.2a1b.2a.2	2021-10-13	SIAT1/SIAT1	<	<	<	<	160	147	320	423	1280	1043	<	<	<	<	D53N, K83E, Y94N, N96S(+cho), V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, I192F, F193S, Y195F
A/England/214100499/2021	3C.2a1b.2a.2	2021-10-07	MDCK1/SIAT1	<	<	<	<	160	143	320	434	1280	960	<	<	<	<	D32X, D53N, K83E, Y94N, N96S(+cho), V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, I192F, F193S, Y195F
A/Switzerland/11712/2021	3C.2a1b.2a.2	2021-12-22	MDCK1/SIAT1	<	<	<	<	80	118	320	311	640	771	<	<	<	<	D53N, K83E, Y94N, N96S(+cho), V106A, I140M, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, I192F, F193S, Y195F
A/Muttrah/72128411/2021	3C.2a1b.2a.2	2021-11-09	SIAT1	<	<	<	<	160	160	640	480	640	910	<	<	<	<	D53N, K83E, Y94N, N96S(+cho), V106A, I140M, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, I192F, F193S, Y195F, Y302X
A/Switzerland/06630/2021	3C.2a1b.2a.2	2021-11-02	MDCK1/SIAT1	<	<	<	<	80	76	160	160	320	392	<	<	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Switzerland/51230/2021	3C.2a1b.1a	2021-12-02	MDCK1/SIAT1	<	<	80	72	<	<	<	<	<	<	40	48	<	<	I51M, V106A, T128A(-cho), T135K(-cho), A138S, G186D, D190N, I192F, F193S, S198P
A/South Africa/R13526/2021	3C.2a1b.1b	2021-09-07	MDCK1/SIAT1	<	<	80	111	<	<	<	<	<	<	40	55	<	<	E50K, V106A, T128A(-cho), K131T, T135K(-cho), S137F, A138S, T160X(-cho), K171N, F193S, K310R
A/South Africa/R13867/2021	3C.2a1b.1b	2021-09-16	MDCK1/SIAT1	<	<	320	286	<	<	<	<	<	<	160	158	<	<	E50K, V106A, T128A(-cho), K131T, T135K(-cho), S137F, A138S, K171N, F193S, T248X(-cho), K310R, P324X
A/Mayotte/555/2021	3C.2a1b.1b	2021-10-01	MDCK2/SIAT1	40	47	320	314	<	<	<	<	<	<	320	252	<	<	E50K, V106A, T128A(-cho), K131T, T135K(-cho), S137F, A138S, F193S, K310R

[1] Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Figure 17. HA phylogeny of A(H3N2) viruses used in VN assays



Influenza B virus analyses.

Influenza B viruses with collection dates after 2021-08-31 were received from thirteen countries in three WHO Regions: African (Cameroon, Madagascar and South Africa), the Eastern Mediterranean (Oman and Qatar) and the European (Germany, Ireland, the Netherlands, Portugal, Romania, Spain, Switzerland and the United Kingdom). They represented 9% of the samples received with collection dates after 2021-08-31 (**Table 1**) and, of those ascribed to a lineage, all bar four were B/Victoria-lineage. Viruses were successfully propagated for 65% of the samples received as clinical specimens and for 97% of the samples provided as virus isolates (**Table 3**). A summary of the HI results obtained for these viruses is shown in **Table 4**.

B/Victoria-lineage.

HI analyses

The collection details and summary results for the 44 viruses with collection dates after 2021-08-31 that were assayed by HI, up to 2022-02-09, are shown in **Table 11**.

HI results for all propagated viruses analysed since the September 2021 report are shown in **Tables 12-1** to **12-5**. Test viruses in each table are sorted by date of collection and those with collection dates after 2021-08-31 are indicated in **Tables 12-2 to 12-5 (**below the red horizontal lines). The HA genetic group is indicated. A summary of the results is shown in **Table 12-6**. **Tables 12-1** and **12-4** include the results for six viruses received from CDC, **Table 12-4** also includes eight viruses from CNIC, and **Table 12-5** includes the results of a virus sent by NIID, Tokyo.

Ferret antisera used were raised against the former vaccine viruses egg-propagated B/Colorado/06/2017 (clade V1A.1) and egg-propagated B/Washington/02/2019 (clade V1A.3), and two different cultivars of the egg-propagated vaccine virus B/Austria/1359417/2021 (subclade V1A.3a.2) – one with an HA sequence identical to the cell culture-propagated cultivar of B/Austria/1359417/2021 and one with a HA1 **G141R** substitution. Antisera used were also raised against cell culture-propagated cultivars of B/Austria/1359417/2021, B/Croatia/7889/2019 (subclade V1A.3a), B/Cote d'Ivoire/948/2020 (subclade V1A.3a.1) and B/Paris/9878/2020 (subclade V1A.3a.2).

The majority of test viruses fell in subclade V1A.3a.2 (n = 38) and a minority in clade V1A.3 (n = 5). In the HI analyses eleven test viruses in subclade V1A.3a.2 (29%) were recognised at titres within 4-fold of the homologous titre by the antiserum raised against egg-propagated B/Colorado/06/2017. Five test viruses in clade V1A.3 (11%), all from Madagascar, were recognised at titres within 4-fold of the homologous titre by the antiserum raised against B/Washington/02/2019. The five test viruses in clade V1A.3 were poorly recognised by all other antisera.

The antiserum raised against cell culture-propagated B/Croatia/7789/2019 recognised all but one of the 38 viruses in subclade V1A.3a.2 at titres within 4-fold of the homologous titre, and 35 of the 44 test viruses (80%) at titres within 2-fold. The HA gene of B/Croatia/7789/2019 encoded the **150K** group substitutions in **HA1** (**N150K, G184E, N197D** (resulting in loss of a glycosylation site) and **R279K**) without additional substitutions. In contrast, an antiserum raised against B/Cote d'Ivoire/948/2020, with additional **HA1 V220M** and **P241Q** substitutions recognised only ten (26%) of the test viruses in subclade V1A.3a.2 at titres within 4-fold of the homologous titre and none within 2-fold.

The antiserum raised against the cell culture-propagated cultivar of B/Austria/1359417/2021 recognised all subclade V1A.3a.2 viruses at titres within 4-fold of the homologous titre, 37 of 38 at titres within 2-fold, and the antiserum raised against the egg-propagated cultivar of B/Austria/1359417/2021 with an identical HA sequence recognised all 38 of these viruses at titres within 2-fold of the homologous titre, as did the antiserum raised against cell culture-propagated B/Paris/9878/2020. The antiserum raised against egg-propagated B/Austria/1359417/2021 with the **HA1 G141R** substitution had a higher homologous HI titre and recognised none of the test viruses at titres within 4-fold of the homologous titre. The latter antiserum also recognised the test viruses in subclade V1A.3a.2 at absolute titres typically 2- to 4-fold lower than the absolute titres seen with the other two antisera raised against B/Austria/1359417/2021.

Of the eight viruses received from CNIC a similar pattern of reactivity was seen for six viruses with the panel of antisera (**Table 12-4**) but two were poorly recognised by antisera raised against B/Austria/1359417/2021. These two viruses were recognised well, at a titre equal to the homologous titre, by the antiserum raised against B/Cote d'Ivoire/948/2019.

Genetic analysis

Figure 18 shows a phylogenetic tree for HA genes of representative influenza B/Victoria-lineage viruses with collection dates after 2021-08-31, as deposited in GISAID at the time of preparing this report. **Figure 19** shows a tree for viruses which were sequenced at the Francis Crick Institute. An alignment of the HA glycoproteins of a selection of viruses is shown in **Figure 20**.

All recently circulating viruses analysed carried HA genes that fell into genetic clade **V1A.3** which encodes an HA with a three-amino acid deletion in **HA1** (**Δ162-164**) and a **HA1 K136E** substitution. But for a small number of viruses from East Africa and Madagascar and some viruses from the Americas**,** most viruses had HA genes encoding the substitutions **N150K, G184E, N197D** (resulting in the loss of a glycosylation site at residues 197-199, formerly usually lost in egg-propagated influenza B viruses) and **R279K** in **HA1**.

Within this group two main subclades can be seen: one with HA genes encoding the **HA1** substitutions **V220M** and **P241Q** (subclade V1A.3a.1, e.g. B/Cote d'Ivoire/948/2020 and some recent viruses from China), and a major group encoding the **HA1** substitutions **A127T, P144L** and **K203R** (subclade V1A.3a.2, e.g. B/Austria/1359417/2021) a small subset of which also encode the **HA1** substitutions **T182A, D197E** and **T221A** (e.g. B/Paris/9878/2020). Subclade V1A.3a.2 viruses from China fall in a large genetic cluster encoding an **HA1 H122Q** substitution. Some subclade V1A.3a.2 viruses have the additional substitution **A202V** in **HA1** and another cluster has HA genes encoding the substitution **D197E** in **HA1**.

Figure 21 shows the locations of amino acid substitutions that define viruses in subclades V1A.3a.1 and V1A.3a.2 on a B/Brisbane/60/2008 HA structure.

Figure 22 shows a phylogenetic tree for NA genes of representative influenza B/Victoria lineage viruses with collection dates after 2021-08-31, as deposited in GISAID at the time of preparing this report. **Figure 23** shows a tree for viruses which were sequenced at the Francis Crick Institute (*cf*. **Figure 19**). An alignment of the NA glycoproteins of a selection of viruses is shown in **Figure 24**.

The NA gene phylogenies is largely congruent with the HA phylogenies and the vast majority of viruses in clade V1A.3 have NA genes that fall into two main groups. Some reassortment of HA and NA genes is evident: viruses from China with HA genes encoding **HA1 H122Q** amino acid substitution have NA genes encoding **NA** substitutions **D53N, N59S** and **G233E**. A large group of viruses from locations around the world, including B/Paris/9878/2020 and B/Austria/1359417/2021, have NA genes encoding the **NA** substitutions **P42Q**, **V71A**, **K343E**, **A395V** and **V401R**. Some viruses in this group have an NA with an insertion of an amino acid in the stalk domain of the molecule (**ins 74L**). NA substitutions **T49A** and **G233R** have been seen in most viruses from South Africa that were analysed.

B/Yamagata-lineage.

Four B/Yamagata-lineage viruses with collection dates after August 2021 have been received at the Crick. These were supplied in lysis buffer and gene sequencing of the samples has not been successful to date.

Antiviral Analysis.

Susceptibility to the neuraminidase inhibitors (NAIs) oseltamivir and zanamivir was assessed phenotypically and by full NA gene sequencing, for 46 B/Victoria-lineage viruses detected and received within the course of the 2021-2022 influenza season (**Table 13**). All showed normal inhibition (NI: no reduced susceptibility) by both antivirals.

All of the 37 B/Victoria-lineage full-length PA gene sequences generated at the Crick, from individual viruses detected in the course of the 2021-2022 influenza season, encoded isoleucine at position 38 (**I38**) and showed no evidence of other substitutions associated with reduced susceptibility to baloxavir marboxil.

Conclusion.

Viruses of the B/Victoria-lineage dominated and all B/Victoria-lineage viruses sequenced had HAs with a three amino acid deletion in **HA1** (**∆162-164**) with the additional **HA1 K136E** amino acid substitution (clade **V1A.3**). The great majority of recent viruses had **HA1** amino acid substitutions **N150K, G184E, N197D** (resulting in the loss of a glycosylation site) and **R279K**, and most of the viruses with collection dates after 2021-08-31 received at the Francis Crick Institute (in shipments from National Influenza Centres) also encoded **HA1** substitutions **A127T, P144L** and **K203R**. These recently collected viruses were poorly recognised by antisera raised against the egg-propagated cultivar of B/Washington/02/2019 but were recognised very well by antisera raised against the cell culture-propagated cultivar of B/Austria/1359417/2021 and the egg-propagated cultivar of B/Austria/1359417/2021 with an identical HA sequence to the cell culture-propagated cultivar.

No viruses of the B/Yamagata-lineage with collection dates after 2021-08-31 have yet been characterised.

All influenza B/Victoria-lineage viruses showed normal inhibition by the NAIs oseltamivir and zanamivir and those viruses yielding full-length PA gene sequences (n = 37) all encoded proteins likely to be well inhibited by baloxavir marboxil.

Table 11. Summary of influenza B (Victoria-lineage) viruses analysed, collected after 2021-08-31

MONTH			B Victoria lineage											
			Post infection ferret antisera raised against											
			Clade V1A.3a.2 virus			Clade V1A.3a.2 virus			Clade V1A.3a.2 virus			Clade V1A.3 virus		
Country	Number received	Number propagated[1,2]	≤2 fold[3]	4 fold[3]	≥8 fold[3]	≤2 fold[4]	4 fold[4]	≥8 fold[4]	≤2 fold[5]	4 fold[5]	≥8 fold[5]	≤2 fold[6]	4 fold[6]	≥8 fold[6]
2021														
SEPTEMBER														
Madagascar	1	0												
Netherlands	1	1		1		1					1			1
Qatar	3	3	3			3					3			3
South Africa	16	16	16			16					16			16
OCTOBER														
Madagascar	3	0												
Oman	1	1	1			1					1			1
Qatar	2	2	2			2					2			2
South Africa	4	4	4			4					4			4
NOVEMBER														
Cameroon	2	0												
Germany	1	0												
Ireland	1	0												
Madagascar	7	2			2			2			2	2		
Oman	3	1	1			1					1			1
Qatar	2	1	1			1					1			1
Romania	1	1	1			1					1			1
South Africa	6	6	6			6					6			6
Spain	1	0												
Switzerland	2	2	2			2					2			2
DECEMBER														
Madagascar	4	3			3			3			3	1	2	
Portugal	2	1	1			1					1			1
Spain	1	0												
	64	44	38	1	5	39	0	5	0	0	44	3	2	39
12 Countries			86.4%	2.3%	11.4%	88.6%	0.0%	11.4%	0.0%	0.0%	100.0%	6.8%	4.5%	88.6%

1 Numbers shown in red indicate that not all of the specimens received had been propagated at the time this report was prepared
2 Propagated to sufficient titre to perform HI assay
3 Compared to homologous titres for ferret antisera raised against tissue culture-propagated B/Austria/1359417/2021 - Clade V1A.3a.2 N150K gp + A127T, P144L, K203R
4 Compared to homologous titres for ferret antisera raised against egg-propagated B/Austria/1359417/2021 - Clade V1A.3a.2 N150K gp +A127T, P144L, K203R (G141)
5 Compared to homologous titres for ferret antisera raised against egg-propagated B/Austria/1359417/2021 - Clade V1A.3a.2 N150K gp +A127T, P144L, K203R (G141R)
6 Compared to homologous titres for ferret antisera raised against egg-propagated B/Washington/02/2019 - Clade V1A.3

As of 2022-02-09

Figure 18. Phylogenetic comparison of B/Victoria HA genes (non-WIC sequences)

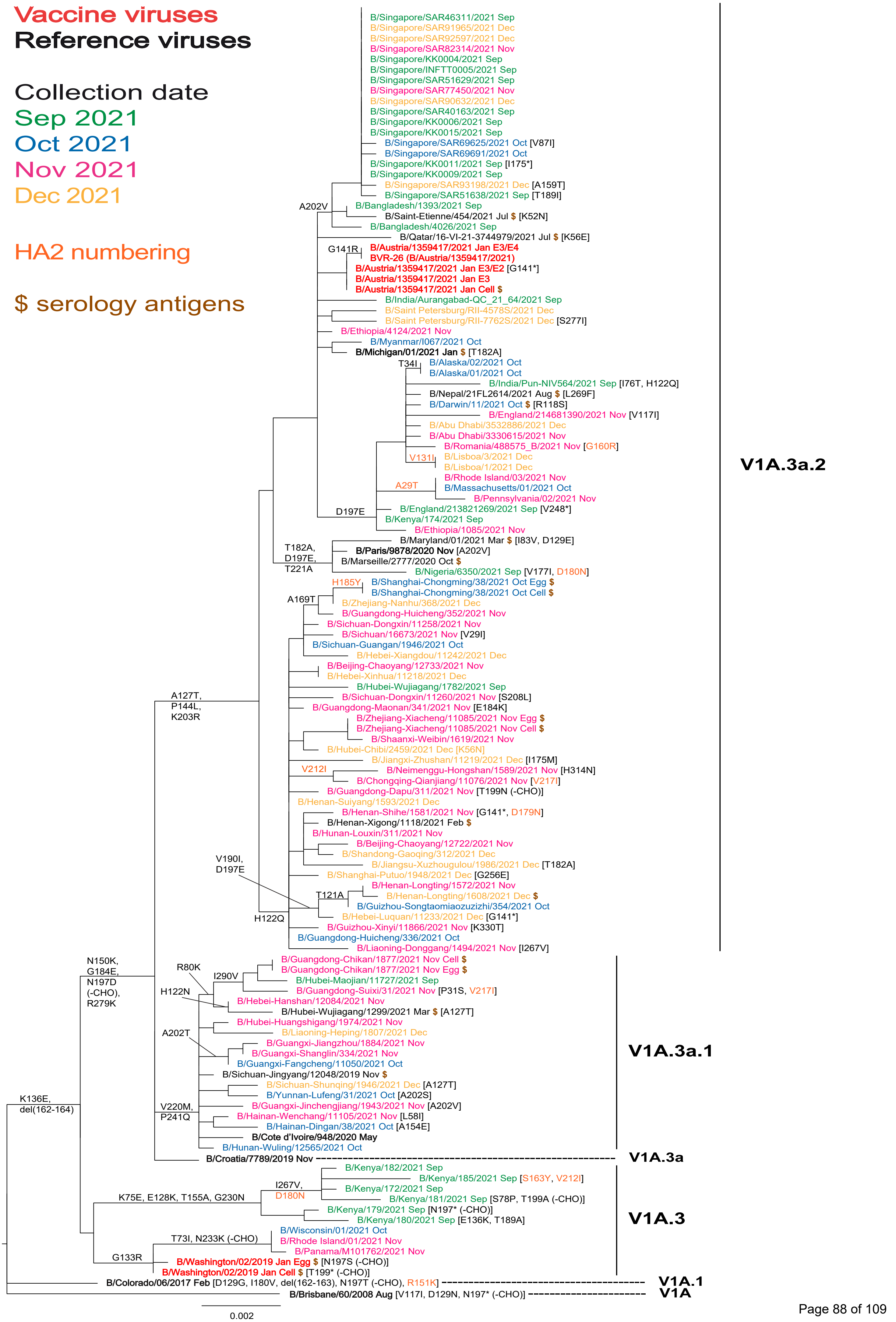


Figure 19. Phylogenetic comparison of B/Victoria HA genes (WIC sequences)

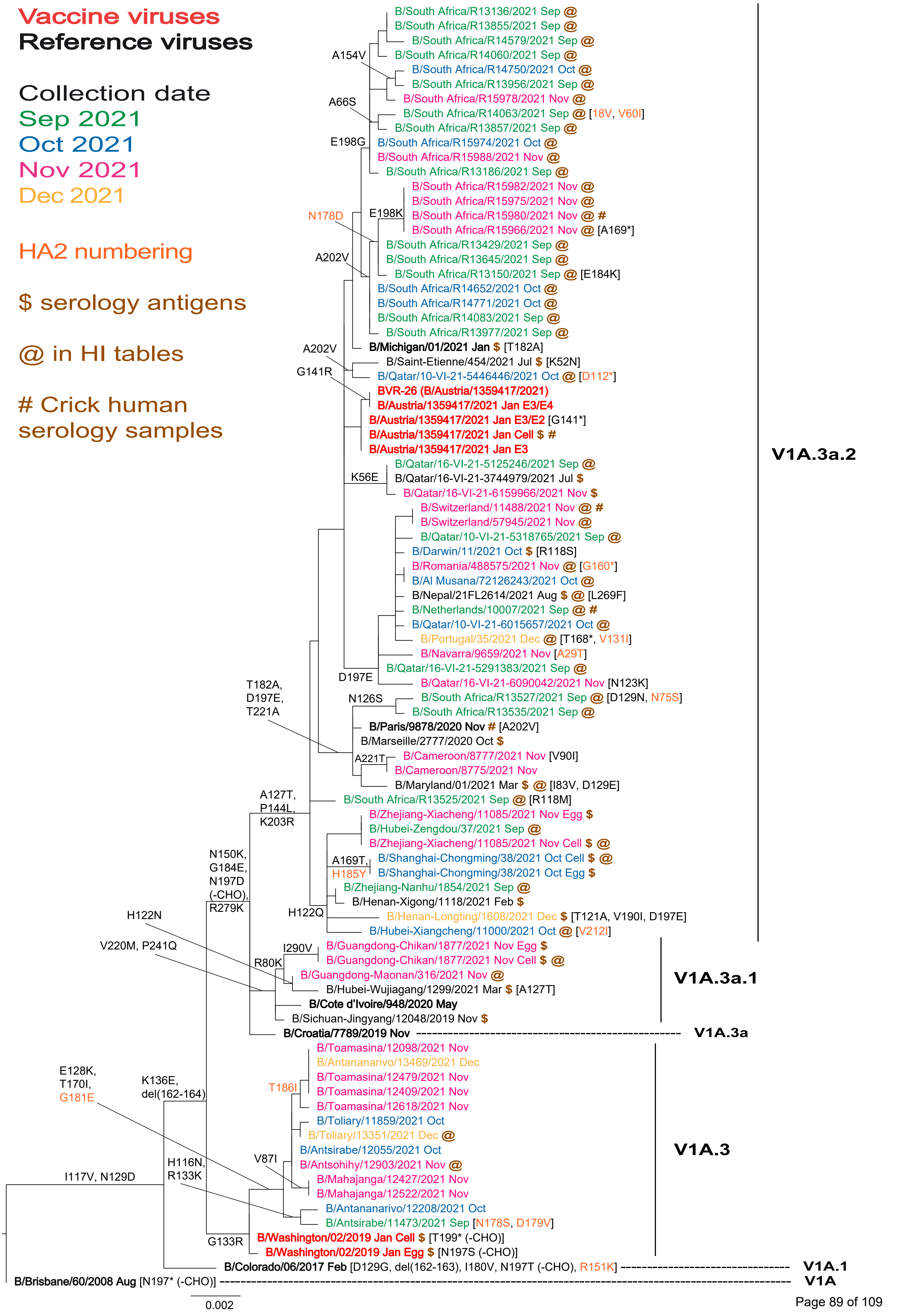


Figure 20. HA protein alignment for B (Victoria-lineage) viruses

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

```
                                  10        20        30        40        50        60        70        80        90       100
                         ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Brisbane/60/2008             DRICTGITSSNSPHVVKTATQGEVNVTGVIPLTTTPTKSHFANLKGTETRGKLCPKCLNCTDLDVALGRPKCTGKIPSARVSILHEVRPVTSGCFPIMHD V1A
B/Kenya/180/2021               ..........................................................................E......................... V1A.3
B/Washington/02/2019E          .................................................................................................... V1A.3
B/Panama/M101762/2021          ........................................................................I........................... V1A.3
B/Antananarivo/12208/2021      .................................................................................................... V1A.3
B/Mahajanga/12427/2021         ......................................................................................I............. V1A.3
B/Toamasina/12409/2021         .................................................................................................... V1A.3
B/Guangdong-Chikan/1877/2021   ...............................................................................K.................... V1A.3a.1
B/Liaoning-Heping/1807/2021    .................................................................................................... V1A.3a.1
B/South_Africa/R13525/2021     .................................................................................................... V1A.3a.2
B/Zhejiang-Xiacheng/11085/2021 .................................................................................................... V1A.3a.2
B/Henan-Longting/1608/2021     .................................................................................................... V1A.3a.2
B/Shanghai-Chongming/38/2021   .................................................................................................... V1A.3a.2
B/Cameroon/8777/2021           .........................................................................................I.......... V1A.3a.2
B/South_Africa/R13527/2021     .................................................................................................... V1A.3a.2
B/Qatar/16-VI-21-6090042/2021  .................................................................................................... V1A.3a.2
B/Pennsylvania/02/2021         .................................................................................................... V1A.3a.2
B/Portugal/35/2021             .................................................................................................... V1A.3a.2
B/Switzerland/57945/2021       .................................................................................................... V1A.3a.2
B/Romania/488575/2021          .................................................................................................... V1A.3a.2
B/South_Africa/R14771/2021     .................................................................................................... V1A.3a.2
B/South_Africa/R15980/2021     .................................................................................................... V1A.3a.2
B/South_Africa/R13857/2021     .................................................................S.................................. V1A.3a.2
B/South_Africa/R14750/2021     .................................................................................................... V1A.3a.2
B/St Petersburg/RII-7762S/2021 .................................................................................................... V1A.3a.2
B/Austria/1359417/2021         .................................................................................................... V1A.3a.2
BVR-26(B/Austria/1359417/2021) .................................................................................................... V1A.3a.2
B/Qatar/16-VI-21-5125246/2021  .......................................................E............................................ V1A.3a.2
B/Qatar/10-VI-21-5446446/2021  .................................................................................................... V1A.3a.2
B/Bangladesh/4026/2021         .................................................................................................... V1A.3a.2
B/Singapore/SAR93198/2021      .................................................................................................... V1A.3a.2
B/Singapore/SAR90632/2021      .................................................................................................... V1A.3a.2

                                 110       120       130       140       150       160       170       180       190       200
                         ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Brisbane/60/2008             RTKIRQLPNLLRGYEHIRLSTHNVINAENAPGGPYKIGTSGSCPNITNGNGFFATMAWAVPKNDKNKTATNPLTIEVPYICTEGEDQITVWGFHSDXETQ
B/Kenya/180/2021               ................V.........................KD................................A..........---..............................................A.......N...
```

```
B/Kenya/180/2021               ................V..........KD......................A.....---..........................A.......N...
B/Washington/02/2019E          ................V...........D...R..E.....................---...............................S...
B/Panama/M101762/2021          ................V...........D...R..E.....................---...............................N...
B/Antananarivo/12208/2021      ...............NV..........KD...K..E.....................---.....I.........................N...
B/Mahajanga/12427/2021         ................V..........KD...R..E.....................---.....I.........................N...
B/Toamasina/12409/2021         ................V..........KD...R..E.....................---.....I.........................N...
B/Guangdong-Chikan/1877/2021   ................V...........D......E.............K.......---.....................E............D...
B/Liaoning-Heping/1807/2021    ................V...........D......E.............K.......---.....................E............D...
B/South_Africa/R13525/2021     ................VM........T.D......E.......L.....K.......---.....................E............D...
B/Zhejiang-Xiacheng/11085/2021 ................V.........T.D......E.......L.....K.......---.....................E............D...
B/Henan-Longting/1608/2021     ................V...AQ....T.D......E.......L.....K.......---.....................E.....I......E...
B/Shanghai-Chongming/38/2021   ................V....Q....T.D......E.......L.....K.......---....T................E............D...
B/Cameroon/8777/2021           ................V.........T.D......E.......L.....K.......---...................A.E............E...
B/South_Africa/R13527/2021     ................V........ST........E.......L.....K.......---...................A.E............E...
B/Qatar/16-VI-21-6090042/2021  ................V.....K...T.D......E.......L.....K.......---.....................E............E...
B/Pennsylvania/02/2021         ................V.........T.D......E.......L.....K.......---.....................E............E...
B/Portugal/35/2021             ................V.........T.D......E.......L.....K.......---...X.................E............E...
B/Switzerland/57945/2021       ................V.........T.D......E.......L.....K.......---.....................E............E...
B/Romania/488575/2021          ................V.........T.D......E.......L.....K.......---.....................E............E...
B/South_Africa/R14771/2021     ................V.........T.D......E.......L.....K.......---.....................E............D...
B/South_Africa/R15980/2021     ................V.........T.D......E.......L.....K.......---.....................E............DK..
B/South_Africa/R13857/2021     ................V.........T.D......E.......L.....K.......---.....................E............DG..
B/South_Africa/R14750/2021     ................V.........T.D......E.......L.....K...V...---.....................E............DG..
B/St Petersburg/RII-7762S/2021 ................V.........T.D......E.......L.....K.......---.....................E............D...
B/Austria/1359417/2021         ................V.........T.D......E.......L.....K.......---.....................E............D...
BVR-26(B/Austria/1359417/2021) ................V.........T.D......E....R..L.....K.......---.....................E............D...
B/Qatar/16-VI-21-5125246/2021  ................V.........T.D......E.......L.....K.......---.....................E............D...
B/Qatar/10-VI-21-5446446/2021  ................V.........T.D......E.......L.....K.......---.....................E............D...
B/Bangladesh/4026/2021         ................V.........T.D......E.......L.....K.......---.....................E............D...
B/Singapore/SAR93198/2021      ................V.........T.D......E.......L.....K........T.---.....................E............D...
B/Singapore/SAR90632/2021      ................V.........T.D......E.......L.....K.......---.....................E............D...

                                 210       220       230       240       250       260       270       280       290       300
                         ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Brisbane/60/2008             MAKLYGDSKPQKFTSSANGVTTHYVSQIGGFPNQTEDGGLPQSGRIVVDYMVQKSGKTGTITYQRGILLPQKVWCASGRSKVIKGSLPLIGEADCLHEKY
B/Kenya/180/2021               .............................N......................................................................
B/Washington/02/2019E          ....................................................................................................
B/Panama/M101762/2021          ................................K...................................................................
B/Antananarivo/12208/2021      ....................................................................................................
B/Mahajanga/12427/2021         ....................................................................................................
B/Toamasina/12409/2021         ....................................................................................................
B/Guangdong-Chikan/1877/2021   ...................M....................Q.....................................K..........V..........
B/Liaoning-Heping/1807/2021    ...................M....................Q.....................................K.....................
B/South_Africa/R13525/2021     ..R...........................................................................K.....................
B/Zhejiang-Xiacheng/11085/2021 ..R...........................................................................K.....................
B/Henan-Longting/1608/2021     ..R...........................................................................K.....................
B/Shanghai-Chongming/38/2021   ..R...........................................................................K.....................
B/Cameroon/8777/2021           ..R...........................................................................K.....................
B/South_Africa/R13527/2021     ..R.................A.........................................................K.....................
B/Qatar/16-VI-21-6090042/2021  ..R........X..................................................................K.....................
B/Pennsylvania/02/2021         ..R...........................................................................K.....................
B/Portugal/35/2021             ..R...........................................................................K.....................
B/Switzerland/57945/2021       ..R...........................................................................K.....................
B/Romania/488575/2021          ..R...........................................................................K.....................
B/South_Africa/R14771/2021     .VR...........................................................................K.....................
B/South_Africa/R15980/2021     .VR...........................................................................K.....................
B/South_Africa/R13857/2021     ..R...........................................................................K.....................
B/South_Africa/R14750/2021     ..R...........................................................................K.....................
B/St Petersburg/RII-7762S/2021 ..R.........................................................................I.K.....................
B/Austria/1359417/2021         ..R...........................................................................K.....................
BVR-26(B/Austria/1359417/2021) ..R...........................................................................K.....................
B/Qatar/16-VI-21-5125246/2021  ..R...........................................................................K.....................
B/Qatar/10-VI-21-5446446/2021  .VR...........................................................................K.....................
B/Bangladesh/4026/2021         .VR...........................................................................K.....................
B/Singapore/SAR93198/2021      .VR...........................................................................K.....................
B/Singapore/SAR90632/2021      .VR...........................................................................K.....................
```

```
                                         310       320       330       340       350       360       370       380       390       400
                                ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Brisbane/60/2008              GGLNKSKPYYTGEHAKAIGNCPIWVKTPLKLANGTKYRPPAKLLKERGFFGAIAGFLEGGWEGMIAGWHGYTSHGAHGVAVAADLKSTQEAINKITKNLN V1A
B/Kenya/180/2021                .................................................................................................... V1A.3
B/Washington/02/2019E           .................................................................................................... V1A.3
B/Panama/M101762/2021           .................................................................................................... V1A.3
B/Antananarivo/12208/2021       .................................................................................................... V1A.3
B/Mahajanga/12427/2021          .................................................................................................... V1A.3
B/Toamasina/12409/2021          .................................................................................................... V1A.3
B/Guangdong-Chikan/1877/2021    .................................................................................................... V1A.3a.1
B/Liaoning-Heping/1807/2021     .................................................................................................... V1A.3a.1
B/South_Africa/R13525/2021      .................................................................................................... V1A.3a.2
B/Zhejiang-Xiacheng/11085/2021  .................................................................................................... V1A.3a.2
B/Henan-Longting/1608/2021      .................................................................................................... V1A.3a.2
B/Shanghai-Chongming/38/2021    .................................................................................................... V1A.3a.2
B/Cameroon/8777/2021            .................................................................................................... V1A.3a.2
B/South_Africa/R13527/2021      .................................................................................................... V1A.3a.2
B/Qatar/16-VI-21-6090042/2021   .................................................................................................... V1A.3a.2
B/Pennsylvania/02/2021          ...........................................................................T........................ V1A.3a.2
B/Portugal/35/2021              .................................................................................................... V1A.3a.2
B/Switzerland/57945/2021        .................................................................................................... V1A.3a.2
B/Romania/488575/2021           .................................................................................................... V1A.3a.2
B/South_Africa/R14771/2021      .................................................................................................... V1A.3a.2
B/South_Africa/R15980/2021      .................................................................................................... V1A.3a.2
B/South_Africa/R13857/2021      .................................................................................................... V1A.3a.2
B/South_Africa/R14750/2021      .................................................................................................... V1A.3a.2
B/St Petersburg/RII-7762S/2021  .................................................................................................... V1A.3a.2
B/Austria/1359417/2021          .................................................................................................... V1A.3a.2
BVR-26(B/Austria/1359417/2021)  .................................................................................................... V1A.3a.2
B/Qatar/16-VI-21-5125246/2021   .................................................................................................... V1A.3a.2
B/Qatar/10-VI-21-5446446/2021   .................................................................................................... V1A.3a.2
B/Bangladesh/4026/2021          .................................................................................................... V1A.3a.2
B/Singapore/SAR93198/2021       .................................................................................................... V1A.3a.2
B/Singapore/SAR90632/2021       .................................................................................................... V1A.3a.2

                                         410       420       430       440       450       460       470       480       490       500
                                ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Brisbane/60/2008              SLSELEVKNLQRLSGAMDELHNEILELDEKVDDLRADTISSQIELAVLLSNEGIINSEDEHLLALERKLKKMLGPSAVEIGNGCFETKHKCNQTCLDRIA
B/Kenya/180/2021                ....................................................................................................
B/Washington/02/2019E           ....................................................................................................
B/Panama/M101762/2021           ....................................................................................................
B/Antananarivo/12208/2021       ....................................................................................................
B/Mahajanga/12427/2021          ....................................................................................................
B/Toamasina/12409/2021          .............X......................................................................................
B/Guangdong-Chikan/1877/2021    ....................................................................................................
B/Liaoning-Heping/1807/2021     ....................................................................................................
B/South_Africa/R13525/2021      ....................................................................................................
B/Zhejiang-Xiacheng/11085/2021  ....................................................................................................
B/Henan-Longting/1608/2021      ....................................................................................................
B/Shanghai-Chongming/38/2021    ....................................................................................................
B/Cameroon/8777/2021            ....................................................................................................
B/South_Africa/R13527/2021      .....................S..............................................................................
B/Qatar/16-VI-21-6090042/2021   ....................................................................................................
B/Pennsylvania/02/2021          ....................................................................................................
B/Portugal/35/2021              .............................................................................I......................
B/Switzerland/57945/2021        ....................................................................................................
B/Romania/488575/2021           ....................................................................................................
B/South_Africa/R14771/2021      ....................................................................................................
B/South_Africa/R15980/2021      ....................................................................................................
B/South_Africa/R13857/2021      ....................................................................................................
B/South_Africa/R14750/2021      ....................................................................................................
B/St Petersburg/RII-7762S/2021  ....................................................................................................
B/Austria/1359417/2021          ....................................................................................................
BVR-26(B/Austria/1359417/2021)  ....................................................................................................
B/Qatar/16-VI-21-5125246/2021   ....................................................................................................
B/Qatar/10-VI-21-5446446/2021   ..........................................................X.........................................
B/Bangladesh/4026/2021          ....................................................................................................
B/Singapore/SAR93198/2021       ....................................................................................................
B/Singapore/SAR90632/2021       ....................................................................................................

                                         510       520       530       540       550       560       570
                                ....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Brisbane/60/2008              AGTFDAGEFSLPTFDSLNITAASLNDDGLDNHTILLYYSTAASSLAVTLMIAIFVVYMVSRDNVSCSICL
B/Kenya/180/2021                ......................................................................
B/Washington/02/2019E           ......................................................................
B/Panama/M101762/2021           ......................................................................
B/Antananarivo/12208/2021       ...........................E..........................................
B/Mahajanga/12427/2021          ...........................E..........................................
B/Toamasina/12409/2021          ...........................E....I.....................................
B/Guangdong-Chikan/1877/2021    ......................................................................
B/Liaoning-Heping/1807/2021     ......................................................................
B/South_Africa/R13525/2021      ......................................................................
B/Zhejiang-Xiacheng/11085/2021  ......................................................................
B/Henan-Longting/1608/2021      ......................................................................
B/Shanghai-Chongming/38/2021    ...............................Y......................................
B/Cameroon/8777/2021            ......................................................................
B/South_Africa/R13527/2021      ......................................................................
B/Qatar/16-VI-21-6090042/2021   ......................................................................
B/Pennsylvania/02/2021          ......................................................................
B/Portugal/35/2021              ................X.....................................................
B/Switzerland/57945/2021        ......................................................................
B/Romania/488575/2021           ......X...............................................................
B/South_Africa/R14771/2021      ......................................................................
B/South_Africa/R15980/2021      ........................D.............................................
B/South_Africa/R13857/2021      ......................................................................
B/South_Africa/R14750/2021      ......................................................................
B/St Petersburg/RII-7762S/2021  ......................................................................
B/Austria/1359417/2021          ......................................................................
BVR-26(B/Austria/1359417/2021)  ......................................................................
B/Qatar/16-VI-21-5125246/2021   ......................................................................
B/Qatar/10-VI-21-5446446/2021   ......................................................................
B/Bangladesh/4026/2021          ......................................................................
B/Singapore/SAR93198/2021       ......................................................................
B/Singapore/SAR90632/2021       ......................................................................
```

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

Figure 21. Amino acid substitutions and deletions in the HA of recently circulating B/Victoria-lineage viruses mapped onto B/Brisbane/60/2008

Substitutions defining subclade 1A(△3)B viruses shown in Black

Viruses displaying low reactivity with antisera raised against B/Washington/02/2019:

The virus group (N150K) with 4 substitutions, has evolved in two directions V220M [hidden] + P241Q and A127T + P144L + K203R + (T182A + D197E + T221A [hidden]) with both subgroups displaying low reactivity

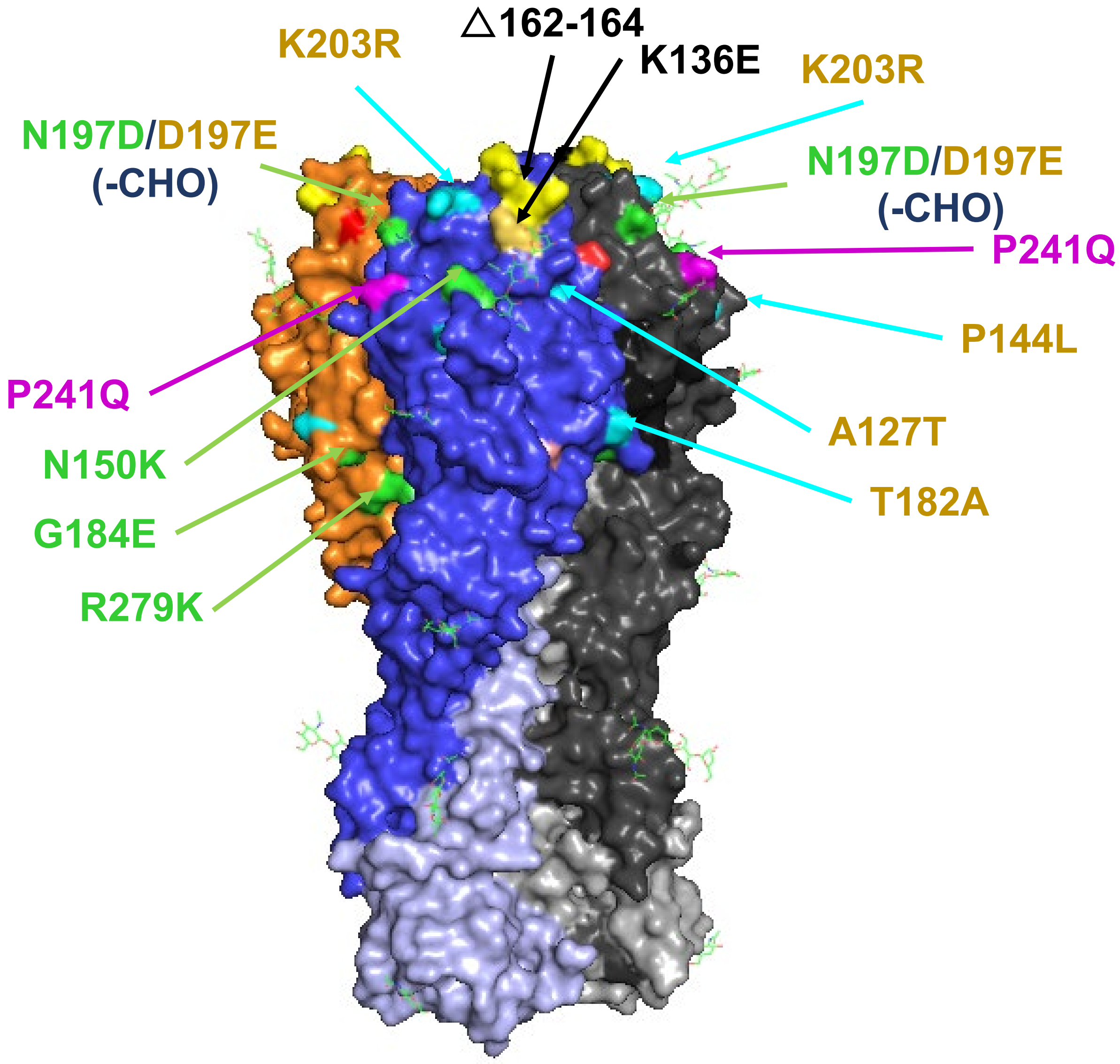


Figure 22. Phylogenetic comparison of B/Victoria NA genes (non-WIC sequences)

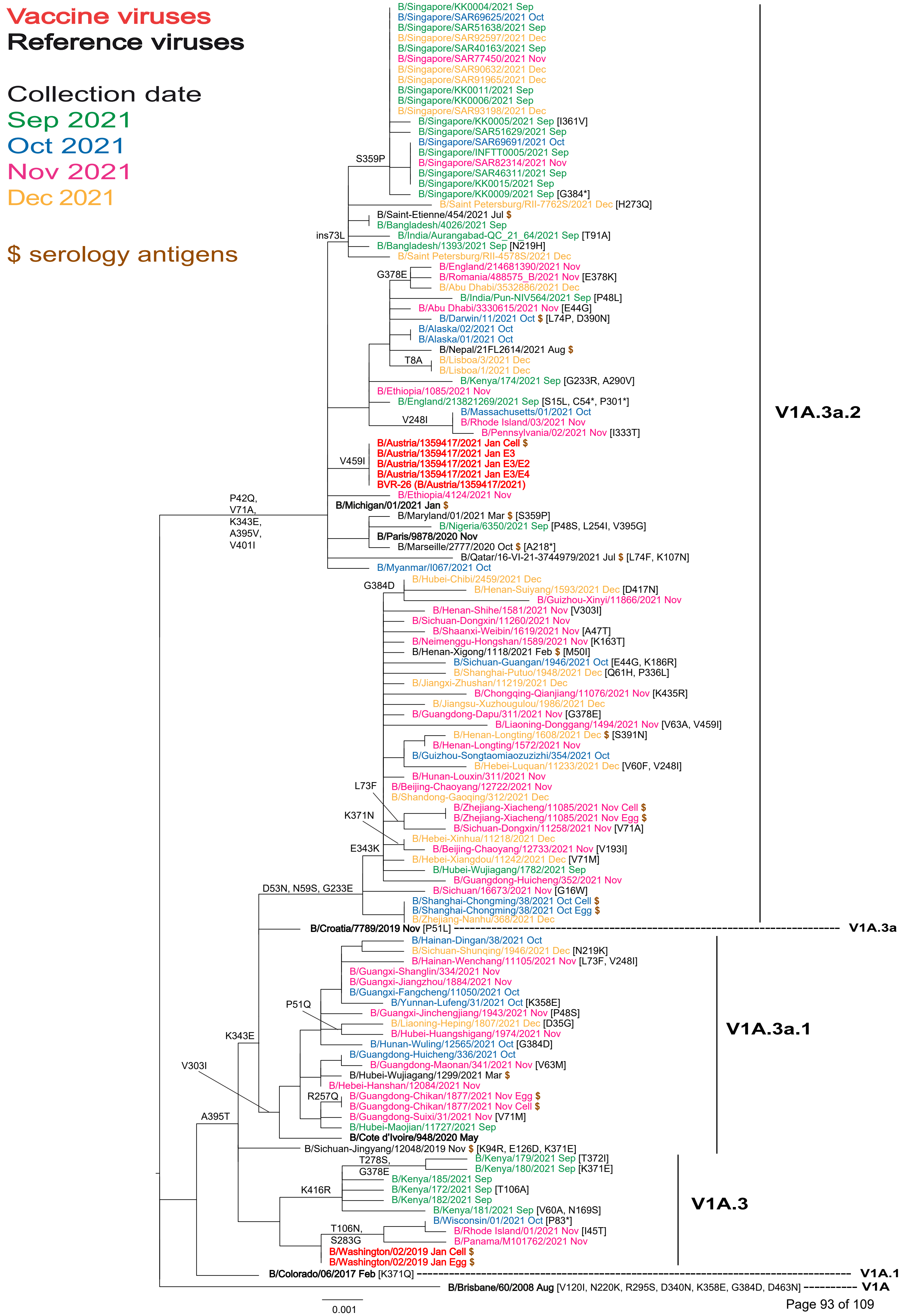


Figure 23. Phylogenetic comparison of B/Victoria NA genes (WIC sequences)

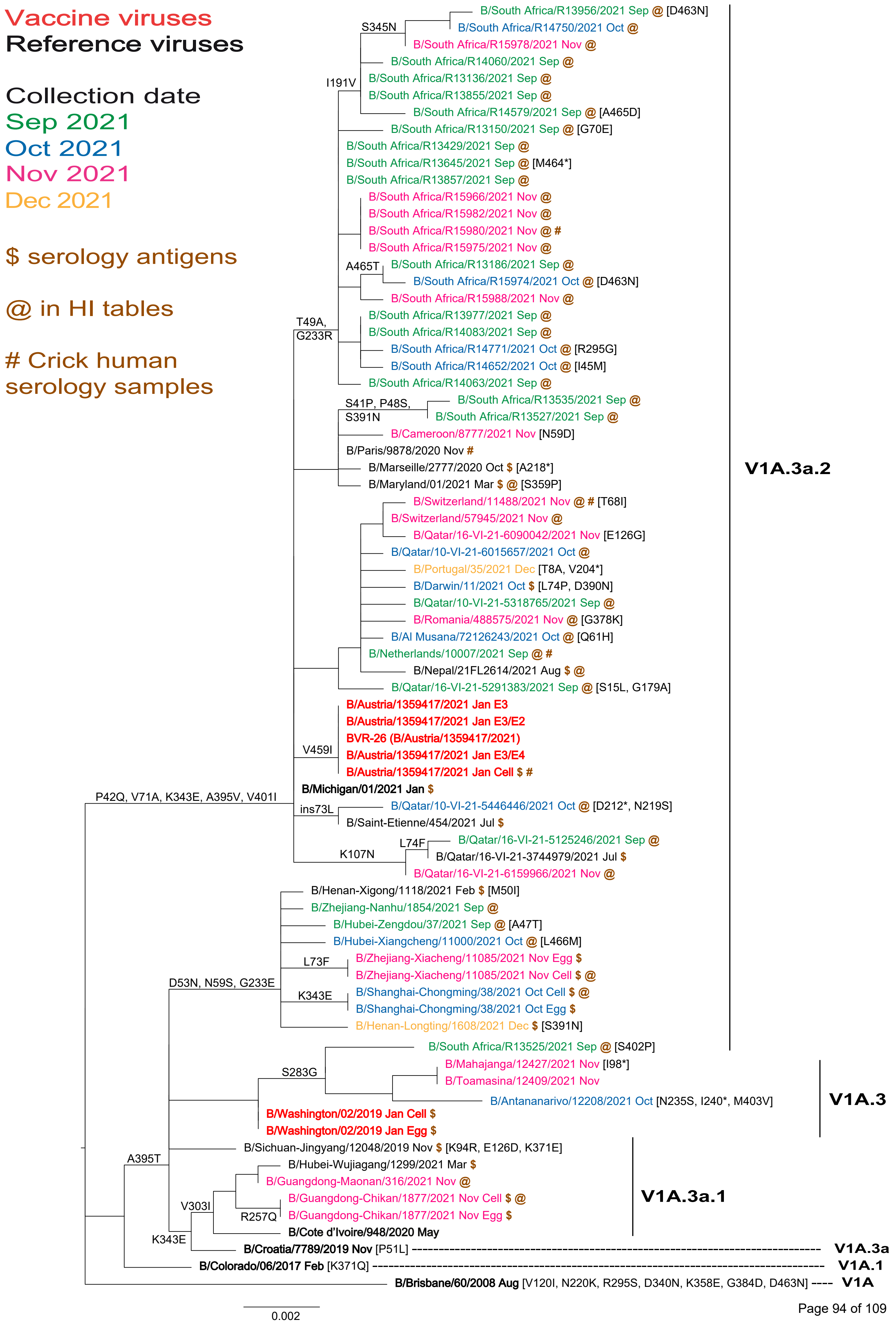


Figure 24. NA protein alignment for B (Victoria-lineage) viruses

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

```
                                                10        20        30        40        50        60        70        80        90       100
                                       ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Brisbane/60/2008                     MLPSTIQTLTLFLTSGGVLLSLYVSASLSYLLYSDILLKFSPTEITAPTMPLDCANASNVQAVNRSATKGVT-LLLPEPEWTYPRLSCPGSTFQKALLIS V1A
B/Kenya/180/2021                       ........................................................................-........................... V1A.3
B/Washington/02/2019E                  ........................................................................-........................... V1A.3
B/Panama/M101762/2021                  ........................................................................-........................... V1A.3
B/Antananarivo/12208/2021              ........................................................................-........................... V1A.3
B/Mahajanga/12427/2021                 ........................................................................-.........................X. V1A.3
B/Toamasina/12409/2021                 ........................................................................-........................... V1A.3
B/Guangdong-Chikan/1877/2021           ........................................................................-........................... V1A.3A.1
B/Liaoning-Heping/1807/2021            ..................................G...............Q.....................-........................... V1A.3A.1
B/South_Africa/R13525/2021             ........................................................................-........................... V1A.3A.2
B/Zhejiang-Xiacheng/11085/2021         ....................................................N.....S.............-F.......................... V1A.3A.2
B/Henan-Longting/1608/2021             ....................................................N.....S.............-........................... V1A.3A.2
B/Shanghai-Chongming/38/2021           ....................................................N.....S.............-........................... V1A.3A.2
B/Cameroon/8777/2021                   .........................................Q................D...........A.-........................... V1A.3A.2
B/South_Africa/R13527/2021             ........................................PQ......S.....................A.-........................... V1A.3A.2
B/Qatar/16-VI-21-6090042/2021          .........................................Q.............................A.-........................... V1A.3A.2
B/Pennsylvania/02/2021                 .........................................Q.............................A.-........................... V1A.3A.2
B/Portugal/35/2021                     .......A.................................Q.............................A.-........................... V1A.3A.2
B/Switzerland/57945/2021               .........................................Q.............................A.-........................... V1A.3A.2
B/Romania/488575/2021                  .........................................Q.............................A.-........................... V1A.3A.2
B/South_Africa/R14771/2021             .........................................Q......A......................A.-........................... V1A.3A.2
B/South_Africa/R15980/2021             .........................................Q......A......................A.-........................... V1A.3A.2
B/South_Africa/R13857/2021             .........................................Q......A......................A.-........................... V1A.3A.2
B/South_Africa/R14750/2021             .........................................Q......A......................A.-........................... V1A.3A.2
B/St Petersburg/RII-7762S/2021         .........................................Q.............................A.L........................... V1A.3A.2
B/Austria/1359417/2021                 .........................................Q.............................A.-........................... V1A.3A.2
BVR-26(B/Austria/1359417/2021)         .........................................Q.............................A.-........................... V1A.3A.2
B/Qatar/16-VI-21-5125246/2021          .........................................Q.............................A.-.F......................... V1A.3A.2
B/Qatar/10-VI-21-5446446/2021          .........................................Q.............................A.L........................... V1A.3A.2
B/Bangladesh/4026/2021                 .........................................Q.............................A.L........................... V1A.3A.2
B/Singapore/SAR93198/2021              .........................................Q.............................A.L........................... V1A.3A.2
B/Singapore/SAR90632/2021 Dec          .........................................Q.............................A.L........................... V1A.3A.2

                                               110       120       130       140       150       160       170       180       190       200
                                       ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Brisbane/60/2008                     PHRFGETKGNSAPLIIREPFIACGPNECKHFALTHYAAQPGGYYNGTRGDRNKLRHLISVKLGKIPTVENSIFHMAAWSGSACHDGKEWTYIGVDGPDNN
B/Kenya/180/2021                       ....................V...............................................................................
B/Washington/02/2019E                  ....................V...............................................................................
B/Panama/M101762/2021                  ......N.............V...............................................................................
B/Antananarivo/12208/2021              ....................V...............................................................................
B/Mahajanga/12427/2021                 ....................V...............................................................................
B/Toamasina/12409/2021                 ....................V...............................................................................
B/Guangdong-Chikan/1877/2021           ....................V...............................................................................
B/Liaoning-Heping/1807/2021            ....................V...............................................................................
B/South_Africa/R13525/2021             ....................V...............................................................................
B/Zhejiang-Xiacheng/11085/2021         ....................V...............................................................................
B/Henan-Longting/1608/2021             ....................V...............................................................................
B/Shanghai-Chongming/38/2021           ....................V...............................................................................
B/Cameroon/8777/2021                   ....................V...............................................................................
B/South_Africa/R13527/2021             ....................V...............................................................................
B/Qatar/16-VI-21-6090042/2021          ....................V.....G.........................................................................
B/Pennsylvania/02/2021                 ....................V...............................................................................
B/Portugal/35/2021                     ....................V...............................................................................
B/Switzerland/57945/2021               ....................V...............................................................................
B/Romania/488575/2021                  ....................V...............................................................................
B/South_Africa/R14771/2021             ....................V...............................................................................
B/South_Africa/R15980/2021             ....................V...............................................................................
B/South_Africa/R13857/2021             ....................V...............................................................................
B/South_Africa/R14750/2021             ....................V......................................................................V........
B/St Petersburg/RII-7762S/2021         ....................V...............................................................................
B/Austria/1359417/2021                 ....................V...............................................................................
BVR-26(B/Austria/1359417/2021)         ....................V...............................................................................
B/Qatar/16-VI-21-5125246/2021          .......N............V...............................................................................
B/Qatar/10-VI-21-5446446/2021          ....................V...............................................................................
B/Bangladesh/4026/2021                 ....................V...............................................................................
B/Singapore/SAR93198/2021              ....................V...............................................................................
B/Singapore/SAR90632/2021 Dec          ....................V...............................................................................

                                               210       220       230       240       250       260       270       280       290       300
                                       ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Brisbane/60/2008                     ALLKVKYGEAYTDTYHSYANKILRTQESACNCIGGNCYLMITDGSASGVSECRFLKIREGRIIKEIFPTGRVKHTEECTCGFASNKTIECACRDNSYTAK
B/Kenya/180/2021                       ..............................................................................S................R....
B/Washington/02/2019E                  ....................N...........................................................................R....
B/Panama/M101762/2021                  ....................N..............................................................G............R....
B/Antananarivo/12208/2021              ....................N..............S....X..........................................G............R....
B/Mahajanga/12427/2021                 ....................N..............................................................G............R....
B/Toamasina/12409/2021                 ....................N..............................................................G............R....
B/Guangdong-Chikan/1877/2021           ....................N....................................Q......................................R....
B/Liaoning-Heping/1807/2021            ....................N...........................................................................R....
B/South_Africa/R13525/2021             ....................N..............................................................G............R....
B/Zhejiang-Xiacheng/11085/2021         ....................N............E..............................................................R....
B/Henan-Longting/1608/2021             ....................N............E..............................................................R....
B/Shanghai-Chongming/38/2021           ....................N............E..............................................................R....
B/Cameroon/8777/2021                   ....................N...........................................................................R....
B/South_Africa/R13527/2021             ....................N...........................................................................R....
B/Qatar/16-VI-21-6090042/2021          ....................N...........................................................................R....
B/Pennsylvania/02/2021                 ....................N...........................I...............................................R....
B/Portugal/35/2021                     ....X...............N...........................................................................R....
B/Switzerland/57945/2021               ....................N...........................................................................R....
B/Romania/488575/2021                  ....................N...........................................................................R....
B/South_Africa/R14771/2021             ....................N............R..............................................................G....
B/South_Africa/R15980/2021             ....................N............R..............................................................R....
B/South_Africa/R13857/2021             ....................N............R..............................................................R....
B/South_Africa/R14750/2021             ....................N............R..............................................................R....
B/St Petersburg/RII-7762S/2021         ....................N....................................................Q...........................R....
B/Austria/1359417/2021                 ....................N...........................................................................R....
BVR-26(B/Austria/1359417/2021)         ....................N...........................................................................R....
B/Qatar/16-VI-21-5125246/2021          ....................N...........................................................................R....
B/Qatar/10-VI-21-5446446/2021          ............X......SN...........................................................................R....
B/Bangladesh/4026/2021                 ....................N...........................................................................R....
B/Singapore/SAR93198/2021              ....................N...........................................................................R....
B/Singapore/SAR90632/2021 Dec          ....................N...........................................................................R....
```

```
                                         310       320       330       340       350       360       370       380       390       400
                                ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Brisbane/60/2008              RPFVKLNVETDTAEIRLMCTDTYLDTPRPNDGSITGPCESNGDKGSGGIKGGFVHQRMESKIGRWYSRTMSKTERMGMGLYVKYDGDPWADSDALAFSGV V1A
B/Kenya/180/2021                ........................................D.................K............E......E.....G..........T.... V1A.3
B/Washington/02/2019E           ........................................D.................K.........................G..........T.... V1A.3
B/Panama/M101762/2021           ........................................D.................K.........................G..........T.... V1A.3
B/Antananarivo/12208/2021       ........................................D.................K.........................G..........T.... V1A.3
B/Mahajanga/12427/2021          ........................................D.................K.........................G..........T.... V1A.3
B/Toamasina/12409/2021          ........................................D.................K.........................G..........T.... V1A.3
B/Guangdong-Chikan/1877/2021    ...I....................................D..E..............K.........................G..........T.... V1A.3A.1
B/Liaoning-Heping/1807/2021     ...I....................................D..E..............K.........................G..........T.... V1A.3A.1
B/South_Africa/R13525/2021      ........................................D.................K.........................G..........T.... V1A.3A.2
B/Zhejiang-Xiacheng/11085/2021  ........................................D.................K.........................G..........T.... V1A.3A.2
B/Henan-Longting/1608/2021      ........................................D.................K.........................G......N...T.... V1A.3A.2
B/Shanghai-Chongming/38/2021    ........................................D..E..............K.........................G..........T.... V1A.3A.2
B/Cameroon/8777/2021            ........................................D..E..............K.........................G..........V.... V1A.3A.2
B/South_Africa/R13527/2021      ........................................D..E..............K.........................G......N...V.... V1A.3A.2
B/Qatar/16-VI-21-6090042/2021   ........................................D..E..............K.........................G..........V.... V1A.3A.2
B/Pennsylvania/02/2021          .................................T......D..E..............K.........................G..........V.... V1A.3A.2
B/Portugal/35/2021              ........................................D..E..............K.........................G..........V.... V1A.3A.2
B/Switzerland/57945/2021        ........................................D..E..............K.........................G..........V.... V1A.3A.2
B/Romania/488575/2021           ........................................D..E..............K...................K....G..........V.... V1A.3A.2
B/South_Africa/R14771/2021      ........................................D..E..............K.........................G..........V.... V1A.3A.2
B/South_Africa/R15980/2021      ........................................D..E..............K.........................G..........V.... V1A.3A.2
B/South_Africa/R13857/2021      ........................................D..E..............K.........................G..........V.... V1A.3A.2
B/South_Africa/R14750/2021      ........................................D..E.N............K.........................G..........V.... V1A.3A.2
B/St Petersburg/RII-7762S/2021  ........................................D..E..............K.........................G..........V.... V1A.3A.2
B/Austria/1359417/2021          ........................................D..E..............K.........................G..........V.... V1A.3A.2
BVR-26(B/Austria/1359417/2021)  ........................................D..E..............K.........................G..........V.... V1A.3A.2
B/Qatar/16-VI-21-5125246/2021   ........................................D..E..............K.........................G..........V.... V1A.3A.2
B/Qatar/10-VI-21-5446446/2021   ........................................D..E..............K.........................G..........V.... V1A.3A.2
B/Bangladesh/4026/2021          ........................................D..E..............K.........................G..........V.... V1A.3A.2
B/Singapore/SAR93198/2021       ........................................D..E..............KP........................G..........V.... V1A.3A.2
B/Singapore/SAR90632/2021 Dec   ........................................D..E..............KP........................G..........V.... V1A.3A.2

                                         410       420       430       440       450       460
                                ....|....|....|....|....|....|....|....|....|....|....|....|..
B/Brisbane/60/2008              MVSMKEPGWYSFGFEIKDKKCDVPCIGIEMVHDGGKETWHSAATAIYCLMGSGQLLWDTVTGVNMAL
B/Kenya/180/2021                ................R..............................................D...
B/Washington/02/2019E           ...............................................................D...
B/Panama/M101762/2021           ...............................................................D...
B/Antananarivo/12208/2021       ...V...........................................................D...
B/Mahajanga/12427/2021          ...............................................................D...
B/Toamasina/12409/2021          ...............................................................D...
B/Guangdong-Chikan/1877/2021    ...............................................................D...
B/Liaoning-Heping/1807/2021     ...............................................................D...
B/South_Africa/R13525/2021      ..P............................................................D...
B/Zhejiang-Xiacheng/11085/2021  ...............................................................D...
B/Henan-Longting/1608/2021      ...............................................................D...
B/Shanghai-Chongming/38/2021    ...............................................................D...
B/Cameroon/8777/2021            .I.............................................................D...
B/South_Africa/R13527/2021      .I.............................................................D...
B/Qatar/16-VI-21-6090042/2021   .I..................X..........................................D...
B/Pennsylvania/02/2021          .I.............................................................D...
B/Portugal/35/2021              .I.............................................................D...
B/Switzerland/57945/2021        .I.............................................................D...
B/Romania/488575/2021           .I.............................................................D...
B/South_Africa/R14771/2021      .I.............................................................D...
B/South_Africa/R15980/2021      .I.............................................................D...
B/South_Africa/R13857/2021      .I.............................................................D...
B/South_Africa/R14750/2021      .I.............................................................D...
B/St Petersburg/RII-7762S/2021  .I.............................................................D...
B/Austria/1359417/2021          .I.........................................................I...D...
BVR-26(B/Austria/1359417/2021)  .I.........................................................I...D...
B/Qatar/16-VI-21-5125246/2021   .I.............................................................D...
B/Qatar/10-VI-21-5446446/2021   .I.............................................................D...
B/Bangladesh/4026/2021          .I.............................................................D...
B/Singapore/SAR93198/2021       .I.............................................................D...
B/Singapore/SAR90632/2021 Dec   .I.............................................................D...
```

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

Table 12-1. Antigenic analyses of influenza B viruses (Victoria lineage) 2021-10-01

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre – Post-infection ferret antisera: B/Bris 60/08	B/Colorado 06/17	B/Wash'ton 02/19	B/Croatia 7889/19	B/CIV 948/20	B/Austria 1359417/21	B/Austria 1359417/21	B/Paris 9878/20	B/Mich 01/21
	Passage history			Egg	Egg	Egg	MDCK	MDCK	MDCK	Egg	MDCK	Cell
	Ferret number			Sh 539, 540, 543, 544, 570, 571, 574[*1,3]	F11/18[*4]	F20/20[*2]	F19/21[*1]	F08/21[*1]	NIB F01/21[*1]	F15/21[*1]	F12/21[*1]	CDC Ferret 2021-102[*1]
	Genetic group			V1A	V1A.1	V1A.3	V1A.3a	V1A.3a.1	V1A.3a.2	V1A.3a.2	V1A.3a.2	V1A.3a.2
REFERENCE VIRUSES												
B/Brisbane/60/2008	V1A	2008-08-04	E4/E4	2560	40	40	160	<	<	<	<	<
B/Colorado/06/2017	V1A.1	2017-02-05	E5/E2	1280	320	80	160	<	<	<	<	<
B/Washington/02/2019	V1A.3	2019-01-19	E3/E2	1280	80	320	160	40	<	<	<	<
B/Croatia/7789/2019	V1A.3a	2019-11-11	MDCKx/MDCK2	1280	160	ND	640	320	160	80	<	80
B/Cote d'Ivoire/948/2020	V1A.3a.1	2020-05-28	MDCK2	320	<	ND	640	640	160	80	<	80
B/Austria/1359417/2021	V1A.3a.2	2021-01-09	SIAT1/MDCK4	640	<	<	320	160	2560	1280	320	640
B/Austria/1359417/2021 Isolate 2	V1A.3a.2	2021-01-09	E3/E2	320	20	10	320	160	2560	1280	320	640
BVR-26 (B/Austria/1359417/2021)	V1A.3a.2	2021-01-09	E3/D9/E1	160	<	<	160	160	1280	640	80	320
B/Paris/9878/2020	V1A.3a.2	2020-11-20	MDCK2	640	80	10	320	160	1280	640	320	640
B/Michigan/01/2021	V1A.3a.2	2021-01-09	C2/MDCK1	640	<	20	640	160	2560	1280	320	640
TEST VIRUSES												
B/Michigan/01/2021	V1A.3a.2	2021-01-09	E3/E1	640	20	20	640	160	2560	640	320	640
B/Maryland/01/2021	V1A.3a.2	2021-03-05	C1/MDCK1	640	<	20	320	160	1280	1280	320	640
B/New Hampshire/01/2021	V1A.3a.2	2021-03-11	C2/MDCK1	640	<	20	640	160	2560	640	320	640
B/New Hampshire/01/2021	V1A.3a.2	2021-03-11	E2/E1	320	20	10	320	320	2560	1280	320	320
B/Qatar/34-VI-21-1991557/2021	V1A.3a.2	2021-04-25	MDCK2	640	<	20	640	160	2560	1280	320	640
B/Qatar/10-VI-21-2014283/2021	V1A.3a.2	2021-04-26	MDCK3	640	20	20	320	160	1280	1280	320	640
B/Qatar/13-VI-21-2113853/2021	V1A.3a.2	2021-05-02	MDCK2	640	20	40	640	160	2560	1280	320	640
B/Qatar/16-VI-21-2216251/2021	V1A.3a.2	2021-05-08	MDCK2	640	<	40	320	160	1280	1280	320	640
B/Qatar/10-VI-21-2307943/2021	V1A.3a.2	2021-05-14	MDCK2	320	<	10	320	80	1280	640	160	320
B/Qatar/47-VI-21-2376814/2021	V1A.3a.2	2021-05-18	MDCK2	640	<	20	640	320	2560	640	320	640
B/Singapore/WUH4618/2021	V1A.3a.2	2021-06-18	E2/E2	320	<	<	320	160	2560	640	160	320
B/Qatar/10-VI-21-3001000/2021	V1A.3a.2	2021-06-23	MDCK2	640	<	10	320	160	1280	640	160	320
B/Qatar/16-VI-21-3268554/2021	V1A.3a.2	2021-07-06	MDCK2	640	<	10	320	160	2560	640	320	640
B/Qatar/16-VI-21-3506748/2021	V1A.3a.2	2021-07-18	MDCK1	640	<	20	320	160	2560	1280	320	640
B/Qatar/15-VI-21-3927143/2021	V1A.3a.2	2021-08-09	MDCK2	640	<	20	320	160	2560	1280	320	640
					Vaccine SH 2019 NH 2019-20	Vaccine SH 2020 NH 2020-21 SH 2021 NH 2021-22				Vaccine SH 2022		

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):
[1] < = <40; [2] < = <10; [3] hyperimmune sheep serum; [4] < = <20; ND = Not Done

Sequences in phylogenetic trees

< 4-fold | 4-fold | 8-fold | > 8-fold | < not recognised by the antiserum | ≥ 160 (when homologous titre ≥ 2560) | ≥ 160 (no homologous titre)

Table 12-2. Antigenic analyses of influenza B viruses (Victoria lineage) 2021-11-11

Viruses	Other information		Collection date	Passage history	B/Bris 60/08	B/Colorado 06/17	B/Wash'ton 02/19	B/Croatia 7889/19	B/CIV 948/20	B/Austria 1359417/21	B/Austria 1359417/21	B/Austria 1359417/21 (NEW)	B/Paris 9878/20
		Passage history			Egg	Egg	Egg	MDCK	MDCK	MDCK	Egg G141	Egg G141R	MDCK
		Ferret number			Sh 539, 540, 543, 544, 570, 571, 574[*1,3]	F11/18[*4]	F20/20[*2]	F19/21[*1]	F08/21[*1]	NIB F01/21[*1]	F15/21[*1]	F44/21[*1]	F12/21[*1]
		Genetic group			V1A	V1A.1	V1A.3	V1A.3a	V1A.3a.1	V1A.3a.2	V1A.3a.2	V1A.3a.2	V1A.3a.2
REFERENCE VIRUSES													
B/Brisbane/60/2008		V1A	2008-08-04	E4/E4	2560	40	40	160	<	<	<	<	<
B/Colorado/06/2017		V1A.1	2017-02-05	E5/E2	1280	320	80	160	<	<	<	<	<
B/Washington/02/2019		V1A.3	2019-01-19	E3/E2	1280	80	320	160	40	<	<	<	<
B/Croatia/7789/2019		V1A.3a	2019-11-11	MDCKx/MDCK2	1280	160	ND	640	320	160	80	40	<
B/Cote d'Ivoire/948/2020		V1A.3a.1	2020-05-28	MDCK2	320	<	ND	640	640	160	80	80	<
B/Austria/1359417/2021		V1A.3a.2	2021-01-09	SIAT1/MDCK4	640	<	<	320	160	2560	1280	320	320
B/Austria/1359417/2021 Isolate 2	G141	V1A.3a.2	2021-01-09	E3	320	20	10	320	160	2560	1280	1280	320
B/Austria/1359417/2021 Isolate 2	G141X	V1A.3a.2	2021-01-09	E3/E2	ND	ND	ND	ND	ND	ND	2560	5120	ND
B/Austria/1359417/2021 Isolate 2	G141R	V1A.3a.2	2021-01-09	E3/E4	ND	ND	ND	ND	ND	ND	1280	5120	ND
B/Paris/9878/2020		V1A.3a.2	2020-11-20	MDCK2	640	80	10	320	160	1280	640	320	320
TEST VIRUSES													
B/Netherlands/10007/2021		V1A.3a.2	2021-09-09	MDCK-MIX2	640	40	10	160	80	640	1280	320	320

Haemagglutination inhibition titre: Post-infection ferret antisera

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):
[1] < = <40; [2] < = <10; [3] hyperimmune sheep serum; [4] < = <20; ND = Not Done

B/Colorado 06/17: Vaccine SH 2019 NH 2019-20
B/Wash'ton 02/19: Vaccine SH 2020 NH 2020-21 SH 2021 NH 2021-22
B/Austria 1359417/21 Egg G141 / Egg G141R: Vaccine SH 2022

Sequences in phylogenetic trees

< 4-fold | 4-fold | 8-fold | > 8-fold | < not recognised by the antiserum | ≥ 160 (when homologous titre ≥ 2560) | ≥ 160 (no homologous titre)

Table 12-3. Antigenic analyses of influenza B viruses (Victoria lineage) 2022-01-19

Viruses	Other information	Genetic group	Collection date	Passage history	B/Bris 60/08 Egg	B/Colorado 06/17 Egg	B/Wash'ton 02/19 Egg	B/Croatia 7889/19 MDCK	B/CIV 948/20 MDCK	B/Austria 1359417/21 MDCK	B/Austria 1359417/21 Egg G141	B/Austria 1359417/21 Egg G141R	B/Paris 9878/20 MDCK
		Ferret number			Sh 539, 540, 543, 544, 570, 571, 574*1,3	F11/18*4	F20/20*2	F19/21*1	F08/21*2	NIB F01/21*1	F15/21*1	F44/21*1	F12/21*1
		Genetic group			V1A	V1A.1	V1A.3	V1A.3a	V1A.3a.1	V1A.3a.2	V1A.3a.2	V1A.3a.2	V1A.3a.2
REFERENCE VIRUSES													
B/Brisbane/60/2008		V1A	2008-08-04	E4/E4	2560	40	40	80	<	<	<	<	<
B/Colorado/06/2017		V1A.1	2017-02-05	E5/E2	2560	160	80	80	<	<	<	<	<
B/Washington/02/2019		V1A.3	2019-01-19	E3/E2	1280	80	160	80	<	<	40	<	<
B/Croatia/7789/2019		V1A.3a	2019-11-11	MDCKx/MDCK3	640	80	80	640	160	160	160	80	40
B/Cote d'Ivoire/948/2020		V1A.3a.1	2020-05-28	MDCK2	640	<	80	640	640	80	160	160	80
B/Austria/1359417/2021		V1A.3a.2	2021-01-09	SIAT1/MDCK4	640	<	<	320	<	1280	2560	640	640
B/Austria/1359417/2021 Isolate 2	G141	V1A.3a.2	2021-01-09	E3	ND	ND	ND	ND	ND	1280	1280	640	640
B/Austria/1359417/2021 Isolate 2	G141R	V1A.3a.2	2021-01-09	E3/E4	320	10	<	320	160	1280	2560	5120	640
B/Paris/9878/2020		V1A.3a.2	2020-11-20	MDCK2	1280	40	<	320	<	1280	1280	640	640
TEST VIRUSES													
B/South Africa/VI00241/2021		V1A.3a.2		MDCK1/MDCK1	640	<	<	640	160	2560	1280	640	640
B/South Africa/R11347/2021		V1A.3a.2	2021-07-23	MDCK1/MDCK1	640	<	<	320	<	1280	1280	320	640
B/South Africa/R11951/2021		V1A.3a.2	2021-08-11	MDCK1/MDCK1	640	<	<	320	<	2560	1280	640	640
B/South Africa/R12061/2021		V1A.3a.2	2021-08-12	MDCK1/MDCK1	640	<	<	320	<	1280	1280	640	640
B/South Africa/R12283/2021		V1A.3a.2	2021-08-13	MDCK1/MDCK1	640	<	<	320	<	1280	1280	640	640
B/South Africa/R12597/2021		V1A.3a.2	2021-08-23	MDCK1/MDCK1	640	<	<	160	<	1280	1280	320	320
B/South Africa/R12796/2021		V1A.3a.2	2021-08-24	MDCK1/MDCK1	640	<	<	320	<	1280	1280	640	320
B/South Africa/R13233/2021		V1A.3a.2	2021-08-31	MDCK1/MDCK1	640	<	<	320	<	1280	1280	320	320
B/South Africa/R12113/2021		V1A.3a.2	2021-08-11	MDCK1/MDCK1	640	40	<	160	<	640	1280	320	640
B/South Africa/R13171/2021		V1A.3a.2	2021-08-31	MDCK1/MDCK1	1280	40	<	160	<	1280	2560	640	640
B/South Africa/R13150/2021		V1A.3a.2	2021-09-02	MDCK1/MDCK1	320	<	<	320	<	1280	1280	640	320
B/South Africa/R13136/2021		V1A.3a.2	2021-09-02	MDCK1/MDCK1	320	<	<	320	<	1280	1280	320	320
B/South Africa/R13186/2021		V1A.3a.2	2021-09-03	MDCK1/MDCK1	320	<	<	160	<	640	1280	320	320
B/South Africa/R13525/2021		V1A.3a.2	2021-09-07	MDCK1/MDCK1	640	<	<	320	<	1280	1280	640	320
B/South Africa/R13857/2021		V1A.3a.2	2021-09-14	MDCK1/MDCK1	640	<	<	320	<	1280	1280	320	320
B/South Africa/R14063/2021		V1A.3a.2	2021-09-16	MDCK1/MDCK1	320	<	<	320	<	1280	1280	320	320
B/South Africa/R14060/2021		V1A.3a.2	2021-09-16	MDCK1/MDCK1	320	<	<	320	<	1280	1280	320	320
B/South Africa/R13855/2021		V1A.3a.2	2021-09-16	MDCK1/MDCK1	640	<	<	320	<	1280	1280	640	320
B/South Africa/R14771/2021		V1A.3a.2	2021-10-07	MDCK1/MDCK1	320	<	<	320	<	1280	1280	640	320
B/South Africa/R15982/2021		V1A.3a.2	2021-11-02	MDCK1/MDCK1	320	<	<	320	<	1280	1280	320	320
B/South Africa/R15980/2021		V1A.3a.2	2021-11-02	MDCK1/MDCK1	640	<	<	320	<	1280	1280	640	320
B/South Africa/R15975/2021		V1A.3a.2	2021-11-02	MDCK1/MDCK1	320	<	<	320	<	1280	1280	640	320
B/South Africa/R15966/2021		V1A.3a.2	2021-11-02	MDCK2/MDCK1	320	<	<	320	<	1280	1280	320	320
B/South Africa/R14750/2021		V1A.3a.2	2021-10-07	MDCK1/MDCK1	640	20	<	320	<	1280	1280	320	320
B/South Africa/R15974/2021		V1A.3a.2	2021-10-29	MDCK1/MDCK1	640	20	<	320	<	1280	1280	320	640
B/South Africa/R15988/2021		V1A.3a.2	2021-11-02	MDCK1/MDCK1	640	20	<	320	<	1280	1280	640	640
B/South Africa/R15978/2021		V1A.3a.2	2021-11-03	MDCK1/MDCK1	640	20	<	320	<	1280	1280	640	640
B/Switzerland/57945/2021		V1A.3a.2	2021-11-24	MDCK1/MDCK1	640	40	<	160	<	1280	1280	640	640
B/Switzerland/11488/2021		V1A.3a.2	2021-11-29	SIAT1/MDCK1	640	40	<	320	<	1280	2560	640	640
B/South Africa/R13527/2021		V1A.3a.2	2021-09-07	MDCK1/MDCK1	640	40	<	320	40	1280	2560	640	640
B/South Africa/R13429/2021		V1A.3a.2	2021-09-07	MDCK1/MDCK1	640	<	<	320	40	1280	1280	640	320
B/South Africa/R13535/2021		V1A.3a.2	2021-09-08	MDCK1/MDCK1	1280	40	<	320	40	1280	2560	640	640
B/South Africa/R13645/2021		V1A.3a.2	2021-09-10	MDCK1/MDCK1	640	<	<	320	40	2560	1280	640	320
B/South Africa/R13956/2021		V1A.3a.2	2021-09-13	MDCK1/MDCK1	640	<	<	320	40	2560	2560	640	320
B/South Africa/R13977/2021		V1A.3a.2	2021-09-16	MDCK1/MDCK1	640	<	<	320	40	1280	1280	640	320
B/South Africa/R14083/2021		V1A.3a.2	2021-09-20	MDCK1/MDCK1	640	<	<	640	160	2560	1280	640	640
B/South Africa/R14579/2021		V1A.3a.2	2021-09-28	MDCK1/MDCK1	640	20	<	320	160	2560	2560	640	640
B/South Africa/R14652/2021		V1A.3a.2	2021-10-06	MDCK1/MDCK1	640	<	<	320	160	2560	1280	640	640
						Vaccine SH 2019 NH 2019-20	Vaccine SH 2020 NH 2020-21 SH 2021 NH 2021-22				Vaccine SH 2022		

Haemagglutination inhibition titre — Post-infection ferret antisera

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):
[1] < = <40; [2] < = <10; [3] hyperimmune sheep serum; [4] < = <20; ND = Not Done

Sequences in phylogenetic trees

Titre >5120 taken as 5120 for heat-mapping

< 4-fold | 4-fold | 8-fold | > 8-fold | < not recognised by the antiserum | ≥ 160 (when homologous titre ≥ 2560) | ≥ 160 (no homologous titre)

Table 12-4. Antigenic analyses of influenza B viruses (Victoria lineage) 2022-02-02

Viruses	Other information		Collection date	Passage history	Haemagglutination inhibition titre – Post-infection ferret antisera								
					B/Bris 60/08	B/Colorado 06/17	B/Wash'ton 02/19	B/Croatia 7889/19	B/CIV 948/20	B/Austria 1359417/21	B/Austria 1359417/21	B/Austria 1359417/21	B/Paris 9878/20
		Passage history			Egg	Egg	Egg	MDCK	MDCK	MDCK	Egg G141	Egg G141R	MDCK
		Ferret number			Sh 539, 540, 543, 544, 570, 571, 574*[1,3]	F11/18*[4]	F20/20*[2]	F19/21*[1]	F08/21*[5]	NIB F01/21*[1]	F15/21*[1]	F44/21*[1]	F12/21*[1]
		Genetic group			V1A	V1A.1	V1A.3	V1A.3a	V1A.3a.1	V1A.3a.2	V1A.3a.2	V1A.3a.2	V1A.3a.2
REFERENCE VIRUSES													
B/Brisbane/60/2008		V1A	2008-08-04	E4/E4	1280	80	40	80	<	<	<	<	<
B/Colorado/06/2017		V1A.1	2017-02-05	E5/E2	1280	640	80	80	<	<	<	<	<
B/Washington/02/2019		V1A.3	2019-01-19	E3/E2	1280	320	320	160	<	<	<	<	<
B/Croatia/7789/2019		V1A.3a	2019-11-11	MDCKx/MDCK3	640	160	160	640	160	160	160	40	<
B/Cote d'Ivoire/948/2020		V1A.3a.1	2020-05-28	MDCK2	320	40	80	640	640	160	160	160	<
B/Austria/1359417/2021		V1A.3a.2	2021-01-09	SIAT1/MDCK4	640	80	<	320	<	1280	1280	640	640
B/Austria/1359417/2021 Isolate 2	G141	V1A.3a.2	2021-01-09	E3	ND	ND	ND	320	160	2560	1280	640	320
B/Austria/1359417/2021 Isolate 2	G141R	V1A.3a.2	2021-01-09	E3/E4	320	40	<	320	160	1280	1280	5120	640
B/Paris/9878/2020		V1A.3a.2	2020-11-20	MDCK2	640	160	<	320	<	1280	1280	640	640
TEST VIRUSES													
B/Maryland/01/2021		V1A.3a.2	2021-03-05	C2/MDCK1	1280	320	<	320	320	1280	2560	640	1280
B/New Hampshire/01/2021		V1A.3a.2	2021-03-11	C2/MDCK1	640	80	<	320	<	1280	1280	320	320
B/Guangdong-Zhenjiang/1516/2021		V1A.3a.2	2021-06-28	C1/C1/MDCK1	1280	160	<	640	320	2560	2560	640	640
B/Zhejiang-Nanhu/1854/2021		V1A.3a.2	2021-09-06	C1/C1/MDCK1	640	80	<	640	160	2560	2560	320	640
B/Hubei-Zengdou/37/2021		V1A.3a.2	2021-09-14	C1/C1/MDCK1	640	40	<	320	<	1280	640	160	320
B/Hubei-Xiangcheng/11000/2021		V1A.3a.2	2021-10-20	C1/C1/MDCK1	640	80	<	320	<	1280	1280	320	320
B/Shanghai-Chongming/38/2021		V1A.3a.2	2021-10-26	C1/C1/MDCK1	640	80	<	320	160	2560	1280	320	640
B/Zhejiang-Xiacheng/11085/2021		V1A.3a.2	2021-11-01	C1/C1/MDCK1	640	80	<	320	160	1280	1280	320	320
B/Guangdong-Maonan/316/2021		V1A.3a.1	2021-11-01	C1/C1/MDCK1	320	40	40	640	640	80	80	80	<
B/Guangdong-Chikan/1877/2021		V1A.3a.1	2021-11-01	C1/C1/MDCK1	320	40	20	640	640	80	80	80	<
B/Qatar/47-VI-21-4381753/2021		V1A.3a.2	2021-08-26	MDCK1	1280	320	<	320	160	1280	1280	640	640
B/Qatar/16-VI-21-4499116/2021		V1A.3a.2	2021-08-29	MDCK1	1280	320	<	320	160	1280	2560	640	640
B/Qatar/16-VI-21-6159966/2021		V1A.3a.2	2021-11-04	MDCK3	320	40	<	320	<	1280	1280	320	320
B/Portugal/35/2021	T168X(-CHO)	V1A.3a.2	2021-12-15	MDCK3	640	160	<	80	80	640	640	320	320
B/Qatar/16-VI-21-5125246/2021		V1A.3a.2	2021-09-22	MDCK1	640	80	<	320	160	1280	1280	640	640
B/Qatar/16-VI-21-5291383/2021		V1A.3a.2	2021-09-28	MDCK1	1280	320	<	640	160	1280	2560	640	640
B/Qatar/10-VI-21-5318765/2021		V1A.3a.2	2021-09-29	MDCK1	1280	160	<	320	160	1280	1280	320	640
B/Qatar/10-VI-21-5446446/2021		V1A.3a.2	2021-10-04	MDCK1	1280	160	<	640	160	2560	1280	640	640
B/Al Musana/72126243/2021		V1A.3a.2	2021-10-05	MDCK1	640	320	<	320	160	1280	1280	640	640
B/Qatar/10-VI-21-6015657/2021		V1A.3a.2	2021-10-29	MDCK1	640	160	<	320	160	1280	1280	640	640
B/Romania/488575/2021		V1A.3a.2	2021-11-10	SIAT2/MDCK1	1280	320	<	320	160	1280	2560	320	640
B/Jalan Bani Bu Ali/72129017/2021		no sequence	2021-11-20	MDCK1	640	160	<	320	160	1280	1280	640	640
						Vaccine SH 2019 NH 2019-20	Vaccine SH 2020 NH 2020-21 SH 2021 NH 2021-22				Vaccine SH 2022		

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):
[1] < = <40; [2] < = <10; [3] hyperimmune sheep serum; [4] < = <20; [5] < = <80; ND = Not Done

Sequences in phylogenetic trees

Titre >5120 taken as 5120 for heat-mapping

< 4-fold | 4-fold | 8-fold | > 8-fold | < not recognised by the antiserum | ≥ 160 (when homologous titre ≥ 2560) | ≥ 160 (no homologous titre)

Table 12-5. Antigenic analyses of influenza B viruses (Victoria lineage) 2022-02-09

Viruses	Other information		Collection date	Passage history	Haemagglutination inhibition titre								
					Post-infection ferret antisera								
					B/Bris 60/08	B/Colorado 06/17	B/Wash'ton 02/19	B/Croatia 7889/19	B/CIV 948/20	B/Austria 1359417/21	B/Austria 1359417/21	B/Austria 1359417/21	B/Paris 9878/20
		Passage history			Egg	Egg	Egg	MDCK	MDCK	MDCK	Egg G141	Egg G141R	MDCK
		Ferret number			Sh 539, 540, 543, 544, 570, 571, 574[*1,3]	F11/18[*4]	F20/20[*4]	F19/21[*1]	F08/21[*5]	NIB F01/21[*1]	F15/21[*1]	F44/21[*1]	F12/21[*1]
		Genetic group			V1A	V1A.1	V1A.3	V1A.3a	V1A.3a.1	V1A.3a.2	V1A.3a.2	V1A.3a.2	V1A.3a.2
REFERENCE VIRUSES													
B/Brisbane/60/2008		V1A	2008-08-04	E4/E4	1280	80	40	80	<	<	<	<	<
B/Colorado/06/2017		V1A.1	2017-02-05	E5/E2	1280	640	80	80	<	<	<	<	<
B/Washington/02/2019		V1A.3	2019-01-19	E3/E2	1280	320	160	160	<	<	<	<	<
B/Croatia/7789/2019		V1A.3a	2019-11-11	MDCKx/MDCK3	640	160	160	640	160	160	160	40	<
B/Cote d'Ivoire/948/2020		V1A.3a.1	2020-05-28	MDCK2	320	40	80	640	640	160	160	160	<
B/Austria/1359417/2021		V1A.3a.2	2021-01-09	SIAT1/MDCK4	640	80	<	320	<	1280	1280	640	640
B/Austria/1359417/2021 Isolate 2	G141	V1A.3a.2	2021-01-09	E3	ND	ND	ND	320	160	2560	1280	640	320
B/Austria/1359417/2021 Isolate 2	G141R	V1A.3a.2	2021-01-09	E3/E4	320	40	<	320	160	1280	1280	5120	640
B/Paris/9878/2020		V1A.3a.2	2020-11-20	MDCK2	640	160	<	320	<	1280	1280	640	640
TEST VIRUSES													
B/Nepal/21FL2614/2021		V1A.3a.2	2021-08-10	hCK3/MDCK1	640	320	<	160	160	1280	1280	640	640
B/Antsohihy/12903/2021		V1A.3	2021-11-26	MDCK1	40	10	80	<	<	<	<	<	<
B/Toamasina/12786/2021		V1A.3	2021-11-28	MDCK1	160	20	80	<	<	<	<	<	<
B/Mahajanga/12999/2021		V1A.3	2021-12-06	MDCK1	80	40	80	<	<	<	<	<	<
B/Toliary/13351/2021		V1A.3	2021-12-07	MDCK2	80	20	40	<	<	<	<	<	<
B/Toamasina/13670/2021		V1A.3	2021-12-15	MDCK1	80	20	40	<	<	<	<	<	<
						Vaccine SH 2019 NH 2019-20	Vaccine SH 2020 NH 2020-21 SH 2021 NH 2021-22				Vaccine SH 2022		

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):
[1] < = <40; [2] < = <10; [3] hyperimmune sheep serum; [4] < = <20; [5] < = <80; ND = Not Done

Sequences in phylogenetic trees

Titre >5120 taken as 5120 for heat-mapping

< 4-fold | 4-fold | 8-fold | > 8-fold | < not recognised by the antiserum | ≥ 160 (when homologous titre ≥ 2560) | ≥ 160 (no homologous titre)

Table 12-6. Antigenic analyses of influenza B viruses (Victoria lineage) Summary

Viruses		Haemagglutination inhibition titre								
		Post-infection ferret antisera								
		B/Bris 60/08	B/Colorado 06/17	B/Wash'ton 02/19	B/Croatia 7889/19	B/CIV 948/20	B/Austria 1359417/21	B/Austria 1359417/21	B/Austria 1359417/21	B/Paris 9878/20
	Passage history	Egg	Egg	Egg	MDCK	MDCK	MDCK	Egg G141	Egg G141R	MDCK
	Ferret number	Sh 539, 540, 543, 544, 570, 571, 574[*1,3]	F11/18[*4]	F20/20[*4]	F19/21[*1]	F08/21[*5]	NIB F01/21[*1]	F15/21[*1]	F44/21[*1]	F12/21[*1]
	Genetic group	V1A	V1A.1	V1A.3	V1A.3a	V1A.3a.1	V1A.3a.2	V1A.3a.2	V1A.3a.2	V1A.3a.2
REFERENCE VIRUSES										
B/Brisbane/60/2008	V1A	1280	80	40	80	<	<	<	<	<
B/Colorado/06/2017	V1A.1	1280	640	80	80	<	<	<	<	<
B/Washington/02/2019	V1A.3	1280	320	160	160	<	<	<	<	<
B/Croatia/7789/2019	V1A.3a	640	160	160	640	160	160	160	40	<
B/Cote d'Ivoire/948/2020	V1A.3a.1	320	40	80	640	640	160	160	160	<
B/Austria/1359417/2021	V1A.3a.2	640	80	<	320	<	1280	1280	640	640
B/Austria/1359417/2021 Isolate 2	V1A.3a.2	ND	ND	ND	320	160	2560	1280	640	320
B/Austria/1359417/2021 Isolate 2	V1A.3a.2	320	40	<	320	160	1280	1280	>5120	640
B/Paris/9878/2020	V1A.3a.2	640	160	<	320	<	1280	1280	640	640
TEST VIRUSES										
Total Number tested		44	44	44	44	44	44	44	44	44
No with titre reduction ≤2-fold		10	3	3	35		38	39		39
%		*22.7*	*6.9*	*6.9*	*79.5*		*86.4*	*88.6*		*88.6*
No with titre reduction =4-fold		20	9	2	3	11	1			
%		*45.5*	*20.4*	*4.5*	*6.9*	*25.00*	*2.2*			
No with titre reduction ≥8-fold		14	32	39	6	33	5	5	44	5
%		*31.8*	*72.7*	*88.6*	*13.6*	*75.0*	*11.4*	*11.4*	*100*	*11.4*
Number tested	V1A.3a.2	38	38	38	38	38	38	38	38	38
No with titre reduction ≤2-fold		9	3		34		37	38		38
%		*23.7*	*7.8*		*89.5*		*97.3*	*100*		*100*
No with titre reduction =4-fold		20	8		3	10	1			
%		*52.6*	*21.1*		*7.8*	*26.3*	*2.7*			
No with titre reduction ≥8-fold		9	27	38	1	28			38	
%		*23.7*	*71.1*	*100*	*2.7*	*73.7*			*100*	
Number tested	V1A.3	5	5	5	5	5	5	5	5	5
No with titre reduction ≤2-fold				3						
%				*60.0*						
No with titre reduction =4-fold				2						
%				*40.0*						
No with titre reduction ≥8-fold		5	5		5	5	5	5	5	5
%		*100*	*100*		*100*	*100*	*100*	*100*	*100*	*100*
			Vaccine SH 2019 NH 2019-20	Vaccine SH 2020 NH 2020-21 SH 2021 NH 2021-22				Vaccine SH 2022		

Reference virus results are taken from an individual table as an example. Summaries for each antiserum are based on fold-reductions observed on the days that HI assays were performed.

Figure 25. Phylogenetic comparison of influenza B (Yamagata-lineage) HA genes

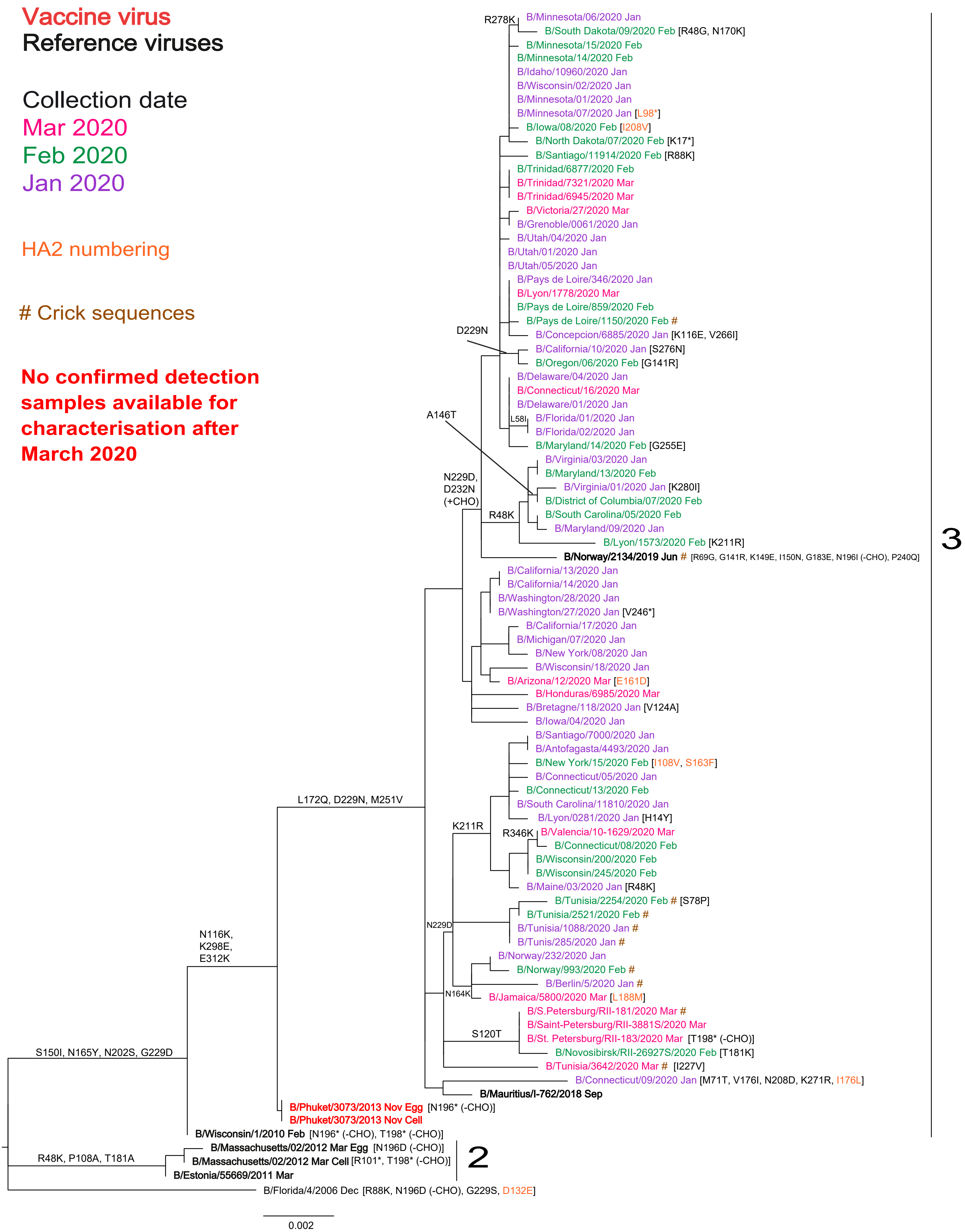


Figure 26. HA protein alignment for B (Yamagata-lineage) viruses

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

```
                                        10        20        30        40        50        60        70        80        90       100
                                ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Phuket/3073/2013 Nov Egg      DRICTGITSSNSPHVVKTATQGEVNVTGVIPLTTTPTKSYFANLKGTRTRGKLCPDCLNCTDLDVALGRPMCVGTTPSAKASILHEVRPVTSGCFPIMHD 3
B/Novosibirsk/RII-26927S/2020   .................................................................................................... 3
B/S.Petersburg/RII-181/2020 Ma  .................................................................................................... 3
B/Jamaica/5800/2020 Mar         .................................................................................................... 3
B/Norway/993/2020 Feb           .................................................................................................... 3
B/Tunisia/1088/2020 Jan         .................................................................................................... 3
B/Tunisia/2254/2020 Feb         .............................................................................P...................... 3
B/Valencia/10-1629/2020 Mar     .................................................................................................... 3
B/Santiago/7000/2020 Jan        .................................................................................................... 3
B/Lyon/0281/2020 Jan            .............Y...................................................................................... 3
B/Connecticut/05/2020 Jan       .................................................................................................... 3
B/Mauritius/I-762/2018 Sep      .................................................................................................... 3
B/Connecticut/09/2020 Jan       ......................................................................T............................. 3
B/Honduras/6985/2020 Mar        .................................................................................................... 3
B/Bretagne/118/2020 Jan         .................................................................................................... 3
B/Arizona/12/2020 Mar           .................................................................................................... 3
B/Lyon/1573/2020 Feb            ...............................................K.................................................... 3
B/Maryland/13/2020 Feb          ...............................................K.................................................... 3
B/Minnesota/14/2020 Feb         .................................................................................................... 3
B/South_Dakota/09/2020 Feb      ...............................................G.................................................... 3
B/Iowa/08/2020 Feb              .................................................................................................... 3
B/Trinidad/7321/2020 Mar        .................................................................................................... 3
B/Maryland/14/2020 Feb          .................................................................................................... 3
B/Florida/02/2020 Jan           .........................................................I.......................................... 3
B/Connecticut/16/2020 Mar       .................................................................................................... 3
B/Santiago/11914/2020 Feb       .......................................................................................K............ 3
B/Oregon/06/2020 Feb            .................................................................................................... 3
B/Victoria/27/2020 Mar          .................................................................................................... 3
B/Concepcion/6885/2020 Jan      .................................................................................................... 3
B/Lyon/1778/2020 Mar            .................................................................................................... 3

                                       110       120       130       140       150       160       170       180       190       200
                                ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Phuket/3073/2013 Nov Egg      RTKIRQLPNLLRGYEKIRLSTQNVIDAEKAPGGPYRLGTSGSCPNATSKIGFFATMAWAVPKDNYKNATNPLTVEVPYICTEGEDQITVWGFHSDXKTQM
B/Novosibirsk/RII-26927S/2020   ...................T...................................................Q........K..............N....
B/S.Petersburg/RII-181/2020 Ma  ...................T...................................................Q.......................N....
B/Jamaica/5800/2020 Mar         ...............................................................K.......Q.......................N....
B/Norway/993/2020 Feb           ...............................................................K.......Q.......................N....
B/Tunisia/1088/2020 Jan         .......................................................................Q.......................N....
B/Tunisia/2254/2020 Feb         .......................................................................Q.......................N....
B/Valencia/10-1629/2020 Mar     .......................................................................Q.......................N....
B/Santiago/7000/2020 Jan        .......................................................................Q.......................N....
B/Lyon/0281/2020 Jan            .......................................................................Q.......................N....
B/Connecticut/05/2020 Jan       .......................................................................Q.......................N....
B/Mauritius/I-762/2018 Sep      .......................................................................Q.......................N....
B/Connecticut/09/2020 Jan       .......................................................................Q...I...................N....
B/Honduras/6985/2020 Mar        .......................................................................Q.......................N....
B/Bretagne/118/2020 Jan         .......................A...............................................Q.......................N....
B/Arizona/12/2020 Mar           .......................................................................Q.......................N....
B/Lyon/1573/2020 Feb            .......................................................................Q.......................N....
B/Maryland/13/2020 Feb          .......................................................................Q.......................N....
B/Minnesota/14/2020 Feb         .......................................................................Q.......................N....
B/South_Dakota/09/2020 Feb      .....................................................................K.Q.......................N....
B/Iowa/08/2020 Feb              .......................................................................Q.......................N....
B/Trinidad/7321/2020 Mar        .......................................................................Q.......................N....
B/Maryland/14/2020 Feb          .......................................................................Q.......................N....
B/Florida/02/2020 Jan           .......................................................................Q.......................N....
B/Connecticut/16/2020 Mar       .......................................................................Q.......................N....
B/Santiago/11914/2020 Feb       .......................................................................Q.......................N....
B/Oregon/06/2020 Feb            ........................................R..............................Q.......................N....
B/Victoria/27/2020 Mar          .......................................................................Q.......................N....
B/Concepcion/6885/2020 Jan      ...............E.......................................................Q.......................N....
B/Lyon/1778/2020 Mar            .......................................................................Q.......................N....

                                       210       220       230       240       250       260       270       280       290       300
                                ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Phuket/3073/2013 Nov Egg      KSLYGDSNPQKFTSSANGVTTHYVSQIGDFPDQTEDGGLPQSGRIVVDYMMQKPGKTGTIVYQRGVLLPQKVWCASGRSKVIKGSLPLIGEADCLHEEYG
B/Novosibirsk/RII-26927S/2020   ............................N.....................V.................................................
B/S.Petersburg/RII-181/2020 Ma  ............................N.....................V.................................................
B/Jamaica/5800/2020 Mar         ..................................................V.................................................
B/Norway/993/2020 Feb           ..................................................V.................................................
B/Tunisia/1088/2020 Jan         ..................................................V.................................................
B/Tunisia/2254/2020 Feb         ..................................................V.................................................
B/Valencia/10-1629/2020 Mar     ..........R.......................................V.................................................
B/Santiago/7000/2020 Jan        ..........R.......................................V.................................................
B/Lyon/0281/2020 Jan            ..........R.......................................V.................................................
B/Connecticut/05/2020 Jan       ..........R.......................................V.................................................
B/Mauritius/I-762/2018 Sep      ............................N.....................V.................................................
B/Connecticut/09/2020 Jan       .......D....................N.....................V...................R.............................
B/Honduras/6985/2020 Mar        ............................N.....................V.................................................
B/Bretagne/118/2020 Jan         ............................N.....................V.................................................
B/Arizona/12/2020 Mar           ............................N.....................V.................................................
B/Lyon/1573/2020 Feb            ..........R....................N..................V.................................................
B/Maryland/13/2020 Feb          ...............................N..................V.................................................
B/Minnesota/14/2020 Feb         ...............................N..................V.................................................
B/South_Dakota/09/2020 Feb      ...............................N..................V..........................K......................
B/Iowa/08/2020 Feb              ...............................N..................V.................................................
B/Trinidad/7321/2020 Mar        ...............................N..................V.................................................
B/Maryland/14/2020 Feb          ...............................N..................V...E.............................................
B/Florida/02/2020 Jan           ...............................N..................V.................................................
B/Connecticut/16/2020 Mar       ...............................N..................V.................................................
B/Santiago/11914/2020 Feb       ...............................N..................V.................................................
B/Oregon/06/2020 Feb            ............................N..N..................V.................................................
B/Victoria/27/2020 Mar          ...............................N..................V.................................................
B/Concepcion/6885/2020 Jan      ...............................N..................V..............I..................................
B/Lyon/1778/2020 Mar            ...............................N..................V.................................................
```

```
                                          310       320       330       340       350       360       370       380       390       400
                                  ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Phuket/3073/2013 Nov Egg        GLNKSKPYYTGKHAKAIGNCPIWVKTPLKLANGTKYRPPAKLLKERGFFGAIAGFLEGGWEGMIAGWHGYTSHGAHGVAVAADLKSTQEAINKITKNLNS
B/Novosibirsk/RII-26927S/2020     ....................................................................................................
B/S.Petersburg/RII-181/2020 Ma    ....................................................................................................
B/Jamaica/5800/2020 Mar           ....................................................................................................
B/Norway/993/2020 Feb             ....................................................................................................
B/Tunisia/1088/2020 Jan           ....................................................................................................
B/Tunisia/2254/2020 Feb           ....................................................................................................
B/Valencia/10-1629/2020 Mar       .............................................K......................................................
B/Santiago/7000/2020 Jan          ....................................................................................................
B/Lyon/0281/2020 Jan              ....................................................................................................
B/Connecticut/05/2020 Jan         ....................................................................................................
B/Mauritius/I-762/2018 Sep        ....................................................................................................
B/Connecticut/09/2020 Jan         ....................................................................................................
B/Honduras/6985/2020 Mar          ....................................................................................................
B/Bretagne/118/2020 Jan           ....................................................................................................
B/Arizona/12/2020 Mar             ....................................................................................................
B/Lyon/1573/2020 Feb              ....................................................................................................
B/Maryland/13/2020 Feb            ....................................................................................................
B/Minnesota/14/2020 Feb           ....................................................................................................
B/South_Dakota/09/2020 Feb        ....................................................................................................
B/Iowa/08/2020 Feb                ....................................................................................................
B/Trinidad/7321/2020 Mar          ....................................................................................................
B/Maryland/14/2020 Feb            ....................................................................................................
B/Florida/02/2020 Jan             ....................................................................................................
B/Connecticut/16/2020 Mar         ....................................................................................................
B/Santiago/11914/2020 Feb         ....................................................................................................
B/Oregon/06/2020 Feb              ....................................................................................................
B/Victoria/27/2020 Mar            ....................................................................................................
B/Concepcion/6885/2020 Jan        ....................................................................................................
B/Lyon/1778/2020 Mar              ....................................................................................................
```

```
                                          410       420       430       440       450       460       470       480       490       500
                                  ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Phuket/3073/2013 Nov Egg        LSELEVKNLQRLSGAMDELHNEILELDEKVDDLRADTISSQIELAVLLSNEGIINSEDEHLLALERKLKKMLGPSAVDIGNGCFETKHKCNQTCLDRIAA
B/Novosibirsk/RII-26927S/2020     ....................................................................................................
B/S.Petersburg/RII-181/2020 Ma    ....................................................................................................
B/Jamaica/5800/2020 Mar           ....................................................................................................
B/Norway/993/2020 Feb             ....................................................................................................
B/Tunisia/1088/2020 Jan           ....................................................................................................
B/Tunisia/2254/2020 Feb           ....................................................................................................
B/Valencia/10-1629/2020 Mar       ....................................................................................................
B/Santiago/7000/2020 Jan          ....................................................................................................
B/Lyon/0281/2020 Jan              ....................................................................................................
B/Connecticut/05/2020 Jan         ....................................................................................................
B/Mauritius/I-762/2018 Sep        ....................................................................................................
B/Connecticut/09/2020 Jan         ....................................................................................................
B/Honduras/6985/2020 Mar          ....................................................................................................
B/Bretagne/118/2020 Jan           ....................................................................................................
B/Arizona/12/2020 Mar             ....................................................................................................
B/Lyon/1573/2020 Feb              ....................................................................................................
B/Maryland/13/2020 Feb            ....................................................................................................
B/Minnesota/14/2020 Feb           ....................................................................................................
B/South_Dakota/09/2020 Feb        ....................................................................................................
B/Iowa/08/2020 Feb                ....................................................................................................
B/Trinidad/7321/2020 Mar          ....................................................................................................
B/Maryland/14/2020 Feb            ....................................................................................................
B/Florida/02/2020 Jan             ....................................................................................................
B/Connecticut/16/2020 Mar         ....................................................................................................
B/Santiago/11914/2020 Feb         ....................................................................................................
B/Oregon/06/2020 Feb              ....................................................................................................
B/Victoria/27/2020 Mar            ....................................................................................................
B/Concepcion/6885/2020 Jan        ....................................................................................................
B/Lyon/1778/2020 Mar              ....................................................................................................
```

```
                                          510       520       530       540       550       560
                                  ....|....|....|....|....|....|....|....|....|....|....|....|....
B/Phuket/3073/2013 Nov Egg        GTFNAGEFSLPTFDSLNITAASLNDDGLDNHTILLYYSTAASSLAVTLMLAIFIVYMVSRDNVSCSICL
B/Novosibirsk/RII-26927S/2020     .....................................................................
B/S.Petersburg/RII-181/2020 Ma    .....................................................................
B/Jamaica/5800/2020 Mar           .................................M...................................
B/Norway/993/2020 Feb             .....................................................................
B/Tunisia/1088/2020 Jan           .....................................................................
B/Tunisia/2254/2020 Feb           .....................................................................
B/Valencia/10-1629/2020 Mar       .....................................................................
B/Santiago/7000/2020 Jan          .....................................................................
B/Lyon/0281/2020 Jan              .....................................................................
B/Connecticut/05/2020 Jan         .....................................................................
B/Mauritius/I-762/2018 Sep        .....................................................................
B/Connecticut/09/2020 Jan         ...................................................L.................
B/Honduras/6985/2020 Mar          .....................................................................
B/Bretagne/118/2020 Jan           .....................................................................
B/Arizona/12/2020 Mar             ......D..............................................................
B/Lyon/1573/2020 Feb              .....................................................................
B/Maryland/13/2020 Feb            .....................................................................
B/Minnesota/14/2020 Feb           .....................................................................
B/South_Dakota/09/2020 Feb        .....................................................................
B/Iowa/08/2020 Feb                .....................................................V...............
B/Trinidad/7321/2020 Mar          .....................................................................
B/Maryland/14/2020 Feb            .....................................................................
B/Florida/02/2020 Jan             .....................................................................
B/Connecticut/16/2020 Mar         .....................................................................
B/Santiago/11914/2020 Feb         .....................................................................
B/Oregon/06/2020 Feb              .....................................................................
B/Victoria/27/2020 Mar            .....................................................................
B/Concepcion/6885/2020 Jan        .....................................................................
B/Lyon/1778/2020 Mar              .....................................................................
```

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

Figure 27. Phylogenetic comparison of influenza B (Yamagata-lineage) NA genes

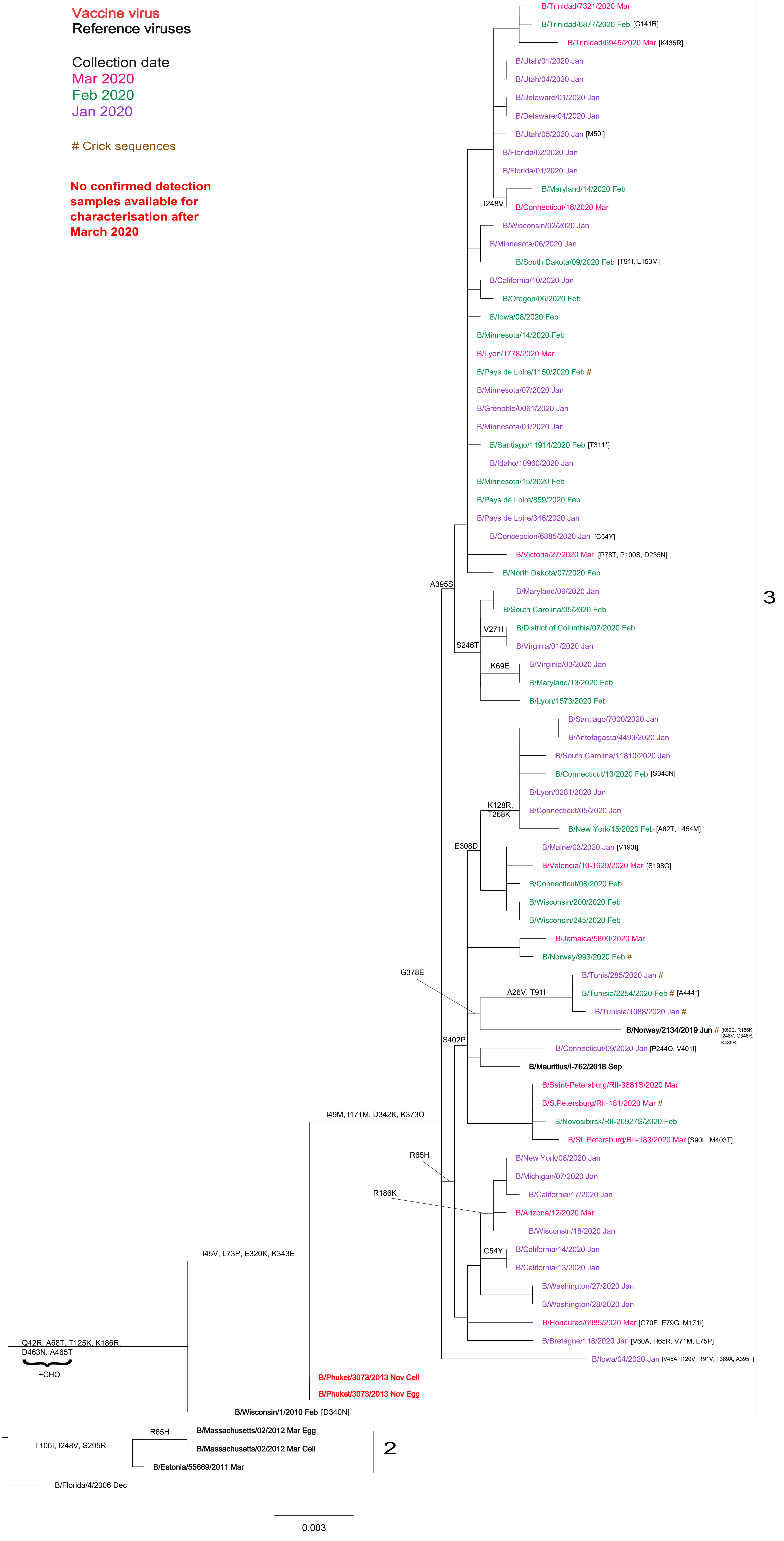


Figure 28. NA protein alignment for B (Yamagata-lineage) viruses

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

```
                                        10        20        30        40        50        60        70        80        90       100
                               ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Phuket/3073/2013 Nov Egg     MLPSTIQTLTLFLTSGGVLLSLYVSASLSYLLYSDILLKFSRTEVTAPIMPLDCANASNVQAVNRSATKGVTPLLPEPEWTYPRLSCPGSTFQKALLISP 3
B/Novosibirsk/RII-26927S/2020  ................................................M...............H................................... 3
B/S.Petersburg/RII-181/2020 Ma ................................................M...............H................................... 3
B/Jamaica/5800/2020 Mar        ................................................M...............H................................... 3
B/Norway/993/2020 Feb          ................................................M...............H................................... 3
B/Tunisia/1088/2020 Jan        .........................V......................M...............H.........................I......... 3
B/Tunisia/2254/2020 Feb        .........................V......................M...............H.........................I......... 3
B/Valencia/10-1629/2020 Mar    ................................................M...............H................................... 3
B/Santiago/7000/2020 Jan       ................................................M...............H................................... 3
B/Lyon/0281/2020 Jan           ................................................M...............H................................... 3
B/Connecticut/05/2020 Jan      ................................................M...............H................................... 3
B/Mauritius/I-762/2018 Sep     ................................................M...............H................................... 3
B/Connecticut/09/2020 Jan      ................................................M...............H................................... 3
B/Honduras/6985/2020 Mar       ................................................M...............H....E........G..................... 3
B/Bretagne/118/2020 Jan        ................................................M..........A..........M...P......................... 3
B/Arizona/12/2020 Mar          ................................................M...............H................................... 3
B/Lyon/1573/2020 Feb           ................................................M................................................... 3
B/Maryland/13/2020 Feb         ................................................M...................E............................... 3
B/Minnesota/14/2020 Feb        ................................................M................................................... 3
B/South_Dakota/09/2020 Feb     ................................................M.........................................I......... 3
B/Iowa/08/2020 Feb             ................................................M................................................... 3
B/Trinidad/7321/2020 Mar       ................................................M................................................... 3
B/Maryland/14/2020 Feb         ................................................M................................................... 3
B/Florida/02/2020 Jan          ................................................M................................................... 3
B/Connecticut/16/2020 Mar      ................................................M................................................... 3
B/Santiago/11914/2020 Feb      ................................................M................................................... 3
B/Oregon/06/2020 Feb           ................................................M................................................... 3
B/Victoria/27/2020 Mar         ................................................M............................T.....................S 3
B/Concepcion/6885/2020 Jan     ................................................M....Y.............................................. 3
B/Lyon/1778/2020 Mar           ................................................M................................................... 3

                                       110       120       130       140       150       160       170       180       190       200
                               ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Phuket/3073/2013 Nov Egg     HRFGETKGNSAPLIIREPFIACGPKECKHFALTHYAAQPGGYYNGTREDRNKLRHLISVKLGKIPTVENSIFHMAAWSGSACHDGREWTYIGVDGPDSNA
B/Novosibirsk/RII-26927S/2020  ......................................................................M.............................
B/S.Petersburg/RII-181/2020 Ma ......................................................................M.............................
B/Jamaica/5800/2020 Mar        ......................................................................M.............................
B/Norway/993/2020 Feb          ......................................................................M.............................
B/Tunisia/1088/2020 Jan        ......................................................................M.............................
B/Tunisia/2254/2020 Feb        ......................................................................M.............................
B/Valencia/10-1629/2020 Mar    ......................................................................M..........................G..
B/Santiago/7000/2020 Jan       ...........................R..........................................M.............................
B/Lyon/0281/2020 Jan           ...........................R..........................................M.............................
B/Connecticut/05/2020 Jan      ...........................R..........................................M.............................
B/Mauritius/I-762/2018 Sep     ......................................................................M.............................
B/Connecticut/09/2020 Jan      ......................................................................M.............................
B/Honduras/6985/2020 Mar       ....................................................................................................
B/Bretagne/118/2020 Jan        ......................................................................M.............................
B/Arizona/12/2020 Mar          ......................................................................M..............K..............
B/Lyon/1573/2020 Feb           ......................................................................M.............................
B/Maryland/13/2020 Feb         ......................................................................M.............................
B/Minnesota/14/2020 Feb        ......................................................................M.............................
B/South_Dakota/09/2020 Feb     ....................................................M.................M.............................
B/Iowa/08/2020 Feb             ......................................................................M.............................
B/Trinidad/7321/2020 Mar       ......................................................................M.............................
B/Maryland/14/2020 Feb         ......................................................................M.............................
B/Florida/02/2020 Jan          ......................................................................M.............................
B/Connecticut/16/2020 Mar      ......................................................................M.............................
B/Santiago/11914/2020 Feb      ......................................................................M.............................
B/Oregon/06/2020 Feb           ......................................................................M.............................
B/Victoria/27/2020 Mar         ......................................................................M.............................
B/Concepcion/6885/2020 Jan     ......................................................................M.............................
B/Lyon/1778/2020 Mar           ......................................................................M.............................

                                       210       220       230       240       250       260       270       280       290       300
                               ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Phuket/3073/2013 Nov Egg     LLKIKYGEAYTDTYHSYAKNILRTQESACNCIGGDCYLMITDGPASGISECRFLKIREGRIIKEIFPTGRVKHTEECTCGFASNKTIECACRDNSYTAKR
B/Novosibirsk/RII-26927S/2020  ....................................................................................................
B/S.Petersburg/RII-181/2020 Ma ....................................................................................................
B/Jamaica/5800/2020 Mar        ....................................................................................................
B/Norway/993/2020 Feb          ....................................................................................................
B/Tunisia/1088/2020 Jan        ....................................................................................................
B/Tunisia/2254/2020 Feb        ....................................................................................................
B/Valencia/10-1629/2020 Mar    ....................................................................................................
B/Santiago/7000/2020 Jan       ...................................................................K................................
B/Lyon/0281/2020 Jan           ...................................................................K................................
B/Connecticut/05/2020 Jan      ...................................................................K................................
B/Mauritius/I-762/2018 Sep     ....................................................................................................
B/Connecticut/09/2020 Jan      ...........................................Q........................................................
B/Honduras/6985/2020 Mar       ....................................................................................................
B/Bretagne/118/2020 Jan        ....................................................................................................
B/Arizona/12/2020 Mar          ....................................................................................................
B/Lyon/1573/2020 Feb           .............................................T......................................................
B/Maryland/13/2020 Feb         ..............X..............................T......................................................
B/Minnesota/14/2020 Feb        ....................................................................................................
B/South_Dakota/09/2020 Feb     ....................................................................................................
B/Iowa/08/2020 Feb             ....................................................................................................
B/Trinidad/7321/2020 Mar       ....................................................................................................
B/Maryland/14/2020 Feb         ...............................................V....................................................
B/Florida/02/2020 Jan          ....................................................................................................
B/Connecticut/16/2020 Mar      ...............................................V....................................................
B/Santiago/11914/2020 Feb      ....................................................................................................
B/Oregon/06/2020 Feb           ....................................................................................................
B/Victoria/27/2020 Mar         ..................................N.................................................................
B/Concepcion/6885/2020 Jan     ....................................................................................................
B/Lyon/1778/2020 Mar           ....................................................................................................
```

```
                                        310       320       330       340       350       360       370       380       390       400
                                ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Phuket/3073/2013 Nov Egg      PFVKLNVETDTAEIRLMCTKTYLDTPRPNDGSITGPCESDGDEGSGGIKGGFVHQRMASKIGRWYSRTMSKTKRMGMGLYVKYDGDPWTDSEALALSGVM
B/Novosibirsk/RII-26927S/2020   .........................................K..............................Q...........................
B/S.Petersburg/RII-181/2020 Ma  .........................................K..............................Q...........................
B/Jamaica/5800/2020 Mar         .........................................K..............................Q...........................
B/Norway/993/2020 Feb           .........................................K..............................Q...........................
B/Tunisia/1088/2020 Jan         .........................................K..............................Q....E......................
B/Tunisia/2254/2020 Feb         .........................................K..............................Q....E......................
B/Valencia/10-1629/2020 Mar     .......D.................................K..............................Q...........................
B/Santiago/7000/2020 Jan        .......D.................................K..............................Q...........................
B/Lyon/0281/2020 Jan            .......D.................................K..............................Q...........................
B/Connecticut/05/2020 Jan       .......D.................................K..............................Q...........................
B/Mauritius/I-762/2018 Sep      .........................................K..............................Q...........................
B/Connecticut/09/2020 Jan       .........................................K..............................Q...........................
B/Honduras/6985/2020 Mar        .........................................K..............................Q...........................
B/Bretagne/118/2020 Jan         .........................................K..............................Q...........................
B/Arizona/12/2020 Mar           .........................................K..............................Q...........................
B/Lyon/1573/2020 Feb            .........................................K..............................Q.....................S.....
B/Maryland/13/2020 Feb          .........................................K..............................Q.....................S.....
B/Minnesota/14/2020 Feb         .........................................K..............................Q.....................S.....
B/South_Dakota/09/2020 Feb      .........................................K..............................Q.....................S.....
B/Iowa/08/2020 Feb              .........................................K..............................Q.....................S.....
B/Trinidad/7321/2020 Mar        .........................................K..............................Q.....................S.....
B/Maryland/14/2020 Feb          .........................................K..............................Q.....................S.....
B/Florida/02/2020 Jan           .........................................K..............................Q.....................S.....
B/Connecticut/16/2020 Mar       .........................................K..............................Q.....................S.....
B/Santiago/11914/2020 Feb       ..........X..............................K..............................Q.....................S.....
B/Oregon/06/2020 Feb            .........................................K..............................Q.....................S.....
B/Victoria/27/2020 Mar          .........................................K..............................Q.....................S.....
B/Concepcion/6885/2020 Jan      .........................................K..............................Q.....................S.....
B/Lyon/1778/2020 Mar            .........................................K..............................Q.....................S.....

                                        410       420       430       440       450       460
                                ....|....|....|....|....|....|....|....|....|....|....|....|.
B/Phuket/3073/2013 Nov Egg      VSMEEPGWYSFGFEIKDKKCDVPCIGIEMVHDGGKTTWHSAATAIYCLMGSGQLLWDTVTGVNMTL
B/Novosibirsk/RII-26927S/2020   .P................................................................
B/S.Petersburg/RII-181/2020 Ma  .P................................................................
B/Jamaica/5800/2020 Mar         .P................................................................
B/Norway/993/2020 Feb           .P................................................................
B/Tunisia/1088/2020 Jan         .P................................................................
B/Tunisia/2254/2020 Feb         .P................................................................
B/Valencia/10-1629/2020 Mar     .P................................................................
B/Santiago/7000/2020 Jan        .P................................................................
B/Lyon/0281/2020 Jan            .P................................................................
B/Connecticut/05/2020 Jan       .P................................................................
B/Mauritius/I-762/2018 Sep      .P................................................................
B/Connecticut/09/2020 Jan       IP................................................................
B/Honduras/6985/2020 Mar        ..................................................................
B/Bretagne/118/2020 Jan         ..................................................................
B/Arizona/12/2020 Mar           ..................................................................
B/Lyon/1573/2020 Feb            ..................................................................
B/Maryland/13/2020 Feb          ..................................................................
B/Minnesota/14/2020 Feb         ..................................................................
B/South_Dakota/09/2020 Feb      ..................................................................
B/Iowa/08/2020 Feb              ..................................................................
B/Trinidad/7321/2020 Mar        ..................................................................
B/Maryland/14/2020 Feb          ..................................................................
B/Florida/02/2020 Jan           ...................................X..............................
B/Connecticut/16/2020 Mar       ..................................................................
B/Santiago/11914/2020 Feb       ..................................................................
B/Oregon/06/2020 Feb            ..................................................................
B/Victoria/27/2020 Mar          ..................................................................
B/Concepcion/6885/2020 Jan      ..................................................................
B/Lyon/1778/2020 Mar            ..................................................................
```

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

Table 13. Antiviral testing of influenza viruses with collection dates after 2021-08-31*

A. Neuraminidase inhibitors (phenotypic assay)

Month of Collection	Number of viruses tested for phenotype[1]								
	A(H1N1)pdm09		A(H3N2)		Influenza B-Victoria		Influenza B-Yamagata		
	NI	RI/HRI	NI	RI/HRI	NI	RI/HRI	NI	RI/HRI	Totals
2021									
September	14		58	1	20				93
October	21		37		7				65
November	13		86		15				114
December	11		60		4				75
2021									
January	7		30						37
Totals	66	0	271	1	46	0	0	0	384

[1] Phenotype: NI (normal inhibition), RI (reduced inhibition), HRI (highly reduced inhibition) as defined in WHO Wkly. Epidemiol. Rec. (2012) 87, 369-374

Virus name	Subtype/ Lineage	Collection date	Fold difference		NA substitution
			Oseltamivir	Zanamivir	
A/Tripoli/5646/2021	A(H3N2)	2021-09-07	NI	11.89 (RI)	A246V (NXT)

B. Baloxavir marboxil (PA genetic screening)

Virus	Clinical specimens	Virus isolates	CS/Virus pairs	Totals
A(H1N1)pdm09	6	40	1	47
A(H3N2)	104	142	0	246
B/Victoria	9	28	0	37
B/Yamagata	0	0	0	0
Total	119	210	1	330

None showed markers of reduced susceptibility to baloxavir marboxil

* As of 2022-02-15